IEEE Access
Multidisciplinary ⁝ Rapid Review ⁝ Open Access Journal

Received 10 June 2024, accepted 11 July 2024, date of publication 15 July 2024, date of current version 26 July 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3428553

RESEARCH ARTICLE

Improving Plant Disease Classification With Deep-Learning-Based Prediction Model Using Explainable Artificial Intelligence

NATASHA NIGAR [1], HAFIZ MUHAMMAD FAISAL [2], MUHAMMAD UMER [3], OLUKAYODE OKI [4], AND JOSE MANAPPATTUKUNNEL LUKOSE [4]

[1]Department of Computer Science (RCET), University of Engineering and Technology, Lahore 54000, Pakistan
[2]School of Engineering and Design, Technical University of Munich, 80333 Munich, Germany
[3]TEKHQS INC., Irvine, CA 92620, USA
[4]Department of Information Technology, Walter Sisulu University, Mthatha 5117, South Africa

Corresponding author: Natasha Nigar (natasha@uet.edu.pk)

ABSTRACT Plant diseases can have profound effects on the economy, impacting both local and global scales. These diseases can lead to substantial losses in agricultural productivity, affecting crop yields and quality. In this context, deep learning algorithms are widely acknowledged as effective solutions. However, the use of these black-box approaches raises concerns about trust in interpreting and validating the decisions generated by the models. This study proposes an explainable artificial intelligence (XAI) based plant disease classification system to classify and identify distinct ailments with improved accuracy. The system correctly identifies 38 different plant diseases with accuracy, precision, and recall as 99.69%, 98.27%, and 98.26%, respectively. These predictions are subjected to additional analysis employing the local interpretable model-agnostic explanations (LIME) framework to produce visual explanations aligning with prior beliefs and adhering to established best practices in explanations. This system will serve as a promising avenue for revolutionizing disease detection, fostering informed decision-making, and ultimately contributing to global food security.

INDEX TERMS Plant disease detection, deep learning, explainable artificial intelligence, prediction model.

I. INTRODUCTION

With the increasing global population and changing climatic conditions, the challenges faced by farmers have been intensified. One of the critical challenges worldwide is the identification and management of plant diseases [1]. These diseases, if undetected, can lead to substantial reductions in crop yield and quality, posing severe economic losses and threatening food security on a larger scale [2]. The traditional methods of disease diagnosis often involve manual inspection by experts, which is both time-consuming and subjective [1]. Moreover, by the time symptoms are visible to the naked eye, the disease might have already spread extensively, making mitigation efforts less effective.

The associate editor coordinating the review of this manuscript and approving it for publication was Gustavo Olague.

In recent years, agriculture has witnessed a significant transformation driven by technological advancements [3], offering solutions that can revolutionize traditional farming practices. Furthermore, as the global population continues to rise, leveraging technology in agriculture is not just an innovative approach but a necessary one to ensure a sustainable and secure food supply for the future. Image recognition using machine learning (ML), has shown promise in detecting plant diseases at early stages, providing an opportunity to curb their spread and minimize damage [4]. This early assessment also depends on the plant species and the diseases. It varies from disease to disease, some disease takes around 1 week whereas some takes 2 or more weeks.

Numerous computer-aided diagnosis (CAD) systems [5] have been created to aid farmers. These systems effectively tackle prevailing challenges and improve the accuracy, efficiency, and objectivity of the diagnostic process. In this regard, deep learning (DL) algorithms have emerged as particularly promising, demonstrating significant potential for image processing and data analysis. The integration of deep learning techniques, particularly convolutional neural networks (CNNs), holds promise for accurate and efficient disease identification in plants [5].

Nevertheless, the ambiguity surrounding the processing cycle involved in model learning and feature encoding in these CAD systems raises doubts about their reliability [6]. The DL model without a rational explanation is a barrier in accurate decision making. Hence, these models are of black-box nature which makes them not to use with 100% confidence. Therefore, there is a need to develop robust methods that can help better understand the decisions made by the black box. These methods are often known as interpretable deep learning or XAI [6]. The incorporation of XAI ensures transparency and interpretability in the decision-making process, addressing the crucial need for trust and understanding in agricultural practices.

In this study, the state-of-the-art model, namely, EfficientNetB0 [7] is applied for classifying 38 plant diseases. The explanation method employed is LIME, incorporating enhanced explanations to improve both interpretability and accuracy. The EfficientNetB0 with 237 layers is utilized for feature extraction and model training. The model loads pre-trained weights from a local file to enhance efficiency and accuracy. At first, EfficientNetB0 predicts the disease or no disease; after that LIME is applied to generate explanations for the predictions of the classifier. LIME typically provides feature importance scores, highlighting the regions or aspects of the input that influenced the model's decision. The significance of this study lies in its potential to empower farmers with a robust tool that not only identifies plant diseases but also provides insights into the underlying factors contributing to the classification decisions. The combination of deep learning and explainability enhances the system's reliability and enables farmers to make informed decisions regarding disease management strategies. In summary, we introduce a robust model that exhibits improved accuracy, achieved through the integration of XAI techniques for plant disease detection. The main contributions of this study include:

i) **New System Design:** A novel system is designed using DL and XAI for accurate and quick diagnosis of 38 distinct plant diseases with improved accuracy. The system also provides farmers with actionable insights including treatment suggestions
ii) **Comparative Study of State-of-the-art Algorithms:** Various cutting-edge ML algorithms are compared for the diagnosis of plant diseases.
iii) **Mobile Application Development:** A user-friendly mobile application named as 'PlantCare' is also developed for farmers assistance.
iv) **Data Set:** The proposed approach is tested on both real-world and benchmark data instances from Kaggle.

The paper is organised into 6 section. Section II overviews the background and related works. The proposed approach is in Section III followed by experimental design and results in section IV. Section V presents the limitations of this work. This study concludes in section VI with future directions.

II. LITERATURE REVIEW

A. BACKGROUND

Machine learning for plant disease detection has emerged as a transformative technology in agriculture, offering efficient and accurate solutions to identify and combat crop diseases [4]. By leveraging advanced algorithms, ML models can analyze datasets to detect subtle signs of diseases that may go unnoticed by the human eye [8]. In this context, we assess the effectiveness of cutting-edge algorithms for diagnosing plant diseases. The rationale behind choosing these algorithms is their simplicity in implementation, coupled with the fact that a majority of them are open source.

The convolutional neural networks (CNNs) [9] are a class of deep neural networks designed specifically for image recognition and processing tasks. Inspired by the visual processing in the human brain, CNNs excel in capturing hierarchical features and patterns within images. A basic CNN model consists of an input layer, multiple convolutional layers followed by pooling layers, and fully connected (dense) layers leading to the output layer. The convolutional layers apply filters to the input to extract features, while the pooling layers downsample the spatial dimensions, leading to more efficient processing. The dense layers then perform classification based on the extracted features [10]. CNNs have proven highly effective in computer vision applications, such as image classification, object detection, and facial recognition.

Residual network (ResNet) [11] is one of the most powerful deep neural networks, demonstrating outstanding performance in addressing classification problems. Built upon CNN architecture, ResNet is designed to accommodate hundreds or even thousands of convolutional layers.

EfficientNetB0 [7] is a state-of-the-art CNN architecture designed to achieve impressive performance with high efficiency in terms of both computational resources and model size. This architecture is particularly notable for its ability to outperform larger and computationally expensive models on various computer vision tasks, including image classification [7].

MobileNet [12] is a lightweight CNN architecture designed specifically for efficient and high-performance image classification on mobile and embedded devices. MobileNet focuses on achieving a balance between accuracy and computational efficiency, making it well-suited for applications with limited computational resources [12].

LIME [13] is a method used in the field of ML for explaining the predictions of complex models in a human-understandable way. The LIME is designed to provide insights into the decision-making process of black-box machine learning models by approximating their behavior locally. It focuses on local interpretability, aiming to explain individual predictions rather than the entire model [13]. This makes it particularly useful for understanding specific instances where model predictions may be unclear. The explanations are generated by perturbing input data and observing the model's response.

B. RELATED WORKS

Plant disease detection plays a crucial role in agriculture productivity with significant impact on the economy [28]. Timely identification and management of plant diseases helps in preventing crop losses, ensuring a stable food supply and supporting the livelihoods of farmers [22]. This section presents an overview of recent ML model-based approaches for plant disease detection.

Ferentinos et al. [4] presented the development of CNN models with variants (AlexNet, AlexNetOWTBn, GoogleNet, Overfeat, VGG) for plant disease detection and diagnosis using images of simple leaves from healthy and diseased plants. The models were trained on an extensive database of 87,848 images, encompassing 25 plant species in 58 unique [plant, disease] combinations. The best-performing model achieved an impressive 99.53% success rate in accurately identifying the [plant, disease] combination or healthy plants.

Mehedi et al. [14] developed a transfer learning approach with three pre-trained models (EfficientNetV2L, MobileNetV2, ResNet152V2). The study detected 38 leaf diseases across 14 different plants. The dataset was taken from Kaggle [15]. EfficientNetV2L demonstrated the highest accuracy at 99.63%. The integration of XAI through LIME enhances model interpretability.

Mohanty et al. [16] utilized a public dataset [17] containing 54,306 images of diseased and healthy plant leaves, deep CNNs (AlexNet and GoogLeNet) were trained to identify 14 crop species and 26 diseases with an impressive accuracy of 99.35% on a held-out test set.

Jasim et al. [18] explored the application of DL models in the early detection and classification of plant diseases, highlighting the potential for increased accuracy compared to traditional ML approaches. The Plant Village [17] dataset was used with 20,636 images as three plants, namely, tomato, pepper, and potato crops were chosen because of the most famous types of plants. The CNN classifier achieved 98.029% accuracy, with the potential for further improvement with a larger training dataset.

Ramesh et al. [19] employed Random Forest for classifying healthy and diseased leaves based on leaf images. The proposed methodology involved dataset creation, feature extraction using Histogram of an Oriented Gradient (HOG), classifier training, and classification. In testing with 160 papaya leaf images, the Random Forest classifier achieved approximately 70% accuracy.

Harakannanavar et al. [20] focused on addressing plant diseases in tomato crops. Employing machine learning and image processing, the study introducesd a robust algorithm for early detection of leaf diseases. The proposed model employs Support Vector Machine (SVM) [29], K-Nearest Neighbor (K-NN) [30], and CNN for classification, achieving high accuracy rates of 88%, 97%, and 99.6% respectively. Benito Fernández et al. [21] used CNN model leveraging XAI methods, specifically LIME [13], SHAP [31], and Grad-CAM [32], to enhance the interpretability of CNN outputs.

Kinger et al. [22] addressed the challenge of making deep learning models for plant disease detection more interpretable for human users. The VGG16 was employed with achieved accuracy rate of 98.15%. They used XAI method Grad-CAM, to provide human-understandable explanations for the model's decisions.

Nahiduzzaman et al. [23] introduced an XAI-based CNN model for classifying mulberry leaf diseases. They utilized a novel lightweight CNN model and achieved impressive accuracy of 95.05 $\pm$ 2.86% for three-class classifications and 96.06 $\pm$ 3.01% for binary classifications. The model outperformed well-known deep transfer learning models, offering better accuracy, fewer parameters, layers, and overall size. The interpretability of the model was ensured through SHAP explanations.

Arsenovic et al. [1] addressed the significant agricultural challenge of plant diseases by leveraging DL methods for accurate detection. They introduced a novel dataset comprising 79,265 diverse leaf images, employing both traditional augmentation and state-of-the-art generative adversarial networks for dataset expansion. Experimental evaluations demonstrate the model's effectiveness in identifying plant diseases under various conditions, achieving an accuracy of 93.67%.

Khattak et al. [24], addressed the crucial issue of less citrus fruit yield due to diseases, by proposing an integrated approach using CNN. The CNN model is designed to differentiate healthy citrus fruits and leaves from those affected with common diseases and achieved a remarkable test accuracy of 94.55%. The proposed model outperformed other classifiers, achieving a high accuracy of 95.65% in classifying citrus fruit/leaf diseases, establishing its potential as a valuable decision support tool for farmers.

Singh et al. [26] addressed the significant issue of plant disease-related crop yield loss in India by proposing a novel solution using computer vision. The authors introduced a comprehensive dataset comprising 2,598 data points across 13 plant species and up to 17 disease classes, annotated with approximately 300 human hours of effort. This study demonstrates the efficacy of the dataset by training three models for plant disease classification, revealing a notable increase of up to 31% in classification accuracy compared to existing datasets.

Besides these, some studies [2], [3], [5], [28], [33] review the application of DL models in visualizing and detecting plant diseases. Various DL architectures and visualization

TABLE 1. Research matrix of related works.

Study	Dataset(s)	Methodology	Results	XAI Method	Mobile App	Classes	Species	Instances
Ferentinos et al. [4]	Open Database	CNN and Variants	accuracy: 99.53%	No	No	58	25	87,848
Mehedi et al. [14]	Kaggle [15]	EfficientNetV2L, MobileNetV2, ResNet152V2	accuracy: 99.63%	Lime	No	38	14	54,305
Mohanty et al. [16]	PlantVillage [17]	AlexNet and GoogLeNet	accuracy: 99.35%	No	No	26	14	54,306
Jasim et al. [18]	PlantVillage [17]	CNN	accuracy: 98.029%	No	Yes	15	3	20,636
Ramesh et al. [19]	Random	Random Forest	accuracy: 70%	No	No	2	1	170
Harakannanavar et al. [20]	PlantVillage [17]	SVN, K-NN, CNN	accuracy: 99.6%	No	No	2	1	-
Fernandez et al. [21]	PlantVillage [17]	CNN	Lime is better	LIME, SHAP, Grad-CAM	No	15	-	20,639
Kinger et al. [22]	PlantVillage [17]	VGG16	98.15%	Grad-CAM	No	38	14	>70,000
Nahiduzzaman et al. [23]	Real	Proposed CNN	accuracy: 95.05 ± 2.86%	SHAP	No	3	1	6,327
Arsenovic et al. [1]	PlantVillage [17]	Proposed: PlantDiseaseNet	accuracy: 93.67%	No	No	42	12	79,265
Khattak et al. [24]	Citrus [26], PlantVillage [17]	Proposed CNN	accuracy: 94.55 %	No	No	5	1	2,293
Singh et al. [26]	Google Images + Ecosia [27]	MobileNet, Faster-RCNN	accuracy: increase by 31%	No	No	17	13	2,598

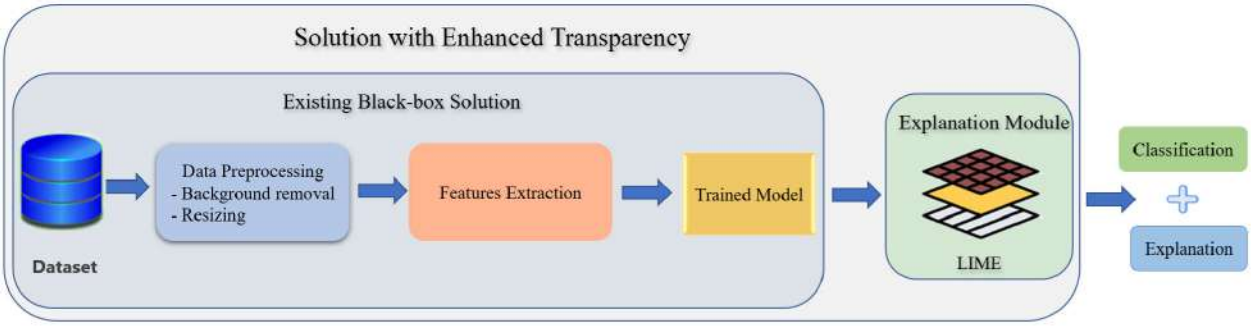


FIGURE 1. The proposed methodology workflow.

techniques were discussed; however, the review identifies research gaps, including the limited diversity in datasets used for evaluation, emphasizing the need for more realistic environmental considerations.

Table 1 provides a summary of these studies. It is evident that a significant number of researchers have not integrated XAI methods. This observation motivates our research, which seeks to develop a robust XAI-based model, with the goal of improving transparency, traceability, and the overall efficacy of plant disease classification in AI models.

III. PROPOSED METHODOLOGY

This section describes the proposed methodology in detail as shown in Fig. 1.

A. DATA PRE-PROCESSING

In this step, background is removed from the images to separate the region of interest (ROI). The background can introduce noise into the data, leading to potential misdiagnoses. For instance, a leaf's shadow or the surrounding environment might be mistaken for a disease symptom, skewing the results. The images are also separated into distinct classes based on different crops to enhance model training efficiency. This step overall enhances the optimal accuracy and reliability of the trained classifier in disease detection, while decreasing the corresponding computation time. Some results before and after pre-processing are shown in Fig. 2 and 3 respectively.

FIGURE 2. Randomly selected plant images before preprocessing.

B. FEATURES EXTRACTION

This step involves selecting a subset of the most important features from the existing data for developing a ML model. It is a crucial step in the ML workflow, given that the performance of a model can be significantly influenced by the quality and quantity of features. The default feature extraction technique is transfer learning with the EfficientNetB0 model, where the base model's layers are frozen to prevent training. The model is then extended with global average pooling and a dense layer for multi class classification. It is employed to generate multiple features, enhancing the capabilities of disease detection.

C. MODEL TRAINING AND VALIDATION

To ensure the robustness of our model, the dataset is split into an 80:20 ratio, with 80% of the images being used for training and the remaining 20% for validation.

TABLE 2. Dataset description.

Plant	Disease	No of Samples	Plant	Disease	No of Samples
Apple	Apple Scab	2520	Potato	Early Blight	2424
Apple	Black Rot	2484	Potato	Late Blight	2424
Apple	Cedar Apple Rust	2200	Potato	Healthy	2280
Apple	Healthy	2510	Raspberry	Healthy	2226
Blueberry	Healthy	2270	Soybean	Healthy	2527
Cherry	Powdery Mildew	2104	Squash	Powdery Mildew	2170
Cherry	Healthy	2282	Strawberry	Leaf Scorch	2218
Corn (Maize)	Cercospora Leaf Spot / Gray Leaf Spot	2052	Strawberry	Healthy	2280
Corn (Maize)	Common Rust	2384	Tomato	Bacterial Spot	2127
Corn (Maize)	Northern Leaf Blight	2385	Tomato	Early Blight	2400
Corn (Maize)	Healthy	2324	Tomato	Late Blight	2314
Grape	Black Rot	2360	Tomato	Leaf Mold	2352
Grape	Esca (Black Measles)	2400	Tomato	Septoria Leaf Spot	2181
Grape	Leaf Blight (Isariopsis Leaf Spot)	2152	Tomato	Spider Mites / Two-Spotted Spider Mite	2176
Grape	Healthy	2115	Tomato	Target Spot	2284
Orange	Haunglongbing (Citrus Greening)	2513	Tomato	Yellow Leaf Curl Virus	2451
Peach	Bacterial Spot	2297	Tomato	Mosaic Virus	2238
Peach	Healthy	2160	Tomato	Healthy	2407
Pepper, Bell	Bacterial Spot	2391			
Pepper, Bell	Healthy	2485			

FIGURE 3. Randomly selected plant images after preprocessing.

D. PREDICTION EXPLANIBILITY

During this phase, the LIME framework is utilized. LIME is a method employed to explain individual predictions by using a local interpretable model to approximate any black box ML-based model.The process involves perturbing the original data points, inputting them into a black box model, and observing the resulting outcomes. Following this, the method allocates weights to the supplementary data points depending on their proximity to the initial location. Subsequently, these sample weights are employed to train a surrogate model on the dataset, such as linear regression. The resultant explanation model, once trained, can then be applied to each of the original data points.

IV. EXPERIMENTAL ANALYSIS

A. EXPERIMENTAL SETUP

1) DATASET

In this paper, the dataset named as 'New Plant Diseases' is taken from Kaggle as described in Table 2. This dataset is a rich collection of 87,000 images, providing a diverse range of plant diseases 38 distinct classes across 14 plant species. Determining the economic relevance of diseases per plant depends on various factors such as the crop's economic importance, the severity and prevalence of the disease, the cost of control measures, and the potential yield losses. However, some insights are provided for diseases which are most economically relevant, per plant as shown in 'red' color in Table 2 and healthy instances are shown in 'green' color. Moreover, this dataset is imbalanced. It means that number of images in each class are not equal.

2) PERFORMANCE MEASURES

This section presents the quantitative metrics employed to assess classifier performance. The classifier is attempting to identify what kind of disease it is given plant species. It can also identify the leaf as healthy leaf which is the case in Blueberry, Raspberry, and Soyabean. In classification problems where results are categorized into positive or negative classes, the evaluation involves four potential states, often referred to as the confusion matrix [34].

i) True positive (TP): Correctly identifying instances of the positive class
ii) True negative (TN): Correctly identifying instances of the negative class
iii) False positive (FP): Incorrectly classifying instances as belonging to the positive class
iv) False negative (FN): Incorrectly classifying instances as belonging to the negative class

The performance of the results is assessed through accuracy, precision, and recall, computed as follows:

$$Accuracy = \frac{TP + TN}{FP + TN + TP + FN} \tag{1}$$

TABLE 3. Hyperparameters for classifiers.

Parameters	Values
Input Size	224*224
Batch Size	32
Error Function	Categorical Cross Entropy
Activation	ReLU
Optimizer	ADAM
Learning Rate	0.0001
Dropout	0.5
Train/Validation	80% / 20%

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall/Sensivity = \frac{TP}{TP + FN} \tag{3}$$

B. PARAMETER SETTINGS

The hyperparameter settings employed for the classification of plant diseases are outlined in Table 3.

1) EXPERIMENTAL ENVIRONMENT

The experiments are performed on GPU-enabled TensorFlow [35] with TF Lite [36] for converting the trained TensorFlow model into a format suitable for mobile applications; using Python programming language running on a personal computer with Apple M1 Chip, and 8 GB RAM.

C. RESULTS AND DISCUSSION

1) COMPARISON OF STATE-OF-THE-ART METHODS

The proposed methodology uses four distinct state-of-the-art models namely, CNN, MobileNetV2, EfficientNetB0, and ResNet-50 to detect the plant diseases. The results presented in Table 4 show that EfficientNetB0 outperforms four other models in terms of accuracy, precision, and recall. After EfficientNetB0, MobileNetV2 is performing well with the highest classification performance as 96.89%. ResNet-50 is performing least as compared to its peers with accuracy rate of 79.83%. These results demonstrate the effectiveness of the EfficientNetB0 model, achieving the highest accuracy among other models. The confusion matrix is shown in Fig. 4. The graphical representation of model accuracy is presented in Fig. 5. It is clear that there is variation in training and validation accuracy's values as the epochs count increases.

We have utilized XAI techniques, specifically leveraging the Lime framework, to enhance the interpretability of our machine learning models. This framework enabled us to explain the predictions of our model, providing valuable insights into how the model arrived at its decisions as shown in Fig. 6, which indicates LIME explanations for predicted black-box model. These are Pepper,Bell with Bacterial Spot disease images with LIME explanations. Moreover, 100% confidence means model predicted the disease with high probability. A user-friendly mobile application is also developed to assist the farmers as shown in Fig. 7.

TABLE 4. Performance metrics of CNN models.

Model	Accuracy (%)	Precision (%)	Recall (%)
CNN	96.44	94.22	94.22
MobileNetV2	96.89	96.12	96.14
EfficientNetB0	**99.69**	**98.27**	**98.26**
ResNet-50	79.83	75.21	75.21

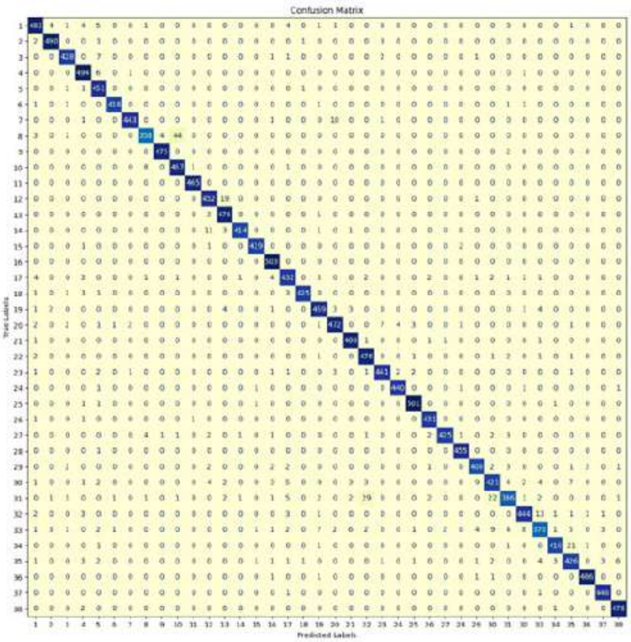


FIGURE 4. Confusion matrix of proposed model.

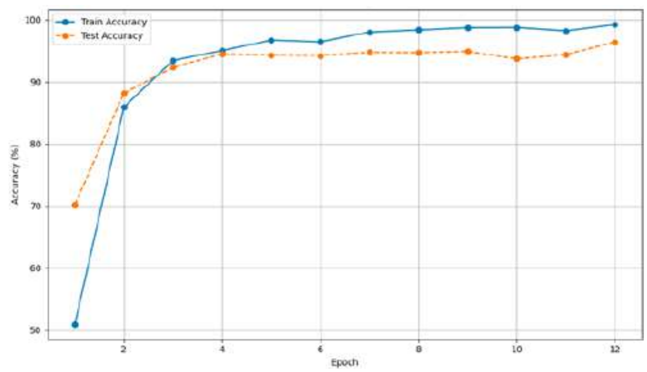
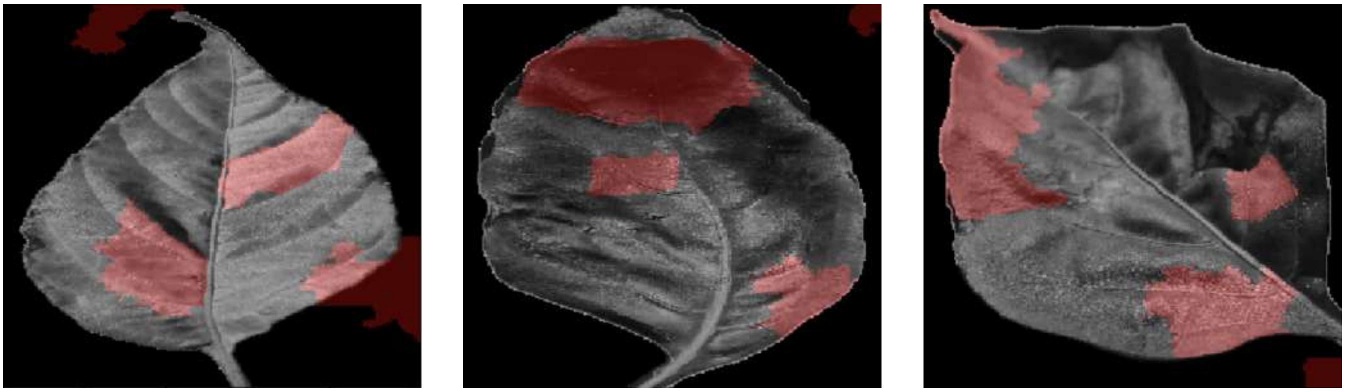


FIGURE 5. Accuracy Curve for plant disease detection.

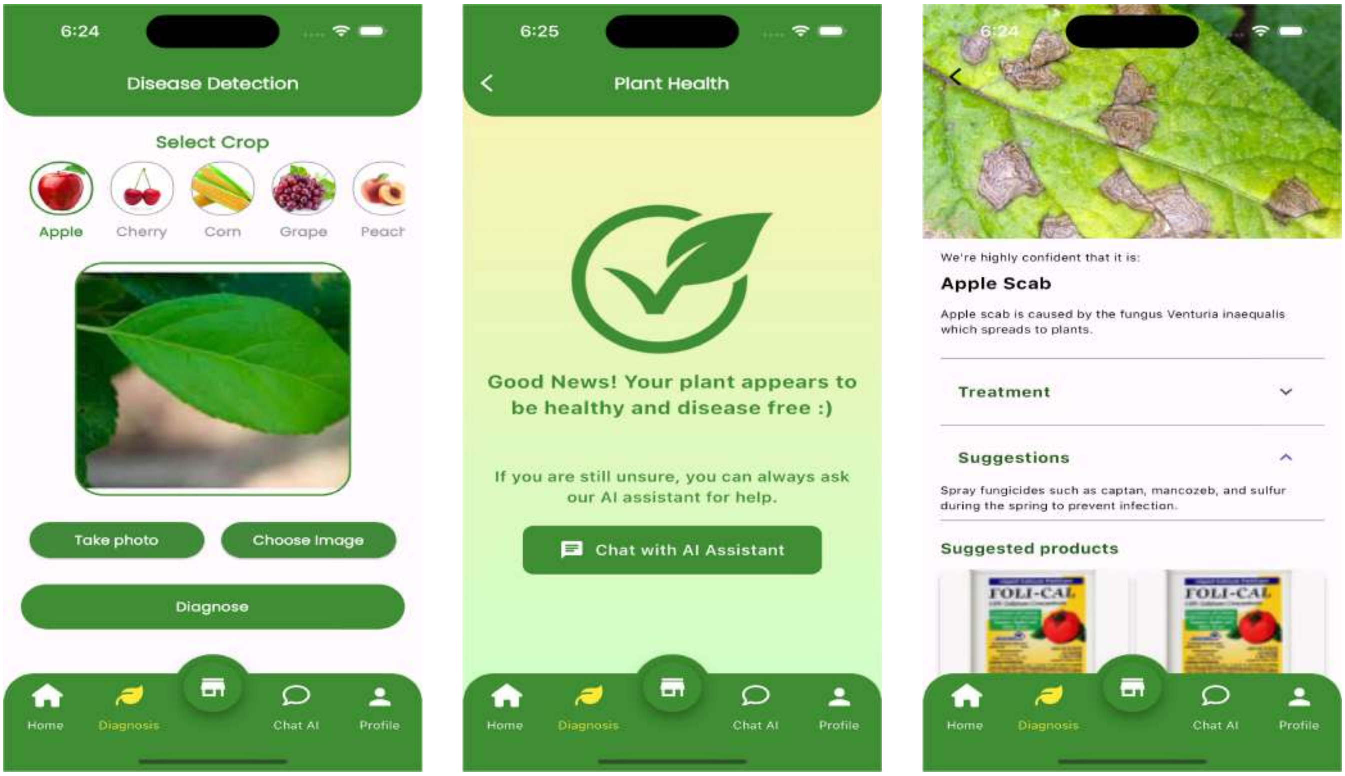
2) STATISTICAL ANALYSIS

In this section, an analysis of variance (ANOVA) test [37] is conducted to evaluate the statistical significance of the machine learning models. ANOVA serves as a statistical method employed to analyse variations among group means within a sample. It assesses whether the variations between the means are statistically significant or if they could occur by chance. ANOVA is particularly useful when comparing the means of three or more groups, providing insights into whether there are significant differences among them. The



(a) (b) (c)

FIGURE 6. Infected images detection by our developed model with 100% confidence.



(a) Main Interface (b) Plant Health Diagnosis (c) Suggestions and Recommendations

FIGURE 7. Plant disease detection - mobile application.

TABLE 5. ANOVA test results for model accuracy.

Statistical Measure	Value
F-statistic	20.8237
p-value	0.0002

F-statistic and p-value obtained from the ANOVA test are presented in Table 5.

Since the p-value is less than the chosen significance level ($\alpha = 0.05$), therefore, we reject the null hypothesis. These results suggest that the variations in disease detection accuracy across different models are statistically significant based on our model's performance.

3) VISUALIZATION

We also performed an analysis using Power BI to gain insights from our dataset, presenting them in various types of charts and graphs. In Fig. 8, number of samples per plant are shown, whereas Fig. 9 represents accuracy achieved by each plant. These values may also depends on the quality of the

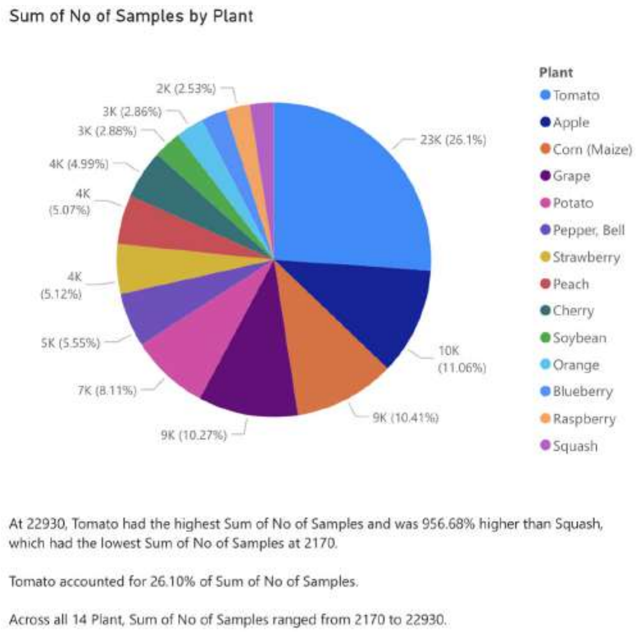


FIGURE 8. Number of samples per plant.

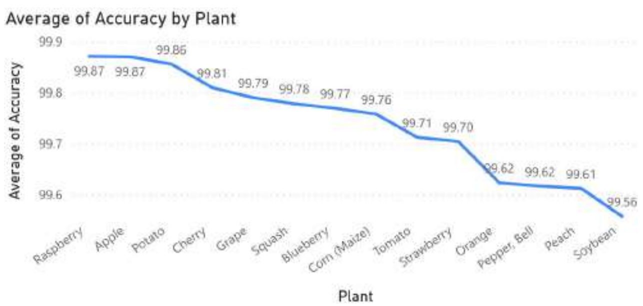


FIGURE 9. Achieved average accuracy for each plant.

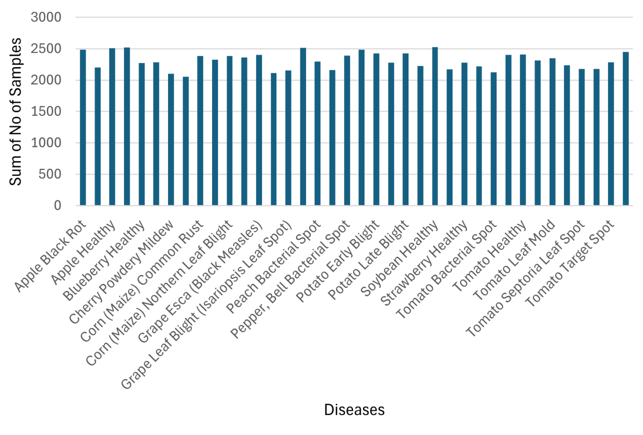


FIGURE 10. Number of samples by disease.

images in data. The number of samples by disease are shown in Fig. 10.

V. THREATS TO VALIDITY

The primary objective of this study is to assist farmers in the early assessment of plant diseases. Nonetheless, this study has certain constraints. Initially, we did not account for extensive or diverse datasets. Secondly, our approach relied on four pre-trained models for comparsion. Extending the model to integrate more advanced pre-trained models may enhance classification performance. Lastly, employing a larger set of training data could potentially yield superior results.

VI. CONCLUSION

Plant diseases pose a significant threat to our economy, causing substantial losses in agricultural productivity and impacting the livelihoods of farmers. Addressing plant diseases is not just a matter of agricultural concern but also a strategic move for economic prosperity. In this regard, this research presents the effectiveness of a deep learning-based plant disease detection system employing explainable artificial intelligence (XAI). The integration of advanced deep learning models not only enhances the accuracy of disease identification but also provides interpretability through the incorporation of XAI techniques.

This paper employs EfficientNetB0 to develop the ML model, utilizing a dataset consisting of 87,000 images. The developed model demonstrates proficiency in accurately categorizing 38 distinct types of diseases, achieving accuracy, precision, and recall rates of 99.69%, 98.27%, and 98.26%, respectively. Further, the LIME framework is employed to provide meaningful explanations that supports informed decision-making. The visual explanations not only showcase the model's effective generalization but also reveal biases acquired from outlier images. These insights empower researchers and field experts to gain a deeper understanding of the rationale behind the classification of plant diseases, shedding light on the inner workings of the black-box model. Additionally statistical analyses, notably ANOVA, showcases significant models' performance.

In future, there is potential for the creation of a more resilient model that takes into account diseases affecting various plant species. Moreover, the compilation of written reports documenting disease observations in both technical and non-technical languages stands as an additional task that contributes to the model's adoption. Furthermore, the integration of Internet of Things (IoT) devices will contribute to the fully automated disease detection systems on farms.

ACKNOWLEDGMENT

The authors acknowledge the effort of Mahnoor Khalid, Faizan, and Mehroz for developing a mobile application-based system to demonstrate the potential and scope of this research.

REFERENCES

[1] M. Arsenovic, M. Karanovic, S. Sladojevic, A. Anderla, and D. Stefanovic, "Solving current limitations of deep learning based approaches for plant disease detection," *Symmetry*, vol. 11, no. 7, p. 939, Jul. 2019.

[2] M. Nagaraju and P. Chawla, "Systematic review of deep learning techniques in plant disease detection," *Int. J. Syst. Assurance Eng. Manag.*, vol. 11, pp. 547–560, May 2020.

[3] L. Li, S. Zhang, and B. Wang, "Plant disease detection and classification by deep learning—A review," *IEEE Access*, vol. 9, pp. 56683–56698, 2021.
[4] K. P. Ferentinos, "Deep learning models for plant disease detection and diagnosis," *Comput. Electron. Agricult.*, vol. 145, pp. 311–318, Feb. 2018.
[5] M. H. Saleem, J. Potgieter, and K. M. Arif, "Plant disease detection and classification by deep learning," *Plants*, vol. 8, no. 11, p. 468, 2019.
[6] N. Nigar, M. Umar, M. K. Shahzad, S. Islam, and D. Abalo, "A deep learning approach based on explainable artificial intelligence for skin lesion classification," *IEEE Access*, vol. 10, pp. 113715–113725, 2022.
[7] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.
[8] N. Nigar, A. Jaleel, S. Islam, M. K. Shahzad, and E. A. Affum, "IoMT meets machine learning: From edge to cloud chronic diseases diagnosis system," *J. Healthcare Eng.*, vol. 2023, no. 1, 2023, Art. no. 9995292.
[9] K. O'Shea and R. Nash, "An introduction to convolutional neural networks," 2015, *arXiv:1511.08458*.
[10] Z. Li, F. Liu, W. Yang, S. Peng, and J. Zhou, "A survey of convolutional neural networks: Analysis, applications, and prospects," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 6999–7019, Dec. 2022.
[11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[12] D. Sinha and M. El-Sharkawy, "Thin MobileNet: An enhanced MobileNet architecture," in *Proc. IEEE 10th Annu. Ubiquitous Comput., Electron. Mobile Commun. Conf. (UEMCON)*, Oct. 2019, pp. 0280–0285.
[13] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why should i trust you?' Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2016, pp. 1135–1144.
[14] M. H. K. Mehedi, A. K. M. S. Hosain, S. Ahmed, S. T. Promita, R. K. Muna, M. Hasan, and M. T. Reza, "Plant leaf disease detection using transfer learning and explainable AI," in *Proc. IEEE 13th Annu. Inf. Technol., Electron. Mobile Commun. Conf. (IEMCON)*, Oct. 2022, pp. 0166–0170.
[15] S. Bhattarai. *New Plant Diseases Dataset*. [Online]. Available: https://www.kaggle.com/datasets/vipoooool/new-plant-diseases-dataset
[16] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers Plant Sci.*, vol. 7, p. 1419, Sep. 2016.
[17] S. Mohanty. *PlantVillage Dataset*. [Online]. Available: https://github.com/salathegroup/plantvillage_deeplearning_paper_dataset
[18] M. A. Jasim and J. M. Al-Tuwaijari, "Plant leaf diseases detection and classification using image processing and deep learning techniques," in *Proc. Int. Conf. Comput. Sci. Softw. Eng. (CSASE)*, Apr. 2020, pp. 259–265.
[19] S. Ramesh, R. Hebbar, M. Niveditha, R. Pooja, N. Shashank, and P. V. Vinod, "Plant disease detection using machine learning," in *Proc. Int. Conf. Design Innov. 3Cs Compute Communicate Control (ICDI3C)*, Apr. 2018, pp. 41–45.
[20] S. S. Harakannanavar, J. M. Rudagi, V. I. Puranikmath, A. Siddiqua, and R. Pramodhini, "Plant leaf disease detection using computer vision and machine learning algorithms," *Global Transitions Proc.*, vol. 3, no. 1, pp. 305–310, Jun. 2022.
[21] M. de Benito Fernández, D. L. Martínez, A. González-Briones, P. Chamoso, and E. S. Corchado, "Evaluation of XAI models for interpretation of deep learning techniques' results in automated plant disease diagnosis," in *Proc. Sustain. Smart Cities Territories Int. Conf.* Cham, Switzerland: Springer, 2023, pp. 417–428.
[22] S. Kinger and V. Kulkarni, "Explainable ai for deep learning based disease detection," in *Proc. 13th Int. Conf. Contemp. Comput. (IC)*, 2021, pp. 209–216.
[23] M. Nahiduzzaman, M. E. H. Chowdhury, A. Salam, E. Nahid, F. Ahmed, N. Al-Emadi, M. A. Ayari, A. Khandakar, and J. Haider, "Explainable deep learning model for automatic mulberry leaf disease classification," *Frontiers Plant Sci.*, vol. 14, Sep. 2023, Art. no. 1175515.
[24] A. Khattak, M. U. Asghar, U. Batool, M. Z. Asghar, H. Ullah, M. Al-Rakhami, and A. Gumaei, "Automatic detection of citrus fruit and leaves diseases using deep neural network model," *IEEE Access*, vol. 9, pp. 112942–112954, 2021.
[25] H. T. Rauf, B. A. Saleem, M. I. U. Lali, M. A. Khan, M. Sharif, and S. A. C. Bukhari, "A citrus fruits and leaves dataset for detection and classification of citrus diseases through machine learning," *Data Brief*, vol. 26, Oct. 2019, Art. no. 104340.
[26] D. Singh, N. Jain, P. Jain, P. Kayal, S. Kumawat, and N. Batra, "PlantDoc: A dataset for visual plant disease detection," in *Proc. 7th ACM IKDD CoDS 25th COMAD*, 2020, pp. 249–253.
[27] Ecosia. (2019). *Search Engine*. [Online]. Available: https://www.ecosia.org/?c=en
[28] V. K. Vishnoi, K. Kumar, and B. Kumar, "Plant disease detection using computational intelligence and image processing," *J. Plant Diseases Protection*, vol. 128, pp. 19–53, Aug. 2021.
[29] S. Suthaharan and S. Suthaharan, "Support vector machine," in *Machine Learning Models and Algorithms for Big Data Classification: Thinking With Examples for Effective Learning*. Boston, MA, USA: Springer, 2016, pp. 207–235.
[30] G. Guo, H. Wang, D. Bell, Y. Bi, and K. Greer, "KNN model-based approach in classification," in *Proc. Move Meaningful Internet Syst., CoopIS, DOA, ODBASE: OTM Confederated Int. Conf., CoopIS, DOA, ODBASE*. Sicily, Italy: Springer, 2003, pp. 986–996.
[31] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–10.
[32] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.
[33] A. Abade, P. A. Ferreira, and F. de Barros Vidal, "Plant diseases recognition on images using convolutional neural networks: A systematic review," *Comput. Electron. Agricult.*, vol. 185, Jun. 2021, Art. no. 106125.
[34] S. Visa, B. Ramsay, A. L. Ralescu, and E. Van Der Knaap, "Confusion matrix-based feature selection," in *Proc. MAICS*, 2011, vol. 710, no. 1, pp. 120–127.
[35] M. Abadi, P. Barham, J. Chen, Z. Chen, A. Davis, J. Dean, M. Devin, S. Ghemawat, G. Irving, M. Isard, and M. Kudlur, "TensorFlow: A system for large-scale machine learning," in *Proc. 12th USENIX Symp. Operating Syst. Design Implement. (OSDI)*, 2016, pp. 265–283.
[36] R. David, J. Duke, A. Jain, V. Janapa Reddi, N. Jeffries, J. Li, N. Kreeger, I. Nappier, M. Natraj, T. Wang, P. Warden, and R. Rhodes, "TensorFlow lite micro: Embedded machine learning for tinyml systems," in *Proc. Mach. Learn. Syst.*, vol. 3, 2021, pp. 800–811.
[37] L. Stahle and S. Wold, "Analysis of variance (ANOVA)," *Chemometrics Intell. Lab. Syst.*, vol. 6, no. 4, pp. 259–272, Nov. 1989.

NATASHA NIGAR received the Ph.D. degree from the School of Computer Science, University of Birmingham, U.K., in 2021. She is currently an Assistant Professor with the Department of Computer Science, University of Engineering and Technology Lahore, Pakistan. From 2008 to 2009, she was a Laboratory Engineer with the National University of Computer and Emerging Sciences, Lahore, Pakistan. She was a Software Engineer with Palmchip Pvt. Ltd., Lahore, from 2009 to 2011; and a Senior Software Quality Assurance Engineer with Netsol Technologies, Lahore, from 2011 to 2013. Her research interests include computational intelligence, optimization in dynamic and uncertain environments, and machine learning. She was awarded the Faculty Development Program Scholarship to pursue the Ph.D. studies.

HAFIZ MUHAMMAD FAISAL received the B.Sc. degree from the Department of Electrical Engineering, University of Engineering and Technology, Lahore, Pakistan, in 2020. He is currently pursuing the degree with the Technical University of Munich, Germany. He was a Power System Protection Engineer with the Electrical and Instrument Department, Fatima Fertilizer Company, Sadiqabad, Pakistan. From 2021 to 2022, he was an Area Electrical Engineer with Calcium Ammonium Nitrate Plant, Fatima Fertilizer Company, Sadiqabad. He was also an Intern with the Electrical and Instrument Department, Engro Fertilizer Ltd., in 2019. His research interests include power system protection, smart grids, machine learning, and power electronics. He was awarded the Higher Education Commission (HEC) Scholarship for the bachelor's studies.

MUHAMMAD UMER received the M.Sc. degree from the Department of Computer Science, University of Engineering and Technology, Lahore, Pakistan, in 2020. He was a Software Engineer at B.I.S.E, Lahore, Pakistan, from 2017 to 2018; and Ebryx Pvt. Ltd., Lahore, Pakistan, from 2016 to 2017. From 2018 to 2020, he was a Senior Software Engineer at NETSOL Technologies Inc., Lahore, Pakistan. He is currently a Software Developer at TEKHQS INC., Irvine, CA, USA. His research interests include artificial intelligence, machine learning, and cloud computing. He has been awarded the Outstanding Performance Award and the Certificate of Appreciation during his professional career.

OLUKAYODE OKI received the Ph.D. degree from the University of Zululand, South Africa, in 2019. He is currently a Lecturer with the Department of Information Technology, Walter Sisulu University, South Africa. He has authored more than 40 articles. His research interests include biologically inspired computation, ICT4D, communication networks, the IoT, machine learning, data analytics, and climate-smart agriculture. He has received several grants both for research and development and to attend conferences. He was a recipient of the South Africa National Research Foundation (NRF) Rated Researcher Award, an Honorary Rosalind Member of *London Journal* press, and a member of the IEEE South Africa Subsection. subsection.

JOSE MANAPPATTUKUNNEL LUKOSE received the Ph.D. degree from Fort Hare University, South Africa. He is currently a Senior Lecturer and the HoD of the Department of Information Technology, Walter Sisulu University, South Africa. His research interests include, but not limited to ICT in education and ICT for sustainable development.

IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 2 September 2023, accepted 15 September 2023, date of publication 29 September 2023, date of current version 12 October 2023.

Digital Object Identifier 10.1109/ACCESS.2023.3320686

RESEARCH ARTICLE

Plant Disease Classifier: Detection of Dual-Crop Diseases Using Lightweight 2D CNN Architecture

HASIBUL ISLAM PEYAL[1,2,3], MD. NAHIDUZZAMAN[3,4], MD. ABU HANIF PRAMANIK[3], MD. KHALID SYFULLAH[3], SALEH MOHAMMED SHAHRIAR[3], ABIDA SULTANA[2,3], MOMINUL AHSAN[5], JULFIKAR HAIDER[6], AMITH KHANDAKAR[4], (Senior Member, IEEE), AND MUHAMMAD E. H. CHOWDHURY[4], (Senior Member, IEEE)

[1]Department of Computer Science and Engineering, Rajshahi University of Engineering and Technology, Rajshahi 6204, Bangladesh
[2]Department of Computer Science and Engineering, Uttara University, Dhaka 1230, Bangladesh
[3]Department of Electrical and Computer Engineering, Rajshahi University of Engineering and Technology, Rajshahi 6204, Bangladesh
[4]Department of Electrical Engineering, Qatar University, Doha, Qatar
[5]Department of Computer Science, University of York, YO10 5GH York, U.K.
[6]Department of Engineering, Manchester Metropolitan University, M15 6BH Manchester, U.K.

Corresponding authors: Muhammad E. H. Chowdhury (mchowdhury@qu.edu.qa) and Mominul Ahsan (mominul.ahsan2@gmail.com)

This work was supported by Qatar National Library (QNL).

ABSTRACT Tomatoes are the most widely grown crop in the world, and they may be found in a variety of forms in every kitchen, regardless of cuisine. It is, after potato and sweet potato, the most widely farmed crop on the planet. Cotton is another essential cash crop because most farmers grow it in huge quantities. However, many diseases reduce the quality and quantity of tomato and cotton crops, resulting in a significant loss in production and productivity. It is critical to detect these disorders at an early stage of diagnosis. The purpose of this work is to categorize 14 classes for both cotton and tomato crops, with 12 diseased classes and two healthy classes using a deep learning-based lightweight 2D CNN architecture and to implement the model in an android application named ''Plant Disease Classifier'' for smartphone-assisted plant disease diagnosis system, the results of the experiments reveal that the proposed model outperforms the pre-trained models VGG16, VGG19 and InceptionV3 despite having fewer parameters. With slightly larger parameters than MobileNet and MobileNetV2, proposed model also attains considerably larger accuracy than these models. The classification accuracy varies between 57% and 92% for these models, and the proposed model's average accuracy is 97.36%. Also, the precision, recall, F1-score of the proposed model is 97 % and Area Under Curve (AUC) score of the model is 99.9% which is an indicator of the very good performance of the model. Class activation maps were shown using the Gradient Weighted Class Activation Mapping (Grad-CAM) technique to visually explain the disease detected by the proposed model, and a heatmap was produced to indicate the responsible region for classification. The app works very impressively and classified the correct disease in a shorter period of time of about 4.84 ms due to the lightweight nature of the model.

INDEX TERMS Convolution neural network (CNN), lightweight 2D CNN, android application, plant disease diagnosis system, gradient weighted class activation mapping (Grad-CAM), tomato, cotton.

I. INTRODUCTION

Agriculture is a crucial term in many developing countries' economies, and the primary source of food and revenue. Agriculture supports approximately 60% of the world's population, as it is a vital aspect of many people's livelihood and subsistence [1]. Tomatoes and cotton are two of the most diverse and important commercial crops in Bangladesh, with the majority of people relying on agriculture for their income and survival. Tomato ranks as the fourth most widely cultivated vegetable in the nation. In Bangladesh, approximately 4.48 lakh tons of tomatoes were produced

The associate editor coordinating the review of this manuscript and approving it for publication was Massimo Cafaro.

in 2021 [2]. Bangladesh's national economy gains significantly from tomato production, which has positive economic effects. Around one hundred million dollars in tomatoes were exported from Bangladesh in the year 2020 and also Millions of people are employed in this sector [3]. Cotton is also a major cash crop in Bangladesh. It is the second-largest cotton importer and the fourth-largest consumer in the world. In 2021, Bangladesh exported $210 million in Raw Cotton, making it the 84th largest exporter of Raw Cotton in the world. In the same year, Raw Cotton was the 407th most exported product in Bangladesh [4]. In 2022, the country's domestic cotton production was around 0.2 million bales [5]. The Cotton Development Board of Bangladesh is taking the necessary steps to increase cotton production to meet the uprising cotton demand in Bangladesh as it is the second-largest apparel exporter in the world [6]. These statistics prove that cotton and tomato are significant crops in terms of the agricultural situation in Bangladesh. Plant diseases, on the other hand, have a negative impact on agricultural production. Organic agents such as fungi, bacteria, algae, and others cause plant diseases. Temperature imbalances, chemical toxicity, incorrect fertilizer, rainfall, nutrient inadequacy, and other nonliving variables, on the other hand, induce plant disorders [7].

Currently, these plant diseases are manually recognized with the naked eye, which is a time-consuming technique and frequently fails to effectively detect infections [8]. Farmers' incorrect diagnosis of diseases can result in the overuse or misuse of pesticides, which can harm plants and their yield. Plant disease classification is an important step that can help with the early diagnosis of pests and insects, disease treatment, and productivity increases, among other advantages. Disease symptoms are mostly visible on the leaves of cotton and tomato plants. For this, image processing techniques are applied to plant leaves in order to classify the diseases quickly, accurately, and appropriately. Consequently, utilizing deep learning to identify the pests and diseases in plants can dramatically speed up the detection process, reducing economic losses and saving time with even more accuracy.

A novel method using 2D CNN is proposed for identifying diseases in tomato and cotton crops after analyzing leaf photos in this research. Because 2D CNN outperforms other current methods in terms of localized feature extraction, efficiency, robustness, adaptability, and end-to-end learning capabilities, it is chosen for disease identification. These distinctive features place CNNs as the cutting-edge approach for image analysis and understanding and make them the superior option for a variety of computer vision jobs. Unquestionably, they have contributed to the field's advancement and encouraged innovation. The tomato and cotton leave images were collected from the publicly available PlantVillage dataset [9] and publicly available 'cotton-leaf-infection' dataset [39] for this experiment and classified using a lightweight 2D CNN model. The model's performance was evaluated using several metrics such as training accuracy, validation accuracy, precision, recall, F1-score, ROC curve, precision Vs recall curve and the number of trainable parameters.

The novel contributions of this work are:

(1) A novel lightweight 2D CNN model is constructed for extracting the most relevant features from the plant leaf images.

(2) Multiclass disease classification has been carried out for two different crops for an imbalanced large dataset using a 2D CNN architecture that has fewer parameters and is compared with other pre-trained models.

(3) The disease detection made by the trained model is visually explained by plotting class activation maps using the Grad-CAM technique and a heatmap was generated to identify and explain the characteristics region for the classification.

(4) An android application named "Plant Disease Classifier" has been developed using the trained lightweight 2D CNN model for exhibiting classified diseases and their information.

This technique will help farmers to identify plant diseases without having to rely on plant scientists and control the plant's disease in a timely manner, therefore enhancing the quality and quantity of the cotton and tomato crops produced and, as a result, providing economic benefits to the growers.

The paper has been organized as follows: Section II discusses findings from recent research on this topic. Section III detailed the proposed methodology of the model development. The results of the proposed model and a comparison with state-of-the-art (SOTA) models along with visualization using Grad-Cam and details of the developed Android application are presented in Section IV. Finally, the important conclusions from this study are provided in Section V.

II. RELATED WORKS

A detailed literature review has revealed that the automatic identification of plant diseases is carried out broadly by machine learning and deep learning-based methods. The following sections present a review of recent literature on this topic.

Basavaiah and Arlene Anthony [10] outlined an innovative strategy based on the extraction and concatenation of features to detect five classes of tomato leaf diseases for 500 images employing decision trees and Random Forest (RF) classifiers. The accuracy score for the suggested method was 94% for the RF classifier and 90% for the decision tree classifier. Chen et al. [11] proposed a method for detecting five tomato leaf diseases using a combination of the Artificial Bee Colony Algorithm-Binary Wavelet Transform combined with Retinex (ABCK-BWTR) and Both-channel Residual Attention Network (BARNet) models, with an overall detection accuracy of 89% for 5 classes and 8616 images from PlantVillage dataset. Agarwal et al. [12] utilized a CNN-based technique to detect and classify 9 tomato diseases and one healthy class and 91.2 % accuracy has been achieved for 10000 images of PlantVillage dataset. Cengil and Çınar [13] proposed a hybrid-based CNN model where feature extraction is done using AlexNet, ResNet50, and VGG16 and after the concatenation of the features, machine learning algorithms were used for classification, which gave 98.3% accuracy for the same PlantVillage dataset and 96.3% accuracy for a dataset having 6 classes and 4976 images. Khan and Narvekar [14] proposed a method for the automatic detection and classification of four classes of tomato diseases using superpixel-based optimized segmentation for the combined dataset of around 733 images and attained an average accuracy of 93.18%. Zhao et al. [15] proposed a tomato disease detection based on improved CNN by attention module which gave 96.81% accuracy for PlantVillage dataset having 10 classes and 4585 images and after enhancement 4 times the total resultant images were 22,925. Al-Gaashani et al. [16] proposed method for tomato disease detection where features from the images were extracted using pre-trained models and reduced using Kernel Principal Component Analysis (PCA) and multinomial logistic regression was used for classification which achieved an average accuracy of 97% for 1152 images from PlantVillage dataset for six classes. Wspanialy and Moussa [17] proposed a tomato disease detection & severity estimation method using ResNet & modified U-Net deep learning architecture which gave 97% accuracy for PlantVillage dataset having 10 classes. Chowdhury et al. [18] proposed a work in which classification is done on 18,162 images from PlantVillage dataset for ten classes where DenseNet201 achieved an accuracy of 98.05% for ten classes & 97.99% for six classes. For the same amount of data from PlantVillage dataset, Jasim et al. [19] proposed a method using CNN model and attained an average accuracy of 97.39% and Özbilge et al. [38] proposed a method using Compact CNN model and attained an average accuracy of 99.70% for ten classes. Chen et al. [20] proposed a method to implement the function of AlexNet modification architecture-based CNN on the Android platform to predict tomato diseases which gave 98% accuracy for a dataset having 10 classes and 22,930 images. Karthik et al. [21] proposed a method based on residual learning and attention mechanism using two different deep architectures to predict tomato diseases which gave 98% accuracy for a dataset having 4 classes and 120,000 images. Agarwal et al. [22] proposed a work in which eight hidden layers made up the compressed CNN model. For ten classes of tomato crops in the PlantVillage dataset, there are 14,529 training images and 3,631 validation images. The proposed work achieves an accuracy of 98.4%. Trivedi et al. [23] proposed a method where the classification of ten classes of tomato leaf diseases is done using CNN for 3000 images where the achieved average accuracy is 98.49%. Bhujel et al. [24] proposed a lightweight convolutional neural network by incorporating different attention modules for 11 classes of 19,510 tomato leaf images which is a combination of PlantVillage dataset and collected tomato leaf images from greenhouse and convolutional block attention module had attained the best accuracy of 99.34%. Indu and Priyadharsini [25] proposed an automatic tomato leaf disease identification system by using crossover-based wind-driven optimization (CROWDO) algorithm optimized Alexnet for 4 classes of 9584 tomato leaf images and attained 99.86% accuracy. But Alexnet has 60 million parameters which are very large and consumes very large disk space. Abbas et al. [26] proposed a deep learning-based method for tomato disease detection that utilizes the Conditional Generative Adversarial Network (C-GAN) to generate synthetic images of tomato plant leaves and DenseNet121 is used for the classification of 5 classes, 7 classes, and 10 classes and the achieved accuracy is 99.51%, 98.65%, and 97.11% respectively for PlantVillage dataset. In [27], the work proposed a method for the classification of ten classes of tomato diseases using GoogleNet for 18160 images and attained an average accuracy of 99.39%. For the same dataset, Tan et al. [28] proposed a method using ResNet34 where the achieved average accuracy is 99.7%. Nandhini and Ashokkumar [29] proposed a method for the classification of five classes of tomato leaf diseases using InceptionV3 and VGG16 utilizing the ICRMBO-CNN approach for optimization using PlantVillage dataset comprising 6208 images. But pre-trained networks have a very large number of parameters & the optimized VGG16 and Inception V3 provide a final classification accuracy of 99.98% and 99.94% respectively.

Based on the literature, the following gaps have been identified.

1. Using conventional image processing methods, it is difficult to locate and extract important information from tomato leaves that can help distinguish between the characteristics of various diseases. Since the features of these diseases vary greatly, it is necessary to thoroughly investigate their patterns using a variety of datasets in an automated manner.

2. The characteristics of the manually chosen, handcrafted features are the only factors that affect how well the machine learning-based models perform. Therefore, feature extraction needs to be automated in order to choose and learn the best collection of features for classification.

3. Some deep learning models make use of widely used and tested architectures, such as VGG16, GoogleNet, ResNet34 etc. As a result, it uses millions of parameters for classification. The computational complexity and accuracy must be balanced in order to deploy these models in real time.

4. Disease detection for the combination of two types of crops can be done using models having fewer parameters than disease detection for one type of crop.

5. Additionally, a high number of samples must be used to train the deep learning network in order to ensure improved feature generalization.

III. MATERIALS AND METHODS

A. PROPOSED METHODOLOGY

With the application of 2D CNN, a complete overview of the overall detection methodology of tomato & cotton leaf diseases is provided in this study. 2D CNN has been chosen because traditional approaches call for handcrafted features, whereas 2D CNN learns hierarchical representations over numerous layers, which enables it to capture increasingly complicated and abstract characteristics. The ability to extract hierarchical features improves the accuracy with which visual data can be understood and interpreted. Additionally, parameter sharing is used by CNNs, which greatly reduces the number of trainable parameters. As a result, CNNs are more effective than other techniques at handling enormous datasets and challenging tasks. Because of their resistance to changes in object position and translation invariance, CNNs are very good at tasks like object localization and detection. The benefit of CNNs is end-to-end learning, where the model gains knowledge directly from the input data. CNNs can identify complex patterns and connections that might not be visible using conventional techniques since they automatically learn features and representations. Following the mentioned reasons, two datasets (tomato and cotton) are merged and a large number of data are used, which influenced the authors to choose 2D CNN over the other conventional methods as it is suitable for detecting tomato and cotton leaf diseases. A flowchart depicting the workflow of the whole study is displayed in Fig. 1. The conducted approach is outlined step by step in the following subsections.

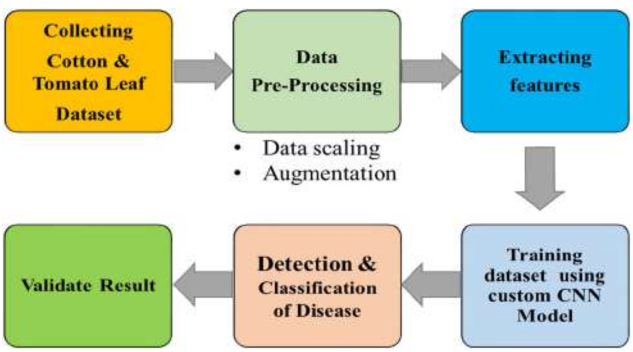


FIGURE 1. **Overall research methodology overview.**

B. DATASET DESCRIPTION

The dataset is at the heart of deep learning research. Tomato and cotton leaf images represent different types of plant diseases as well as healthy leaf images. Tomato dataset was obtained from the publicly available 'PlantVillage' dataset [9] and cotton dataset was obtained from publicly available 'cotton-leaf-infection' dataset [39]. The PlantVillage dataset has become a key tool in the field of agricultural computer Vision. It has sparked studies and innovation in the automated identification and diagnosis of plant diseases due to its large collection of marked photos, related metadata, and various plant species and diseases. As the cotton leaf dataset is not included in the PlantVillage dataset, it was collected separately, and merged for this research work. The merged tomato and cotton datasets contain a total of 11,366 training and 1,327 testing images organized into 14 classes. Therefore, the split ratio of the training and testing dataset is approximately 9:1. Fig. 2 depicts example images from all the classes of the dataset.

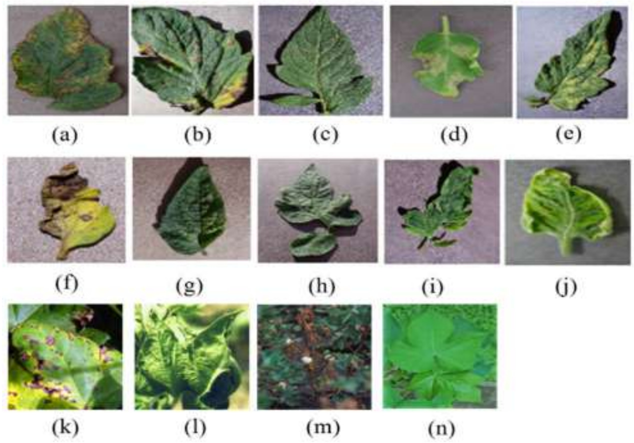


FIGURE 2. **Sample leaf images from the dataset representing 14 plant disease classes: (a) Bacterial Spot, (b) Early Blight, (c) Healthy (Tomato), (d) Late Blight, (e) Leaf Mold (f) Septoria Leaf Spot, (g) Spider Mites, (h) Target Spot, (i) Mosaic Virus (j) Yellow Leaf Curl Virus, (k) Bacterial Blight, (l) Curl Virus,(m) Fussarium Wilt, (n) Healthy (Cotton).**

TABLE 1. **Dataset details.**

Plant	Disease Type	Training	Testing
Tomato	Bacterial Spot	1000	100
	Early Blight	1000	100
	Healthy (Tomato)	1000	100
	Late Blight	1000	100
	Leaf Mold	1000	100
	Septoria Leaf Spot	1000	100
	Spider Mites	1000	100
	Target Spot	1000	100
	Mosaic Virus	1000	100
	Yellow Leaf Curl Virus	1000	100
Cotton	Bacterial Blight	358	90
	Curl Virus	335	84
	Fussarium Wilt	336	84
	Healthy (Cotton)	341	85
	Total	11,366	1,327

Table 1 displays the class-wise dataset's details for both training and testing.

C. DATA PRE-PROCESSING

Image preprocessing of the diseased tomato leaves is a crucial step before classification as the quality of images has a considerable impact on the classification outcomes. It is done in the following steps:

1) DATA SCALING/RESIZING

When dealing with 2D CNN, data scaling is a recommended stage of pre-processing. The technique equalizes all of the images in the datasets and transforms the image size to fit the model developed. The data resizing would minimize the memory to be used when the images are loaded into the training stage. For the proposed model, the image data was resized 150 × 150 and for transfer learning models the image data size was 224 × 224. as it is the default image size for implementing transfer learning models.

2) IMAGE AUGMENTATION

Image augmentation is a technique for artificially extending the number of training images in a dataset. This strategy aids in the model's performance enhancement capability and reduces overfitting issues [21]. This approach can manipulate the training images in a variety of ways, including zooming, rotating, shifting, and flipping. In this case, six types of augmentations were applied e.g., rotation, width shifting, height shifting, shearing, zooming and horizontal flipping as shown in Fig. 3. The value of the rotation range parameter was set to 40 degrees. As a result, randomly rotated images with a rotation range of 1 degree to 40 degrees were obtained. Similarly, the parameters for width shifting, height shifting, shearing and zooming were set to 0.2. By setting the horizontal flip parameter to true, horizontally flipped images were obtained. Image augmentation was only applied to the training images. The total amount of training images was 11,366. After augmentation, the images were 6 times greater than they were before. As a result, a total of 68,196 training images were attained.

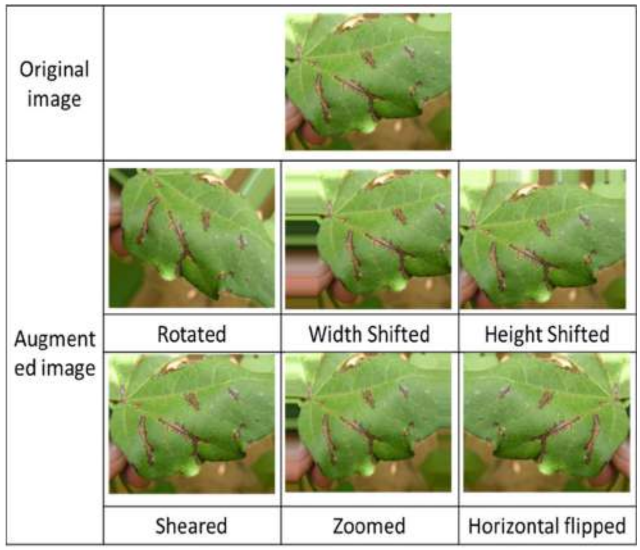


FIGURE 3. Sample augmented images from original input tomato leaf image.

TABLE 2. Summary of proposed simple 2D CNN model.

Layer (Type)	Output Shape	Params
Input Layer	(150,150,3)	
L1 (Conv2D)	(None, 148, 148, 16)	448
max_pooling2d_15 (MaxPooling)	(None, 74, 74, 16)	0
L2 (Conv2D)	(None, 72, 72, 32)	4640
max_pooling2d_16 (MaxPooling)	(None, 36, 36, 32)	0
L3 (Conv2D)	(None, 34, 34, 64)	18496
max_pooling2d_17 (MaxPooling)	(None, 17, 17, 64)	0
flatten_5 (Flatten)	(None, 18496)	0
dense1 (Dense)	(None, 512)	9470464
batch_normalization_4 (Batch)	(None, 512)	2048
dropout_6 (Dropout)	(None, 512)	0
dense_5 (Dense)	(None, 14)	7182
Total params: 9,503,278 Trainable params: 9,502,254 Non-trainable params: 1,024		

D. MODEL BUILDING

A simple 2D CNN has been proposed in this part to extract the most important features. If the relevant distinctive features between the various leaf diseases are retrieved, the model's classification performance will improve. These extracted attributes can be effectively used to classify leaf diseases. The proposed 2D CNN model architecture is represented in Fig. 4. Three convolution and max-pooling layers were included in the proposed 2D CNN design. A max-pooling layer was incorporated after each convolutional layer. Batch normalization was used since it accelerated and improved the model's performance by re-centering and re-scaling the layers' inputs [30]. The most significant features were extracted using max-polling, which was utilized to select the largest value from each cluster's whole neuron. The next step is to flatten the pooled featured map once it has been acquired. The entire pooled feature map matrix is flattened into a single column, which is then provided to the neural network for processing. In this scenario, dropout was used to reduce overfitting by frequently skipping training all nodes in each layer during the training phase, resulting in a significant increase in training speed [20].

The proposed 2D CNN model is described in Table 2 and the values of the hyperparameters of the models (2D CNN, VGG-16, VGG-19, Inception-V3, MobileNet and MobileNetV2) used in this work are shown in Table 3. After numerous training attempts, the values of the other model's hyperparameters was kept identical to have the best

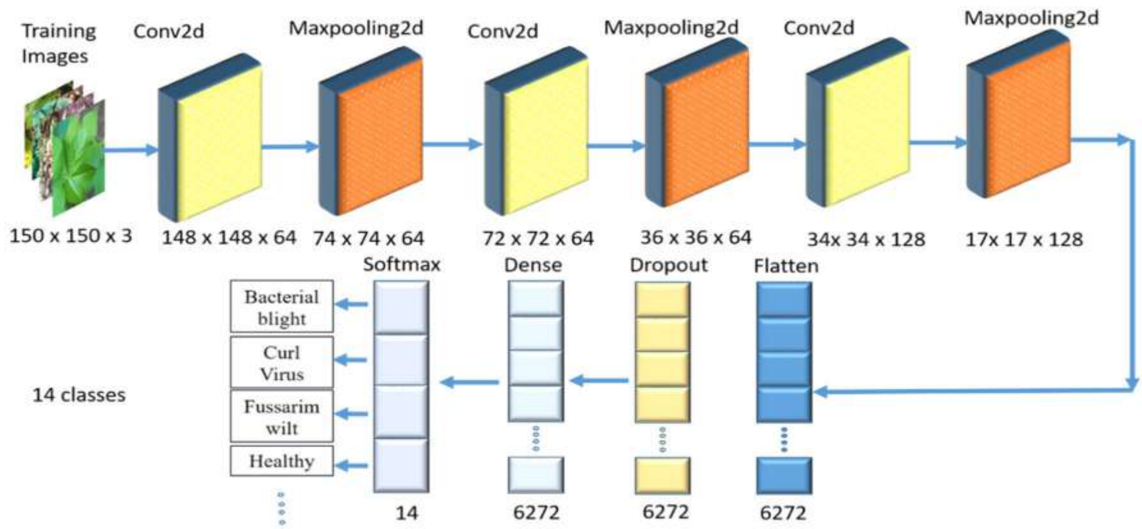


FIGURE 4. Lightweight 2D CNN architecture for tomato leaf disease classification.

optimized model. The proposed model contained a total number of 9,503,278 trainable parameters and 1,024 non-trainable parameters. Categorical cross-entropy was used as a loss function as multiclass data need to be detected. Adam optimizer was used to ensure reliable performance for the bigger dataset used [31].

TABLE 3. Hyperparameters used for the models.

Parameter Name	Attribute
Learning rate	0.001
Activation function	ReLU
Batch size	32
Loss function	Categorical cross-entropy
Optimizer	Adam

IV. EXPERIMENTS AND RESULTS

A. EXPERIMENTAL RESULTS & PERFORMANCE METRICS

The experiment was carried out on Kaggle platform (www.Kaggle.com) in order to take advantage of its available Nvidia P100GPU. Specification of the GPU is: GPU Memory: 16 GB, GPU Memory Clock: 1.32 GHz, Performance: 9.3 TFLOPS.

Several assessment metrics have been utilized such as accuracy (ACC), precision (P), recall (R), F1-score, and area under the curve (AUC), to quantitatively assess the performances of the developed model. The percentage of accurately detected images among all the images is known as accuracy (ACC). It represented the degree to which the classification method was successful in identifying leaf diseases. True positive (TP) means a disease-infected leaf is detected as infected, true negative (TN) means a non-diseased leaf is detected as non-infected, false positive (FP) means a non-diseased leaf was incorrectly detected as infected leaf was incorrectly detected as non-diseased [32]. disease-infected, and false negative (FN) means a disease Equation 1 to Equation 4 defines various performance measure metrics.

$$Accuracy = \frac{T_P + T_N}{T_P + T_N + F_P + F_N} \tag{1}$$

$$Precision = \frac{T_P}{T_P + F_P} \tag{2}$$

$$Recall = \frac{T_P}{T_P + F_N} \tag{3}$$

$$F1\_score = \frac{2 \times Recall \times Precision}{Recall + Precision} \tag{4}$$

B. MULTICLASS CLASSIFICATION RESULTS

1) ACCURACY GRAPHS

The efficacy of the models was validated in a series of studies using healthy and infected tomato and cotton leaf images using VGG-16, VGG-19, Inception-V3, MobileNet, MobileNetV2 and Proposed 2D CNN. The training and validation accuracy graphs for VGG-16, VGG-19, Inception-V3, MobileNet, MobileNetV2 and Proposed 2D CNN are shown in Fig. 5(a), Fig. 5(b), Fig. 5(c), Fig. 5(d), Fig. 5(e) and Fig. 5(f) respectively.

From Table 4, it can be seen that the average validation accuracy for VGG-16, VGG-19, Inception-V3, MobileNet, MobileNetV2 and proposed model are 81.39%, 82.97%, 92.31%, 83.65%, 57.50% and 97.36% respectively. Therefore, it is evident that the evaluation metrics accuracy, precision, recall and F1-score of the proposed model is significantly higher than the transfer learning models which is a very good indicator on the reliability of the proposed model's performance in classifying 14 categories of plant leaf diseases from a huge dataset.

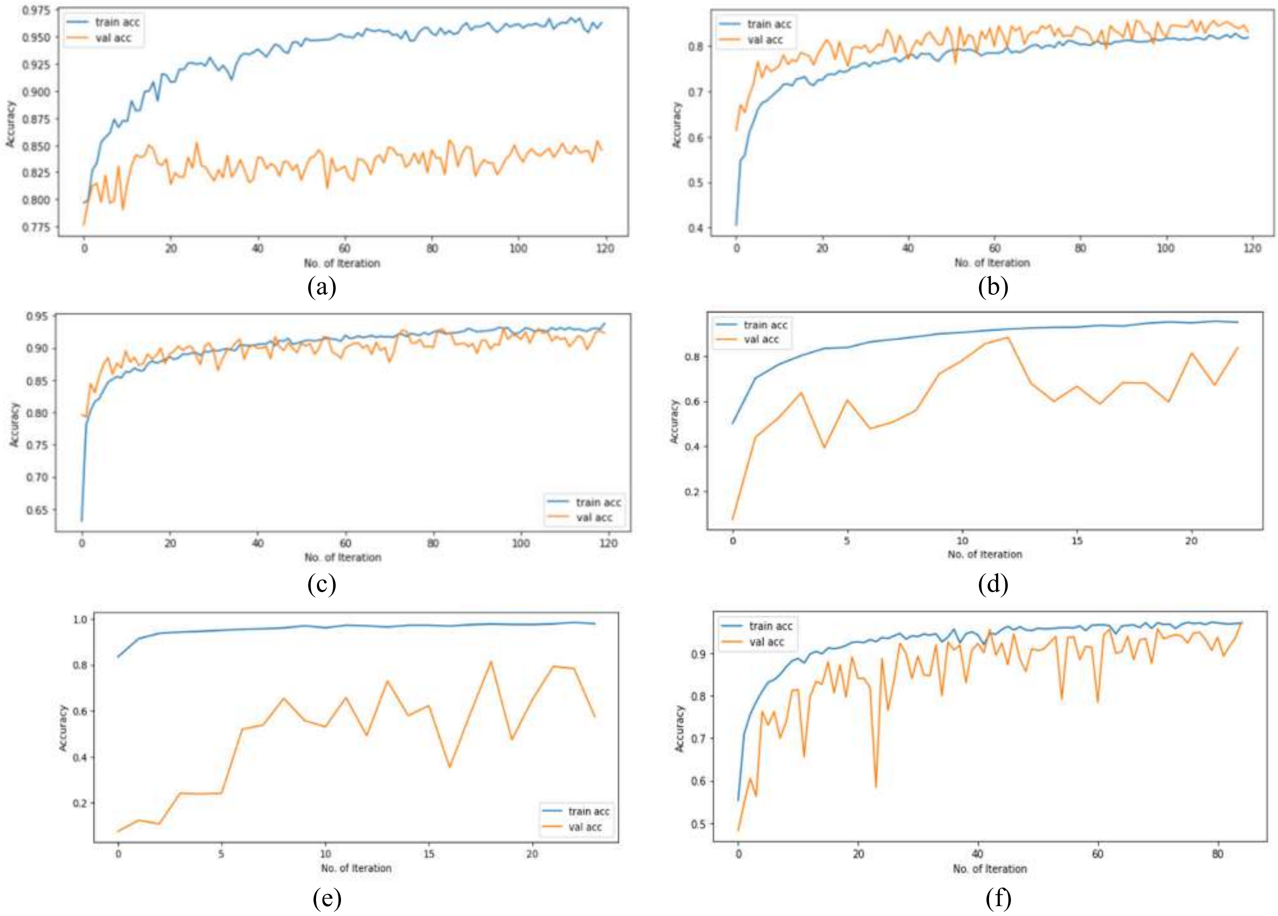


FIGURE 5. Accuracy graphs for training and validation accuracy for (a) VGG16 model, (b) VGG19 model, (c) InceptionV3 model, (d) MobileNet model, (e) MobileNetV2 model and (f) Proposed model.

TABLE 4. Evaluation metrics comparison with transfer learning models.

Models	Accuracy	Precision	Recall	F1-Score
VGG16	81.39%	86%	85%	85%
VGG19	82.97%	86%	83%	82%
Inception V3	92.31%	93%	92%	92%
MobileNet	83.65%	90%	88%	88%
MobileNetV2	57.50%	86%	81%	82%
Proposed CNN model	97.36%	97%	97%	97%

2) CONFUSION MATRIX

The confusion matrix, which is important in determining the performance of architectures, is also taken into account in this research as shown in Fig. 6. It displays the degree to which the classification method is successful in identifying the diseases in plant leaf images. This study considered a 14-class problem, which consisted of 2 healthy classes of cotton and tomato leaves and 12 different unhealthy classes of the both cotton and tomato leaves. It is noticed that out of 1327 images, 98, 242, 118, 156, 246 and 51 images were misclassified for VGG16, VGG19, Inception V3, MobileNet, MobileNetV2 and the proposed model respectively. So, it is clear that the proposed model can classify 14 numbers of classes accurately rather than the existing transfer learning models with a lightweight structure.

3) ROC CURVES

From Table 5 it is noticeable that the AUC score for the proposed model is almost nearly one and also it has surpassed the other transfer learning model's AUC.ROC curve is a metric for assessing and fine-tuning the performance of a deep learning model and ROC curves for different models are presented in Fig. 7. The precision vs. recall curves for the four models are shown in Fig. 8.

The precision-recall curve shows the trade-off between precision and recall for different thresholds. A low false positive rate is associated with good precision, whereas

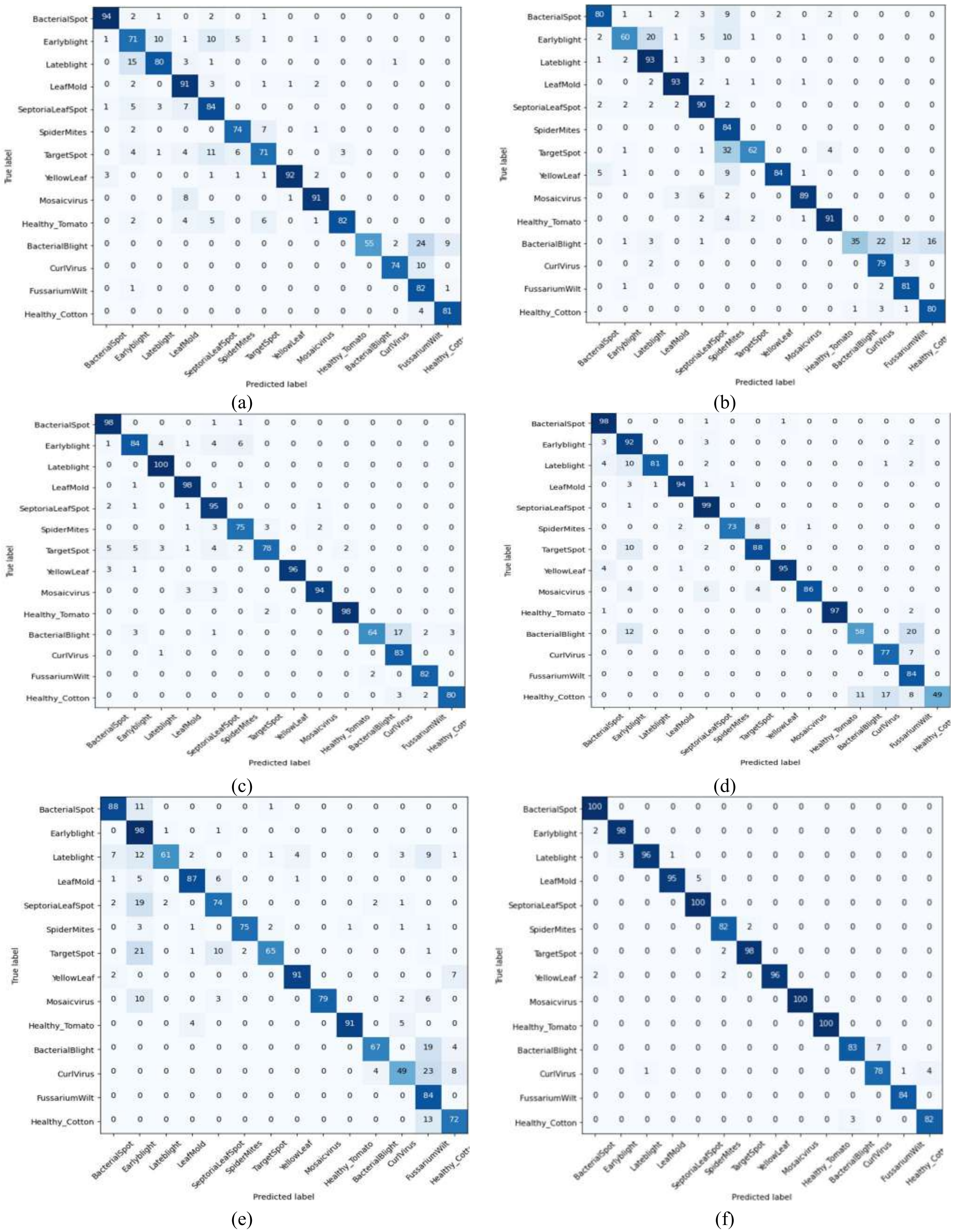


FIGURE 6. **Confusion matrix for (a) VGG16 model, (b) VGG19 model, (c) InceptionV3 model, (d) MobileNet model, (e) MobileNetV2 model and. (f) Proposed model.**

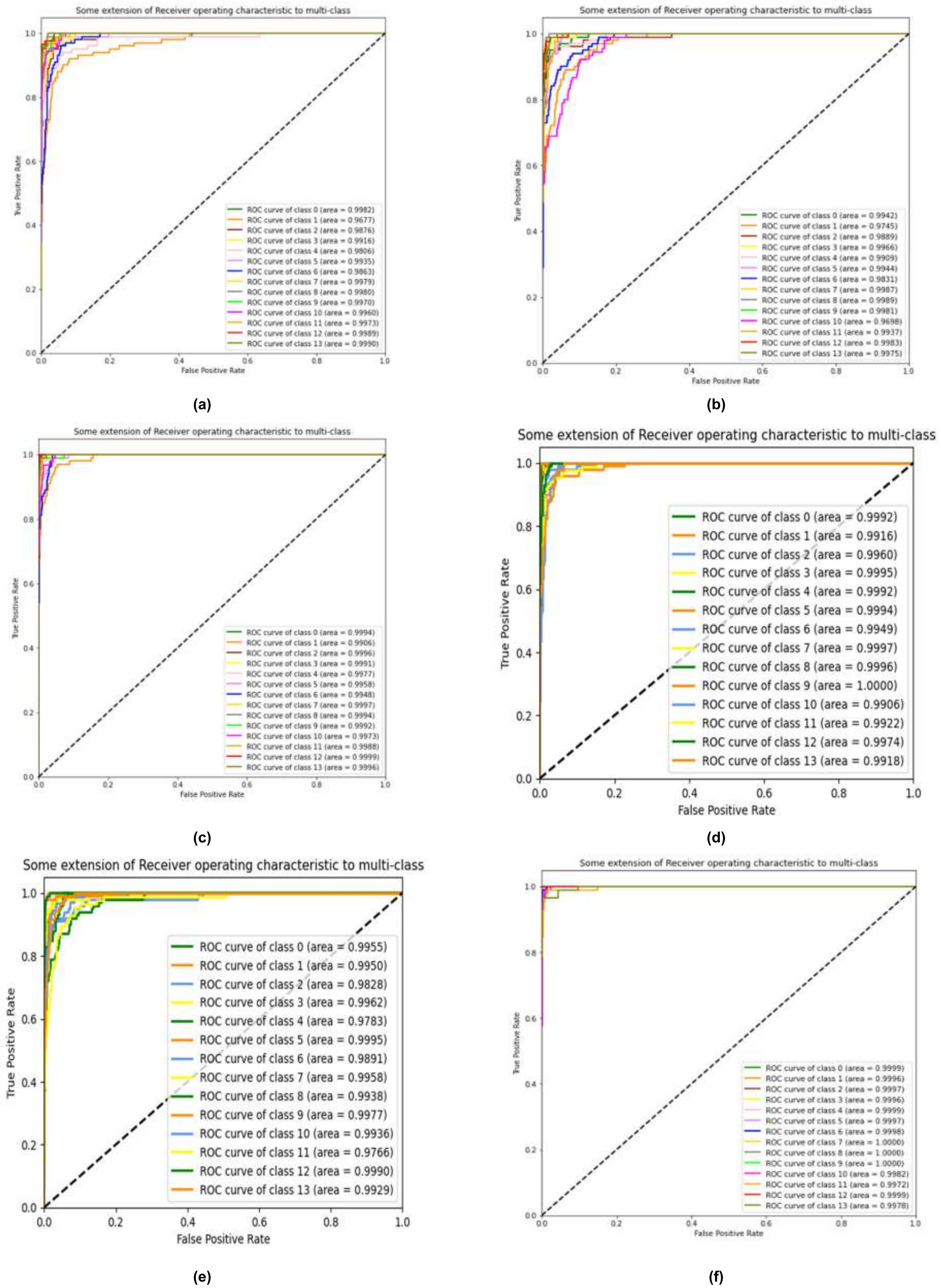


FIGURE 7. ROC Curve for (a) VGG16 model, (b) VGG19 model, (c) InceptionV3 model, (d) MobileNet model, (e) MobileNetV2 model and (f) Proposed model.

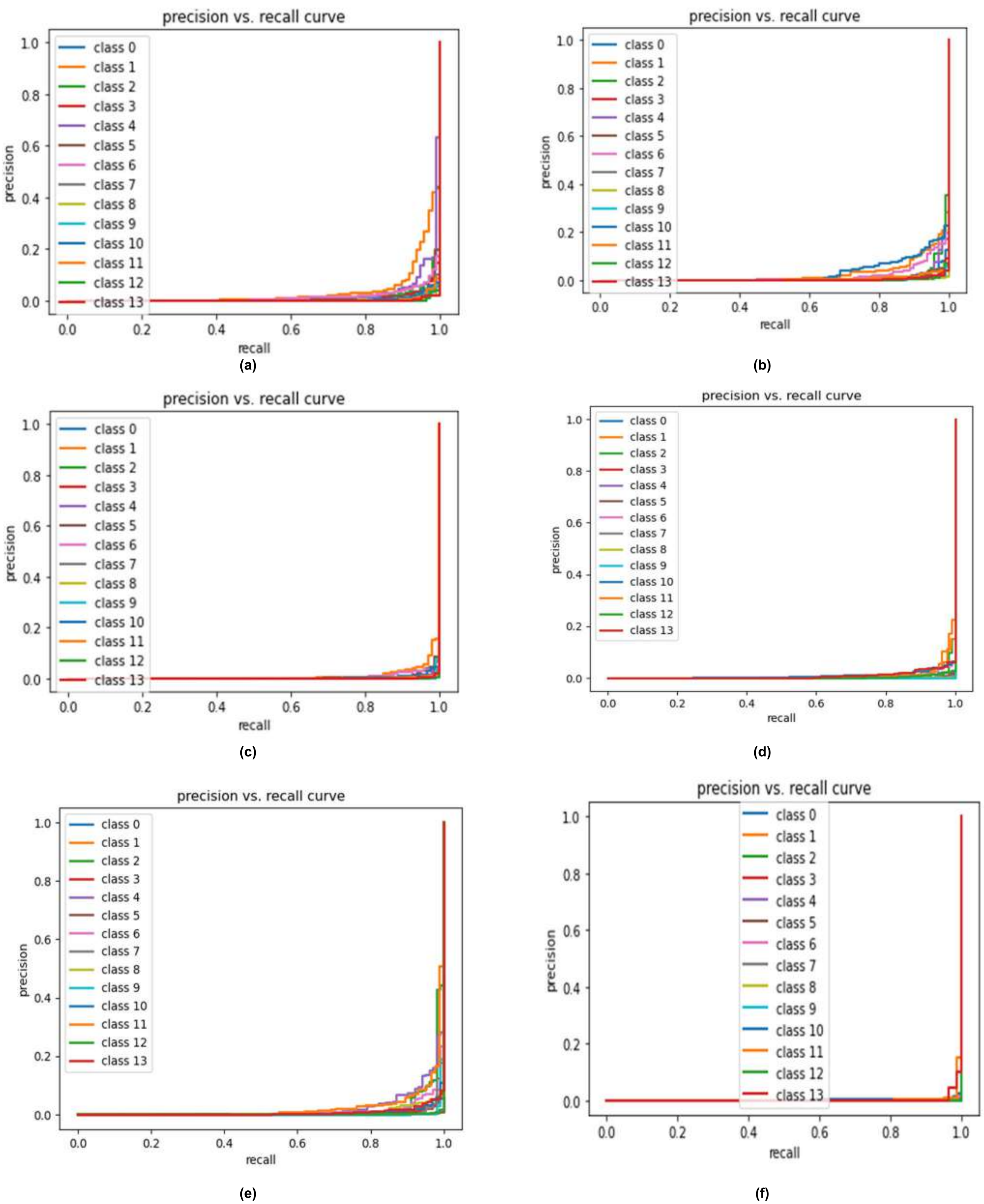


FIGURE 8. Precision Vs Recall Curves for (a) VGG16 model, (b) VGG19 model, (c) InceptionV3 model, (d) MobileNet model, (e) MobileNetV2 model and (f) Proposed model.

TABLE 5. AUC score comparison with transfer learning models.

Models	AUC SCORE
VGG16	99.23%
VGG19	99.12%
Inception V3	99.79%
MobileNet	99.63%
MobileNetV2	99.18%
Proposed 2D CNN model	99.93%

a low false negative rate is associated with strong recall. Great recall and high precision is both indicated by a high area under the curve. The precision-recall curve is utilized because ROC curves are appropriate for datasets that have a uniform distribution of observations across classes, whereas precision-recall curves are appropriate for imbalanced datasets that have been used in this experiment. So the evidence of the enhanced performance of the proposed model is clearly visible from this comparison.

4) COMPARATIVE ANALYSIS

The proposed model is said to be lightweight because it contained significantly less trainable and non-trainable parameters than the transfer learning models. From Table 6, the lightweightness of the proposed CNN model has been proved against some existing transfer learning models. Compared to the popular pretrained models such as VGG16, VGG19 and InceptionV3, the proposed 2D CNN has comparably much lower architectural complexity and parameter count. From Table 6, it is clear that size of the proposed model appeared almost 3 times smaller than VGG16 and VGG19 and almost 4 times smaller than Inception V3. In future, smartphones and portable devices with higher computation power will further normalize the issue. The proposed model's size on the disk is greater than the size of the MobileNet & MobileNetV2 models, but in terms of accuracy, the proposed 2D CNN model has outperformed the mentioned lightweight MobileNet family. The training time and epochs required for the proposed model is also lesser compared to the heavyweight transfer learning models (VGG16, VGG19 and Inception V3). Transfer learning models have more trained weights and it is trained on millions of images. It can perform better for very high number of classes, but in this research, the compared transfer learning models failed to impress against the proposed 2D CNN model. The 2D CNN model has worked better for an imbalanced dataset with a very small number of parameters than the existing transfer learning models.

The lightweightness of the proposed 2D CNN model has also been analyzed against the existing research works mentioned in Table 7. By looking at some of the previous works, it was noticed that some methods proposed in references [19], [20], [22] and [23] used very lightweight CNN models that have very similar accuracies and parameter count is lower than that of the proposed model. However, they are incapable of classifying 14 different classes with high accuracy and precision unlike the proposed method. Even though, 9.5 million+ parameter count may sound like a contradictory proposition for this method, the model has the capability to classify 14 different classes unlike most of the literatures in Table 7. Besides, the proposed 2D CNN only requires high computational power for training purposes, but model evaluation or sample prediction time is comparably still low as demonstrated. One common trend of most of the mentioned literatures in this field is that many methods can show testing accuracies greater than 97% in 2- or 3- class predictions, but then gradually decrease to around 90% as the classes start to increase [12], [33]. The reason behind this issue could be due to a comparably higher parameter count (9.5M+) in the proposed 2D CNN model, it can extract feature maps at higher resolution scales in the Conv2D layers. Because it comprised much less trainable and non-trainable parameters than the transfer learning models, the suggested model is referred to as being a lightweight model.

From Table 7, the performance of the proposed model has also been compared with existing works in terms of evaluation metrics like accuracy, precision, recall, F1 – score and number of classes. It is clear that the proposed model has better results in these evaluation metrics than the existing works for two types of crops and 14 classes whereas most of the existing compared works are based on one type of crop (Tomato). Among the mentioned works, the highest number of classes is 10 for the tomato plant whereas in the proposed work, these 10 classes are merged with 4 cotton classes to form the merged dataset with a view to forming a multi-crop disease dataset. During training, various types of image augmentation techniques are applied to this dataset to increase the dataset for better training.

In [13], transfer learning architectures AlexNet, ResNet50, and VGG16 are employed for the extraction of features which has more layers than the proposed model as well as more parameters.

In [16] accuracy and precision are less than the proposed work. This work deals with 6 classes which is two times fewer than the classes used in the proposed work.

In [17], the work utilizes Modified UNET and ResNet used to have larger parameters [35] for the detection of tomato leaf diseases and attained an average accuracy of 97% for 10 classes of the PlantVillage dataset.

In [18], DenseNet201 was used to achieve the best accuracy, but this architecture has 20.2M parameters which is very large and consume very large disk space.

In [23], the attained accuracy is 98.49% for 10 classes of tomato diseases and the training dataset contains 3000 images. Here accuracy is a bit higher than the proposed model but training has been done on a lesser number of training data.

TABLE 6. Parameter comparison with transfer learning models.

Models	Total Params	Trainable Params	Non-Trainable Params	Size (MB)	Time Required (s)
VGG16	41,469,774	26,755,086	14,714,688	362	17184.55
VGG19	46,779,470	26,755,086	20,024,384	382	18257.30
Inception V3	47,525,806	38,550,542	8,975,264	476	10464.20
MobileNet	3,254,158	3,232,270	21,888	37	4153.15
MobileNetV2	2,275,918	17,934	2,257,984	26	4555.46
Proposed 2D CNN model	9,503,278	9,502,254	1,024	108	6840.23

TABLE 7. Performance comparison with previous works.

Ref. No.	Method	Accuracy	Precision	Recall	AUC	F1-score	Classes	Plant
[12]	CNN	91.2%	90%	92%	-	91%	10	Tomato
[13]	Hybrid-based CNN	96.3%	-	-	-	-	6	Tomato
		98.3%	-	-	-	-	10	
[15]	CNN with attention mechanism	96.81%	96.77%	96.81%	-	96.79%	10	Tomato
[16]	Transfer learning with features concatenation	97%	96.5%	97.33%	-	98.16%	6	Tomato
[17]	ResNet	97%	-	-	-	-	10	Tomato
[18]	InceptionV3	97.99%	97.99%	97.99%	-	97.98%	6	Tomato
	DenseNet201	98.05%	98.03%	98.03%	-	98.03%	10	
[19]	CNN	97.39%	-	-	-	-	10	Tomato
[20]	AlexNet modification architecture-based CNN	96%	98%	95%	-	97%	10	Tomato
[22]	CNN	98.4%	-	-	100%	92.59%	10	Tomato
[23]	CNN	98.49%	-	-	-	-	10	Tomato
[26]	DenseNet121	97.11%	97%	97%	-	97%	10	Tomato
[27]	GoogleNet	99.39%	99.29%	99.12%	99.72%	99.20%	10	Tomato
[28]	ResNet34	99.7%	99.6%	99.7%	100%	99.7%	10	Tomato
[33]	Neural network classifier	92.5%	-	-	-	-	4	Cotton & Tomato
[34]	ResNet	97.28%	-	-	-	-	8	Tomato
Proposed Model	Lightweight 2D CNN Model	97.36%	97%	97%	99.93%	97%	14	Cotton & Tomato

In [26], DenseNet121 was used which has 8.1M parameters which is less compared to the proposed model but the attained accuracy and other evaluation metrics are less than the proposed model.

In [27], the attained accuracy was good using GoogleNet and this network has seven million parameters but this network is much deeper and wider, with 22 total layers. The processing time for the classification is 133.15 minute which is very time consuming than the proposed work's processing time (114 minute). This work had an AUC of 99.72% which is lesser then the AUC of the proposed work. Similarly, in [28] ResNet34 has almost 60.5M parameters which is very large and can consume more space and processing time.

From Table 8, the classwise accuracy comparison has been demonstrated with two of the existing works. In [13], though the average classification accuracy is higher (98.3%) than the proposed model (97.3%), the proposed model has higher classification accuracy (100%) for classes named healthy, septoria leaf spot. In [17], the individual classification accuracy for spider mites and target spot is 78.8% and 77.4% whereas in the proposed work classification accuracy for these two classes is 97.61% and 98%, so the probability of

TABLE 8. Classwise accuracy comparison with existing works.

Disease Class Names for tomato leaves	Accuracy (%)		
	[13]	[17]	Proposed Model
Bacterial Spot	99.9	99.4	100
Early Blight	99.2	99.7	98
Healthy (Tomato)	99.6	-	100
Late Blight	99.9	96.1	96
Leaf Mold	99.8	99.7	95
Septoria Leaf Spot	99.6	99.7	100
Spider Mites	99.9	78.8	82
Target Spot	100	77.4	98
Mosaic Virus	99.9	99.7	100
Yellow Leaf Curl Virus	100	99.8	96

misclassification for the mentioned classes is clearly less in the proposed model.

So, it is worth mentioning that the proposed work deals with multiclass classification for two different crops for an imbalanced dataset with large diversity instead the existing works deal with a dataset of tomato plant disease only, which has less diversity than two merged datasets. Also, most of the existing models have greater parameters. The superiority of the proposed model is that it can detect multiple crop diseases with fewer parameters and comparable accuracy to the existing models. Moreover, in this work, the model is implemented on an Android application for disease detection purposes which hasn't been done for all the works except the work in [20].

To the best of our knowledge, the combination of cotton and tomato crops is not too much used. Work is found for this combination [33] which classifies only 4 classes and attained an accuracy of 92.5% for less amount of data which is smaller than the proposed work's accuracy of 97.36% for the comparatively large number of classes.

C. EXPLAINING THE MODEL USING GRAD-CAM

To make deep learning more comprehensible and explainable, a lot of effort is being done. It is critical to make the deep learning model more interpretive in various deep learning applications connected to plant disease imaging. The Gradient Weighted Class Activation Mapping (Grad-CAM) approach, developed by Selvaraju et al. [36], provides an explainable view of the deep learning models. Grad-CAM creates an observable explanation for any deeply linked Neural Network, which aids in understanding more about the model while executing detection or prediction tasks.

Grad-CAM has been used in this study to evaluate whether the leaf sections in the input image have a significant impact on the diagnosis or to see the evidence of diagnosis visually. To determine the degree of importance of each map, the Grad-CAM calculates a gradient of the target class on each feature map and averages them. Calculating a weighted sum of each

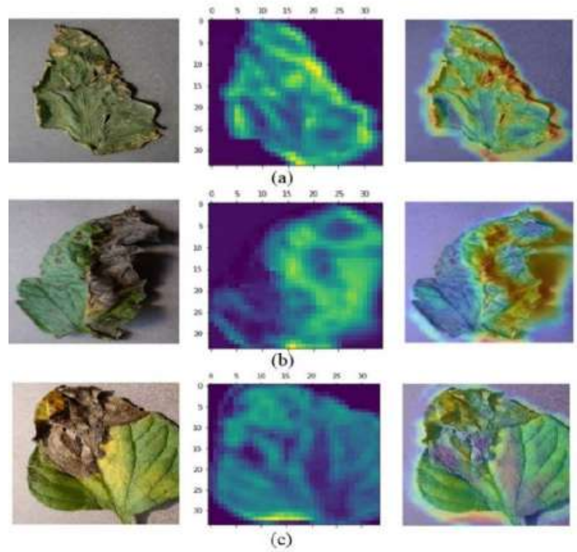


FIGURE 9. Input, Heatmap and superimposed image on input images of (a) Bacterial Spot, (b) Early Blight, (c) Late Blight.

FIGURE 10. Images of the android application activities (a) Opening activity, (b) Launching Activity, (c) Checking Activity, (d) Gallery Images.

feature map activation with the significance corresponding to the input image yields visualization. Grad-CAM is a suitable approach that does not sacrifice execution speed because it does not require extra specially designed components [37]. As shown in Fig. 9, a simple image was taken as an input by the Grad-CAM and applied detection techniques using the proposed model. In most cases, the last convolution layer was the one that will be utilized to apply the Grad-CAM. Grad-CAM utilizes the gradients of the target of the final convolutional layer and creates a heat map that highlights the key areas of an image. When this heatmap is superimposed on the original image, the regions based on which classification is done can be identified.

D. ANDROID APPLICATION

To demonstrate the implementation of the established model in a practical application, a serviceable prototype of an Android App was built and developed. The trained 2D CNN model was translated to TFLite model format in order to create the Android app with a user interface (UI) for classifying

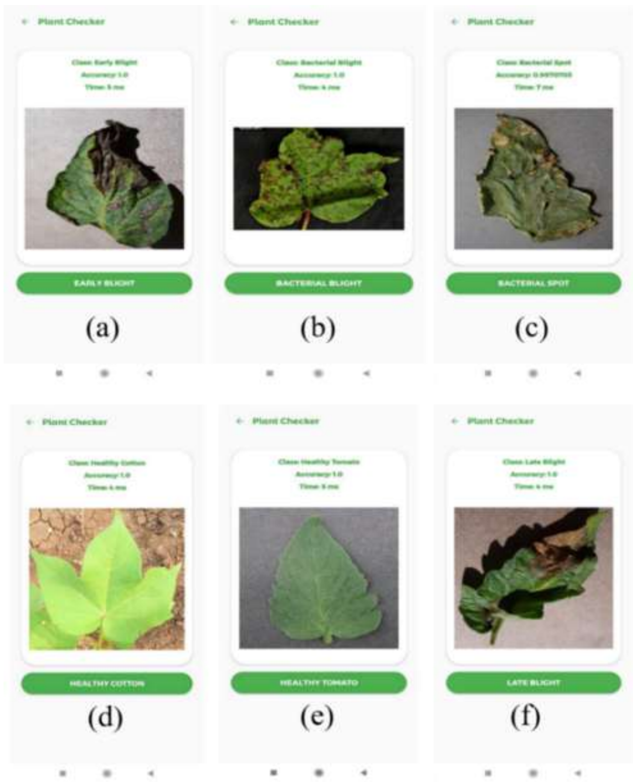


FIGURE 11. Sample Images of Android application TFLite based image classification of (a) Early Blight (Tomato), (b) Bacterial Blight (Cotton), (c) Bacterial Spot (Tomato), (d) Healthy (Cotton), (e) Healthy (Tomato), (f) Late Blight (Tomato).

the diseased leaf images on an Android device (Fig. 10). The Android app first loaded the model into memory before allocating tensors and creating a TFLite Interpreter depending on the loaded model. The Interpreter then imported the input leaf image into the input tensor, which was scaled to 150 × 150 pixels and normalized by using TFLite Image Processor API. The Interpreter then performed inference and output categorization outcomes into the output tensor. The TFLite model can identify a total of 12 disease classes as well as two healthy classes. Fig. 11 depicts the test accuracy and inference runtime outputs. All of the tests were ran using GPU acceleration and required between 4 to 7 milliseconds to run inference on the example leaf images and to diagnose cotton and tomato leaf diseases with great accuracy and faster than a plant diseases specialist. GPU acceleration though was not compulsory to classify an image. All testing were carried out using a Xiaomi Poco F3 with Android 11 and a Qualcomm SM8250-AC Snapdragon 870 5G (7 nm). The Android app is able to detect 14 different forms of plant leaf diseases. The Android app would take considerably less time (4 ms to 7 ms) to diagnose cotton and tomato leaf diseases with great accuracy than a plant doctor. The average detection accuracy is approximately 4.84 ms which can be considered almost real-time disease detection. From the test images, it is observable that the detection accuracy is pretty excellent in the application and the range is between 0.997-1.0 for the mentioned classes. This is a very good aspect of this research. This detection time can be further lessened by using a smartphone that has a processor greater than the mobile's processor used in this research.

As a result, when several plant leaf images have to be examined one after another, this app can save a huge amount of time. Furthermore, the android application will be a guiding tool for the farmers who have no idea about the disease but accurately detecting the disease can lead them to take further steps against the disease and can help them economically. By collecting new leaf photos for different cropping, the model's number of disease classes and accuracy can be increased in the future. The application is now only available for Android devices, although an iOS version could be created in the future.

V. CONCLUSION

In the proposed work, a 2D CNN-based model has been constructed to detect the 12 classes of diseases and two healthy classes in tomato and cotton crops. The proposed 2D CNN-based architecture has three convolutional and three max-pooling layers, which are followed by two fully connected layers. Due to this type of shallow structure, the suggested model has fewer parameters and less storage space consumption than transfer learning models. The average testing accuracy of the model is 97.3%, which has surpassed heavy-weight transfer learning architectures (VGG16, VGG19 and Inception V3) and lightweight transfer learning architectures (MobileNet and MobileNetV2) which are having an average accuracy range from 57% to 92%. The model's performance has also been evaluated using the confusion matrix, ROC curve, and AUC score with the transfer learning models. From the result, it can be revealed that the model has attained an excellent performance that can help the farmers as well as plant doctors to precisely detect various cotton and tomato plant diseases, which can save money for the farmers and aid plant doctors to take accurate measures against the disease. Moreover, this can help the country's economy. The storage space needed by the proposed model is approximately 3 and 4 times lesser than the transfer learning models as the model has very lesser parameters. This lightweight structure can help this model to be implemented on mobile and other devices conveniently. To utilize this research work practically, the model has been implemented in an Android application named "Plant Disease Classifier" for a smartphone-aided plant disease diagnosis system. The average disease detection time is 4.84 ms for the Android application which can be considered almost real-time. To visualize the detection achieved by the trained model, class activation maps were created using the Grad-CAM technique, and a heatmap was created to represent the responsible region for categorization. Nevertheless, deploying a model in practical settings can present various challenges and limitations. One major factor to be mentioned is the computational requirements. The proposed 2D CNN model is fairly lightweight for an Intel/AMD-based computational system. However, for mobile devices, it may require significant computational resources, such as large RAM, CPU/GPU processing power, and fast UFS/NVMe storage. As per several tests, it has been observed that devices that have these types of specifications has very less amount of disease detection time and the detection can be considered almost real-time whereas for devices without such high-quality specifications, the detection can't be considered as real-time. Still, the app's performance and responsiveness are very little affected due to the mobile device's low specification and the diseases can be detected accurately. Collecting continuous image data to keep the model updated is another challenging task because data will be greatly varied due to the environmental or regional variations in cotton and tomato species, diseases prevalent, lighting conditions, and other factors. As new cotton and tomato leaf diseases emerge or the existing ones evolve, the app's average detection accuracy may decrease from 97.3% and inference runtime may increase over 4.84 ms. Also, managing future Plant Disease Classifier model updates and delivering them to end-users can be complex, due to the lack of a system for model versioning, deployment, and maintenance. Regardless of the challenges, the proposed model has outperformed the previous works in terms of precision, recall and F1- score despite having more classes.

In the future, authors have intended to modify the model with a greater number of images with some other crops. Moreover, the accuracy of the model will be improved, and the number of parameters will be reduced to decrease the model's size on the disk.

REFERENCES

[1] T. T. Mim, M. H. Sheikh, R. A. Shampa, M. S. Reza, and M. S. Islam, "Leaves diseases detection of tomato using image processing," in *Proc. 8th Int. Conf. Syst. Model. Advancement Res. Trends (SMART)*, 2019, pp. 244–249.

[2] D. Roy and J. Chakma. (Feb. 20, 2023). "Tomato farmers face peak season plight," The Daily Star. Accessed: Jun. 25, 2023. [Online]. Available: https://www.thedailystar.net/business/economy/news/tomato-farmers-face-peak-season-plight-3252416

[3] *Bangladesh Bureau of Statistics-Government of the People's Republic of Bangladesh*. Accessed: Jun. 25, 2023. [Online]. Available: https://www.bbs.gov.bd/

[4] OEC—The Observatory of Economic Complexity. (2021). *Raw Cotton in Bangladesh | OEC*. [Online]. Available: https://oec.world/en/profile/bilateral-product/raw-cotton/reporter/bgd

[5] *Cotton Development Board*. Accessed: Jun. 25, 2023. [Online]. Available: https://www.cdb.gov.bd/

[6] (Jun. 2013). *Current scenario of Bangladesh Textiles Industry | Scenario of Bangladesh Textiles*. [Online]. Available: https://www.fibre2fashion.com

[7] S. S. Chouhan, A. Kaul, U. P. Singh, and S. Jain, "Bacterial foraging optimization based radial basis function neural network (BRBFNN) for identification and classification of plant leaf diseases: An automatic approach towards plant pathology," *IEEE Access*, vol. 6, pp. 8852–8863, 2018.

[8] J. Shijie, J. Peiyi, and H. Siping, "Automatic detection of tomato diseases and pests based on leaf images," in *Proc. Chin. Autom. Congr. (CAC)*, 2017, pp. 2510–2537.

[9] *PlantVillage Dataset*. Accessed: Aug. 17, 2023. [Online]. Available: https://www.kaggle.com/datasets/emmarex/plantdisease

[10] J. Basavaiah and A. A. Anthony, "Tomato leaf disease classification using multiple feature extraction techniques," *Wireless Pers. Commun.*, vol. 115, no. 1, pp. 633–651, Nov. 2020.

[11] X. Chen, G. Zhou, A. Chen, J. Yi, W. Zhang, and Y. Hu, "Identification of tomato leaf diseases based on combination of ABCK-BWTR and B-ARNet," *Comput. Electron. Agricult.*, vol. 178, Nov. 2020, Art. no. 105730.

[12] M. Agarwal, A. Singh, S. Arjaria, A. Sinha, and S. Gupta, "ToLeD: Tomato leaf disease detection using convolution neural network," *Proc. Comput. Sci.*, vol. 167, pp. 293–301, 2020.

[13] E. Cengil and A. Çınar, "Hybrid convolutional neural network based classification of bacterial, viral, and fungal diseases on tomato leaf images," *Concurrency Comput., Pract. Exper.*, vol. 34, no. 4, p. e6617, Feb. 2022.

[14] S. Khan and M. Narvekar, "Novel fusion of color balancing and superpixel based approach for detection of tomato plant diseases in natural complex environment," *J. King Saud Univ. Comput. Inf. Sci.*, vol. 34, no. 6, pp. 3506–3516, Jun. 2022.

[15] S. Zhao, Y. Peng, J. Liu, and S. Wu, "Tomato leaf disease diagnosis based on improved convolution neural network by attention module," *Agriculture*, vol. 11, no. 7, p. 651, Jul. 2021.

[16] M. S. Al-Gaashani, F. Shang, M. S. A. Muthanna, M. Khayyat, and A. A. A. El-Latif, "Tomato leaf disease classification by exploiting transfer learning and feature concatenation," *IET Image Process.*, vol. 16, no. 3, pp. 913–925, Feb. 2022.

[17] P. Wspanialy and M. Moussa, "A detection and severity estimation system for generic diseases of tomato greenhouse plants," *Comput. Electron. Agricult.*, vol. 178, Nov. 2020, Art. no. 105701.

[18] M. E. H. Chowdhury, "Tomato leaf diseases detection using deep learning technique," in *Technology in Agriculture*, F. Ahmad and M. Sultan, Eds. London, U.K.: IntechOpen, 2021.

[19] Y. A. Jasim, "High-performance deep learning to detection and tracking tomato plant leaf predict disease and expert systems," *Adv. Distrib. Comput. Artif. Intell. J.*, vol. 10, no. 2, pp. 97–122, Feb. 2021.

[20] H.-C. Chen, A. M. Widodo, A. Wisnujati, M. Rahaman, J. C.-W. Lin, L. Chen, and C.-E. Weng, "AlexNet convolutional neural network for disease detection and classification of tomato leaf," *Electronics*, vol. 11, no. 6, p. 951, Mar. 2022.

[21] R. Karthik, M. Hariharan, S. Anand, P. Mathikshara, A. Johnson, and R. Menaka, "Attention embedded residual CNN for disease detection in tomato leaves," *Appl. Soft Comput.*, vol. 86, Jan. 2020, Art. no. 105933.

[22] M. Agarwal, S. K. Gupta, and K. K. Biswas, "Development of efficient CNN model for tomato crop disease identification," *Sustain. Comput., Informat. Syst.*, vol. 28, Dec. 2020, Art. no. 100407.

[23] N. K. Trivedi, V. Gautam, A. Anand, H. M. Aljahdali, S. G. Villar, D. Anand, N. Goyal, and S. Kadry, "Early detection and classification of tomato leaf disease using high-performance deep neural network," *Sensors*, vol. 21, no. 23, p. 7987, Nov. 2021.

[24] A. Bhujel, N.-E. Kim, E. Arulmozhi, J. K. Basak, and H.-T. Kim, "A lightweight attention-based convolutional neural networks for tomato leaf disease classification," *Agriculture*, vol. 12, no. 2, p. 228, Feb. 2022.

[25] V. T. Indu and S. S. Priyadharsini, "Crossover-based wind-driven optimized convolutional neural network model for tomato leaf disease classification," *J. Plant Diseases Protection*, vol. 129, no. 3, pp. 559–578, Jun. 2022.

[26] A. Abbas, S. Jain, M. Gour, and S. Vankudothu, "Tomato plant disease detection using transfer learning with C-GAN synthetic images," *Comput. Electron. Agricult.*, vol. 187, Aug. 2021, Art. no. 106279.

[27] V. Maeda-Gutiérrez, C. E. Galván-Tejada, L. A. Zanella-Calzada, J. M. Celaya-Padilla, J. I. Galván-Tejada, H. Gamboa-Rosales, H. Luna-García, R. Magallanes-Quintanar, C. A. G. Méndez, and C. A. Olvera-Olvera, "Comparison of convolutional neural network architectures for classification of tomato plant diseases," *Appl. Sci.*, vol. 10, no. 4, p. 1245, Feb. 2020.

[28] L. Tan, J. Lu, and H. Jiang, "Tomato leaf diseases classification based on leaf images: A comparison between classical machine learning and deep learning methods," *AgriEngineering*, vol. 3, no. 3, pp. 542–558, Jul. 2021.

[29] S. Nandhini and K. Ashokkumar, "Improved crossover based monarch butterfly optimization for tomato leaf disease classification using convolutional neural network," *Multimedia Tools Appl.*, vol. 80, no. 12, pp. 18583–18610, May 2021.

[30] S. Santurkar, D. Tsipras, A. Ilyas, and A. Ma, "How does batch normalization help optimization?" in *Proc. 32nd Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2018, pp. 1–12.

[31] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[32] H. I. Peyal, S. M. Shahriar, A. Sultana, I. Jahan, and M. H. Mondol, ''Detection of tomato leaf diseases using transfer learning architectures: A comparative analysis,'' in *Proc. Int. Conf. Automat., Control Mechtron. Ind. 4.0 (ACMI)*, 2021, pp. 1–6.

[33] C. U. Kumari, S. J. Prasad, and G. Mounika, ''Leaf disease detection: Feature extraction with K-means clustering and classification with ANN,'' in *Proc. 3rd Int. Conf. Comput. Methodologies Commun. (ICCMC)*, 2019, pp. 1095–1098.

[34] K. Zhang, Q. Wu, A. Liu, and X. Meng, ''Can deep learning identify tomato leaf disease?'' *Adv. Multimedia*, vol. 2018, pp. 1–10, Sep. 2018.

[35] Ö. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, ''3D U-Net: Learning dense, volumetric segmentation from sparse annotation,'' in *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2016*, vol. 9901, S. Ourselin, L. Joskowicz, M. R. Sabuncu, G. Unal, and W. Wells, Eds. Cham, Switzerland: Springer, 2016, pp. 424–432.

[36] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, ''Grad-CAM: Visual explanations from deep networks via gradient-based localization,'' in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.

[37] E. Fujita, H. Uga, S. Kagiwada, and H. Iyatomi, ''A practical plant diagnosis system for field leaf images and feature visualization,'' *Int. J. Eng. Technol.*, vol. 7, nos. 4–11, pp. 49–54, Oct. 2018.

[38] E. Özbilge, M. K. Ulukök, Ö. Toygar, and E. Ozbilge, ''Tomato disease recognition using a compact convolutional neural network,'' *IEEE Access*, vol. 10, pp. 77213–77224, 2022.

[39] *Cotton-Leaf-Infection*. Accessed: Apr. 24, 2022. [Online]. Available: https://www.kaggle.com/raaavan/cottonleafinfection

HASIBUL ISLAM PEYAL received the B.Sc. degree in electrical and computer engineering from the Rajshahi University of Engineering and Technology, Rajshahi, Bangladesh, in 2022. He is currently a Lecturer with the Department of Computer Science and Engineering, Uttara University. He became the Winner and the Finalist in two of the IEEE-based international project competitions for developing an AI-based android application regarding COVID detection along with emergency blood donation and a multipurpose quadcopter with robotic hands. He has published several IEEE conference papers and one Q1-ranked journal on *Sensors* (MDPI) in the relevant fields. His research interests include deep learning and its applications in disease detection and agriculture.

MD. NAHIDUZZAMAN received the B.Sc. degree in computer science and engineering from the Rajshahi University of Engineering and Technology (RUET), Rajshahi, Bangladesh, in 2018. He is currently a Lecturer with the Department of Electrical and Computer Engineering, RUET. He is also a Research Assistant with the Qatar University Machine Learning Research Group. He has several peer-reviewed journal publications. His research interests include machine learning and its applications in disease detection, power sector, and agriculture. He is a reviewer for two prestigious journals, such as *BioMedical Engineering Online* and *Multimedia Systems*.

MD. ABU HANIF PRAMANIK received the bachelor's (Hons.) and M.Sc. degrees in electrical and electronic engineering from the Rajshahi University of Engineering and Technology (RUET), Bangladesh, in 2018 and 2022, respectively. After completing the bachelor's degree, he was a Lecturer with Eastern University, Dhaka, Bangladesh. He is currently a Lecturer with RUET. His research interests include machine learning, dynamic stability of power systems, renewable energy integration, smart grids, and nonlinear control theory.

MD. KHALID SYFULLAH is currently pursuing the degree in electrical and computer engineering with the Rajshahi University of Engineering and Technology. He has a year of experience as a Junior Software Developer and is interested in AI, automation, the IoT, back-end development, and mobile app development. He has won prizes for two IEEE-based international project competitions and has worked on various mobile, web, and IoT projects, including EPUB3 Reader Development, mobile Book Shop Portal, Muslim Virtual Aid System, telemedicine Portal, English Grammar App, and the IoT-Based Home Automation With Security Monitoring. He is skilled in Java and Python programming. His research interests include medical imaging, agriculture, and cryptography.

SALEH MOHAMMED SHAHRIAR received the B.Sc. degree in electrical and computer engineering from the Rajshahi University of Engineering and Technology (RUET), Bangladesh, in 2022. He was the Chief of Electrical of the Formula SAE Team, RUET. His research interests include electric vehicles, machine learning, and the application of machine learning in embedded systems. He has published several journal and conference papers on these topics.

ABIDA SULTANA received the B.Sc. degree in electrical and computer engineering from the Rajshahi University of Engineering and Technology (RUET), Rajshahi, Bangladesh, in 2022. She received a number of honors, including scholarships and merit-based fellowships, which are awards for exceptional performance during the studies. She has published journal and conference papers in relevant fields. Her research interest includes machine learning and its uses in disease detection and agriculture.

MOMINUL AHSAN received the B.Sc. degree from the Department of Computer Science and Engineering, State University of Bangladesh, Dhaka, Bangladesh, the M.Eng. degree (Research) from the Faculty of Engineering and Computing, Dublin City University, Dublin, Ireland, and the Ph.D. degree from the School of Computing and Mathematical Sciences, University of Greenwich, London, U.K. He was a Postdoctoral Researcher with the Department of Engineering, Manchester Met University, before joining the University of York. He is currently an Associate Lecturer with the Department of Computer Science, University of York, U.K. He has published and presented more than 50 technical papers in international journals, conferences, and books. His research interests include prognostics, data analytics, artificial intelligence, machine learning, reliability, power electronics, wireless communication, and wearable technology. He is a member of the Institution of Engineering and Technology (MIET), U.K.; a fellow of the Higher Education Academy (FHEA), U.K.; and an Associate Member of the Bangladesh Computer Society. He was a recipient of the M.Eng. Stipend from Dublin City University, the Ph.D. VC Scholarship from the University of Greenwich, and the Excellent Poster Award from the IEEE International Spring Seminar on Electronic Technology, Sofia, Bulgaria, in 2017.

JULFIKAR HAIDER is currently a Senior Lecturer in mechanical engineering with the Department of Engineering, Manchester Metropolitan University, U.K. He has received funding from Innovate U.K. to conduct eight knowledge transfer partnership projects with the industry worth more than a million pounds. He has published and presented more than 150 technical papers in international journals, conferences, and books. His main research interests include materials processing, thin film coating, composite materials, finite element analysis, and artificial intelligence; and it is in these areas that he is renowned for the work both nationally and internationally justified by the research outputs. He has been supervising research associates and Ph.D. students in the above areas. He is acting as an Executive Editor for an international journal *Advances in Materials and Processing Technologies* (Taylor and Francis) and conferences.

AMITH KHANDAKAR (Senior Member, IEEE) received the B.Sc. degree in electronics and telecommunication engineering from North South University, Bangladesh, the master's degree in computing (networking concentration) from Qatar University, in 2014, and the Ph.D. degree in biomedical engineering from Universiti Kebangsaan Malaysia (UKM), Malaysia, in 2023. He has currently around 100 journal publications, ten book chapters, and three registered patents under his name. His research interests include sensors and instrumentation, electronics, engineering education, biomedical engineering, and machine learning applications. He is a certified Project Management Professional and the Cisco Certified Network Administrator. He graduated as the Valedictorian (President Gold Medal Recipient) of North South University.

MUHAMMAD E. H. CHOWDHURY (Senior Member, IEEE) received the Ph.D. degree from the University of Nottingham, U.K., in 2014. He was a Postdoctoral Research Fellow with the Sir Peter Mansfield Imaging Centre, University of Nottingham. He is currently an Assistant Professor with the Department of Electrical Engineering, Qatar University. He is running NPRP, UREP, and HSREP grants from the Qatar National Research Fund (QNRF) and internal grants (IRCC and HIG) from Qatar University along with academic projects from HBKU and HMC. He has filed several patents and published more than 140 peer-reviewed journal articles, more than 30 conference papers, and several book chapters. His current research interests include biomedical instrumentation, signal processing, wearable sensors, medical image analysis, machine learning and computer vision, embedded system design, and simultaneous EEG/fMRI. He is a member of British Radiology, ISMRM, and HBM. He has recently won the COVID-19 Dataset Award, the AHS Award from HMC, and the National AI Competition awards for the contribution to the fight against COVID-19. His team is the gold medalist at the 13th International Invention Fair in the Middle East (IIFME). He is serving as the Guest Editor for *Polymers*, an Associate Editor for IEEE Access, and a Topic Editor and a Review Editor for *Frontiers in Neuroscience*. He has been listed among the Top 2% of scientists in the World List–2022, published by Stanford University.

IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 15 September 2023, accepted 4 October 2023, date of publication 16 October 2023, date of current version 20 October 2023.

Digital Object Identifier 10.1109/ACCESS.2023.3324722

TOPICAL REVIEW

Machine Learning and Deep Learning for Plant Disease Classification and Detection

VASILEIOS BALAFAS, EMMANOUIL KARANTOUMANIS, MALAMATI LOUTA, (Senior Member, IEEE), AND NIKOLAOS PLOSKAS

Department of Electrical and Computer Engineering, University of Western Macedonia, 50100 Kozani, Greece

Corresponding author: Nikolaos Ploskas (nploskas@uowm.gr)

This work was supported by the European Regional Development Fund of the European Union and Greek national funds through the Operational Program Competitiveness, Entrepreneurship and Innovation under Project MIS 5047196. The publication of the article in OA mode was financially supported by HEAL-Link.

ABSTRACT Precision agriculture is a rapidly developing field aimed at addressing current concerns about agricultural sustainability. Machine learning is the cutting edge technology underpinning precision agriculture, enabling the development of advanced disease detection and classification methods. This paper presents a review of the application of machine learning and deep learning techniques in precision agriculture, specifically for detecting and classifying plant diseases. We propose a novel classification scheme that categorizes all relevant works in the associated classes. We separate the studies into two main categories depending on the methodology that they use (i.e., classification or object detection). In addition, we present the available datasets for plant disease detection and classification. Finally, we perform an extensive computational study on five state-of-the-art object detection algorithms on PlantDoc dataset to detect diseases present on the leaves, and eighteen state-of-the-art classification algorithms on PlantDoc dataset to predict whether or not there is a disease in a leaf. Computational results show that object detection accuracy is high with YOLOv5. For the image classification task, the networks ResNet50 and MobileNetv2 have the most optimal trade-off on accuracy and training time.

INDEX TERMS Classification, deep learning, disease detection, machine learning, object detection, precision agriculture.

I. INTRODUCTION

Any nation's economic development depends significantly on agriculture. Meeting the population's current food needs has become a difficult problem because of the growing population, frequent changes in weather, and limited resources. On top of the aforementioned challenges, crop diseases have been increasing in severity and scale. Crop diseases cause production losses that can be mitigated with continuous monitoring. Researchers from the Food and Agriculture Organization of the United Nations predicted that plant diseases alone cost the global economy about US$220 billion annually [1]. Developing new methods to detect diseases on plants or leaves at an early stage can significantly increase yield potential.

The associate editor coordinating the review of this manuscript and approving it for publication was Pasquale De Meo.

Precision agriculture is a rapidly developing field aimed at addressing current concerns about agricultural sustainability. Machine learning (ML) is the cutting edge technology underpinning precision agriculture, allowing the machine to learn without having to be programmed directly, and in conjunction with Internet of Things (IoT) enabled farm equipment, is the future of agriculture. There are several studies that have employed or proposed ML methods to detect or classify plant diseases. The majority of these works take as input a plant/leaf image and detect whether or not there is a disease. These works treat the problem as a classification one, either binary classification (healthy or diseased plant/leaf) or multi-class classification (where various diseases are targeted). Classical ML methods, like Random Forest (RF) and Deep Learning (DL) ones, have been utilized for this purpose. On the other hand, there are fewer works that aim to detect both the type of the disease and the diseased regions of the input image, i.e., treating the problem as an object detection one. The latter problem is more important than the classification problem in cases where either there exist multiple plant diseases in the input image or we want to know the exact region of the diseased plants in an image that covers a large part of the crop (such images are taken by Unmanned Aerial Vehicles (UAVs)). Moreover, the object detection problem is more difficult than the classification problem and the DL methods used to detect objects do not have good results in uncontrolled places, i.e., in images where the objects are placed in a noisy environment.

In this paper, we aim to review ML and DL methods that have been applied to either classify or detect plant diseases. Various papers reviewed ML and DL methods in precision agriculture. Below, we present those papers that include works focused on plant disease detection. For example, Mekonnen et al. [2] provided a study of how different ML techniques are used in sensor data analytics in the agricultural sector. The authors highlighted the benefits of Artificial Intelligence (AI) in agriculture and emphasized that IoT and AI are increasingly becoming essential components for enhancing agricultural productivity and efficiency. They concluded that although clustering is the most common learning model for such problems, there is no general prescription for selecting algorithms, and most of the time, it is a trial and error process in doing so. Finally, they examined some of the ML methods used in agriculture such as regression algorithms, Decision Trees (DT), ensemble learning, Bayesian models, Support Vector Machines (SVM), and Artificial Neural Networks (ANNs).

Benos et al. [3] presented a review of various works aimed at exploring ML in agriculture. Initially, they described the four basic categories in which ML algorithms are involved. Crop management is subdivided into five categories (yield prediction, disease detection, weed detection, crop recognition, and crop quality), water management, soil management, and livestock management. They performed an extensive search on various search engines and selected articles for the period from 2018 to 2020. Finally, they concluded that: (i) a large percentage of research articles were about crop management, (ii) ANNs were the most effective ML models, (iii) the most investigated crops in the papers were maize, wheat, rice, and soybean, and finally, (iv) the most utilized input data was mainly RGB images and then weather, soil, water, and crop quality.

Li et al. [4] presented a review of the research progress of DL methods for crop leaf disease identification. Initially, they reviewed the relevant metrics, the datasets used in the computational experiments, and the data enhancement methods. Then, they presented works based on DL methods in order to predict diseases in crop leaves. Finally, they concluded that: (i) most DL frameworks are not robust, achieving good results only on specific datasets, (ii) most works use synthetic images generated in the laboratory, and (iii) studies focused on early detection of disease are limited.

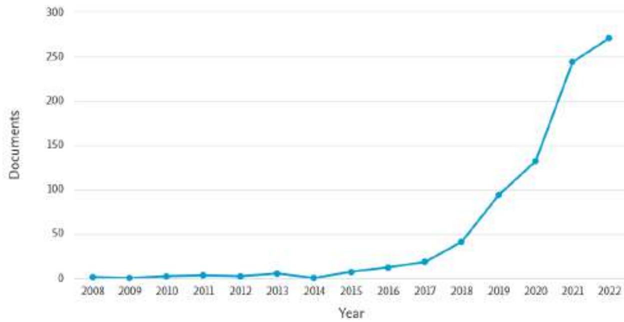


FIGURE 1. Publications per year for papers including the words "plant", "disease", "detection", and "machine learning" either on their title or abstract or keywords (source: Scopus).

Liu and Wang [5] reviewed works focusing on methods for detecting plant diseases and pests published during the period from 2014 to 2020. They presented various methods proposed in the literature and discussed their advantages and disadvantages. They also presented the publicly available datasets for plant diseases and pest detection methods. Finally, they concluded that: (i) the majority of research findings are limited to the laboratory setting and are solely relevant to plant disease and pest images obtained during that period, (ii) early diagnosis is very difficult, and thus, only a few works deal with this, and (iii) less time- and memory-consuming DL models are in need for real-time prediction of diseases and pests.

Yuan et al. [6] examined the progress in technologies utilized for agricultural disease image recognition such as DL and transfer learning. They also analyzed key challenges that require attention to progress research in this field, including creating image datasets, choosing big data auxiliary domains, and improving transfer learning techniques. It is essential to create image datasets captured under actual production conditions to develop a feasible image recognition system for agricultural diseases.

Although there are numerous review papers discussing ML and DL techniques used in precision agriculture, there are only a few review papers that focus on plant disease detection methods, and none of them discusses the two main techniques used in the literature, classification and object detection methods. Typically, the review papers devote a section for plant disease detection methods. In addition, all reviews present works published until 2020. As shown in Figure 1, the publications that deal with ML and DL methods for plant disease detection have doubled since 2020. Therefore, there is a need for a comprehensive review on ML and DL methods for plant disease detection and classification.

In addition, computational comparisons of algorithms used for plant disease detection and classification are missing in the literature. Most papers in this area compare their proposed algorithm with a single state-of-the-art method. In this work, we present a comparison of five state-of-the-art object detection algorithms for identifying diseases on plants and eighteen state-of-the-art classification algorithms for plant disease classification on a widely used dataset.

Our contributions can be summarized as follows:

- We present a detailed review of ML and DL algorithms for plant disease detection and classification. Most studies review works until 2020 and we also include recent works (published until June 2022). We distinguish these works based on the goal of the methods used, i.e., whether classification or object detection is targeted. We present a thorough review on 65 works for classification and 14 works for object detection.
- We propose a novel classification scheme that categorizes all relevant works in the associated classes. By classifying relevant papers according to these classes, we can easily provide an overview of different techniques used in the field.
- We provide a summary of the existing datasets that are being used for plant disease detection and classification tasks. We include details about the types of data they contain, as well as the classes and labels available within these datasets.
- We perform extensive computational analysis on five state-of-the-art object detection algorithms for plant disease detection and eighteen state-of-the-art classification algorithms for plant disease classification on a widely used dataset. This is the first time that such a systematic study has been carried out.
- We discuss gaps in the existing literature and future directions for this field.

The structure of the paper is as follows. Section II reviews research works that proposed/utilized ML and DL algorithms for predicting/recognizing plant diseases. In this section, we separate the papers according to whether they used classification or object detection techniques. In Section III, we propose a classification scheme, categorize all relevant works in the associated classes and present statistics of the most common methods/algorithms/datasets/metrics used by the reviewed papers. Section IV provides an overview of the datasets that are commonly used in the literature for plant disease detection and classification using both object detection and classification techniques. Section V includes the computational comparison of five state-of-the-art object detection algorithms for plant disease detection and eighteen state-of-the-art classification algorithms for plant disease classification. In Section VI, we highlight the practical implication of our findings. Section VII includes the open challenges that need to be addressed, while Section VIII includes the future directions for addressing these open challenges. Finally, Section IX summarizes the work of this paper.

II. LITERATURE REVIEW

This section provides a detailed literature review of disease detection techniques. All papers have been published prior to October 2022. To identify relevant studies on ML and DL methods for plant disease detection and classification, we conducted a search using various search engines, including Scopus, ScienceDirect, Scholar, and Web of Science. Keywords like ‘‘machine learning’’, ‘‘deep learning’’, ‘‘classification’’, ‘‘disease detection’’, ‘‘healthy plant’’, and ‘‘diseased plant’’, were used. Once a relevant work was identified, its references were scanned in order to find relevant works that were not found in the initial search procedure. The abstract of each paper was reviewed in order for all co-authors to decide its appropriateness and inclusion in the paper. All papers that did not have a focus on the ML algorithms were excluded. For example, there exist papers that refer to plant disease detection using ML methods but their main focus is the data gathering part, e.g., through UAVs or Unmanned Ground Vehicles (UGVs). In addition, non-English texts, technical reports, review papers, and Master's and Doctoral theses were excluded. Overall, 79 papers (65 for classification and 14 for object detection) were discovered.

A. CLASSIFICATION

Mohanty et al. [7] trained AlexNet [8] and GoogLeNet [9] for detecting 26 diseases on 14 crop species using the PlantVillage dataset [10] with 54, 306 images. In the preprocessing step, they downsized images to 256×256. They also experimented with the original PlantVillage dataset, with a gray-scaled version of this dataset, and a version in which the leaves were segmented. The models were trained from scratch as well as by utilizing pre-trained models on the ImageNet dataset through transfer learning. The authors reported an accuracy of 99.35% and 85.53% using the GoogLeNet and AlexNet models, respectively.

A plant disease detection approach based on Convolutional Neural Network (CNN) using the Caffe DL framework [11] was proposed by Sladojevic et al. [12]. They collected images from various sources and they used data augmentation techniques (affine transformation, perspective transformation, and rotation) to generate more images. In the preprocessing step, they downsized images to 256×256 and removed the duplicates. In addition, they did not consider images with smaller resolution than 500px. They modified the CaffeNet architecture by altering the last layer and the output of the softmax layer. They achieved an accuracy of 96.3% on average.

In their work, Amara et al. [13] applied the LeNet architecture [14] to classify diseases in banana leaves, belonging to three different categories. In the preprocessing step, they downsized images to 60×60 and converted to grayscale. They tested the proposed architecture on the PlantVillage dataset, achieving an accuracy between 92% and 99%.

Brahimi et al. [15] used two CNN models, AlexNet and GoogleNet, on the PlantVillage dataset for predicting diseases on tomato leaves. They pre-trained the models on ImageNet and then fine-tuned them by replacing the output layer with a new one having nine classes, as the number of the diseases that they consider in this study. In the preprocessing step, they downsized images to 256×256. GoogleNet achieved an accuracy from 97.71% to 99.19%, while AlexNet achieved an accuracy from 94.35% to 98.66%.

Cruz et al. [16] implemented a CNN network based on the LeNet architecture for detecting olive quick decline symptoms. They initially trained the network on the PlantVillage dataset, then retrained on a custom dataset. They improved the performance of the network by providing information about the edge patterns and the shape information. In the preprocessing step, they downsized images to 256×256. The authors reported an accuracy of 99%.

DeChant et al. [17] proposed a three-stage approach using CNN models for identifying NLB-infected maize plants. A three-stage approach was utilized in this study by initially training various CNNs to detect lesions in small patches of the images. Then, the CNNs generated heat maps indicating regions in the images that may be affected. Finally, they trained the CNN models using as input the heat maps of the original images. They generated a custom dataset of 1, 796 images. Their approach resulted in a classification accuracy of 96.7%.

In their study, Liu et al. [18] employed the AlexNet architecture to detect apple leaf diseases, utilizing a dataset of 13, 689 images that contained four diseases: mosaic, brown spot, rust, and Alternaria leaf spot. Prior to classification, they performed image preprocessing by rotating and sharpening the images. Their approach yielded an overall accuracy of 97.62

Lu et al. [19] introduced a multi-stage CNN architecture that was based on AlexNet, aimed at identifying diseases in rice plants. The authors collected images for their study from an agricultural pest and insect pests database, as well as from a book that contained images of diseased plants. They preprocessed the images by resizing them to 512×512 pixels and then applied the ZCA-Whitening technique to remove the correlation between data. The accuracy of the proposed model was 95.48%.

Oppenheim and Shani [20] used a CNN based on VGG [21] to classify potatoes into five classes, four diseased ones and a healthy potato class. They acquired 400 images of contaminated potatoes using three simple digital cameras and augmented data by flipping and cropping the images. In the preprocessing step, they downsized images to 224×224 and converted to grayscale. They experimented with different ratios of train-test sets and concluded that the CNN can achieve an accuracy between 83% for the model trained with only 10% of the data to 96% for the model trained with 90% of the data.

Wang et al. [22] trained and fine-tuned four CNN models, i.e., VGG16 [21], VGG19 [21], InceptionV3 [23], and ResNet50 [24], for detecting apple leaf black rot. The authors utilized the PlantVillage dataset, which contained two classes of images: healthy apple leaves and apple leaf black rot. The images were further categorized by botanists into four stages: healthy, early stage, middle stage, and end-stage. In the preprocessing step, they downsized images to various ratios according to the CNN that they used. They evaluated several models, and the VGG16 model produced the best results, achieving an accuracy of 90.4%.

Abdulridha et al. [25] demonstrated a non-destructive remote sensing approach for detecting the Laurel Wilt disease in infected avocado trees. They used a portable spectral data collection system to classify the data into healthy or not based on causes that elicit comparable symptoms, such as iron and nitrogen deficiency. Two sets of images with spectral resolutions of 10nm and 40nm were gathered in order to feed the DT and MultiLayer Perceptron (MLP) neural network model. They concluded that MLP has a near-perfect classification accuracy (100% at the early stage and 91% at the late stage) compared to DT (82% at the early stage and 82% at the late stage).

Barbedo [26] investigated the major factors influencing the design and performance of CNNs for plant disease recognition. To anticipate corn diseases, they used the PDDB dataset [27] that includes 50,000 images and 171 diseases. They also studied nine factors that influence disease detection in maize fields. They trained the model with four different datasets and the best accuracy was 87% with a subdivided dataset.

Ferentinos [28] employed simple leaf images to perform plant disease detection and analysis using CNN models. They used the PlantVillage dataset with 87,848 images. This dataset has 25 plants and 58 distinct classes. Furthermore, five different CNN algorithms were used: AlexNet, AlexNetOWTBn [29], GoogLeNet, Overfeat [30], and VGG. In the preprocessing step, they downsized images to 256×256. The best results were obtained by the VGG algorithm with 99.48% success rate at the original image dataset and 98.87% success rate at the preprocessed image dataset.

Lu et al. [31] studied the possibility of detecting multiple diseases on tomato leaves at different stages using a portable high-resolution spectral sensor. They utilized the K-Nearest Neighbor (KNN) algorithm to classify the sensor data into four categories: healthy, asymptomatic, early stage, and late stage. Principal Component Analysis (PCA) was used to analyze 57 spectral vegetation indices. They used six principal components and two spectral vegetation indices compositions. The healthy leaves had an accuracy between 85.7% and 100%, the asymptomatic leaves between 86.4% and 100%, the early stage leaves between 73.5% and 93.0%, and the late stage leaves between 77.1% and 100%, depending on the principal component.

In their study, Fuentes et al. [32] proposed a diagnosing system based on a refinement filter bank for detecting diseases and pests on tomato plants. They aimed to overcome the issues of class imbalance and false positives by constructing a refinement filter bank framework. The primary diagnosis unit was the bounding box generator, which was used to create bounding boxes containing the diseased area's position. The secondary diagnosis unit was the CNN filter bank, which was used to validate the results. The proposed method achieved a recognition rate of around 96% on a set of 5, 000 images collected from various tomato farms in South Korea.

Kerkech et al. [33] presented a method that uses a CNN, namely LeNet, and histograms to forecast diseased areas of grapevines and identify symptoms in vineyards. In order to overcome memory constraints, the streaming technique was utilized in the learning process for loading a small amount of data in each iteration. They collected the data through visible-domain images taken by UAVs. They conducted experiments by utilizing various color spaces, vegetation indices, and their combination, to detect plant diseases. The approach achieved an accuracy of 95.8

Pineda et al. [34] predicted the disease caused by the bacterium Dickeya dadantii in melon leaves. They used three ML algorithms to train models and they used thermography and fluorescence imaging techniques to produce more accurate results. The ML algorithms that they used were Logistic Regression Analysis (LRA), SVM, and ANNs. They found that ANNs had better accuracy than the other models (99.1%) when classifying images as entire leaves.

Sharif et al. [35] proposed a two-phase hybrid technique for detecting and categorizing diseases in plants, specifically citrus fruits and leaves. The first phase involved identifying lesion areas, while the second phase focused on classifying the type of disease present. To select the most relevant features, the authors used a hybrid feature selection method that included PCA score, entropy, and a skewness-based covariance vector. This information is provided as input into the MultiClass SVM (M-SVM) for disease classification. They used images from open databases (PlantVillage and Citrus Diseases Database [36]) and a custom-collected image database. The proposed method achieved an accuracy of 97%.

Zhang et al. [37] enhanced maize leaf disease recognition accuracy and reduced the number of network parameters. They gathered 500 images from several sources, including the PlantVillage dataset, and divided the data into eight categories for diseased maize leaves and one category for healthy leaves. To increase the sample of images, they rotated the images by 90°, 180°, and 270°. They used Cifar10 [38] and GoogLeNet, changing the base learning rate to impact the network's identification accuracy. They additionally tweaked hyperparameter settings and enhanced the models. In the preprocessing step, they downsized images to 224×224. They achieved an accuracy of 98.9% for the GoogLeNet model and 98.8% for the Cifar10 model.

An early disease detection technique for avocado trees to identify Laurel Wilt was proposed by Abdulridha et al. [39]. They classified the data into five main categories: Laurel Wilt disease, healthy trees, trees infected by phytophthora root rot, and trees with iron and nitrogen deficiencies. To produce better results, they used image acquisition and preprocessing techniques, and feature extraction techniques. The classification was performed with two classification methods, MLP and KNN. They used a custom dataset acquired through two sensing systems. The best results to predict Laurel Wilt was achieved from the MLP method with an accuracy of 99%.

Al-Saddik et al. [40] studied the optimal spectral bands for the development of a multispectral camera used on a UAV for recognizing diseased grapevine fields with the disease Flavescence dorée. The specific disease is a contagious one that is incurable and can lead to severe production loss. For selecting the optimum spectral bands capable of distinguishing infected from healthy leaves, two spectral analysis methodologies were presented. The first one performs a feature selection strategy, utilizing the successive project algorithm with several spectral preprocessing approaches. The second method looks at a variety of classic vegetation metrics. The two classifiers employed in this research are the SVM and discriminant analysis. The most accurate models were calculated as a function of the grapevine variety in question. The successive projection algorithm approach was better when it came to common vegetation metrics, with an accuracy in classification greater than 96%.

A new dataset consisting of a large number of labeled images of leaves captured in real environments was introduced by Arsenovic et al. [41]. They also proposed an augmentation technique that utilizes Generative Adversarial Networks (GANs). A two-stage CNN algorithm was then proposed for plant disease detection, and the proposed model achieved an accuracy of 93.67%.

The focus of Barbedo [42] was on a distinct issue related to disease detection. They pointed out that current databases lacked the necessary images to accurately represent the various conditions and characteristic symptoms, which could not be reproduced using existing data augmentation techniques. As a result, they looked at using particular lesions and areas for the task, instead of looking at the entire leaf. They managed to expand the existing database because each region has its own characteristics. They created a dataset with images taken from typical cameras, smartphones, or DSLR cameras. In the preprocessing step, they downsized images to $224 \times 224 \times 3$. They achieved 12% higher accuracy than in the original image dataset. Four categories were predicted: healthy leaves with an accuracy of 89%, leaves with mild disease with an accuracy of 31%, leaves with moderate disease with an accuracy of 87%, and leaves with severe disease with an accuracy of 94%.

Coulibaly et al. [43] developed a feature extraction-based methodology that was predicated on the CNN model VGG16, which has been pre-trained on ImageNet. They used 124 images of pearl millet from the ImageNet dataset, with mildew diseases and healthy plants. Additionally, a data augmentation technique was utilized to augment the dataset by increasing the number of images. In the preprocessing step, they downsized images to 150×150. Transfer learning was employed with feature extraction, resulting in an accuracy of 95%.

Dhingra et al. [44] presented methods for recognizing the diseases on the leaves of basil using digital image processing techniques. They used nine different classifiers to classify the diseases. The researchers gathered images from an herb garden and standardized the surface condition by removing any non-uniform distribution of dust across the leaves, ensuring consistency across all leaf categories. The nine classifiers used were DT, SVM, linear models, Naives Bayes, KNN, AdaBoost, discriminant analysis, RF, and ANNs. The classification separated the images into two categories, diseased and healthy. The dataset includes 200 healthy and 200 diseased leaves. Out of the nine classifiers used, RF achieved the highest accuracy of 98.4%.

Hu et al. [45] presented a low-shot learning method to predict diseases in tea leaves. They created their image dataset by using UAVs. The tea disease spots in the images were separated using the SVM method. Conditional Deep Convolutional Generative Adversarial Networks (CDCGAN) [46] were used to supplement disease spot samples, while VGG16 DL networks are used to identify disease spots. They compared five different methods: SVM, DT, RF, and two different CNN methods, CDCGAN and VGG16, and they concluded that CNN achieved better results. The results obtained for the prediction of tea red scab, tea red leaf spot, and tea leaf blight, reached an average accuracy of 90% for the CNN methods.

The goal of Huang et al. [47] was to develop a decision support system for spraying machines that could identify Helminthosporium leaf spots in wheat fields using remote sensing data from a UAV. The dataset was divided into four categories based on disease severity: normal, light, medium, and heavy. In the preprocessing step, they extracted 100×100 samples from the images. A CNN was used for classification, and the proposed method achieved an accuracy of 91.43%.

In their study, Abdulridha et al. [48] aimed to detect two diseases in tomato crops using hyperspectral imaging captured by a UAV in the range of 380–1020nm. The tomato leaves were classified into four categories: healthy, asymptomatic, early, and late disease stages. They generated 35 spectral vegetation indices to identify the best indicators for disease detection and diagnosis and used MLP and Stepwise Discriminant Analysis (STDA) classification methods. They found that the blue band (408–420nm), red band (630–650 nm), and red edge (730–750 nm) were the most effective wavebands for disease detection. The spectral signatures of the two diseases were significantly different at all stages of disease progression, despite having similar symptoms in the early stages. Both laboratory and field tests showed satisfactory results for detecting both diseases in asymptomatic, early, and late stages. The MLP algorithm outperformed STDA, achieving an accuracy of 97% to 99% for all stages.

Abdulridha et al. [49] proposed nondestructive methods to detect diseases that affect tomato crops in two different datasets, a dataset created at the laboratory (benchtop scanning) and a dataset with images captured in the field (using a UAV). These images consisted of images from healthy and diseased plants. The diseases that they targeted were: bacterial spot, target spot, and tomato yellow leaf curl. The diseased plants were identified and classified using several Vegetative Indices and the M statistic technique. They utilized two classification methods, STDA and RBF. For all diseases, both in the symptomatic and asymptomatic stages (i.e., before symptoms become obvious to direct visual observations), the classification results showed extremely accurate discrimination between healthy and diseased plants. In the field, the best classification results were achieved for tomato yellow leaf curl, bacterial spot, and target spot with accuracies of 100%, 98%, and 96%, respectively.

In their study, Abdulridha et al. [50] aimed to detect Powdery mildew in squash at various stages of disease development, including asymptomatic, early, intermediate, and late stages. They utilized hyperspectral imaging (380–1020nm) and machine learning (ML) algorithms to perform the detection, collecting data both in the lab and in the field using a UAV. The radial basis function (RBF) was used as the classification method, and the most significant bands for distinguishing between healthy and diseased stages were identified (388nm, 591nm, 646nm, 975nm, and 1012nm). The RBF method was able to diagnose Powdery mildew disease even in the asymptomatic phases, achieving a classification accuracy of 82% and 89% for the asymptomatic and early stages, respectively, in both laboratory and field settings. The best classification results were obtained in the very late stages of disease progression, with 96% and 99% accuracy in laboratory and field settings, respectively.

Agarwal et al. [51] introduced a CNN model with eight hidden layers that is lightweight and designed for classifying nine types of diseases in tomato crops. The model outperformed traditional ML techniques and pre-trained models with an accuracy of 98.4% on the publicly available PlantVillage dataset, which had 200–1400 images in different classes. The study used various assessment metrics such as accuracy, precision, recall, and F1-score to evaluate the performance of the proposed algorithm. The authors also employed image preprocessing and augmentation techniques to enhance the performance of the CNN. In the preprocessing step, they downsized images to 128×128. Overall, the proposed model achieved a high accuracy of 98.4% on the PlantVillage dataset.

Anagnostis et al. [52] proposed a CNN model that can categorize images of walnut tree leaves based on whether or not they are infected with anthracnose. For this purpose, a set of grayscale and RGB images were used and Fourier transformation was applied to extract features. A CNN architecture was chosen on the basis of its performance. Feature extraction is significantly impacted by the Fourier transformation in grayscale images, as it emphasizes the abrupt changes and edges of the leaves. Infected leaves are more edgy in comparison to healthy ones. In the preprocessing step, they downsized images to

256 × 256. The proposed CNN architecture has better or similar performance to widely-used CNN architectures such as ResNet50, VGG16, InceptionV3, and DenseNet121 [53], which are often used as benchmarks in image classification challenges. The accuracy achieved by the proposed CNN architecture was 99%.

Chen et al. [54] compared Lasso, RF, gradient boosting, and generalized linear models in predicting the possibility of a high rate of occurrence and severity of grape downy mildew using a dataset of nine years of observations. The algorithms evaluated in this study use the date of initial infection, as well as average monthly temperatures and precipitation as input parameters. Among these algorithms, Lasso, RF, and gradient boosting perform better than generalized linear models. Precipitation is found to have a larger impact on accuracy than temperature among weather inputs, while the date of disease onset has a greater impact on accuracy than weather inputs. The best-performing algorithm is selected based on results for high incidence and severity on leaves and bunches and is used to assess the influence of various climatic scenarios on grape downy mildew risk levels. The study found that reduced rainfall and higher temperatures in April-May decrease the probability of grape downy mildew during bunch closure. The highest area under the ROC curve achieved in this study was 86%.

An image processing technique for identifying plant diseases was proposed by Cristin et al. [55] that is efficient. The input image undergoes preprocessing to eliminate noise and artifacts. The preprocessed image is then segmented using a piecewise fuzzy C-means clustering method to obtain the segments. Texture features such as information gain, histogram of oriented gradients, and entropy are extracted from each segment during the feature extraction stage. A Deep Belief Network (DBN) is employed in the classification step, which was trained with the Rider-CSA algorithm, combining the Rider Optimization Algorithm (ROA) with the Cuckoo search method. The Rider-CSA-based DBN method proposed in the study outperformed previously used methods, with a maximum accuracy of 87%, sensitivity of 86%, and specificity of 87% in the experiments.

Lee et al. [56] used four CNN models, i.e., VGG16, InceptionV3, GoogLeNet, and GoogLeNetBN [57], for plant disease identification. They utilized three datasets, PlantVillage, IPM [7], and Bing [7]. They compared the performance trained from scratch and using transfer learning from existing pre-trained models. In the preprocessing step, they downsized and cropped images to different ratios according to the type of CNN. The VGG16 network performed better than the other CNN architectures, achieving an accuracy of 99% on seen crops and 65% on unseen crops.

Giraddi et al. [58] presented a DL system with image processing techniques to detect fungal diseases in maize leaves. They used image classification techniques along with AlexNet, a deep CNN. Their model can detect three different classes of fungal disease, Northern blight, Southern Blight, and Rust. They experimented with two different problems. The first problem is with three classes: healthy, common rust, and Northern blight images, and the second problem is with four classes: healthy, common rust, Northern blight, and multiple diseases in one image. For the first problem, three distinct disease detection models were generated, trained, and evaluated. In the preprocessing step, they downsized images to make them the same size. They used 300 images in the training set, 100 for each class. They also used 150 images in the testing set, 50 for each class. The training set for the second problem consists of 400 images, 100 for each class, while the testing set includes 200 images, 50 for each class. They obtained an accuracy of 98% on the three-class classification problem and an accuracy of 70.5% on the four-class classification problem.

A CNN-based approach for plant disease diagnosis and recognition is proposed by Guo et al. [59]. The model is designed to improve accuracy, generality, and training efficiency. The approach first uses the Region Proposal Network (RPN) [60] to detect and locate the leaves in a challenging environment. Then, the Chan-Vese (CV) method [61] is used to segment the images based on the results of the RPN algorithm, which captures the features of the symptoms. Finally, the segmented leaves are fed into the transfer learning model, which is trained using the PBPC [62] of diseased leaves in a simple environment. The approach is tested for black rot, bacterial plaque, and rust disease, achieving an accuracy of 83.75%, which is higher than conventional methods, and can help minimize the impact of diseases on agricultural production.

Habib et al. [63] introduced an agromedical expert system for papaya disease identification that relies on machine vision and analyzes images taken using portable devices. The proposed system focuses on two main objectives: disease detection and disease classification. The effectiveness of the system was tested through several experiments. Initially, they defined a set of features that can help differentiate between the different diseases. Then, they used the K-means clustering algorithm to segment the disease-affected area in the collected images and extracted the relevant features to classify the diseases using the support vector machine (SVM) algorithm. In the preprocessing step, they downsized images to 300 × 300. The system achieved an accuracy of 90.15%.

Karadağ et al. [64] proposed a method for identifying fusarium disease on peppers utilizing a spectroradiometer from the reflections obtained from pepper leaves in a laboratory. Four sets of pepper leaves were grown in a confined setting to obtain reflections. They proposed a two-step disease detection framework. Subsequently, the extracted feature vector was utilized in the classification process, employing ANNs, Naive Bayes (NB), and KNN. The obtained accuracy for KNN was 100%, 97.5% for ANN, and 90% for NB.

Karlekar and Seal [65] proposed a computer vision method to identify and categorize leaf diseases present in soybean crops. Initially, the proposed approach separates the leaf section by removing the complex background from the entire image. Following this, a deep learning Convolutional Neural Network (CNN) known as SoyNet, trained on a 16-class dataset called PDDB [27], categorizes the segmented leaf images. In the preprocessing step, they downsized images to 100 × 100. The proposed model attains an identification accuracy of 98.14%, with high precision, recall, and F1-score.

Karthik et al. [66] proposed two alternative deep neural network (DNN) models to identify the type of infection present in tomato leaves. The first model applies residual learning on top of a feed-forward CNN to learn crucial classification features, while the second model utilizes the strengths of the attention mechanism and residual learning on CNNs. To assess the performance of the proposed models, the PlantVillage dataset was used, which includes three diseases: early blight, late blight, and leaf mold. They reported an accuracy of 95% and 98% for the two architectures, respectively.

The detection of mildew disease in vineyards using a CNN on UAV images was proposed by Kerkech et al. [67]. Additionally, they introduced a new object recognition technique that aligns visible and infrared images, allowing data from the two sensors to be combined. A fully CNN technique is then used to classify each pixel into different categories, such as ground, shadow, symptom, and healthy. The proposed method was able to detect over 92% of infections at the grapevine level and 87% at the leaf level, indicating that automatic disease detection in vineyards has a promising future.

Khalili et al. [68] experimented with six different ML algorithms, namely MLP, Gradient Tree Boosting (GBT), L1 regularization (LR-L1), L2 regularization (LR-L2), SVM, and RF in order to forecast charcoal rot disease in soybean on a custom dataset [69] of 2,000 healthy and diseased plants. The importance of the features considered by various ML methods was also analyzed. They applied a 10-fold cross-validation to the training set and all ML models achieved an accuracy of more than 96.79%.

The use of an optimized deep neural network for identifying and classifying paddy leaf infections in rice harvests was proposed by Ramesh et al. [70]. The Jaya algorithm was utilized for this study. Images of rice plant leaves from farm fields with normal, sheath rot, blast diseases, brown spots, and bacterial blight were collected. In preprocessing, RGB images were converted to HSV images to eliminate the background, and then binary images based on the hue and saturation sections were retrieved to determine whether or not a disease occurred in a region. Several metrics, such as accuracy, precision, and F1-score, were used to evaluate the results, and the outcomes were compared to those of Denoising Auto-Encoder (DAE), DNN, and ANN classifiers. In the preprocessing step, they downsized images to 300 × 450. The proposed method achieved high accuracy rates of 98.9% for blast-affected images, 95.7% for bacterial blight, 94% for brown spot, 92% for sheath rot, and 90.57% for normal leaf images, outperforming the other classifiers.

Verma et al. [71] trained and compared the CNN models SqueezeNet [72], InceptionV3, and AlexNet, for determining the severity of late blight infection in tomatoes. For this purpose, they utilized the PlantVillage dataset for transfer learning and feature extraction. AlexNet outperformed the other two networks with an accuracy of 93.4%.

Velásquez et al. [73] proposed a technique to detect coffee leaf rust by utilizing data from various sources, including sensor data in JSON format, as well as RGB, RGN, and RE images from UAV multispectral cameras. The proposed approach involved the integration of remote sensing and DL algorithms, specifically MLP and CNN. The sub-models were trained separately, and the resulting F1-scores for each sub-model and the composite model were calculated. The composite model achieved an F1-score of 77.5%.

Yan et al. [74] presented an enhanced version of the VGG16 model to detect diseases in apple leaves. They utilized a dataset that consisted of 10 plant types with 27 different disease categories obtained from the "2008 'AI Challenger' Global Challenge" [75], as well as additional images. The authors employed transfer learning to shorten the learning time. The results demonstrate that the proposed model achieved an overall accuracy of 99.01%.

Zhang et al. [76] presented a structured technique for wheat lodging detection in experimental plots that included aerial imagery collected from UAVs. For classifying wheat lodging, ML algorithms, and CNNs were utilized. For the ML algorithms, 320 extracted features were fed into RF, NN, and SVM. For the CNNs, they used VGGNet and GoogLeNet. GoogLeNet had significantly higher accuracy than VGGNet, about 93%, while SVM had an accuracy of 92%.

Abbas et al. [77] proposed a DL approach for detecting diseases on tomato plant leaves. They utilized the Conditional Generative Adversarial Network (CGAN) [78] in order to generate synthetic images. The classification of ten types of diseases on tomato leaves was carried out using the DenseNet121 model trained on both synthetic and real images through transfer learning. In the preprocessing step, they downsized images to 224 × 224. The study employed the PlantVillage dataset, and the obtained accuracy was 97.11%.

An efficient automated diagnosis system for corn plants was proposed by Akanksha et al. [79]. The proposed method consists of four stages: (i) preprocessing, (ii) feature extraction, (iii) classification, and (vi) segmentation. The preprocessing techniques convert images to RGB and remove noise. They utilized an Optimized Probabilistic Neural Network (OPNN) that was improved by using the Artificial Jelly Optimization (AJO) algorithm for classification. The system achieved an accuracy of 95.5%.

The automatic detection of bacterial spot diseases in peach plants was proposed by Bedi and Gole [80] using a hybrid model that combines a Convolutional Autoencoder (CAE) network with a CNN. They used the CAE for dimensionality reduction to reduce the number of training parameters, and the CNN was used for classification. The model was trained and evaluated on the PlantVillage dataset, achieving an accuracy of 99.35% on the training set and 98.38% on the test set with 9, 914 training parameters.

According to the study conducted by Chowdhury et al. [81], tomato leaf images were classified for tomato diseases using the EfficientNet CNN architecture [82]. The PlantVillage dataset, containing 18,162 tomato images, was used to fine-tune and train the model to detect healthy and diseased tomato leaf images. In the preprocessing step, they downsized images to 224 × 224. The results indicate that the proposed model outperformed various contemporary DL algorithms. Two segmentation models, the original U-net [83] and modified U-net [84], were also trained, validated, and tested for the segmentation of tomato leaf images. The Modified U-net was found to be superior for segmenting leaf images from the background, and EfficientNetB7 was superior at extracting discriminative features from images. The performance of the models generally improved with the number of parameters used. The trained models have potential applications in automatic detection of plant diseases at an early stage, and the study achieved an accuracy of 98.6% for the U-net model, 99.9% for the EfficientNetB7, and 99.8% for the EfficientNetB4.

Dwivedi et al. [85] presented a disease detection network for grapes. This network was based on Faster RCNN [60]. During the evaluation stage, the proposed disease detection mechanism was experimented on the PlantVillage dataset, and it was found that the network was more efficient in detecting infected/diseased regions than existing methods. The results showed an overall accuracy of 99.93% for identifying esca, black rot, and isariopsis diseases.

Mishra et al. [86] developed an automated plant disease detection model using median filter preprocessing techniques and a pixel-level feature extraction technique. In this technique, the statistical features are extracted from an image. Afterward, the researchers utilized the Sine Cosine Algorithm-based Rider Neural Network (SCA-RideNN) as the classifier for detecting plant diseases. They conducted experiments with the PlantVillage dataset and reported an accuracy of around 91.56%.

For guava disease detection, Mostafa et al. [87] utilized five neural network models, namely AlexNet, SqueezeNet, GoogLeNet, ResNet50, and ResNet101 [24]. They started by preprocessing the images using the color histogram and unsharp masking method, followed by augmentation using the affine transformation method. In the preprocessing step, they downsized images to be of the same size. The results indicated that ResNet101 achieved the highest classification accuracy of 97.74%.

The aim of Patil and Patil [88] was to develop a DL method to detect a diseased cotton plant from its leaf images. They used an IoT-based platform to collect various sensor data to detect climatic changes. Their deep CNN architecture comprised convolutional layers, ReLU layers, fully connected layers, pooling layers, and activation layers. They used a custom dataset and applied image preprocessing, augmentation, and fine-tuning techniques. In the preprocessing step, they downsized images to 256 × 256. The proposed method yielded an accuracy of about 98%.

The aim of Sambasivam and Opiyo [89] was to detect cassava infections using a CNN model trained on a dataset containing 10,000 annotated images from [90]. The dataset included five fine-grained cassava leaf disease categories; however, there was a significant class imbalance issue with the data. The authors employed several techniques such as class weight, focus loss, and SMOTE to counter the imbalance and improve the model's performance. After tuning several parameters, including the learning rate, number of epochs, batch size, input shape, optimizer, and number of neurons in the hidden layer, they found that a vector input shape of (448, 448, 3) yielded the best accuracy of 93%.

Sujatha et al. [91] assessed and compared the performance of various ML and DL algorithms for the detection of diseases in citrus leaf images as described in [92]. The algorithms evaluated included SVM, RF, stochastic gradient descent, InceptionV3, VGG16, and VGG19. The results showed that the VGG16 model achieved the highest classification accuracy of 89.5%, while the RF model had the lowest accuracy of 76.8%.

A CNN-based approach proposed by Tiwari et al. [93] was used for the detection and classification of plant diseases from leaf images captured in different resolutions. The study included six crops (apple, potato, tomato, bean, citrus, and rice) and 27 different disease categories under laboratory and on-field conditions. The authors also utilized five-fold cross-validation and tested the model on unseen data to enhance the accuracy of the results. Images were gathered from several databases, such as PlantVillage, iBean leaf image dataset [94], citrus leaf images, and rice leaf images [95]. In the preprocessing step, they downsized images to 224 × 224. The cross-validation accuracy was found to be 99.58%, and the average test accuracy was 99.12% for images with complex environmental situations.

The use of a double GAN for balancing datasets by generating images of diseased plant leaves was proposed by Zhao et al. [96]. Initially, they used healthy leaf images from the PlantVillage dataset as inputs for the Wasserstein GAN to obtain a pre-trained model. Subsequently, diseased leaves were used to generate 64 × 64 pixel images of diseased leaves using the pre-trained model. To expand the unbalanced dataset, a superresolution GAN was used to obtain 256 × 256 pixel images. The authors observed that the generated images from the double GAN were clearer than those from DCGAN. They achieved an accuracy of 99.80% and 99.53% for the double GAN and DCGAN, respectively.

Kathole and Munot [97] evaluated and compared the performance of various CNN models, including VGG architecture from scratch, VGG architecture with pre-trained weights, GoogLeNet, and AlexNet, for detecting plant diseases in tomato crops. The experiments were conducted on six different disease classes and a healthy class, and the data were obtained from PlantVillage. The GoogLeNet model achieved the highest accuracy of 97.94%.

Pan et al. [98] employed multiple CNN models including AlexNet, GoogleNet, VGG16, and VGG19 to detect Northern corn leaf blight in maize crops. The researchers utilized a dataset consisting of 985 images of healthy and infected maize leaves and applied various data augmentation techniques to augment their dataset. Their results showed that the GoogleNet model achieved the best performance, with an accuracy of 99.94% in detecting Northern corn leaf blight.

The ResTS (Residual Teacher/Student) architecture, developed by Shah et al. [99], is a CNN structure that incorporates two classifiers (ResTeacher and ResStudent) and a decoder to aid in the diagnosis of plant diseases through visualization and classification. They utilized residual connections and performed batch normalization on all subsystems. The PlantVillage dataset was used to train their model, and they achieved an F1-score of 99.1%.

Two deep feature extraction-based classification models were proposed by Turkoglu et al. [100] using transfer learning on six pre-trained DL networks (AlexNet, DenseNet201, GoogleNet, ResNet18 [24], ResNet50, and ResNet101) for classifying plant disease images. They also proposed a hybrid CNN-SVM model that combined the early fusion of features extracted from deep networks. The images used for training and testing were from the Turkey-PlantDataset [101], which consists of unconstrained images of 15 types of diseases and pests observed in Turkey. The highest accuracy achieved by all the models was 97.56

The automatic plant disease detection technique proposed by Vallabhajosyula et al. [102] utilizes Deep Ensemble Neural Networks (DENN) and employs data augmentation techniques, including image enhancement, rotation, scaling, and translation. The researchers used the PlantVillage dataset and concluded that DENN's performance surpasses that of pre-trained state-of-the-art models, such as ResNet50, ResNet101, InceptionV3, DenseNet 121 and 201, MobileNetV3 [103], and NasNet [104]. The DENN model achieved an accuracy of 99.99

B. OBJECT DETECTION

Fuentes et al. [105] combined three object detection algorithms (Faster RCNN, R-FCN [106], and SSD [107]) with five feature extractors (VGG16, ResNet50, ResNet101, ResNet152 [24], ResNeXt-50) for recognizing tomato plant diseases and pests. They applied the CNN algorithms on a custom dataset of about 5, 000 images using various methods to augment the dataset size. The results show that R-FCN with ResNet50 achieved the best mAP of 86%.

Jiang et al. [108] used a DL approach based on the GoogLeNet Inception structure and the rainbow concatenation for apple leaf disease detection. They generated a custom dataset of 2, 029 images of diseased apple leaves and trained their algorithm to detect five common apple leaf diseases: Alternaria leaf spot, brown spot, mosaic, grey spot, and rust. Various augmentation techniques have been applied in order to generate a more representative dataset. They achieved a detection performance of 78.80% mAP.

Kim et al. [109] proposed a system for field monitoring with a PTZ camera, a motor system, a wireless transceiver, and an image-logging module. This system periodically captures images from the field and sends them to a CNN model designed to detect onion-downy mildew diseases. The model was trained on a dataset of six classes of images captured by the field monitoring system. In the preprocessing step, they extracted 224×224 samples from images. The proposed model achieved an accuracy ranging from 74.1% to 87.2%.

Li et al. [110] presented a technique for disease detection on rice crops using a CNN. They gathered images and videos from mobile phones. They converted videos to a still frame, then transmitted it to a still-image detector for recognition, and then combined the frames into a video. They utilized Faster RCNN as the framework for the still-image detector. The results revealed that the proposed method with the custom backbone was better than ResNet50, ResNet101, YOLOv3 [111], and VGG16 for recognizing diseases and pests for rice crops.

Saleem et al. [112] combined three object detection algorithms (Faster RCNN, R-FCN, and SSD) with four feature extractors (ResNet50, ResNet101, InceptionV2, and Inception-ResNetV2 [113]) and three optimizers (stochastic gradient descent with momentum, Adam, and RMSProp) for plant disease identification. They used the training weights of the Common Objects in Context (COCO) dataset [114]. They compared the algorithms on the PlantVillage dataset. The SSD model with the feature extractor InceptionV2 trained with the Adam optimizer was the best performer with a mAP of 73.07%.

The detection of Northern maize leaf blight disease on maize crops under complex field environments was proposed by Sun et al. [115] using multi-scale feature fusion and the SSD algorithm. For this purpose, a custom dataset consisting of 18,000 images captured by a camera mounted either on a five-meter boom or on a UAV was used. The proposed technique consists of three steps: preprocessing the dataset, fine-tuning the model, and detecting the disease. The NLB dataset was selected because it is calibrated by human plant pathologists and has a high level of accuracy. Data preparation was done mainly to reduce the effect of high-intensity light on image identification and enhance detection accuracy. The proposed model achieved a total accuracy of 91.83%.

Xie et al. [116] presented a DL-based detector for identifying leaf diseases in grape crops. In order to improve the generalization of the model, they developed the grape leaf disease dataset, which included 4,449 original images captured both in the lab and in actual vineyards, and expanded the dataset with an additional 62,286 diseased leaf images. The authors trained the model using data augmentation techniques and utilized the InceptionV1 module [117], Inception-ResNetV2 module, and SE-blocks [118] to enhance the detection performance of multi-scale and smaller diseased spots. The model achieved an overall accuracy of 81.1%.

The detection of tomato diseases and healthy leaves was the objective of the study by Zhang et al. [119]. For this purpose, they proposed an improved version of Faster RCNN, a popular object detection algorithm, which utilized a depth residual network instead of VGG16 for image feature extraction. They also employed the K-means clustering algorithm to cluster the bounding boxes obtained from the network. Three feature extraction networks were tested in the experiments, namely VGG16, Mobile, and Res101. The proposed model achieved an mAP of 98.54% when trained with the Res101 network and k-means clustering.

Roy and Bhaduri [120] proposed a multi-class plant disease DL model using techniques from the field of object detection. The model was designed to achieve a balance between detection speed and accuracy, and it has been used to identify many classes of apple plant diseases in a real-world setting. The detection rate of 56.9 frames per second was achieved, while improving the mAP and F1-score up to 91.2% and 95.9%, respectively.

Selvaraj et al. [121] proposed a pixel-based classification method coupled with ML models for identifying banana crops in complex African environments using multi-level satellite images and UAVs. They used RF for pixel-based classification, combining vegetation indices and PCA, and developed a mixed-model approach using RetinaNet and a customized classifier for simultaneous banana localization and disease classification with UAV-RGB aerial images. The proposed method achieved high accuracy of 99.4%, 92.8%, 93.3%, and 90.8% for banana bunchy top disease, Xanthomonas wilt of banana, healthy banana cluster, and individual banana plants, respectively, according to their mixed-model results.

The PlantVillage dataset was used by Wang et al. [122] to propose three novel plant disease detection techniques, namely Squeeze-and-Excitation SSD (SE_SSD), Deep Block SSD (DB_SSD), and Deep Block Attention (DBA_SSD). The authors compared the performance of their proposed algorithms with YOLOv3, YOLOv4 [123], YOLOv4 tiny, and Faster RCNN. The mAP obtained for the proposed techniques were 90.77% for SE_SSD, 89.93% for DB_SSD, and 92.20% for DBA_SSD.

An enhanced model for plant disease recognition based on the YOLOv5 network model was proposed by Chen et al. [124]. They made changes to the neck module, the SE module, and the loss function. They collected 2, 375 images of rubber tree diseases. Their dataset consists of 1, 203 powdery mildew images and 1, 172 anthracnose images. In the preprocessing step, they downsized images to 640×640. The improved YOLOv5 network model achieved an accuracy of 70%.

The YOLOv4 algorithm was enhanced for detecting diseases in tomato plants by Roy et al. [125]. The researchers utilized DenseNet as the backbone of the modified algorithm and added two new residual blocks to the backbone. They evaluated the proposed model using 1,200 images from the PlantVillage dataset and compared its performance with other object detection methods. The results showed that the proposed model outperformed other state-of-the-art methods with a mAP of 96.29%.

The proposed method by Wang et al. [126] utilizes a lightweight version of the YOLOv5 model and includes an improved attention sub-module to enhance both accuracy and efficiency. Additionally, they employed the Ghostnet structure to reduce computations and replaced the original FPN/PANet structure with the BiFPN structure. The algorithm was evaluated on a custom dataset for plant disease detection, and its performance was compared to other models. The results indicate that the proposed model achieved an accuracy of 92.57%.

III. CLASSIFICATION SCHEME

In this section, we propose a classification scheme and categorize all relevant works in the associated classes. In addition, we present statistics of the most common crops/data types/methods/algorithms/datasets/metrics used by the reviewed papers. The classes that we use in our classification scheme are the following:

- *Crop*: the type of the crop that each study uses as a case study
- *Input data*: the type of the input data used in the ML algorithms
- *Dataset*: the name of the dataset that is used if this information is available in the paper
- *Models/Algorithms*: the ML algorithm that was used
- *Method*: whether classification (C) or object detection is targeted (C)
- *Metrics*: the evaluation metrics employed to measure the performance of the ML algorithms
- *Results*: a brief description of the results obtained by the ML algorithms

Table 7 in Appendix presents all the related work using the above classification scheme. Figures 2–7 show the percentage of papers in the different classes. Most papers have focused on disease detection in tomato plants. There are also various works that consider multiple crops. Corn, apple, grape, and rice are crops that have also attracted much attention in the literature. Regarding the input data type, it is obvious that most researchers use RGB images as input to their models/algorithms since most datasets include only such images. Regarding the datasets used in the studies, the majority of studies use a custom dataset, showing that there is a need to have publicly available datasets that can be utilized for benchmarking purposes. The PlantVillage dataset is also widely used.

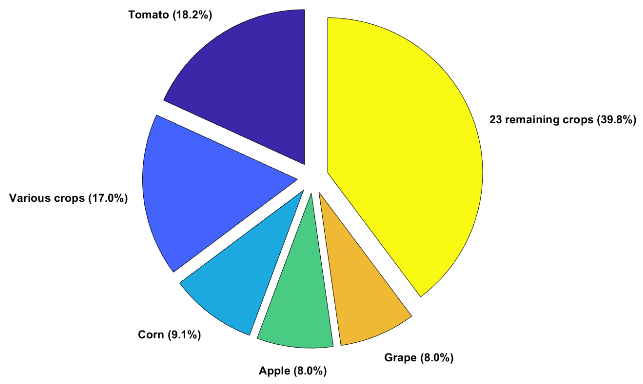


FIGURE 2. Statistics of crops.

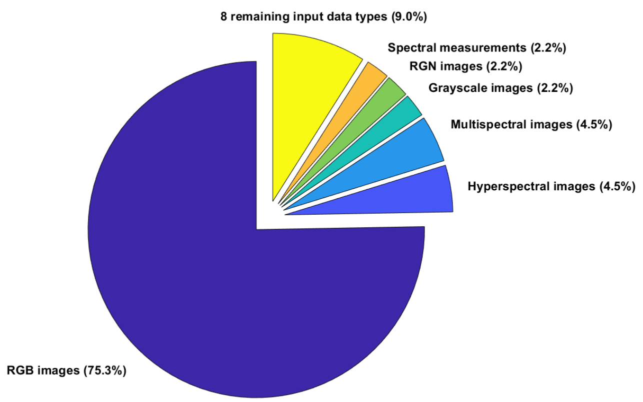


FIGURE 3. Statistics of input data types.

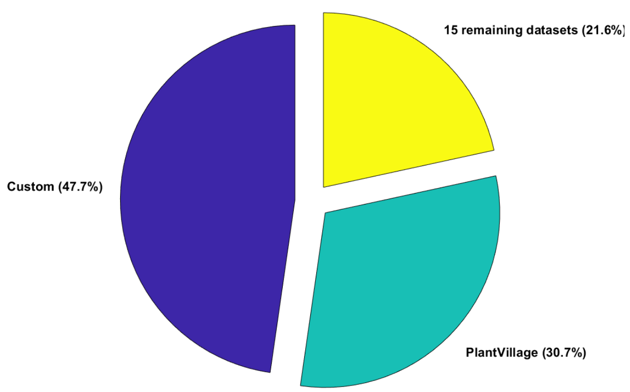


FIGURE 4. Statistics of datasets.

CNN is the most common neural network architecture used throughout the studies. There are also some works dealing with classifying diseases that use SVM, ANN, RF, KNN, and MLP algorithms. Regarding the method that each study targets, classification is the clear winner compared to object detection. Different metrics have been employed in the literature, with accuracy, F1-score, precision, and recall being among the most commonly used ones.

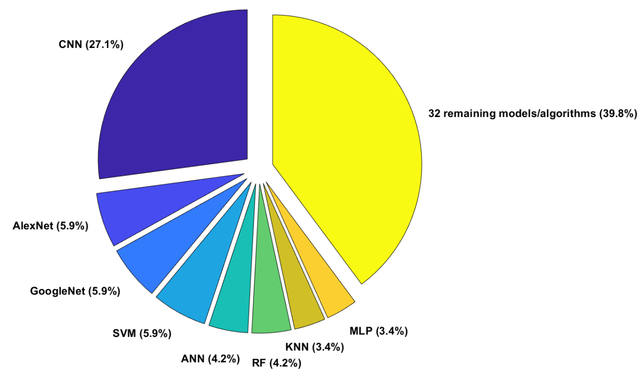


FIGURE 5. Statistics of models/algorithms.

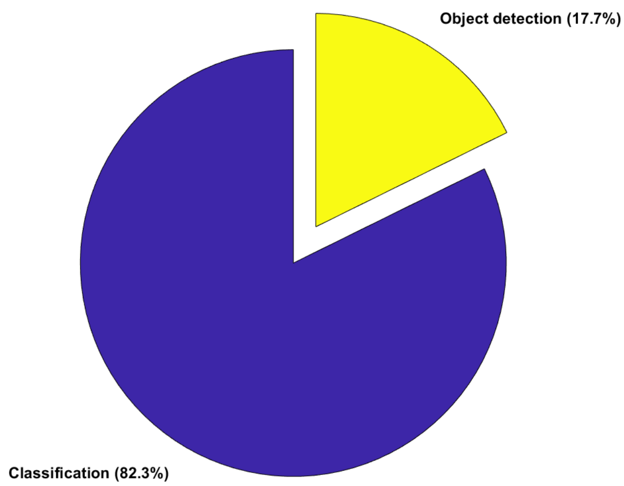


FIGURE 6. Statistics of methods.

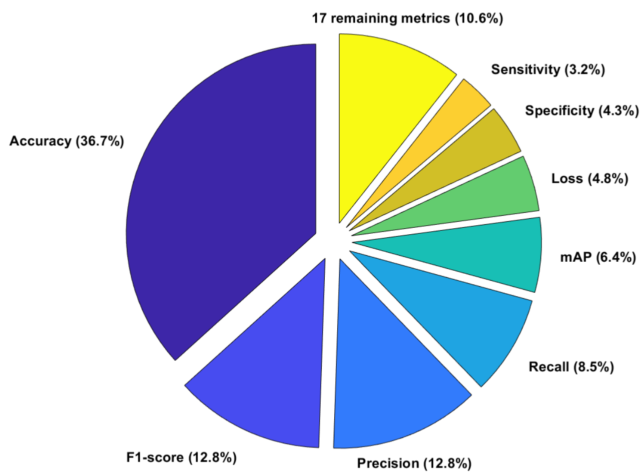


FIGURE 7. Statistics of metrics.

IV. DATASETS

This section provides an overview of publicly available datasets. These datasets serve different purposes; some of them are used for classification to determine if a plant image is healthy or infected with a disease (discussed in Subsection IV-A), while others are used for object detection to identify diseases on plants (discussed in Subsection IV-B).

A. CLASSIFICATION

The PlantVillage dataset [10] comprises 54,303 leaf images, both healthy and diseased, categorized into 38 classes based on species and diseases. The dataset includes images of 14 crop species, such as apple, blueberry, cherry, corn, grape, orange, peach, bell pepper, potato, raspberry, soybean, squash, strawberry, and tomato. It covers 17 fungal-related diseases, four bacterial diseases, two mold (oomycete) diseases, two viral diseases, and one mite-related disease. Additionally, it provides images of disease-free healthy leaves from 12 crop species.

The iBean leaf image dataset [94] comprises images of bean leaves captured from field conditions. The National Crops Resources Research Institute (NaCRRI), the national organization in charge of research in agriculture in Uganda, and the Makerere AI lab collaborated to capture these images in several areas of Uganda. Images were captured from the field or garden using a simple smartphone, which were then analyzed by NaCRRI experts who determined which illness was present in each image. The dataset consists of 1, 296 images and three classes. 428 images are for the healthy class, 432 images are for the angular leaf spot, and the remaining 436 for the bean rust.

The PlantLeaves dataset [127] is comprised of 4, 503 images of plant leaves, both healthy and diseased, categorized into 22 categories based on the species and the state of health. The dataset includes 2, 278 healthy leaf images and 2, 225 diseased ones. The images were captured using a basic digital camera.

The PlantaeK dataset [128] is a leaf database of indigenous plants found in Jammu and Kashmir. It comprises 2, 153 images of healthy and diseased plant leaves, categorized into 16 groups by species and health status. The images feature various crop species, including apple, apricot, cherry, cranberry, grapes, peach, pear, and walnut. The dataset comprises 1, 223 healthy leaf images and 934 diseased leaf images.

The Plant Pathology 2020 challenge dataset [129] is a classification dataset for the foliar disease of apples. The creators of the dataset manually captured 3, 651 real-world symptoms of several apple foliar diseases with varied lighting, angles, surfaces, and noise. The dataset includes 865 healthy leaves, 187 cases of complex diseases, 1, 200 cases of apple scab, and 1, 399 cases of cedar apple rot.

The citrus leaf images dataset [92] contains images of healthy and infected citrus plants with diseases such as black spot, canker, scab, greening, and melanosis. The dataset includes 609 images from citrus leaves, of which 58 are healthy images, and 150 images from citrus fruits, of which 22 are healthy images.

The Kaggle dataset [90] contains 9, 436 annotated images and 12, 595 unlabeled images of cassava leaves. The dataset contains five classes, one is the class for healthy plants and the other four are for diseases (cmd, cgm, cbsd, and cbb). NaCRRI in collaboration with the AI lab at Makerere University captured and annotated these images.

The dataset of Rice leaf images [95] includes 120 images collected from a village in India. The dataset contains 40 images of each disease, for a total of 120 images. The NLB dataset [130] is comprised of 234 images of leaf spot disease in maize crops.

B. OBJECT DETECTION

The PlantDoc dataset [131] includes 2, 345 images. These images contain 13 plant species and 18 classes of diseases. This dataset is publicly available for download and it can also be used as an open dataset for benchmarks. The classes of the PlantDoc dataset are the following: Cherry leaf, Peach leaf, Cherry leaf, Peach leaf, Corn leaf blight, Apple rust leaf, Potato leaf late blight, Strawberry leaf, Corn rust leaf, Tomato leaf late blight, Tomato mold leaf, Potato leaf early blight, Apple leaf, Tomato leaf yellow virus, Blueberry leaf, Tomato leaf mosaic virus, Raspberry leaf, Tomato leaf bacterial spot, Squash Powdery mildew leaf, Grape leaf, Corn Gray leaf spot, Tomato Early blight leaf, Apple Scab Leaf, Tomato Septoria leaf spot, Tomato leaf, Soyabean leaf, Bell pepper leaf spot, Bell pepper leaf, grape leaf black rot, Potato leaf, and Tomato two-spotted spider mites leaf. PlantDoc contains 8, 851 annotations with an average of 3.4 annotations per image. The average image size is 0.53 mp and the distribution of the image sizes starts from 0.01 mp to 24.00 mp. The median image ratio is 800×675. The balance of the classes of PlantDoc is presented in Figure 8. The classes Blueberry leaf, Tomato leaf yellow virus, and Peach leaf are overrepresented with more than 600 images for each class and the classes Tomato leaf late blight, Tomato Early blight leaf, Apple rust leaf, Apple Scab Leaf, grape leaf black rot, Corn rust leaf, Corn Gray leaf spot, Soybean leaf, Potato leaf, and Tomato two-spotted spider mites leaf are under-represented with less than 220 images for each class. A sample of four images with their annotations are shown in Figure 9.

The CropDeep dataset [132] is composed of 31, 147 images containing over 49, 000 annotated instances from 31 different classes. The images were captured in greenhouses under various conditions using different cameras. Additionally, the IP102 dataset [133] is a comprehensive benchmark dataset for recognizing insect pests. It comprises over 10, 000 images divided into 102 categories, with insect pests that mainly target one agricultural product grouped together in the same top-level category. The IP102 dataset has a hierarchical taxonomy.

Table 1 presents the statistics for each dataset including: (i) the name of the dataset, (ii) the type of the crop, (iii) the number of images, (iv) the number of classes, and (v) whether classification (C) or object detection is targeted (C).

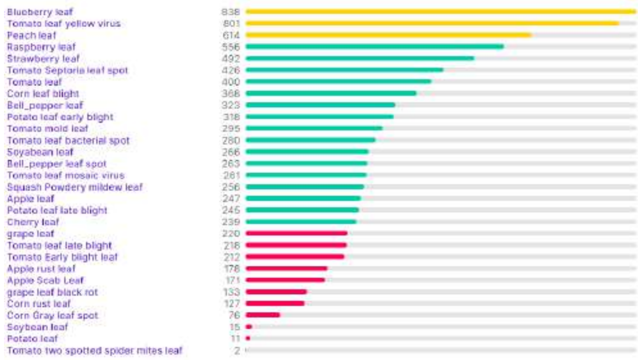


FIGURE 8. Class balance of the PlantDoc dataset.

FIGURE 9. Visualization of the PlantDoc dataset with image annotations.

TABLE 1. Datasets statistics.

Dataset	Crop	Images	Classes	Method
PlantVillage	Various crops	54,303	38	C
iBean	Bean	1,296	5	C
PlantLeaves	Various crops	4,502	22	C
PlantaeK	Various crops	2,153	16	C
Plant Pathology	Apples	3,651	4	C
Citrus leaf images	Citrus fruits	759	6	C
Kaggle	Cassava	22,031	5	C
Rice leaf images	Rice	120	3	C
NLB	Maze	234	1	C
PlantDoc dataset	Various crops	2,345	18	O
CropDeep	Various crops	31,147	31	O
IP102	Corn	10,000	102	O

V. COMPUTATIONAL STUDY

In this computational study, we train five state-of-the-art object detection algorithms and eighteen state-of-the-art classification algorithms on the PlantDoc dataset. Before starting the training process, we applied a prepossessing step. At the PlantDoc dataset, there was a total of 28 annotations that needed to be fixed. The bounding boxes were out of frame in some images, thus they were trimmed to match the image's edges. Some images had bounding zero pixels and were afterwards dropped. The training set contained 25 of these, whereas the test set contained three. After that, we applied image resizing, auto-orientation, and image denoising.

All computational experiments were performed on an Intel Xeon Silver 4214 CPU processor with 128 GB of main memory and 48 cores, a clock of 2.2 GHz, running under Ubuntu 20.10 64-bit, and on a NVIDIA Tesla V100 GPU with 32 GB GDDR5 and a core clock of 1683 MHz.

For the training process, we utilized Fastai's learn.lr_find() [134] method to find the optimal learning rate for each algorithm. The idea of this method comes from the Train Learner over a few iterations [135]. This method starts with a very low start_lr and changes it at each mini-batch until it reaches a very high end_lr. We record the loss at each iteration and check how it relates to the learning rate to determine the optimal value before the loss diverges.

For the object detection problem, we trained and evaluated the performance and the accuracy of the following algorithms:

1) *EfficientDet* [136] includes a collection of eight highly efficient and accurate algorithms developed by Google. EfficientDet detectors are single-shot detectors like SSD and RetinaNet. The EfficientNet, trained in the ImageNet [137] dataset, serves as the network's backbone. The model utilizes a BiFPN as the feature network, while a separate class and box network is responsible for generating the class and bounding box predictors. On the COCO dataset, the overall mAP of EfficientDet-D7 is 52.2%.
2) *Faster RCNN* [60] belongs to the two-stage detector family because its object detection work occurs in two major stages. Faster RCNN is slower than single-stage detectors, but it is more accurate. On the COCO dataset, the overall mAP@.5 is 42.7%.
3) *RetinaNet* [138], a single-stage object detection model, is considered one of the top performers in detecting both dense and small objects. It has two main improvements over other single-stage models: FPN [139] and Focal Loss [138]. On the COCO dataset, RetinaNet's mAP@.5 is 59.1%.
4) *SSD* [107] is based on the VGG16 architecture but lacks fully connected layers. For object detection, it employs a single deep neural network. The VGG architecture was employed as the foundation, and several layers were utilized for prediction at various scales. On the COCO dataset, the mAP@.5 is 46.5%.
5) YOLOv5 [140] is a notable upgrade from YOLO [141], an object detection algorithm that partitions images into a grid system, with each grid cell responsible for detecting objects within it. YOLOv5 is a significant improvement over its predecessor. YOLOv5 uses the PANet structure that includes various layers. On the COCO dataset, the average accuracy is 43.5% AP.

Table 2 provides a summary of the hyperparameters used for the trained object detection algorithms in our study.

The hyperparameters include the learning rate (LR), batch size (BS), optimizer, and backbone architecture. The choice of hyperparameters for our trained object detection algorithms played a pivotal role in configuring and effectively training our models for leaf disease detection. Rather than relying on default values, we meticulously considered each hyperparameter based on its impact on model performance. For the EfficientDet algorithm, a learning rate (LR) of 5e-2 was chosen with a batch size (BS) of 8. The Adam optimizer was utilized, and the EfficientNet served as the backbone architecture. This configuration likely leverages the efficient architecture and large learning rate to achieve rapid convergence. In the case of Faster RCNN, a lower learning rate of 2e-4 was employed along with a batch size of 16. Stochastic Gradient Descent (SGD) [142] was selected as the optimizer, and ResNet-50 served as the backbone. The lower learning rate and larger batch size indicate a more stable and gradual training process. For RetinaNet, a smaller learning rate of 7e-5 was used with a batch size of 8. Similar to EfficientDet, the Adam [143] optimizer was chosen, and ResNet-50 [24] was employed as the backbone architecture. This configuration may favor a balance between rapid convergence and fine-tuning. The SSD algorithm used a learning rate of 7e-5 with a larger batch size of 32. The Adam optimizer was also selected, but the VGG16 [21] architecture was chosen as the backbone. This combination of parameters aims at accommodating a larger batch size while leveraging the SSD architecture. Lastly, YOLOv5 used a relatively high learning rate of 3e-3 with a batch size of 32. Similar to the other models, Adam was used as the optimizer, and the new CSP-Darknet53 [144] architecture served as the backbone, a modification of the Darknet architecture used in previous versions [123]. This configuration may achieve rapid convergence and adapt to the YOLO architecture.

TABLE 2. Hyperparameters of the algorithms.

Algorithm	LR	BS	Optimizer	Backbone
EfficientDet	5e-2	8	Adam	EfficientNet
Faster RCNN	2e-4	16	SGD	ResNet-50
RetinaNet	7e-5	8	Adam	ResNet-50
SSD	7e-5	32	Adam	VGG16
YOLOv5	3e-3	32	Adam	CSPDarknet53

Table 3 presents the mean average precision (mAP) results of the leaf disease detection models after training.

To evaluate the overall accuracy of the object detector models, we utilized several COCO challenge metrics, namely (i) AP, (ii) $AP^{IoU=.50}$, (iii) $AP^{IoU=.75}$, (iv) AP small, (v) AP medium, and (vi) AP large. The assessment results for the COCO metrics are presented in Table 4.

Based on the results, it is evident that YOLOv5 exhibits superior accuracy compared to the other algorithms, as it outperforms them in terms of detecting objects of all sizes in the images. This superiority is reflected not only in the strict AP metric of the COCO challenge but also in other evaluation metrics such as small, medium, and large object detection. EfficientDet, Faster RCNN, RetinaNet, and SSD have almost the same accuracy for this dataset. From the visualization of the results, it is shown that all algorithms were able to identify and localize most of the leaf species and diseases. However, there are images that show that the detection results are suboptimal, as shown in Figure 10.

TABLE 3. mAP of the trained leaf disease detection models.

Algorithm	mAP
EfficientDet	0.256552
Faster RCNN	0.276112
RetinaNet	0.285083
SSD	0.273251
YOLOv5	0.393168

TABLE 4. Average precision on COCO metrics.

Algorithm	AP	$AP_{IoU=.50}$	$AP_{IoU=.75}$	AP small	AP medium	AP large
EfficientDet	0.257	0.415	0.275	0.050	0.159	0.346
Faster RCNN	0.259	0.424	0.278	0.026	0.158	0.370
RetinaNet	0.285	0.431	0.310	0.050	0.180	0.372
SSD	0.273	0.467	0.288	0.031	0.152	0.355
YOLOv5	0.393	0.560	0.430	0.000	0.323	0.459

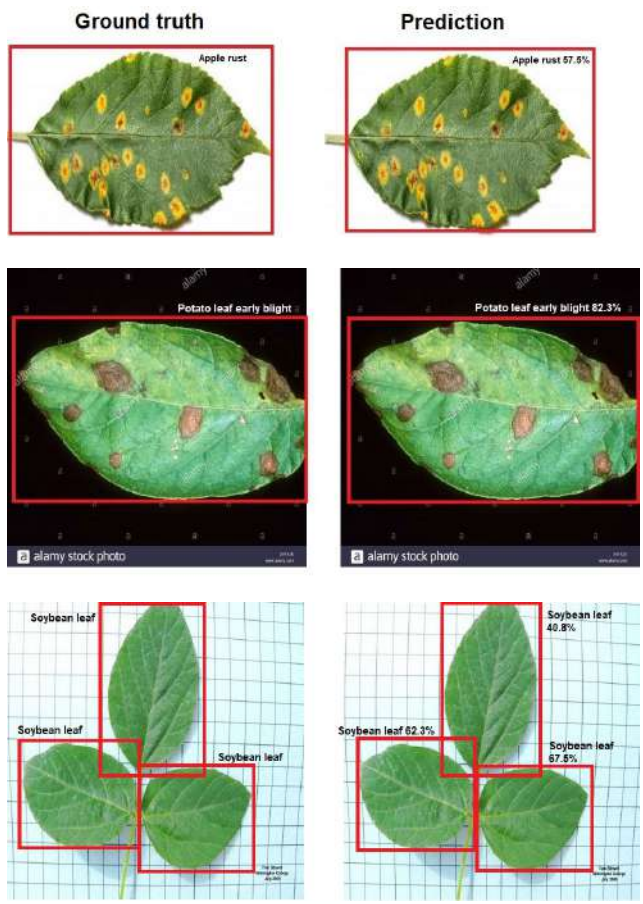


FIGURE 10. YOLOv5 ground truth vs prediction visualization.

TABLE 5. Hyperparameters for classification models.

Model	LR	BS	Optimizer
AlexNet	0.001	64	Adam
DenseNet121	0.001	32	SGD
DenseNet161	0.001	32	SGD
DenseNet169	0.001	32	SGD
DenseNet201	0.001	32	SGD
MobileNetV2	0.001	64	Adam
ResNet50	0.001	32	SGD
ResNet101	0.001	32	SGD
ResNet152	0.001	32	SGD
ResNeXt50_32x4d	0.001	32	SGD
ResNeXt101_32x8d	0.001	32	SGD
ShuffleNet_v2_x2_0	0.001	64	Adam
VGG16	0.001	32	SGD
VGG16_bn	0.001	32	SGD
VGG19	0.001	32	SGD
VGG19_bn	0.001	32	SGD
WideResNet50_2	0.001	32	SGD
WideResNet101_2	0.001	32	SGD

TABLE 6. Comparison of classification models in terms of accuracy and training time.

Model	Accuracy	Training time
AlexNet	0.444915	52m 25s
DenseNet121	0.610169	77m 9s
DenseNet161	0.601695	126m 2s
DenseNet169	0.576271	83m 29s
DenseNet201	0.559322	92m 52s
MobileNetV2	0.572034	67m 5s
ResNet50	0.559322	73m 26s
ResNet101	0.580508	89m 22s
ResNet152	0.601695	105m 50s
ResNeXt50_32x4d	0.576271	84m 40s
ResNeXt101_32x8d	0.593220	147m 20s
ShuffleNet_v2_x2_0	0.203390	64m 27s
VGG16	0.580508	96m 46s
VGG16_bn	0.567797	106m 54s
VGG19	0.525424	112m 1s
VGG19_bn	0.572034	123m 11s
WideResNet50_2	0.601695	148m 47s
WideResNet101_2	0.567797	122m 45s

We trained and evaluated eighteen CNN model architectures that are currently considered state-of-the-art for the image classification problem. AlexNet, DenseNet (121, 161, 169, 201), MobileNetV2, ResNet (50, 101, 151), ResNeXt (50, 101) [145], ShuffleNet [146], VGG, and WideResNet (50, 101) [147]. Eighteen state-of-the-art CNN model architectures pre-trained on the ImageNet dataset were trained and evaluated for the image classification problem. The models were trained using the same hyperparameters for 200 epochs and were assessed based on their accuracy and training time. The Stochastic Gradient Descent optimizer and the softmax cross entropy loss function were utilized by all models.

The hyperparameters presented in Table 5 for the various classification models were selected after careful experimentation and tuning to achieve optimal model performance. These hyperparameters, including the learning rate (LR), batch size (BS), and optimizer, play a crucial role in determining the effectiveness of the training process and ultimately impact the model's accuracy and training times. The choice of a learning rate of 0.001 is a common starting point for many deep-learning models. It is often used as a default value and then fine-tuned during experimentation. The batch size of 32 or 64 is also a result of experimentation. Smaller batch sizes may provide more stable convergence, but larger batch sizes can lead to faster training times. As for the optimizer, models like AlexNet and MobileNetV2 benefit from the Adam optimizer, which adapts learning rates individually for each parameter. On the other hand, models like DenseNet and ResNet variants perform well with the Stochastic Gradient Descent (SGD) optimizer. The choice of optimizer depends on the model architecture and its convergence characteristics.

Table 6 presents the results. The primary evaluation metric in this computational study is accuracy, which is the ratio of the number of correct predictions to the total number of predictions.

According to the study's results, the DenseNet121 model outperformed all other models by achieving the highest accuracy of 61.01%. DenseNet161, ResNet152, and WideResNet50_2 also performed well, achieving an accuracy of 60.16%. On the other hand, ShuffleNet_v2_x2_0 had the worst accuracy of 20.33% on the PlantDoc dataset.

The validation accuracy curve of all the models from epoch 0 to epoch 200 is depicted in Figure 11, whereas Figure 12 displays the training accuracy curve of all the models. All of the networks achieved a training accuracy of more than 97%, except for AlexNet which was not able to reach more than 85% of training accuracy. Finally, we compared all of the training times in Figure 13. MobileNetv2, ResNet50, and DenseNet121 had the shortest training times of 67m, 73m, and 77m, respectively, for an accuracy of more than 55%. The ideal balance between accuracy and training time was achieved from DenseNet121 by achieving the highest accuracy and one of the fastest training times. The reason for DenseNet's superior performance is that it requires fewer parameters than traditional CNNs, eliminating the need to learn redundant feature maps. Moreover, some variations of ResNet models have revealed that several layers do not contribute much and can be discarded. Finally, in the surveyed papers presented in the literature review, there are no algorithms that have been tested in the PlantDoc dataset. However, many studies use the PlantVillage dataset. The algorithms used in the PlantVillage dataset for the classification category, specifically for RGB images, achieve

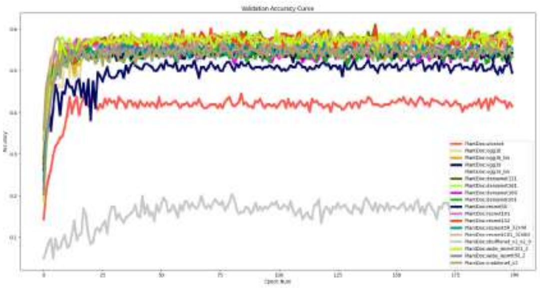


FIGURE 11. Comparison of the validation accuracy of the classification models.

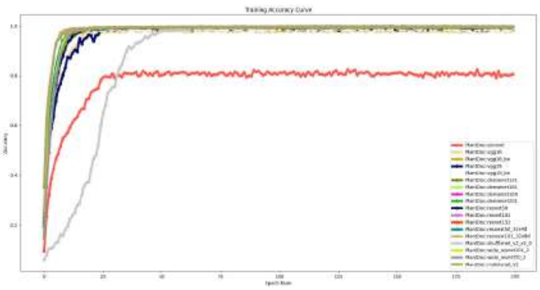


FIGURE 12. Comparison of the training accuracy of the classification models.

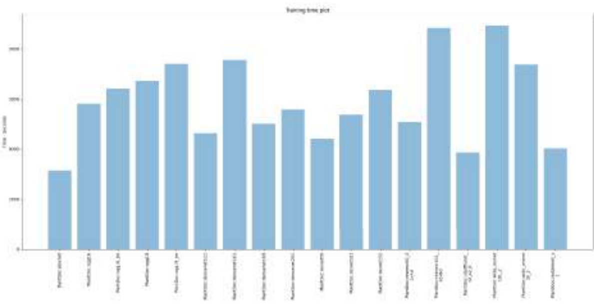


FIGURE 13. Total training time of each classification model.

a performance accuracy of up to 99% [7], [13], [15]. In contrast, the algorithms utilized in the PlantDoc dataset reach a maximum accuracy of 60% for the same category. Moving on to the object detection category, the algorithms employed in the PlantVillage dataset demonstrate an accuracy of up to 96% [112], [122], [125]. However, the algorithms utilized in the PlantDoc dataset achieve a lower accuracy of up to 56% for object detection. The size and diversity of the dataset can significantly impact algorithm performance. The PlantVillage dataset is larger and contains a more diverse set of images, thus it provides the algorithms with a broader range of examples to learn from. A larger dataset can help the algorithms generalize better to new, unseen data. Both in the classification method and in the object detection method we use the algorithms that are state-of-the-art and which are also used in other studies for different datasets.

VI. DISCUSSION

In the field of computer vision, the decision-making process for selecting appropriate methodologies often takes into account the trade-offs between classification and object detection methods. The classification method serves as a foundational concept in machine learning, providing simplicity and interpretability. It allows for the straightforward categorization of inputs into predefined classes. Additionally, its computational efficiency makes it suitable for applications with limited resources. However, the classification method does have its limitations. It lacks fine-grained information, such as spatial details or the ability to handle multiple objects simultaneously. This makes it less suitable for tasks that require precise localization or the identification of multiple objects at once. On the other hand, object detection methods not only provide class labels but also offer precise spatial information, which addresses the challenge of localization. These methods excel in scenarios where multiple objects need to be handled and are versatile for tasks like instance segmentation and keypoint detection. Despite their advantages, object detection methods are more complex to implement. They require bounding box regression and labor-intensive data annotation. Moreover, they often demand significant computational resources, which restricts their deployment in real-time and resource-constrained environments. The choice between classification and object detection depends on the available data and the specific requirements of the task.

High-performing disease detection algorithms are pivotal in precision agriculture, aiding early disease identification, reducing crop losses, and efficient resource allocation. They precisely detect diseases like fungal infections and viral outbreaks, minimizing the need for extensive treatments. Their implementation may require specific hardware, e.g., GPUs, and robust computational resources, with infrastructure and cost considerations being crucial. Moreover, quality labeled datasets, capturing diverse disease manifestations and environmental conditions, are vital for accurate model training. Finally, the integration with existing precision agriculture systems, through standardized interfaces and APIs, and user-friendliness, with intuitive interfaces and interpretable outputs, empowers farmers and practitioners, fostering trust. In practical terms, several projects exemplify the application of these advanced algorithms in precision agriculture:

- PlantVillage [10] utilizes machine learning and computer vision for disease detection in crops. Its user-friendly mobile application enables farmers to take pictures of their crops and receive automated disease diagnoses. This project showcases the practical application of classification algorithms for real-time disease identification, benefiting small-scale farmers.
- John Deere's See & Spray technology [148] exemplifies object detection in precision agriculture. It leverages computer vision and AI to identify and precisely target individual weeds, reducing the need for indiscriminate herbicide application. This project illustrates how object detection algorithms optimize resource usage.
- Blue River Technology [149] employs computer vision and machine learning for their See & Spray system,

autonomously identifying and selectively spraying herbicides only where needed. This project demonstrates the practical implications of object detection algorithms in enhancing the sustainability and cost-effectiveness of crop management.
- IBM's Smart Agriculture solution [150] leverages AI and machine learning to offer insights into crop health and disease detection. By integrating data from various sources, including satellite imagery and IoT sensors, it provides actionable recommendations to farmers. This project showcases the integration of advanced algorithms with a data-driven approach in precision agriculture.

VII. CHALLENGES IN PLANT DISEASE DETECTION

After a detailed review of ML and DL algorithms for plant disease detection and classification and the detailed computational study on five state-of-the-art object detection algorithms for plant disease detection and eighteen state-of-the-art classification algorithms for plant disease classification on a widely-used dataset, we have identified several challenges in practical applications of plant disease detection:

1) There is a lack of models that handle non-image data. Most existing classification and object detection algorithms focus solely on image data, neglecting other relevant information such as temperature and humidity. Developing techniques to incorporate non-image data is essential for more accurate predictions.
2) There are only a few completely annotated open datasets. Many studies rely on the PlantVillage dataset, which was obtained in a controlled laboratory setting. Generating larger datasets under real-world conditions is crucial. Collaborative efforts are needed to create representative datasets.
3) Most works treat the disease detection problem as a classification problem, either binary classification or multi-class classification. While many works treat disease detection as a classification problem, more emphasis should be placed on object detection to identify both the disease type and affected regions in the image.
4) Most papers use a single dataset used to train and test the model. Models trained on a single dataset often perform poorly on different datasets. It is essential to consider diverse datasets to improve model robustness.
5) Overreliance on CNN architectures: While CNNs yield good results, exploring other neural network architectures like recurrent neural networks can enhance disease detection methods.
6) Small leaf and early-stage disease recognition: Current datasets mainly consist of images with large leaves. Annotating datasets for early-stage disease detection and small leaf recognition is necessary.
7) Challenges with illumination and occlusion: Existing algorithms struggle with images under varying lighting conditions and occlusion. More robust methods are needed to address these issues.
8) Computational efficiency: Many models are computationally intensive, hindering real-time applications. Researchers should focus on improving the computational efficiency of their models.

VIII. FUTURE DIRECTIONS IN PLANT DISEASE DETECTION

In addition to the challenges mentioned above, there are several promising directions for future research in plant disease detection:

1) Integration of non-image data: Develop models that can effectively integrate non-image data, such as environmental factors, into disease detection algorithms to improve prediction accuracy.
2) Creation of diverse and real-world datasets: Collaborate with experts to generate large, representative datasets under real-world agricultural conditions to enhance the generalizability of models.
3) Emphasis on object detection: Explore the potential of object detection methods for predicting plant diseases, which can provide more detailed information about disease localization.
4) Robustness across datasets: Develop models that perform consistently well across various datasets to ensure their practical utility.
5) Exploration of alternative neural network architectures: Experiment with different neural network architectures beyond CNNs, such as recurrent neural networks, to uncover their potential in disease detection.
6) Early-stage and small leaf recognition: Annotate datasets specifically for early-stage disease recognition and the identification of diseases on plants or leaves with small sizes.
7) Addressing illumination and occlusion challenges: Implement techniques to enhance the robustness of algorithms in the presence of variable lighting conditions and occluded images.
8) Improved computational efficiency: Focus on optimizing model architectures and algorithms to make them suitable for real-time applications.

IX. CONCLUSION

The aim of this study is to examine existing research that utilizes ML and DL techniques in precision agriculture, with a particular focus on plant disease detection and classification methods. Additionally, a new classification scheme is introduced, which categorizes all relevant works into their respective classes. We separate the studies into two main categories depending on the methodology that they use (i.e. classification and object detection). Furthermore, we present the available datasets for plant disease detection and classification, and provide details about their classes and data, and whether the specific dataset is suitable for classification or object detection.

TABLE 7. Classification of the related works.

Ref	Crop	Input Data	Dataset	Models/Algorithms	Method	Metrics	Results
[7]	Various crops	RGB images	PlantVillage	AlexNet, GoogleNet	C	Accuracy	Accuracy = 99%
[12]	Various crops	Various formats	Custom	CaffeNet	C	Accuracy	Accuracy = 96.3%
[13]	Banana	RGB images	PlantVillage	LeNet	C	Accuracy	Accuracy = 99%
[15]	Tomato	RGB images	PlantVillage	AlexNet, GoogleNet	C	Accuracy	Accuracy = 99%
[16]	Olive	RGB images	PlantVillage and Custom	LeNet	C	Accuracy, precision, F1-score, recall, Matthew's correlation coefficient	Accuracy = 99%
[17]	Corn	RGB images	Custom	CNN	C	Accuracy	Accuracy = 96.7%
[105]	Tomato	RGB images	Custom	CNN	O	Accuracy	Accuracy = 86%
[18]	Apple	RGB images	Custom	AlexNet	C	Accuracy	Accuracy = 97.62%
[19]	Rice	RGN images	Various sources	AlexNet	C	Accuracy, missing report rate, false report rate	Accuracy = 95.48%
[20]	Potato	RGB images	Custom	ResNet50	C	Accuracy	Accuracy = 96%
[22]	Apple	RGB images	PlantVillage	VGG16, VGG19, InceptionV3, ResNet50	C	Accuracy	Accuracy = 90.4%
[25]	Avocado	Hyperspectral images	Custom	MLP, DT	C	Accuracy	Accuracy = 82–100%
[26]	Corn	RGB images	PDDB	CNN	C	Accuracy	Accuracy = 87%
[28]	Various crops	RGB images	PlantVillage	CNN	C	Accuracy, average error	Accuracy = 99.53%
[31]	Tomato	Spectral measurements	Custom	KNN	C	Accuracy	Accuracy = 73.5–100%
[32]	Tomato	RGB images	Custom	CNN	C	Accuracy, loss	Accuracy = 96%
[33]	Grape	RGB images	Custom	CNN	C	Accuracy	Accuracy = 95.8%
[34]	Melon	Thermography, Fluorescence	Custom	LRA, AVM, ANN	C	Accuracy	Accuracy = 99.1%
[35]	Citrus	RGB images	PlantVillage, Citrus Diseases database, Custom	SVM	C	Accuracy	Accuracy = 97%
[37]	Corn	RGB images	PlantVillage and other websites	GoogLeNet, Cifar10	C	Accuracy, loss	Accuracy = 98.9%
[39]	Avocado	RGB and multispectral images	Custom	MLP, KNN	C	Accuracy	Accuracy = 99%
[40]	Grape	Multispectral images	Custom	SVM, LDA	C	Accuracy	Accuracy = 96 %
[41]	Various crops	RGB images	Custom	CNN	C	Accuracy	Accuracy = 93.67%
[42]	Various crops	RGB images	Custom	GoogLeNet	C	Accuracy	Accuracy = 94%
[43]	Millet	RGB images	ImageNet	CNN	C	Accuracy, precision, recall, F1-score	Accuracy = 95%
[44]	Basil	RGB images	Custom	DT, RF, SVM, AdaBoost, GLM, ANN, NB, KNN, LDA	C	Accuracy, recall, precision, error rate	Accuracy = 98.4%
[45]	Tea	RGB images	Custom	GAN	C	Accuracy	Accuracy = 90%
[47]	Wheat	RGB images	Custom	CNN	C	Accuracy, Standard error	Accuracy = 91.43%

TABLE 7. *(Continued.)* **Classification of the related works.**

[108]	Apple	RGB images	Custom	CNN	O	Accuracy, mAP	mAP = 78.80%
[48]	Tomato	Hyperspectral images	Custom	MLP, STDA	C	Accuracy	Accuracy = 99%
[49]	Tomato	Hyperspectral images	Custom	STDA, RBF	C	Accuracy	Accuracy $\geq$ 96%
[50]	Squash	Hyperspectral images	Custom	RBF	C	Accuracy	Accuracy = 82%
[51]	Tomato	RGB images	PlantVillage	CNN	C	Accuracy, precision, recall, F1-score	Accuracy = 98.4%
[52]	Walnut	Grayscale and RGB images	Custom	CNN	C	Accuracy, precision, recall, F1-score	Accuracy = 99%
[54]	Grape	Time-series data	Custom	Lasso, RF, gradient boosting, generalized linear models	C	AUC	AUC = 86%
[55]	Various crops	RGB, grayscaled, and segmented images	PlantVillage	DBN	C	Accuracy, sensitivity, specificity	Accuracy = 87%
[110]	Rice	RGB images, video	Custom	Faster RCNN	O	Precision, recall, F1-score	Precision = 90.9%
[58]	Corn	RGB images	Custom	AlexNet	C	Accuracy	Accuracy = 98%
[59]	Various crops	RGB images	PPBC	CNN	C	Accuracy	Accuracy = 83.75%
[63]	Papaya	RGB images	Custom	K-means, SVM	C	Accuracy, sensitivity, specificity precision	Accuracy = 90.15%
[64]	Pepper	Spectral measurements	Custom	ANN, NB, KNN	C	Accuracy	Accuracy = 100%
[65]	Soybean	RGB images	PDDB	CNN	C	Accuracy, precision, recall, F1-score	Accuracy = 98.14%
[66]	Tomato	RGB images	PlantVillage	CNN	C	Accuracy, loss	Accuracy = 98%
[67]	Grape	Multispectral and RGB images	Custom	CNN	C	Accuracy, recall, precision, F1-score / Dice, RMSE	Accuracy = 92%
[68]	Soybean	Multispectral images	Custom	MLP, GBT, LR-L1, LR-L2, SVM, and RF	C	Accuracy, sensitivity, specificity, precision, negative predictive value, F-measure, Matthews corellation coefficient, AUC	Accuracy = 96.79%
[109]	Onion	RGB images	Custom	CNN	O	mAP, loss	mAP = 87.2%
[56]	Various crops	RGB images	PlantVillage, IPM, Bing	VGG16, InceptionV3, GoogLeNet, GoogLeNetBN	C	Accuracy	Accuracy = 99%
[70]	Rice	RGB images	Custom	ANN with Jaya algorithm	C	Accuracy, Precision, F1-score	Accuracy $\geq$ 90%

TABLE 7. *(Continued.)* Classification of the related works.

[112]	Various crops	RGB images	PlantVillage	CNN	O	mAP	mAP = 73.07%
[121]	Banana	RGB images	Custom	RF	O	Accuracy, Precision, F1-score	Accuracy ≥ 90%
[71]	Tomato	RGB images	PlantVillage	CNN	C	Accuracy	Accuracy = 93.4%
[115]	Corn	RGB images	Corn northern leaf blight dataset	SSD	O	mAP	mAP = 91.83%
[73]	Coffee	Sensor data, RGB, RGN, and RE images	Custom	ANN	C	F1-score	F1-score = 77.5%
[116]	Grape	RGB images	Custom	CNN	O	Accuracy, mAP	Accuracy = 81.1%
[119]	Tomato	RGB images	AIChallenger 2018	Faster RCNN	O	mAP	mAP = 98.54%
[120]	Apple	RGB images	Kaggle PlantPathology Apple	CNN	O	mAP	mAP = 91.2%
[122]	Various crops	RGB images	PlantVillage	SSD	O	mAP	mAP = 92.20%
[74]	Apple	RGB images	2008 'AI Challenger' Global Challenge	CNN	C	Accuracy, precision, recall, F1-score	Accuracy = 99.01%
[76]	Wheat	RGB images	Custom	RF, NN, SVM, CNN	C	Accuracy, precision, recall, F1-score	Accuracy = 93%
[77]	Tomato	RGB images	PlantVillage	GAN	C	Accuracy, precision, recall, F1-score	Accuracy = 97.11%
[79]	Corn	RGB images	PlantVillage	Hybrid OPNN-AJO	C	Accuracy, sensitivity, specificity	Accuracy = 95.5%
[80]	Peach	RGB images	PlantVillage	Hybrid CAE-CNN	C	Accuracy, loss	Accuracy = 99.35%
[81]	Tomato	RGB images	PlantVillage	CNN	C	Accuracy, precision, sensitivity, F1-score, specificity, loss, IoU, dice	Accuracy = 99.9%
[85]	Grape	RGB images	PlantVillage	GLDNN	C	Accuracy, precision, recall, mAP	Accuracy = 99.93%
[86]	Various crops	RGB images	PlantVillage	SCA-RideNN	C	Accuracy, sensitivity, specificity	Accuracy = 91.56%
[87]	Guava	RGB images	Custom	AlexNet, SqueezeNet, GoogLeNet, ResNet50, ResNet101	C	Accuracy, specificity, F1-score, precision, Kappa-Cohen index	Accuracy = 97.74%
[88]	Cotton	RGB images	Custom	CNN	C	Accuracy, F1-score	Accuracy = 98%
[89]	Cassava	RGB images	Kaggle	CNN	C	Accuracy, precision, recall, F1-score, support, MSE	Accuracy = 93%
[91]	Citrus	RGB images	Citrus leaf images	CNN	C	Accuracy, precision, F1-score, AuC	Accuracy = 89.5%
[93]	Apple, potato, tomato, bean, citrus, rice	RGB images	PlantVillage, iBean, citrus leaf images, rice leaf images	CNN	C	Accuracy, specificity, recall, precision, F1-score	Accuracy = 99.58%
	Apple, corn,						

TABLE 7. ***(Continued.)*** **Classification of the related works.**

[96]	grape, potato, tomato	RGB images	PlantVillage	GAN	C	Accuracy	Accuracy = 99.80%
[124]	Rubber tree	RGB images	Custom	CNN	O	Accuracy, mAP	Accuracy = 70%
[97]	Tomato	RGB images	PlantVillage	CNN	C	Accuracy, loss	Accuracy = 97.94%
[98]	Corn	RGB images	Custom	AlexNet, GoogleNet, VGG16, VGG19	C	Accuracy, loss	Accuracy = 99.94%
[125]	Tomato	RGB images	PlantVillage	YOLOv4	O	Accuracy, precision, recall, F1-score, mAP	mAP = 96.29%
[99]	Various crops	RGB images	PlantVillage	CNN	C	F1-score, Training loss, Validation loss	F1-score = 91.1
[100]	Various crops	RGB images	Turkey-PlantDataset	Hybrid CNN-SVM	C	Accuracy, F1-score	Accuracy = 97.56%
[102]	Various crops	RGB images	PlantVillage	DENN	C	Accuracy, Precision, Recall, F1-score	Accuracy = 99.99%
[126]	Various crops	RGB images	Custom	YOLOv5	O	Accuracy, precision, F1-score, recall, mAP, loss	Accuracy = 92.57%

In addition, we perform an extensive computational study on five state-of-the-art object detection algorithms on the PlantDoc dataset to detect the diseases on the leaves, and eighteen state-of-the-art classification algorithms on the PlantDoc dataset, to predict whether or not there is a disease in a leaf and which one it is. For the object detection problem, YOLOv5 is able to achieve a high accuracy not only at the strict AP metric of the COCO challenge but also in the detection of small, medium, and large objects in the images. For the classification problem, the networks ResNet50 and MobileNetv2 had the most optimal trade-off on accuracy and training time. ResNet50 achieved a total accuracy of 61.01% and was trained on about 18 minutes, while MobileNetv2 achieved a total accuracy of 59.74% and was trained for about 16 minutes.

In future work, we plan to study more algorithms for classification and object detection on more datasets to see whether the results are consistent across different datasets. We also intend to study some image preprocessing and data augmentation techniques to see whether the accuracy of the algorithms can be improved with these techniques.

APPENDIX
CLASSIFICATION OF THE RELATED WORKS

See Table 7.

REFERENCES

[1] Food and Agriculture Organization. (2019). *New Standards to Curb the Global Spread of Plant Pests and Diseases*. Accessed: Nov. 8, 2022. [Online]. Available: https://www.fao.org/news/story/en/item/1187738/icode/

[2] Y. Mekonnen, S. Namuduri, L. Burton, A. Sarwat, and S. Bhansali, "Machine learning techniques in wireless sensor network based precision agriculture," *J. Electrochem. Soc.*, vol. 167, no. 3, Jan. 2020, Art. no. 037522.

[3] L. Benos, A. C. Tagarakis, G. Dolias, R. Berruto, D. Kateris, and D. Bochtis, "Machine learning in agriculture: A comprehensive updated review," *Sensors*, vol. 21, no. 11, p. 3758, May 2021.

[4] L. Li, S. Zhang, and B. Wang, "Plant disease detection and classification by deep learning—A review," *IEEE Access*, vol. 9, pp. 56683–56698, 2021.

[5] J. Liu and X. Wang, "Plant diseases and pests detection based on deep learning: A review," *Plant Methods*, vol. 17, no. 1, pp. 1–18, Dec. 2021.

[6] Y. Yuan, L. Chen, H. Wu, and L. Li, "Advanced agricultural disease image recognition technologies: A review," *Inf. Process. Agricult.*, vol. 9, no. 1, pp. 48–59, Mar. 2022.

[7] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers Plant Sci.*, vol. 7, p. 1419, Sep. 2016.

[8] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, 2012, pp. 1097–1105.

[9] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.

[10] Kaggle. (2018). *Plantvillage Dataset*. Accessed: Nov. 8, 2022. [Online]. Available: https://www.kaggle.com/datasets/emmarex/plantdisease

[11] Y. Jia, E. Shelhamer, J. Donahue, S. Karayev, J. Long, R. Girshick, S. Guadarrama, and T. Darrell, "Caffe: Convolutional architecture for fast feature embedding," in *Proc. 22nd ACM Int. Conf. Multimedia*, Nov. 2014, pp. 675–678.

[12] S. Sladojevic, M. Arsenovic, A. Anderla, D. Culibrk, and D. Stefanovic, "Deep neural networks based recognition of plant diseases by leaf image classification," *Comput. Intell. Neurosci.*, vol. 2016, pp. 1–11, May 2016.

[13] J. Amara, B. Bouaziz, and A. Algergawy, "A deep learning-based approach for banana leaf diseases classification," in *Proc. Datenbanksys. Für Bus., Technol. Web (BTW), Workshopband*, CA, USA, Jul. 2017, pp. 1–24.

[14] Y. LeCun, B. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. Hubbard, and L. D. Jackel, "Backpropagation applied to handwritten zip code recognition," *Neural Comput.*, vol. 1, no. 4, pp. 541–551, Dec. 1989.

[15] M. Brahimi, K. Boukhalfa, and A. Moussaoui, "Deep learning for tomato diseases: Classification and symptoms visualization," *Appl. Artif. Intell.*, vol. 31, no. 4, pp. 299–315, Apr. 2017.

[16] A. C. Cruz, A. Luvisi, L. De Bellis, and Y. Ampatzidis, "X-FIDO: An effective application for detecting olive quick decline syndrome with deep learning and data fusion," *Frontiers Plant Sci.*, vol. 8, p. 1741, Oct. 2017.

[17] C. DeChant, T. Wiesner-Hanks, S. Chen, E. L. Stewart, J. Yosinski, M. A. Gore, R. J. Nelson, and H. Lipson, "Automated identification of northern leaf blight-infected maize plants from field imagery using deep learning," *Phytopathology*, vol. 107, no. 11, pp. 1426–1432, Nov. 2017.

[18] B. Liu, Y. Zhang, D. He, and Y. Li, "Identification of apple leaf diseases based on deep convolutional neural networks," *Symmetry*, vol. 10, no. 1, p. 11, Dec. 2017.

[19] Y. Lu, S. Yi, N. Zeng, Y. Liu, and Y. Zhang, "Identification of rice diseases using deep convolutional neural networks," *Neurocomputing*, vol. 267, pp. 378–384, Dec. 2017.

[20] D. Oppenheim and G. Shani, "Potato disease classification using convolution neural networks," *Adv. Animal Biosci.*, vol. 8, no. 2, pp. 244–249, 2017.

[21] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[22] G. Wang, Y. Sun, and J. Wang, "Automatic image-based plant disease severity estimation using deep learning," *Comput. Intell. Neurosci.*, vol. 2017, pp. 1–8, Jul. 2017.

[23] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.

[24] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[25] J. Abdulridha, Y. Ampatzidis, R. Ehsani, and A. I. de Castro, "Evaluating the performance of spectral features and multivariate analysis tools to detect laurel wilt disease and nutritional deficiency in avocado," *Comput. Electron. Agricult.*, vol. 155, pp. 203–211, Dec. 2018.

[26] J. G. A. Barbedo, "Factors influencing the use of deep learning for plant disease recognition," *Biosyst. Eng.*, vol. 172, pp. 84–91, Aug. 2018.

[27] J. G. A. Barbedo, L. V. Koenigkan, B. A. Halfeld-Vieira, R. V. Costa, K. L. Nechet, C. V. Godoy, M. L. Junior, F. R. A. Patricio, V. Talamini, L. G. Chitarra, S. A. S. Oliveira, A. K. N. Ishida, J. M. C. Fernandes, T. T. Santos, F. R. Cavalcanti, D. Terao, and F. Angelotti, "Annotated plant pathology databases for image-based detection and recognition of diseases," *IEEE Latin Amer. Trans.*, vol. 16, no. 6, pp. 1749–1757, Jun. 2018.

[28] K. P. Ferentinos, "Deep learning models for plant disease detection and diagnosis," *Comput. Electron. Agricult.*, vol. 145, pp. 311–318, Feb. 2018.

[29] A. Krizhevsky, "One weird trick for parallelizing convolutional neural networks," 2014, *arXiv:1404.5997*.

[30] P. Sermanet, D. Eigen, X. Zhang, M. Mathieu, R. Fergus, and Y. LeCun, "OverFeat: Integrated recognition, localization and detection using convolutional networks," 2013, *arXiv:1312.6229*.

[31] J. Lu, R. Ehsani, Y. Shi, A. I. de Castro, and S. Wang, "Detection of multi-tomato leaf diseases (late blight, target and bacterial spots) in different stages by using a spectral-based sensor," *Sci. Rep.*, vol. 8, no. 1, pp. 1–11, Feb. 2018.

[32] A. F. Fuentes, S. Yoon, J. Lee, and D. S. Park, "High-performance deep neural network-based tomato plant diseases and pests diagnosis system with refinement filter bank," *Frontiers Plant Sci.*, vol. 9, p. 1162, Aug. 2018.

[33] M. Kerkech, A. Hafiane, and R. Canals, "Deep leaning approach with colorimetric spaces and vegetation indices for vine diseases detection in UAV images," *Comput. Electron. Agricult.*, vol. 155, pp. 237–243, Dec. 2018.

[34] M. Pineda, M. L. Pérez-Bueno, and M. Barón, "Detection of bacterial infection in melon plants by classification methods based on imaging data," *Frontiers Plant Sci.*, vol. 9, p. 164, Feb. 2018.

[35] M. Sharif, M. A. Khan, Z. Iqbal, M. F. Azam, M. I. U. Lali, and M. Y. Javed, "Detection and classification of citrus diseases in agriculture based on optimized weighted segmentation and feature selection," *Comput. Electron. Agricult.*, vol. 150, pp. 220–234, Jul. 2018.

[36] USDA/APHIS/PPQ Center for Plant Health Science and Technology. (2010). *Citrus Diseases Database*. Accessed: Nov. 8, 2022. [Online]. Available: http://idtools.org/id/citrus/diseases/gallery.php

[37] X. Zhang, Y. Qiao, F. Meng, C. Fan, and M. Zhang, "Identification of maize leaf diseases using improved deep convolutional neural networks," *IEEE Access*, vol. 6, pp. 30370–30377, 2018.

[38] A. Krizhevsky and G. Hinton, "Learning multiple layers of features from tiny images," M.S. thesis, Dept. Comput. Sci., Univ. Toronto, Toronto, ON, Canada, 2009.

[39] J. Abdulridha, R. Ehsani, A. Abd-Elrahman, and Y. Ampatzidis, "A remote sensing technique for detecting laurel wilt disease in avocado in presence of other biotic and abiotic stresses," *Comput. Electron. Agricult.*, vol. 156, pp. 549–557, Jan. 2019.

[40] H. Al-Saddik, J. C. Simon, and F. Cointault, "Assessment of the optimal spectral bands for designing a sensor for vineyard disease detection: The case of 'Flavescence dorée,'" *Precis. Agricult.*, vol. 20, no. 2, pp. 398–422, Apr. 2019.

[41] M. Arsenovic, M. Karanovic, S. Sladojevic, A. Anderla, and D. Stefanovic, "Solving current limitations of deep learning based approaches for plant disease detection," *Symmetry*, vol. 11, no. 7, p. 939, Jul. 2019.

[42] J. G. A. Barbedo, "Plant disease identification from individual lesions and spots using deep learning," *Biosyst. Eng.*, vol. 180, pp. 96–107, Apr. 2019.

[43] S. Coulibaly, B. Kamsu-Foguem, D. Kamissoko, and D. Traore, "Deep neural networks with transfer learning in millet crop images," *Comput. Ind.*, vol. 108, pp. 115–120, Jun. 2019.

[44] G. Dhingra, V. Kumar, and H. D. Joshi, "A novel computer vision based neutrosophic approach for leaf disease identification and classification," *Measurement*, vol. 135, pp. 782–794, Mar. 2019.

[45] G. Hu, H. Wu, Y. Zhang, and M. Wan, "A low shot learning method for tea leaf's disease identification," *Comput. Electron. Agricult.*, vol. 163, Aug. 2019, Art. no. 104852.

[46] A. Radford, L. Metz, and S. Chintala, "Unsupervised representation learning with deep convolutional generative adversarial networks," 2015, *arXiv:1511.06434*.

[47] H. Huang, J. Deng, Y. Lan, A. Yang, L. Zhang, S. Wen, H. Zhang, Y. Zhang, and Y. Deng, "Detection of helminthosporium leaf blotch disease based on UAV imagery," *Appl. Sci.*, vol. 9, no. 3, p. 558, Feb. 2019.

[48] J. Abdulridha, Y. Ampatzidis, S. C. Kakarla, and P. Roberts, "Detection of target spot and bacterial spot diseases in tomato using UAV-based and benchtop-based hyperspectral imaging techniques," *Precis. Agricult.*, vol. 21, no. 5, pp. 955–978, Oct. 2020.

[49] J. Abdulridha, Y. Ampatzidis, J. Qureshi, and P. Roberts, "Laboratory and UAV-based identification and classification of tomato yellow leaf curl, bacterial spot, and target spot diseases in tomato utilizing hyperspectral imaging and machine learning," *Remote Sens.*, vol. 12, no. 17, p. 2732, Aug. 2020.

[50] J. Abdulridha, Y. Ampatzidis, P. Roberts, and S. C. Kakarla, "Detecting powdery mildew disease in squash at different stages using UAV-based hyperspectral imaging and artificial intelligence," *Biosyst. Eng.*, vol. 197, pp. 135–148, Sep. 2020.

[51] M. Agarwal, S. K. Gupta, and K. K. Biswas, "Development of efficient CNN model for tomato crop disease identification," *Sustain. Comput., Informat. Syst.*, vol. 28, Dec. 2020, Art. no. 100407.

[52] A. Anagnostis, G. Asiminari, E. Papageorgiou, and D. Bochtis, "A convolutional neural networks based method for anthracnose infected walnut tree leaves identification," *Appl. Sci.*, vol. 10, no. 2, p. 469, Jan. 2020.

[53] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.

[54] M. Chen, F. Brun, M. Raynal, and D. Makowski, "Forecasting severe grape downy mildew attacks using machine learning," *PLoS ONE*, vol. 15, no. 3, Mar. 2020, Art. no. e0230254.

[55] R. Cristin, B. S. Kumar, C. Priya, and K. Karthick, "Deep neural network based rider-cuckoo search algorithm for plant disease detection," *Artif. Intell. Rev.*, vol. 53, no. 7, pp. 4993–5018, Oct. 2020.

[56] S. H. Lee, H. Goëau, P. Bonnet, and A. Joly, "New perspectives on plant disease characterization based on deep learning," *Comput. Electron. Agricult.*, vol. 170, Mar. 2020, Art. no. 105220.
[57] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.
[58] S. Giraddi, S. Desai, and A. Deshpande, "Deep learning for agricultural plant disease detection," in *Proc. ICDSMLA*. Singapore: Springer, 2020, pp. 864–871.
[59] Y. Guo, J. Zhang, C. Yin, X. Hu, Y. Zou, Z. Xue, and W. Wang, "Plant disease identification based on deep learning algorithm in smart farming," *Discrete Dyn. Nature Soc.*, vol. 2020, pp. 1–11, Aug. 2020.
[60] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015.
[61] P. Getreuer, "Chan-vese segmentation," *Image Process. Line*, vol. 2, pp. 214–224, Aug. 2012.
[62] Institute of Botany of the Chinese Academy of Sciences. (2022). *Plant Photo Bank of China*. Accessed: Nov. 8, 2022. [Online]. Available: https://ppbc.iplant.cn/
[63] M. T. Habib, A. Majumder, A. Z. M. Jakaria, M. Akter, M. S. Uddin, and F. Ahmed, "Machine vision based papaya disease recognition," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 32, no. 3, pp. 300–309, Mar. 2020.
[64] K. Karadağ, M. E. Tenekeci, R. Taşaltın, and A. Bilgili, "Detection of pepper fusarium disease using machine learning algorithms based on spectral reflectance," *Sustain. Computing: Informat. Syst.*, vol. 28, p. 100299, 2020.
[65] A. Karlekar and A. Seal, "SoyNet: Soybean leaf diseases classification," *Comput. Electron. Agricult.*, vol. 172, May 2020, Art. no. 105342.
[66] R. Karthik, M. Hariharan, S. Anand, P. Mathikshara, A. Johnson, and R. Menaka, "Attention embedded residual CNN for disease detection in tomato leaves," *Appl. Soft Comput.*, vol. 86, Jan. 2020, Art. no. 105933.
[67] M. Kerkech, A. Hafiane, and R. Canals, "Vine disease detection in UAV multispectral images using optimized image registration and deep learning segmentation approach," *Comput. Electron. Agricult.*, vol. 174, Jul. 2020, Art. no. 105446.
[68] E. Khalili, S. Kouchaki, S. Ramazi, and F. Ghanati, "Machine learning techniques for soybean charcoal rot disease prediction," *Frontiers Plant Sci.*, vol. 11, Dec. 2020, Art. no. 590529.
[69] E. Khalili. (2020). *Soybean Charcoal Rot Disease Prediction Dataset*. Accessed: Nov. 8, 2022. [Online]. Available: https://github.com/Elham-khalili/Soybean-Charcoal-Rot-Disease-Prediction-Dataset-code
[70] S. Ramesh and D. Vydeki, "Recognition and classification of paddy leaf diseases using optimized deep neural network with Jaya algorithm," *Inf. Process. Agricult.*, vol. 7, no. 2, pp. 249–260, Jun. 2020.
[71] S. Verma, A. Chug, and A. P. Singh, "Application of convolutional neural networks for evaluation of disease severity in tomato plant," *J. Discrete Math. Sci. Cryptogr.*, vol. 23, no. 1, pp. 273–282, Jan. 2020.
[72] F. N. Iandola, S. Han, M. W. Moskewicz, K. Ashraf, W. J. Dally, and K. Keutzer, "SqueezeNet: Alexnet-level accuracy with 50x fewer parameters and <0.5 MB model size," 2016, *arXiv:1602.07360*.
[73] D. Velásquez, A. Sánchez, S. Sarmiento, M. Toro, M. Maiza, and B. Sierra, "A method for detecting coffee leaf rust through wireless sensor networks, remote sensing, and deep learning: Case study of the caturra variety in Colombia," *Appl. Sci.*, vol. 10, no. 2, p. 697, Jan. 2020.
[74] Q. Yan, B. Yang, W. Wang, B. Wang, P. Chen, and J. Zhang, "Apple leaf diseases recognition based on an improved convolutional neural network," *Sensors*, vol. 20, no. 12, p. 3535, Jun. 2020.
[75] E&M AI Lab. (2022). *Global AI Challenge for Building E&M Facilities*. Accessed: Nov. 8, 2022. [Online]. Available: https://www.globalaichallenge.com/en/home
[76] Z. Zhang, P. Flores, C. Igathinathane, D. L. Naik, R. Kiran, and J. K. Ransom, "Wheat lodging detection from UAS imagery using machine learning algorithms," *Remote Sens.*, vol. 12, no. 11, p. 1838, Jun. 2020.
[77] A. Abbas, S. Jain, M. Gour, and S. Vankudothu, "Tomato plant disease detection using transfer learning with C-GAN synthetic images," *Comput. Electron. Agricult.*, vol. 187, Aug. 2021, Art. no. 106279.
[78] M. Mirza and S. Osindero, "Conditional generative adversarial nets," 2014, *arXiv:1411.1784*.
[79] E. Akanksha, N. Sharma, and K. Gulati, "OPNN: Optimized probabilistic neural network based automatic detection of maize plant disease detection," in *Proc. 6th Int. Conf. Inventive Comput. Technol. (ICICT)*, Jan. 2021, pp. 1322–1328.
[80] P. Bedi and P. Gole, "Plant disease detection using hybrid model based on convolutional autoencoder and convolutional neural network," *Artif. Intell. Agricult.*, vol. 5, pp. 90–101, Jan. 2021.
[81] M. E. H. Chowdhury, T. Rahman, A. Khandakar, M. A. Ayari, A. U. Khan, M. S. Khan, N. Al-Emadi, M. B. I. Reaz, M. T. Islam, and S. H. M. Ali, "Automatic and reliable leaf disease detection using deep learning techniques," *AgriEngineering*, vol. 3, no. 2, pp. 294–312, May 2021.
[82] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.
[83] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.
[84] V. Petsiuk. (2019). *Lung-Segmentation-2D: Ng Fields Segmentation on CXR Images Using Convotional Neural Networks*. Accessed: Nov. 8, 2022. [Online]. Available: https://github.com/imlab-uiip/lung-segmentation-2d
[85] R. Dwivedi, S. Dey, C. Chakraborty, and S. Tiwari, "Grape disease detection network based on multi-task learning and attention features," *IEEE Sensors J.*, vol. 21, no. 16, pp. 17573–17580, Aug. 2021.
[86] M. Mishra, P. Choudhury, and B. Pati, "Modified ride-NN optimizer for the IoT based plant disease detection," *J. Ambient Intell. Humanized Comput.*, vol. 12, no. 1, pp. 691–703, Jan. 2021.
[87] A. M. Mostafa, S. A. Kumar, T. Meraj, H. T. Rauf, A. A. Alnuaim, and M. A. Alkhayyal, "Guava disease detection using deep convolutional neural networks: A case study of guava plants," *Appl. Sci.*, vol. 12, no. 1, p. 239, Dec. 2021.
[88] B. V. Patil and P. S. Patil, "Computational method for cotton plant disease detection of crop management using deep learning and Internet of Things platforms," in *Evolutionary Computing and Mobile Sustainable Networks*. Singapore: Springer, 2021, pp. 875–885.
[89] G. Sambasivam and G. D. Opiyo, "A predictive machine learning application in agriculture: Cassava disease detection and classification with imbalanced dataset using convolutional neural networks," *Egyptian Informat. J.*, vol. 22, no. 1, pp. 27–34, Mar. 2021.
[90] Kaggle. (2019). *Cassava Disease Classification*. Accessed: Nov. 8, 2022. [Online]. Available: https://www.kaggle.com/c/cassava-disease/overview
[91] R. Sujatha, J. M. Chatterjee, N. Jhanjhi, and S. N. Brohi, "Performance of deep learning vs machine learning in plant leaf disease detection," *Microprocessors Microsyst.*, vol. 80, Feb. 2021, Art. no. 103615.
[92] H. T. Rauf, B. A. Saleem, M. I. U. Lali, M. A. Khan, M. Sharif, and S. A. C. Bukhari, "A citrus fruits and leaves dataset for detection and classification of citrus diseases through machine learning," *Data Brief*, vol. 26, Oct. 2019, Art. no. 104340.
[93] V. Tiwari, R. C. Joshi, and M. K. Dutta, "Dense convolutional neural networks based multiclass plant disease detection and classification using leaf images," *Ecol. Informat.*, vol. 63, Jul. 2021, Art. no. 101289.
[94] AIR Lab Makerere University. (2020). *iBean Leaf Image Dataset*. Accessed: Nov. 8, 2022. [Online]. Available: https://github.com/AI-Lab-Makerere/ibean and https://www.fao.org/news/story/en/item/1187738/icode/
[95] H. B. Prajapati, J. P. Shah, and V. K. Dabhi, "Detection and classification of rice plant diseases," *Intell. Decis. Technol.*, vol. 11, no. 3, pp. 357–373, Aug. 2017.
[96] Y. Zhao, Z. Chen, X. Gao, W. Song, Q. Xiong, J. Hu, and Z. Zhang, "Plant disease detection using generated leaves based on DoubleGAN," *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 19, no. 3, pp. 1817–1826, May/Jun. 2021.
[97] V. Kathole and M. Munot, "Deep learning models for tomato plant disease detection," in *Advanced Machine Intelligence and Signal Processing*. Singapore: Springer, 2022, pp. 679–686.
[98] S.-Q. Pan, J.-F. Qiao, R. Wang, H.-L. Yu, C. Wang, K. Taylor, and H.-Y. Pan, "Intelligent diagnosis of northern corn leaf blight with deep learning model," *J. Integrative Agricult.*, vol. 21, no. 4, pp. 1094–1105, Apr. 2022.

[99] D. Shah, V. Trivedi, V. Sheth, A. Shah, and U. Chauhan, "ResTS: Residual deep interpretable architecture for plant disease detection," *Inf. Process. Agricult.*, vol. 9, no. 2, pp. 212–223, Jun. 2022.
[100] M. Turkoglu, B. Yanikoğlu, and D. Hanbay, "Plantdiseasenet: Convolutional neural network ensemble for plant disease and pest detection," *Signal, Image Video Process.*, vol. 16, no. 2, pp. 301–309, 2022.
[101] M. Turkoglu. (2021). *Turkey-Plantdataset*. Accessed: Nov. 8, 2022. [Online]. Available: https://github.com/mturkoglu23/PlantDiseaseNet
[102] S. Vallabhajosyula, V. Sistla, and V. K. K. Kolli, "Transfer learning-based deep ensemble neural network for plant leaf disease detection," *J. Plant Diseases Protection*, vol. 129, no. 3, pp. 545–558, Jun. 2022.
[103] A. Howard, M. Sandler, B. Chen, W. Wang, L.-C. Chen, M. Tan, G. Chu, V. Vasudevan, Y. Zhu, R. Pang, H. Adam, and Q. Le, "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1314–1324.
[104] B. Zoph, V. Vasudevan, J. Shlens, and Q. V. Le, "Learning transferable architectures for scalable image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8697–8710.
[105] A. Fuentes, S. Yoon, S. Kim, and D. Park, "A robust deep-learning-based detector for real-time tomato plant diseases and pests recognition," *Sensors*, vol. 17, no. 9, p. 2022, Sep. 2017.
[106] J. Dai, Y. Li, K. He, and J. Sun, "R-FCN: Object detection via region-based fully convolutional networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 379–387.
[107] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 21–37.
[108] P. Jiang, Y. Chen, B. Liu, D. He, and C. Liang, "Real-time detection of apple leaf diseases using deep learning approach based on improved convolutional neural networks," *IEEE Access*, vol. 7, pp. 59069–59080, 2019.
[109] W.-S. Kim, D.-H. Lee, and Y.-J. Kim, "Machine vision-based automatic disease symptom detection of onion downy mildew," *Comput. Electron. Agricult.*, vol. 168, Jan. 2020, Art. no. 105099.
[110] D. Li, R. Wang, C. Xie, L. Liu, J. Zhang, R. Li, F. Wang, M. Zhou, and W. Liu, "A recognition method for rice plant diseases and pests video detection based on deep convolutional neural network," *Sensors*, vol. 20, no. 3, p. 578, Jan. 2020.
[111] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.
[112] M. H. Saleem, S. Khanchi, J. Potgieter, and K. M. Arif, "Image-based plant disease identification by deep learning meta-architectures," *Plants*, vol. 9, no. 11, p. 1451, Oct. 2020.
[113] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. A. Alemi, "Inception-v4, inception-resnet and the impact of residual connections on learning," in *Proc. 31st AAAI Conf. Artif. Intell.*, 2017, pp. 4278–4284.
[114] T. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.* Springer, 2014, pp. 740–755.
[115] J. Sun, Y. Yang, X. He, and X. Wu, "Northern maize leaf blight detection under complex field environment based on deep learning," *IEEE Access*, vol. 8, pp. 33679–33688, 2020.
[116] X. Xie, Y. Ma, B. Liu, J. He, S. Li, and H. Wang, "A deep-learning-based real-time detector for grape leaf diseases using improved convolutional neural networks," *Frontiers Plant Sci.*, vol. 11, p. 751, Jun. 2020.
[117] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.
[118] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[119] Y. Zhang, C. Song, and D. Zhang, "Deep learning-based object detection improvement for tomato disease," *IEEE Access*, vol. 8, pp. 56607–56614, 2020.
[120] A. M. Roy and J. Bhaduri, "A deep learning enabled multi-class plant disease detection model based on computer vision," *AI*, vol. 2, no. 3, pp. 413–428, Aug. 2021.
[121] M. G. Selvaraj, A. Vergara, F. Montenegro, H. A. Ruiz, N. Safari, D. Raymaekers, W. Ocimati, J. Ntamwira, L. Tits, A. B. Omondi, and G. Blomme, "Detection of banana plants and their major diseases through aerial images and machine learning methods: A case study in DR Congo and republic of Benin," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 110–124, Nov. 2020.
[122] J. Wang, L. Yu, J. Yang, and H. Dong, "DBA_SSD: A novel end-to-end object detection algorithm applied to plant disease detection," *Information*, vol. 12, no. 11, p. 474, Nov. 2021.
[123] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.
[124] Z. Chen, R. Wu, Y. Lin, C. Li, S. Chen, Z. Yuan, S. Chen, and X. Zou, "Plant disease recognition model based on improved YOLOv5," *Agronomy*, vol. 12, no. 2, p. 365, Jan. 2022.
[125] A. M. Roy, R. Bose, and J. Bhaduri, "A fast accurate fine-grain object detection model based on YOLOv4 deep neural network," *Neural Comput. Appl.*, vol. 34, no. 5, pp. 3895–3921, Mar. 2022.
[126] H. Wang, S. Shang, D. Wang, X. He, K. Feng, and H. Zhu, "Plant disease detection and classification method based on the optimized lightweight YOLOv5 model," *Agriculture*, vol. 12, no. 7, p. 931, Jun. 2022.
[127] S. Siddharth, U. Singh, A. Kaul, and S. Jain, "A database of leaf images: Practice towards plant conservation with plant pathology," Mendeley Data, 2019. [Online]. Available: https://data.mendeley.com/datasets/hb74ynkjcn/1
[128] V. P. Kour and S. Arora, "Plantaek: A leaf database of native plants of jammu and kashmir," in *Recent Innovations in Computing*. Singapore: Springer, 2022, pp. 359–368.
[129] R. Thapa, K. Zhang, N. Snavely, S. Belongie, and A. Khan, "The plant pathology challenge 2020 data set to classify foliar disease of apples," *Appl. Plant Sci.*, vol. 8, no. 9, p. e11390, Sep. 2020.
[130] Science Data Bank. (2019). *Corn Northern Leaf Blight Dataset*. Accessed: Nov. 8, 2022. [Online]. Available: https://www.scidb.cn/en/detail?dataSetId=557575357876666368&dataSetType=project&code=p00001&tID=journalOne
[131] D. Singh, N. Jain, P. Jain, P. Kayal, S. Kumawat, and N. Batra, "PlantDoc: A dataset for visual plant disease detection," in *Proc. 7th ACM IKDD CoDS 25th COMAD*, 2020, pp. 249–253.
[132] Y.-Y. Zheng, J.-L. Kong, X.-B. Jin, X.-Y. Wang, and M. Zuo, "CropDeep: The crop vision dataset for deep-learning-based classification and detection in precision agriculture," *Sensors*, vol. 19, no. 5, p. 1058, Mar. 2019.
[133] H. Wu, T. Wiesner-Hanks, E. L. Stewart, C. DeChant, N. Kaczmar, M. A. Gore, R. J. Nelson, and H. Lipson, "Autonomous detection of plant disease symptoms directly from aerial imagery," *Plant Phenome J.*, vol. 2, no. 1, pp. 1–9, Jan. 2019.
[134] J. Howard and S. Gugger, "Fastai: A layered API for deep learning," *Information*, vol. 11, no. 2, p. 108, Feb. 2020.
[135] L. N. Smith, "A disciplined approach to neural network hyper-parameters: Part 1—Learning rate, batch size, momentum, and weight decay," 2018, *arXiv:1803.09820*.
[136] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10778–10787.
[137] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.
[138] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.
[139] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.
[140] G. Jocher. (2022). *Ultralytics/YOLOv5: V3.1—Bug Fixes and Performance Improvements*. Accessed: Nov. 8, 2022. [Online]. Available: https://github.com/ultralytics/yolov5
[141] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.
[142] S. Ruder, "An overview of gradient descent optimization algorithms," 2016, *arXiv:1609.04747*.
[143] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[144] Ultralytics. *Architecture Summary*. Accessed: Jul. 1, 2023. [Online]. Available: https://docs.ultralytics.com/yolov5/tutorials/architecture_description/
[145] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," 2016, *arXiv:1611.05431*.
[146] X. Zhang, X. Zhou, M. Lin, and J. Sun, "ShuffleNet: An extremely efficient convolutional neural network for mobile devices," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6848–6856.

[147] S. Zagoruyko and N. Komodakis, "Wide residual networks," 2016, *arXiv:1605.07146*.
[148] J. Deere. *See & Spray Ultimate*. Accessed: Sep. 14, 2023. [Online]. Available: https://www.deere.com/en/sprayers/see-spray-ultimate/
[149] A. Yeshe, P. Gourkhede, and P. Vaidya, "Blue river technology: Futuristic approach of precision farming," *Just Agricult.*, vol. 2, no. 7, pp. 1–14, 2022.
[150] IBM. *Agriculture—Environmental Intelligence Suite*. Accessed: Sep. 14, 2023. [Online]. Available: https://www.ibm.com/products/environmental-intelligence-suite/agriculture

VASILEIOS BALAFAS received the B.Sc. (integrated master's) degree from the Department of Electrical and Computer Engineering, University of Western Macedonia, Greece, in 2021, where he is currently pursuing the Ph.D. degree, working on developing optimization and constraint programming algorithms for cloud computing optimization. His Diploma thesis focused on computationally comparing object detection methods. He has published one journal article and two conference papers.

EMMANOUIL KARANTOUMANIS received the B.Sc. (integrated master's) degree from the Department of Informatics and Telecommunications Engineering, University of Western Macedonia, Greece, in 2019. He is currently pursuing the Ph.D. degree with the Department of Electrical and Computer Engineering, University of Western Macedonia, working on developing hybrid optimization and machine learning algorithms in order to optimize black-box problems. His Diploma thesis focused on applying machine learning algorithms in order to predict power consumption in data centers. He has published five conference papers.

MALAMATI LOUTA (Senior Member, IEEE) received the M.Eng. and Ph.D. degrees in electrical and computer engineering, in 1997 and 2000, respectively, and the M.B.A. degree from the National Technical University of Athens, in 2004. She is currently the Director of the Telecommunication Networks and Advanced Services Laboratory and an Associate Professor with the Electrical and Computer Engineering Department, School of Engineering, University of Western Macedonia, Greece. Her research interests include telecommunication networks and advanced services engineering. She is a member of the ACM and the Technical Chamber of Greece. She serves as an associate editor, the general chair, the technical program committee chair, a member, a session organizer, and a reviewer for a number of international conferences and journals.

NIKOLAOS PLOSKAS received the B.Sc., M.Sc., and Ph.D. degrees in applied informatics from the University of Macedonia, Thessaloniki, Greece, in 2007, 2009, and 2014, respectively. He is currently an Associate Professor with the Department of Electrical and Computer Engineering, University of Western Macedonia. His research interests include mathematical programming, optimization, operations research, analysis and design of algorithms, cloud computing, and machine learning. He is also serving on the editorial board for four scientific journals. He has served on the organizing/program committee for 20 international conferences.

IEEE Access
Multidisciplinary ⁝ Rapid Review ⁝ Open Access Journal

Received March 10, 2021, accepted March 24, 2021, date of publication April 8, 2021, date of current version April 19, 2021.

Digital Object Identifier 10.1109/ACCESS.2021.3069646

Plant Disease Detection and Classification by Deep Learning—A Review

LILI LI[1], SHUJUAN ZHANG[2], AND BIN WANG[2]
[1]College of Information Science and Engineering, Shanxi Agricultural University, Jinzhong 030800, China
[2]College of Agricultural Engineering, Shanxi Agricultural University, Jinzhong 030800, China

Corresponding author: Shujuan Zhang (zsujuan1@163.com)

This work was supported in part by the Innovation Project of Shanxi for Postgraduate Education under Grant J202082047.

ABSTRACT Deep learning is a branch of artificial intelligence. In recent years, with the advantages of automatic learning and feature extraction, it has been widely concerned by academic and industrial circles. It has been widely used in image and video processing, voice processing, and natural language processing. At the same time, it has also become a research hotspot in the field of agricultural plant protection, such as plant disease recognition and pest range assessment, etc. The application of deep learning in plant disease recognition can avoid the disadvantages caused by artificial selection of disease spot features, make plant disease feature extraction more objective, and improve the research efficiency and technology transformation speed. This review provides the research progress of deep learning technology in the field of crop leaf disease identification in recent years. In this paper, we present the current trends and challenges for the detection of plant leaf disease using deep learning and advanced imaging techniques. We hope that this work will be a valuable resource for researchers who study the detection of plant diseases and insect pests. At the same time, we also discussed some of the current challenges and problems that need to be resolved.

INDEX TERMS Deep learning, plant leaf disease detection, visualization, small sample, hyperspectral imaging.

I. INTRODUCTION

The occurrence of plant diseases has a negative impact on agricultural production. If plant diseases are not discovered in time, food insecurity will increase [1]. Early detection is the basis for effective prevention and control of plant diseases, and they play a vital role in the management and decision-making of agricultural production. In recent years, plant disease identification has been a crucial issue.

Disease-infected plants usually show obvious marks or lesions on leaves, stems, flowers, or fruits. Generally, each disease or pest condition presents a unique visible pattern that can be used to uniquely diagnose abnormalities. Usually, the leaves of plants are the primary source for identifying plant diseases, and most of the symptoms of diseases may begin to appear on the leaves [2].

In most cases, agricultural and forestry experts are used to identify on-site or farmers identify fruit tree diseases and pests based on experience. This method is not only subjective, but also time-consuming, laborious, and inefficient. Farmers with less experience may misjudgment and use drugs blindly during the identification process. Quality and output will also bring environmental pollution, which will cause unnecessary economic losses. To counter these challenges, research into the use of image processing techniques for plant disease recognition has become a hot research topic.

The associate editor coordinating the review of this manuscript and approving it for publication was Yongming Li.

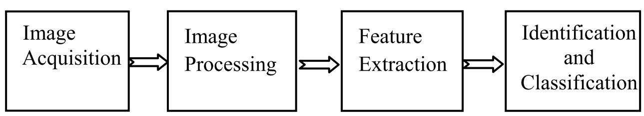


FIGURE 1. Traditional image recognition processing.

The general process of using traditional image recognition processing technology to identify plant diseases is shown in Fig. 1. Dubey and Jalal [3] used the K-means clustering method to segment the lesions regions, and combined the global color histogram (GCH) color coherence vector (CCV) local binary pattern (LBP), and completed local binary pattern (CLBP) was used to extract the color and texture features of apple spots, and three kinds of apple diseases were detected and identified based on improved support vector machine (SVM), and the classification accuracy reached 93%.

Chai *et al.* [4] studied four tomato leaf diseases, including early blight and late blight leaf mildew and leaf spot, and extracted 18 characteristic parameters such as color, texture, and shape information of tomato leaf spot images, using stepwise discriminant and Bayesian discriminant principal component analysis (PCA), respectively. Principal component analysis and fisher discriminant methods were used to extract the characteristic parameters and construct the discriminant model. The accuracy of the two methods reached 94.71% and 98.32%, respectively.

Li and He [5] selected 5 kinds of apple leaf diseases (speckled deciduous disease, yellow leaf disease, round spot disease, mosaic disease, and rust disease) as the research objects. By extracting 8 features of the apple leaf spot image, such as color, texture, and shape. The BP neural network model was used to classify and recognize the diseases, and the average recognition accuracy reached 92.6%.

Guan *et al.* [6] extracted 63 parameters including morphology, color, and texture features of rice leaf disease spots, and applied step-based discriminant analysis and Bayesian discriminant method to classify and recognize three rice diseases (blast, stripe blight, and bacterial leaf blight) with the highest recognition accuracy of 97.2%.

In short, it can be concluded that studies on plant disease recognition based on traditional image processing technology have achieved certain results, with high accuracy of disease recognition, but there are still deficiencies and limitations as follow:

1) The research links and processes are cumbersome, highly subjective, time-consuming and labor-consuming; 2) It is heavily dependent on spot segmentation; 3) It is heavily dependent on artificial feature extraction; 4) It is difficult to test the disease recognition performance of the model or algorithm in more complex environments.

Therefore, it is of great significance to realize intelligent, rapid, and accurate plant leaf disease recognition.

In recent years, deep learning technology in the study of plant disease recognition made more progress. Deep learning (DL) technology in the face of the user is transparent, the researchers of plant protection and statistics professional level is not high, can be automatically extracted image features and classification of plant disease spot, eliminating the traditional image recognition technology of feature extraction and classifier design a lot of work, can express original image characteristics, has the characteristics of the end-to-end. These characteristics make deep learning technology in plant disease recognition-obtained-widespread attention, and it has become a hot research topic. This is due to three factors: the availability of larger datasets, the adaptability of multi-core graphics processing units (GPUs), and the development of training deep neural networks and supporting software libraries, such as the computing unified device architecture (CUDA) from NVIDIA.

Recently, the convolutional neural networks (CNN), a special of deep learning techniques, are quickly becoming the preferred methods [7]. CNN is the most popular classifier for image recognition, and it has shown outstanding ability in image processing and classification [8]. Deep learning approaches were first introduced in plant image recognition based on leaf vein patterns [9]. They used 3-6 layers CNN classified three leguminous plant species: white bean, red bean, and soybean. Mohanty *et al.* [10] trained a deep learning model to recognize 14 crop species and 26 crop diseases. The trained model achieved an accuracy of 99.35% on the test set. Ma *et al.* [11] used a deep CNN to conduct symptom-wise recognition of four cucumber diseases (i.e., downy mildew, anthracnose, powdery mildew, and target leaf spots). The recognition accuracy reached 93.4%. Kawasaki *et al.* [12] introduced a system based on CNN to recognize cucumber leaf disease, which realized an accuracy of 94.9%.

Although very good results have been reported in the literature, however, the diversity of the used datasets is limited. Large datasets (comprised of thousands of images) are required for the training of CNNs. Unfortunately, for plant leaf disease recognition, such large and diverse datasets have not yet been collected for use by researchers. At present, transfer learning is the most effective way to train the robustness of CNN classifiers for plant leaf disease recognition. Transfer learning enables the adaptation of pre-trained CNNs by retraining them with smaller datasets whose distribution is different from the larger datasets previously used to train the network from scratch [13]. Indeed, it is effective that using CNN models pre-trained on the ImageNet dataset and then retraining them for leaf disease recognition. Therefore, the combination of deep learning and transfer learning provides a new way to solve the problem of limited datasets of plant diseases.

There are some research papers previously presented to summarize the research about agriculture (including plant disease recognition) by DL [8], [14], but they lacked some of the recent developments in terms of visualization techniques implemented along with the DL and modified the famous DL models, which were used for plant disease identification.

The article [15] presented many imaging techniques for plant disease detection, and the focus was on imaging techniques. The major techniques presented for plant diseases and classification are SVM, K-means, and KNN.

The article [16] presented many developed/modified DL architectures implemented to detect and classify plant diseases. And provided a comprehensive explanation of DL models used to visualize various plant diseases. But there is no mention of the early detection of the diseases and how to detect and classify plant diseases based on small samples.

In the paper [17], the authors had presented a comprehensive review of recent research work done in plant disease recognition using IPTs, from the perspective of feature extracted based on hand-crafted or using deep learning techniques. And it is concluded that the deep learning techniques have superseded shallow classifiers trained using hand-crafted features. But they lacked some of the recent developments in terms of visualization techniques, and there is no mention of the early detection of the diseases and how to detect and classify plant diseases based on small samples.

This paper aim at the shortcomings of the existing review papers on disease detection, we provide a review of recent studies carried out in the area of plant leaf disease recognition using image processing, hyper-spectral imaging, and deep learning techniques. We hope that this work will be helpful for researchers in the area of plant leaf disease recognition using DL methods.

The rest of this paper is organized as follows. In Section 2, review some basic knowledge including deep learning concept, foundation, framework, development history, model evaluation criteria, the plant leaves disease datasets, and the data enhancement methods, etc. In Section 3, we review research work done so far towards the application of deep learning in crop leaves disease recognition from some aspects. In Section 4, plant disease detection based on small sample data set is discussed. In Section 5, some applications of hyper-spectral imaging in plant disease detection are discussed. Section 6, summarizes and discusses gaps in the existing literature that need to be addressed.

II. BASIC KNOWLEDGE OF DEEP LEARNING

A. THE FRAMEWORK, DEVELOPMENT, AND EVALUATION OF MODELS FOR DEEP LEARNING

1) THE HISTORY OF DEEP LEARNING

Deep Learning (DL) is a subclass of Machine Learning (ML), It was introduced in 1943 [18] and then went into three stages of development. The first generation of neural network-MCP (1943∼1969): originated in 1943, is a linear model, can only deal with linear classification problems.

The second generation of neural network-back propagation (BP) (1986∼1998): Hinton invented the BP algorithm suitable for multi-layer perceptron (MLP) in 1986 and adopted sigmoid function for nonlinear mapping, which effectively solved the problem of nonlinear classification and learning. This method caused the second upsurge of neural networks. However, in 1991, the BP algorithm was pointed out that there was a gradient vanishing problem.

The third generation neural network-DL(2006-present): In 2006, Hinton gradient disappeared in the deep web training are put forward in this problem solution, but because there is no special effective experimental verification and no attention. It was not until 2011 that the ReLU activation function (the activation function that can effectively restrain the gradient disappeared problem) was put forward, then enter the outbreak period in 2012, in the famous ImageNet image recognition contest, the Hinton team used a deep learning model-AlexNet to win, and far more than the second method (SVM). Since then CNN has attracted the attention of many researchers.

After the introduction of AlexNet [19], the DL architecture began to evolve over time as shown in VI. Many advanced DL models/architectures were used for image detection, segmentation, and classification, and these architectures were successively applied to plant disease detection.

2) METRICS

In order to evaluate these algorithms/architectures, top-1%/top-5% error [6], [20]–[22], precision and recall [10], [23], [24], F1 score [24], [25], training/validation accuracy and loss [25], [26], classification accuracy (CA) [27], [28], and mean average precision (mAP) are usually selected as the indicators of judgment.

B. DISEASE DATASETS

Common diseases datasets are: 1) P1antVillage, an open dataset, has now collected 54309 plant leaves disease images, covers 14 kinds of fruit and vegetable crops, such as apple, blueberry cherries, grapes, orange peach bell pepper potato raspberry soybean pumpkin strawberry, and tomatoes, corn contains 26 diseases (17 kinds of fungal disease, 4 kinds of bacteria disease, 2 kinds of mycosis, 2 kinds of viral diseases and 1 kind of diseases caused by mite), also includes 12 healthy crop leaf images. 2) 'Plant Pathology Challenge' for CVPR 2020-FGVC7 (https://www.kaggle.com/c/plantpathology - 2020 fgvc7), it consists of 3,651 high-quality annotated RGB images of 1,200 apple scab and 1,399 cedar apple rust symptoms and 187 complex disease patterns (the leaves with more than one disease in the same leaf) and 865 healthy apple leaves. 3) While others constitute datasets of real images collected by the authors for their research needed(corn, tea, soybeans, cucumbers, apples, grapes). 4) Growing the plants themself and inoculating them with the virus, the method of data acquisition is commonly seen in applications that use hyperspectral images for disease detection.

C. DATA AUGMENTATION

In leaf disease detection, collection and label a large number of disease images require lots of manpower material resources and financial resources. For some certain plant diseases, their onset period is shorter, it is difficult to collect them. In the field of deep learning, the small sample size and dataset imbalance are the key factors leading to the poor recognition effect. Therefore, the deep learning model for leaf disease detection, expand the amount of data is necessary. Data augmentation to meet the requirements for the practical application, and not at liberty to expand (the color is one of the main manifestations of different diseases, for example, when doing image enhancement can't change the color of the original image). There are two common ways to augment the datasets.

1) TRADITIONAL AUGMENTATION

The typical methods are the physical expansion method (tensile rotation adjustment resolution image translation disturbance, etc.), web crawler, variational auto-encoder (VAE), and autoregressive model, etc. The shortcomings of the

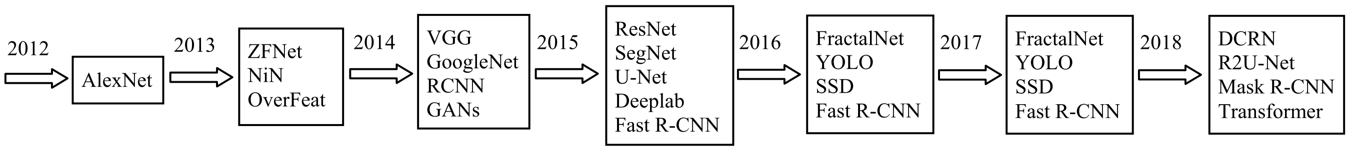


FIGURE 2. The history of DL architectures.

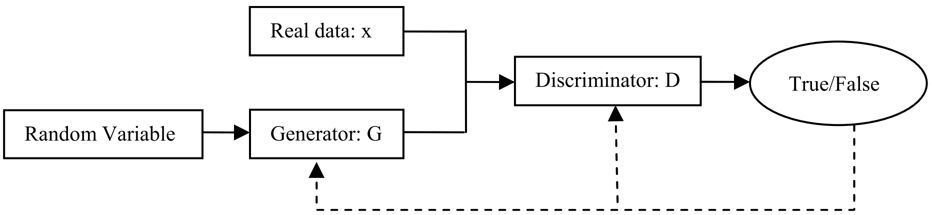


FIGURE 3. Diagram of GANS structure sketch.

produced samples by the traditional expansion method are poor quality, inadequate diversity, and unevenness.

2) GENERATE ADVERSARIAL NETWORKS (GANS)

GANs is a kind of generating model proposed by Goodfellow *et al.* [29] in 2014. Subsequently, many variations of GAN have emerged successively, such as DCGAN, CGAN, PGGAN, LAPGAN, InfoGAN, WGAN, F-GAN, SeqGAN, LeakGAN, etc. The major goal is to generate synthetic samples with the same characteristics as the given training distribution. The GANs models mainly consist of two parts, that is, generator and discriminator. The structure diagram is shown in Fig. 3.

Generative network approaches have been extensively used to generate samples in recent years. Nazki *et al.* [30] is the first work that uses GANs to synthetically augment the dataset to improve the plant disease recognition performance. By optimizing the activation reconstruction loss (ARL) function and put forward an improved AR-GAN, compared with most prominent existing models, the proposed model is introduced into composite images, and nine kinds of tomato on the test data set (2789), the results showed that the classification accuracy is significantly increased (+ 5.2%), compared with the classic way.

Tian *et al.* [31] proposed an approach (CycleGAN) that can generate more apple disease images. Generated images augmented by conditional deep convolutional generative adversarial networks (C-DCGAN) [32] use the segmented tea disease spot image as the input of VGG16. The result showed that the average accuracy is about 28% higher by using C-DCGAN than rotation and translation.

The article [33] generated images by using deep convolutional generative adversarial networks (DC-GAN), and achieved a top-1 average identification accuracy of 94.33% on GoogLeNet. The T-distribution random neighborhood embedding (T-SNE) verified that the image distribution generated by this method was closer to the sample distribution of the real image.

In the paper [34], four different kinds of grape leaf disease images were expanded by a novel Leaf GAN model. The experimental results showed that the Leaf GAN model could make the grape leaf disease images highlight the disease and generate enough grape leaf disease images. It was proved that Leaf GAN was superior to those of the DCGAN and WGAN.

D. VISUALIZATION TECHNIQUE

In recent years, the successful application of deep learning technology in plant disease classification provides a new idea for the research of plant disease classification. However, DL classifiers lack interpretability and transparency. The DL classifiers are often considered black boxes without any explanation or details about the classification mechanism. High accuracy is not only necessary for plant disease classification but also needs to be informed how the detection is achieved and which symptoms are present in the plant. Therefore, in recent years, many researchers have devoted themselves to the study of visualization techniques such as the introduction of visual heat maps and salient maps to better understand the identification of plant diseases. Among them, the works of [35] and [36] are crucial to understanding how CNN recognizes disease from images.

For example, Brahimi *et al.* [35] introduced saliency maps to visualize the symptoms of plant diseases. Mohanty *et al.* [10] used AlexNet and GoogLeNet architectures, through the precision (P), recall (R), F1 score, and the overall accuracy to evaluate the performance of the models on the PlantVillage. Used the three scenarios (color gray and segmentation) to assess the performance of the 2 CNN famous architectures, and come to the conclusion that GoogLeNet outperformed AlexNet, the first layer of the visual results clearly showed the disease spots also. In Cruz *et al.* [37], the improved LeNet model was used to detect olive plant diseases, that is, segmentation and edge maps were used to identify plant diseases. Brahimi *et al.* [38] proposed a new visualization method, that is, a new DL model teacher/student network was introduced to identify the spots of plant diseases, compared with the existing plant disease treatment methods, the new method obtained a clearer visualization effect.

According to the author Dechant *et al.* [39], using different CNN combinations, the visual heat map of maize disease images was used as the inputs, and the probability associated with the occurrence of a particular type of disease was given. The ROC curve was used to evaluate the performance of the model. In addition, the characteristic map of maize diseases was also drawn. Lu *et al.* [40] realized that wheat disease detection by using VGG-FCN and VGG-CNN model and visualized the module features. The results showed that the DMIL-WDDS based on VGG-FCN-VD16 achieved a progressive learning process for fine characteristics of the disease. The feature visualization was a good demonstration of what the DMIL-WDDS was learning. Moreover, the results indicated that Softmax aggregation was a superior choice for DMIL-WDDS to improve the recognition accuracy. Ha *et al.* [41] used the VGG-CNN model to test the blight of radish and used the k-means clustering method to show the disease markers. And the method was able to detect the individual infected areas. That is, the regions of healthy radish and moderate Fusarium wilt of radish were successfully detected by the method. The results showed that the method can also be applied to other crops and plants, including tomato, tobacco, banana, and etc.

Barbedo [42] explored the use of individual lesions and spots for the task, rather than considering the entire leaf, and by using the DL models to identify the plant diseases. The accuracy obtained using the approach was, on average, 12% higher than those achieved using the original images.

Ghosal *et al.* [43], developed a deep CNN framework to identify and classify 8 kinds of soybean stress. And also present an explanation mechanism, used the top-K high-resolution feature maps that isolate the visual symptoms to make predictions. The unsupervised identification of visual symptoms provided a quantitative measure of stress severity, allowing for identification (a type of foliar stress), classification (low, medium, or high stress), without detailed symptom annotation by experts.

Lu *et al.* [44] used CNN to identify rice diseases, early disease detection, and the characteristic maps of disease spots were also obtained. Picon *et al.* [45] proposed an adapted algorithm based on a deep residual neural network to deal with the detection of multiple crop diseases in real conditions for early disease detection. And developed a mobile application in which heat maps were used to identify plant diseases. Obtained results reveal an overall improvement of the balanced accuracy up to 0.87 under exhaustive testing, and the accuracy greater than 0.96 on a pilot test performed in Germany.

Johannes *et al.* [46] used an algorithm based on heat map technology to extract the diseased objects. In addition, each heat map is described by two descriptors, one for evaluating the color information of the disease, and the other for identifying the texture of the heat map. The preliminary hot-spot detection and its ulterior description by color and textural descriptors allow real-time performance as only the suspicious regions are trained and described by the higher level classifiers and descriptors.

Khan *et al.* [47] proposed a new visualization technology using correlation coefficient and DL model (e.g., AlexNet and VGG16 architecture). Kerkech *et al.* [48] variety vegetation indices in color space combined with the LeNet model were used to detect the grape diseases. The article [49] for the reason of interpreting the deep learning model, compared with some of the most popular explanatory methods: significant figure, Smooth-Grad, boot back-propagation, depth Taylor decomposition, integration gradient layered associated transmission, and gradient time input. And trained the DenseNet121 network to identify eight different soybean stresses(biological and non-biological). And concluded that the interpretability methods identified the infected regions of the leaf as important features for some (but not all) of the correctly classified images.

Taken tea leaf diseases images(tea of 261 images of 5 kinds of common disease) in the complex background as the research object, Sun *et al.* [50] proposed a method combining simple linear iterative clustering (SLIC) and support vector machine (SVM), and gain a significant figure accurate tea leaf disease images, with 98.5% accuracy, precision is 96.8%, the recall rate was 98.6%, the F1 score was 97.7%. The results showed that the method can effectively extract tea leaves from complex background significant figure.

Hu *et al.* [51] put forward a kind of new convolution neural network model ARNet (Attention residual network) combining the attention mechanism with the residual idea, and the leaves of five tomato diseases in the early and late periods were studied. The results of the study concluded that, compared with the existed models such as VGG16, the ARNet had a better classification performance. The different layers of (Attention convolutional block, ACB), are visualized in the form of heat maps, and the attention information of the different layers obtained by the module are shown in Fig. 4.

Fig. 4 (a) and (b) represent the output heat maps of late early blight disease and late leaf frost disease in the ACB module at different levels of the ARNET model respectively. Among them, the heat output of each type of image showed in line 1, and line 2 showed that heat superimposed on the original image, from left to right in turn for layer 2, 3, 4, and 5 layers of the last ACB output module. As you can see in Fig. 4, the ACB module can more accurately extract the key feature of each type of disease, shallow ACB module to extract the characteristics of relatively scattered, not as a category, but the deep ACB module to extract the characteristics of more concentrated which the color is more close to red, that is the corresponding place a greater contribution to the final classification decision.

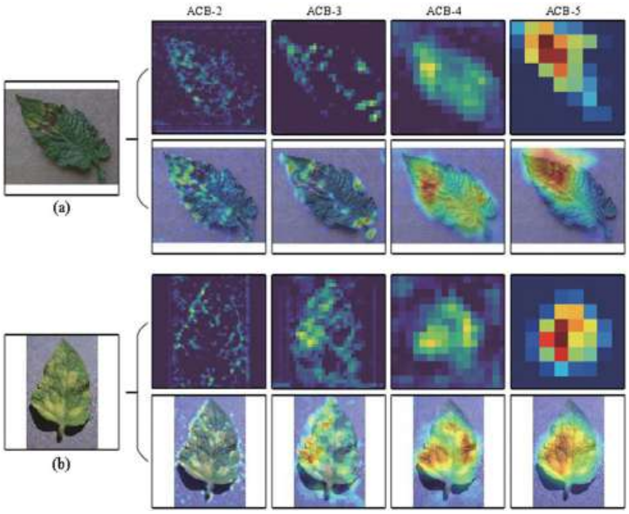


FIGURE 4. The output heat maps of the ACB module in reference [47].

III. LEAF DISEASE DETECTION BY DEEP LEARNING ARCHITECTURES

This section presents the recent researches done by using famous DL architectures for the identification and classification of leaf diseases. Moreover, there are some related works in which modified/improved versions of DL architectures were introduced to achieve better results and software development of disease identification systems.

A. LEAF DISEASE DETECTION BY WELL-KNOWN DEEP LEARNING ARCHITECTURES

1) CLASSIC DEEP LEARNING ARCHITECTURES FOR LEAF-DISEASE DETECTION

Since each disease region has its own characteristics, Barbedo [42] and Lee *et al.* [52], discussed the use of individual lesions and spots rather than considering the whole leaf. The advantages of this method were that occurrence of multiple diseases on the same leaf could be detected and the data can be augmented by cutting up the leaf image into multiple sub-images. The article [55] taken 79 diseases of 14 species of plants in the experimental environment and complex field environment as the research object and used the GoogLeNet model to identify diseases. The overall accuracy of using a single lesion and spot was 94%, which was higher than using the whole image (82%). Lee *et al.* [52] put forward a new view of leaf disease detection that focused on identifying diseases disease area method (i.e. by the common name of disease rather than crops - diseases on the target category), and through the experiments showed that whatever crops, the model training with the common disease were more universal, especially for the new data obtained in different fields or that crops have not been seen.

Qiu *et al.* [53] used the Mask-RCNN whose feature extraction network was ResNet50 or ResNet101 to detect the wheat disease areas, and the average accuracy on the test dataset was 92.01%.

Ahmad *et al.* [54] used four different pretraining convolution neural networks VGG19, VGG16, ResNet, and Inception V3, and the models were trained by fine-tuning parameters. The experimental results showed that the Inception V3 had the best performance on the two datasets(the laboratory dataset and the field dataset). And the average performance superior to 10% to 15% on the laboratory dataset compared with on the field dataset. Bi *et al.* [55] showed that the recognition accuracy rates of apple leaf spot and rust models collected by agricultural experts were 77.65%, 75.59%, and 73.50%, by using ResNet152, Inception V3, and MobileNet, respectively.

Jiang *et al.* [56] used the Mean Shift algorithm to segment four kinds of rice disease spot (red blight stripe disease to rice blast and sheath blight) at first, and then extract shape feature by artificial calculation (put forward three new shape characteristic lesions number N, S lesion area, number of lesions ratio R) and CNN extracts color feature, at last, the SVM classifier was used to identify the diseases, and the results showed that the CNN used segmentation algorithm accuracy was 92.75%, the accuracy was 82.26% without the segmentation algorithm, and the accuracy of the CNN in combination with the SVM model was 96.8%.

Liang *et al.* [57] established a dataset contains 2906 of the positive samples and 2902 of the negative samples to identify rice blasts. And the experimental results showed that the senior characteristics extracted from CNN than the traditional manual extraction of local binary pattern histogram (LBPH) and wavelet transform (Haar-WT) had better identification and effectiveness.

Huang *et al.* [58] put forward a kind of plant leaf image disease recognition method based on the neural structure search algorithm, the method can learn the structure of the neural network to the appropriate depth on the P1antVillage, automatically. According to the results of the studied methods on the dataset of imbalanced and balanced searched out a suitable network structure, and the recognition accuracy of the model was 98.96% and 99.01% respectively. However, if the balance of the gray images was not improved, the accuracy fell to 95.40%.

Long *et al.* [59] used AlexNet for 2 kinds of training, that is, training from scratch and transfer learning from the ImageNet to detect the camellia leaf diseases (4 kinds of diseases and healthy). The results showed that transfer learning can significantly improve the convergence speed and classification performance of the models, and the classification accuracy as high as 96.53%.

Xu *et al.* [60] in order to realize image recognition of corn leaf disease (healthy, leaf blight, rust) in complex field background with small samples, proposed a convolutional neural network model(VGG16) based on transfer learning. The weight parameters of the VGG16 model were trained on ImageNet and transferred to the model, and the average recognition accuracy was 95.33%.

The ResNet50 network pre-trained on ImageNet was used to study 4 types of apple leaf diseases in the Plant Pathology

TABLE 1. Summary of recent research works about the application of DL framework directly.

Reference	Object	DL Frame	Dataset	Sample size	Data enhancement	Metric
[54],2020	Tomato	VGG-16, VGG-19, ResNet, Inception V3	Self-acquired in field and LAB	2681-15216		Average accuracy
[62],2020	Ginkgo biloba	VGG, InceptionV3	Self-acquired in field and LAB	3730-15670	Rotate, Flip	Average accuracy
[55],2020	Apple	ResNet152, InceptionV3 MobileNet	Self-acquired in field	334-2004	Random rotation for cutting and grayscale	Average accuracy, Processing time of each image
[56],2020	Rice	CNN+SVM	Self-acquired in field	6637-8911	Clip	Average accuracy
[57],2019	Rice	CNN	Self-acquired in field	5808	None	Accuracy, ROC, and AUC
[50],2019			Self-acquired in field and LAB	1567-46409	The original image was segmented into single lesions and spots	Detection Rate Classification Accuracy
[52],2020	Plant	VGG16 InceptionV3 GoogLeNet	PlantVillage, IPM, Bing	54305	Random cropping and mirroring	Top-1
[53],2019	Wheat	Mask RCNN	Self-acquired in field	20-2809	Chunking + traditional enhancement	Average accuracy
[60],2020	Corn	VGG-16	Self-acquired in field	600-5400	Rotate, Flip	Average accuracy
[58],2020	Plant	Neural structure search	PlantVillage	54306	None	Average accuracy

2020 Challenge dataset, and the overall test accuracy of the model was 97%. But except for the complex disease pattern category (the combination of several disease symptoms), the recognition accuracy was only 51% [61].

Li *et al.* [62] used VGG16 and Inception V3 models to identify different degrees of Ginkgo biloba diseases, the accuracy of the VGG16 was 98.44% in the laboratory dataset and 92.19% in the field dataset. The accuracy of the Inception V3 model was 92.3% and 93.2%, respectively.

Table 1 offers a brief overview of recent research works about the application of the DL framework directly.

2) NEW/MODIFIED DL ARCHITECTURES FOR LEAF-DISEASE DETECTION

Dechant *et al.* [39] integrated multiple CNN classifiers to study high-resolution corn disease images. The experimental results showed that when a single CNN classifier was used, the accuracy rate was 90.8%, when two first-level classifiers were used, the accuracy rate rise to 95.9%, and when three first-level classifiers were used, the accuracy rate was 97.8%.

Liu *et al.* [63] proposed a new CNN structure to identify the apple leaf disease. The network was formed by cascading an AlexNet-precursor network and an Inception network. The Inception network replaced the fully connected layers in the traditional AlexNet model, significantly reducing the number of trainable parameters, thereby reducing storage requirements. Use Nesterov's accelerated gradient (NAG) optimization algorithm instead of the stochastic gradient descent (SDG) algorithm to update the weights to improve the convergence speed. The performance of this network was compared with SVM, BP, AlexNet, GoogLeNet, ResNet20, and VGG16. The accuracy of these models were 68.73%, 54.63%, 91.19%, 95.69%, 92.76% and 96.32%, while the accuracy of the proposed AlexNet-precursor + Cascade-Inception network was 97.62%.

Picon *et al.* [45] in order to extract the detailed features of the wheat disease symptoms, the first 7×7 convolutional layer of the ResNet50 network was replaced with two 3×3 convolutions and the sigmoid activation function was used instead of the softmax layer for improvement. And used the improved ResNet50 network to detect the early three wheat diseases (septoria, tan spot, and rust), and achieved 96% accuracy on the balanced dataset.

For the existing deep network model existed problems such as a large number of parameters, long training time, high storage cost and computational cost, etc. Wang *et al.* [64] based on the ResNet18, by adding a multi-scale feature extraction module to change the residual layer connection method, decomposes the large convolution kernel and performs group convolution operations, and proposes an improved multi-scale residual (Multi-scale ResNet) model, which significantly reduced the model parameters, storage space and computing overhead. The accuracy rate of 95.95% was achieved on the PlantVillage dataset, and 93.05% was achieved in the self-collected dataset of 7 real environmental diseases.

Aiming at the problem that the current plant leaf disease recognition model is easily interfered with by shadows, occlusions, and light intensity, and the feature extraction is blind and uncertain, Ren *et al.* [65] and others had constructed a deconvolution-guided VGG network (Deconvolution-Guided VGGNet, DGVGGNet) model, which can identify plant leaf disease and segment disease spot. This model had a recognition accuracy of 99.19% for the 10 types of tomato leaf disease images in the PlantVillage dataset. The pixel accuracy and average intersection ratio of disease spots segmentation were 94.66% and 75.36%, respectively.

And it had good robustness in occlusion, low light, and other environments.

Guo *et al.* [66] designed a multi-receptive field recognition model based on AlexNet (Multi-Scale AlexNet) by removing the local response normalization layer of the AlexNet network, modifying its fully connected layers, and setting a multi-scale convolution kernel to extract features. The PlantVillage dataset and self-collected 7 kinds of tomato diseased leaves dataset are the research objects. The model reduced the memory requirements of the original AlexNet by 95.4%, and the average recognition accuracy of tomato leaf diseases and each disease in the early, middle, and late stages was up to 92.7%.

Fan *et al.* [67] added a batch standardization layer to the convolutional layer of the Faster R-CNN model, introduced a central cost function to construct a mixed cost function, and used a stochastic gradient descent algorithm to optimize the training model. They used 9 kinds of corn leaf diseases with complex backgrounds in the field as the research object. Under the same experimental environment, the improved method had an average accuracy increase of 8.86%, and a single image detection time was reduced by 0.139s; compared with the SSD algorithm, the average accuracy was 4.25% higher, and a single image detection time was reduced by 0.018 s.

Wang *et al.* [68] in order to solve the problems of a long time of training, poor segmentation effect, and susceptibility to illumination and background during the image segmentation of cucumber leaf lesions in traditional convolutional neural networks, they proposed a method based on the full convolution neural network (in which the activation function of rectified linear units (RELU) was replaced by the exponential linear unit (ELU), and the batch normalization function was used to stabilize the model training process, and the softmax of the original CNN was replaced with support vector machine (SVM)). The average pixel segmentation accuracy was 80.46% and the average cross-combination ratio was 70.43% on the 6 kinds of cucumber leaf disease dataset.

Hu *et al.* [51] tried to solve the problem of insufficient identification methods for fine-grained tomato diseases. Taken 5 kinds of tomato diseased leaves in the early and late stages as the research objects, and proposed a new convolutional neural network model ARNet based on the combination of attention and residual ideas. Compared with existing models such as VGG16, ARNet had better classification performance, with an average recognition accuracy rate of 88.2%.

Picon *et al.* [69] proposed three different CNN architectures (RESNET-MC-1, RESNET-MC-2, and RESNET-MC-3) to integrate contextual non-image meta-data (such as crop information) into image-based convolutional neural Network. RESNET-MC-1 achieved 98% accuracy in a field environment containing 17 diseases and 5 crops (wheat, barley, corn, rice, and rapeseed).

Chen *et al.* [70] proposed an improved VGG model (INC-VGGN) based on the VGG model framework by introducing two Inception modules, adding a pooling layer, and modifying the activation function. And the average recognition accuracy of corn plant leaf diseases reached 92%.

Zhang *et al.* [71] combined the expansion convolution and global pooling for the problem of the AlexNet model with too many parameters and a single feature scale and proposed a global pooling extended convolutional neural network (GPDCNN) based on the AlexNet model. After the expansion, an accuracy of 95.18% was obtained on the dataset of 6 common cucumber leaf diseases taken in the field.

Due to the problem of low recognition accuracy of grape leaves with different degrees of disease, He *et al.* [72] proposed a Multi-Scale ResNet based on ResNet18 by changing the conv1 layer to a combination of multiple convolution kernels and adding the SENet module to ResNet18 to identify grape leaf disease. The model had an average recognition accuracy of 90.83% for seven grape diseases including different severity.

Agarwal *et al.* [73] developed a CNN model with 3 convolutional layers, 3 maximum pooling layers, and 2 fully connected layers, and each layer had a different number of filters to detect 9 types of tomato leaf diseases. The experimental results showed that the average accuracy of the proposed model on the test set reached 91.2%, and its performance was much better than VGG16, MobileNet, and Inception.

Table 2 briefly introduces the research progress of the improvement of DL in plant disease detection in recent years.

B. TARGET DETECTION OF PLANT DISEASES FOR LEAF DISEASE DETECTION

Fuentes *et al.* [74] used Faster R-CNN, R-FCN, and SSD architectures to locate lesion areas of 9 kinds of tomato leaf diseases and insect pests, and classified them according to the bounding box. And explored the influence of different CNN architectures on the detector. The results showed that ResNet50 as the feature extractor achieved a mean average accuracy (mAP) of 85.98%, and the detection time was about 160 ms per image. Subsequent work [75] refined the Faster R-CNN by introducing a single-class CNN, and the results showed that the mAP increased by 13%.

Jiang *et al.* [76] proposed a novel method that is the SSD with inception module and rainbow concatenation (INAR-SSD). And the VGG16 feature extractor used in the INAR-SSD network was a modification by replacing two convolution layers (Conv4_1 and Conv4_2) with inception modules, fully connected layers of VGG16 were also replaced with 1×1 convolutions. On a dataset of 5 kinds of apple leaf diseases, the proposed INAR-SSD network achieved the highest mAP of 78.8% compared with the Faster R-CNN (73.78%) and SSD (75.82%). Meanwhile, the detection speed of the model was 23.13 FPS.

Li *et al.* [77] took 5 kinds of bitter gourd leaf diseases taken in the field as the research object, modified the Faster R-CNN by increasing the size of the regional suggestion frame and integrating the feature pyramid network (FPN) based on ResNet50. The research results showed that after

TABLE 2. Summary of recent research works about application about the improvement of DL in plant disease detection.

Reference	Object	Dataset	Sample size	Data enhancement	Metric
[64],2020	Tomato, potato, pepper, cucumber,	PlantVillage, AIChallenge2018	22372-134232	Rotate, Clipping Brightness for contrast enhancement, Gamma noise, Flip	Average accuracy, Maximum Accuracy, Training time, Model size
[66],2019	Tomato	Plant Village+ Self-acquired in field	5766-8349	Random cropping, Flip, Rotation, Color dithering, Add noise	Average accuracy
[67],2020	Corn	China Science Data Network, Digipathos Network and Self-acquired in field	1150-1150*8	Flip, Brightness adjustment, Saturation adjustment, and add Gaussian noise	Average accuracy, Average recall, Fl-score, Overall average accuracy
[68],2019	Cucumber	Self-acquired field	255*6-255*6*5	Panning, Rotating, Zooming, Color dithering	Average accuracy in pixel, IOU, Segmentation time
[51],2019	Tomato	AI challenger	8861-44295	Rotate, translation, Scale, Flip horizontally	Accuracy, Confusion matrix
[70],2020	Rice, Corn	Self-acquired in field	966-4000	Rotate, Flip, Scale transformations	Accuracy Sensitivity Specificity
[71],2019	Cucumber	Self-acquired in field	600-30000	Transferring, Scaling, Clipping, Rotation, Strength conversion	Training time, Testing time, Accuracy
[72],2020	Grape	Al Challenger+ Self-acquired in field	1135-3597	Rotation, Brightness adjustment	Average accuracy, Model parameter
[73],2020	Tomato	PlantVillage	20372-134232	Lip, Color dithering, Add noise, translation, Lower brightness	Accuracy in pixel, IOU

integrating the feature pyramid network, the average accuracy of the obtained model was 86.39%, higher than the original model (7.54%), and the accuracy of gray spot detection was improved by 16.56%. The detection time of each image is 0.322s, which can guarantee real-time detection.

Aiming at the problem of difficulty in real-time detection of apple leaf disease images under actual conditions due to the complex background and small lesions, Li *et al.* [78] modified the Faster R-CNN by using the feature pyramid network (FPN) and adopting precise region of interest pooling (PROI Pooling). The research results showed that the improved model can effectively detect five apple leaf diseases under natural conditions, with a mean average accuracy of 82.28%. Compared with Faster R-CNN, YOLOv3, and Mask R-CNN, the mean average accuracy increased by 5.81%, 13.92%, and 4.86%, and the detection time of a single image was reduced by 43ms, respectively.

Li *et al.* [79] proposed a video detection architecture of plant diseases and insect pests based on deep learning and a custom backbone, which can better reflect the quality of video detection in experiments. Experiments showed that compared with VGG16, ResNet50, ResNet101 backbone systems, and YOLOv3, the custom backbone system was more suitable for detecting untrained rice videos. The custom DCNN backbone had eminent detection sensitivity in withered leaves of rice sheath blight and rice stem borer symptoms. And the detection speed was 30 frames per second (FPS).

Aiming at the problem of low segmentation accuracy of traditional convolutional neural networks in crop disease leaf images, Wang *et al.* [80] constructed a regional disease detection network (RD-net) based on the traditional VGG16 model and replaced the fully connected layer with a global pooling layer. Based on the Encoder-Decoder model structure, a regional segmentation network (RS-net) was established, and the multi-scale convolution kernel was used to improve the local receptive field of the original convolution kernel and segment the lesion area accurately. Segmentation experiments were carried out on the field-photographed datasets of corn leaf spot, corn round spot, wheat stripe rust, wheat anthracnose, cucumber target spot disease, and cucumber brown spot. The segmentation accuracy was 87.04% and the recall rate was 78.31%. The comprehensive evaluation index value was 88.22% and the single image segmentation speed was 0.23 s.

Table 3 offers a brief overview of recent research works in target detection of plant diseases.

C. THE SYSTEM OF LEAF-DISEASE DETECTION

In an era when smart agricultural technology is so advanced, mobile phones have become a new type of "farming tool" for farmers, which can help farmers in identifying diseases and insect pests. Currently, researchers develop small programs or mobile apps to help farmers identify crop pests and diseases. The farmer takes pictures and uploads the diseased parts of the crop, and the system will return the recognition result within a few seconds. And provide users with the diagnosis results, similarity, disease characteristics, causes, and prevention and control plans for users, so that farmers can treat diseases and insects in a scientific way and increase crop yields.

Ozguven and Adem [81] modified the Faster R-CNN by increasing the size of the input layer from 32×32 pixels to 600×600 pixels and developed an automatic detection and recognition system for leaf spot disease in 3 levels of sugar beet disease severity (mild, moderate, and severe). The developed Faster R-CNN achieved an accuracy of 95.48% compared to 92.89% achieved by Faster R-CNN.

Aiming at the problem that the classification accuracy of the classification model for the severity of crop diseases and insect pests is not high enough, Yu *et al.* [82] proposed an

TABLE 3. Summary of recent research works about the application in target detection of plant diseases.

Reference	Object	DL frame	Dataset	Sample size	Data enhancement	Average detection time	Metric
[76],2019	Apple	INAR-SSD, SSD	PlantVillage, Self-acquired in field	2029-2029*12	Rotation, horizontal and vertical flip, Gaussian noise		MAP, Confusion matrix, Speed
[77],2020	Bitter gourd	Improved Faster R-CNN	Self-acquired in field	1204-10627	Rotation, horizontal mirror, and vertical mirror	0.322s	MAP, Accuracy, Detection time, Training time
[78],2020	Apple	Improved Faster R-CNN, Faster R-CNN	PlantVillage, Self-acquired in field	2029-2029*10	Changes in rotation, brightness, and contrast	0.43s	Average processing time, Average accuracy
[79],2020	Rice	Faster-RCNN, YOLOv3	Self-acquired in field	5320		30 PFS	Precision, Recall, F1 score
[80],2020	Corn, Wheat, Cucumber	Regional disease detection network (RD-net)	Self-acquired in field	150-150*6		0.23s	Segmentation accuracy, Recall, Single image segmentation speed

TABLE 4. A brief overview of recent research works in the development of plant leaf disease identification system.

Reference	Object	DL frame	Dataset	Sample size	Data enhancement	Metric
[82],2020	Plant	CDCNNv2, ResNet 50	AI Challenger 2018	36261	Brightness change, random rotation, and mirror flip, etc.	Average accuracy, Mean convergence time
[83],2020	Tomato	PARNet, ResNet50	Plant Village	2284-9136	Clip, rotate	Detection rate, Parameter size
[84],2019	Ginger	Improved LeNet-5	China Agricultural Network	400	None	Average accuracy
[85],2019	Apple	Faster R-CNN	Self-acquired in field	2029-24348	Rotate, Horizontal and vertical mirroring, Gaussian blur, etc.	Recall, Average accuracy
[86],2019	Grape	MobileNet, Inception V3	Self-acquired in field	2100	None	Average accuracy
[87],2019	Coffee leaves	AlexNet, VGG19, ResNet50, GoogLeNet	Self-acquired in LAB	1747-2722	Standard expansion, Mix-up	Average accuracy

improved ResNet50 model (CDCNNv2) combined with deep transfer learning and developed a classification system for the severity of crop diseases and insect pests. In addition to real-time and fully automatic detection of crop pests and diseases, the system also implements a series of supporting functions such as prevention and control recommendations and drug recommendations.

Li *et al.* [83] combined the attention mechanism with the residual structure to build the PARNet model and completed the development of the WEB application. The average accuracy of the platform for 5 tomato leaf diseases can reached 96.84%. It was 2.25%∼11.58% higher than other models (VGG16, ResNet50, and SENet).

Jiang *et al.* [84] redesigned and optimized the convolutional neural network structure based on the traditional LeNet-5 network, and proposed a convolutional neural network system for ginger disease recognition based on the four kinds of ginger disease collected in the natural environment. The recognition rate of four kinds of ginger diseases reached 96%.

Zhou [85] identified 5 kinds of apple leaf diseases based on transfer learning and the Faster R-CNN and developed an apple leaf disease detection system based on the Android platform. The detection system had an average recognition accuracy of 76.55% for apple leaf diseases.

Liu *et al.* [86] deployed the MobileNet network on the mobile phone, and the average recognition accuracy of the 6 kinds of grape diseased leaves collected in the field was 87.5%, and the average calculation time for a single image was 134ms.

Based on the ResNet50 architecture, Esgario *et al.* [87] developed a system that can identify and estimate the severity of stress caused by biological agents on coffee leaves. The system had an accuracy of 95.24% for the classification of biological stress on coffee leaves, and an accuracy of 86.51% for estimation of the severity.

Xiong *et al.* [88] proposed an automatic image segmentation algorithm based on the GrabCut algorithm and selected the MobileNet as DL classification model, and designed a crop disease recognition system for mobile smart devices. The system had a recognition accuracy of more than 80% for a total of 27 diseases of 6 crops in the laboratory environment and the field.

Table 4 offers a brief overview of recent research works in the development of plant leaf disease identification systems.

IV. LEAF-DISEASE DETECTION BASED ON SMALL SAMPLES

In practical applications, the incidence of some plant diseases is low and the cost of acquiring disease images is high, resulting in only a few or dozens of disease images collected. The transfer learning method can transfer the knowledge learned from the general large dataset to the professional fields with relatively little data. But for the datasets with only a few or dozens of images, the transfer learning method also has the problem of low recognition accuracy [23]. This is because it is difficult for the deep network to learn different features, which leads to problems that are difficult to converge or over-fitting. Therefore, plant disease datasets with single or small samples can hardly support the training of DL architecture. On the other hand, for the recognition of new classes that do not appear in the training set, the deep learning model needs to be retrained.

Recent advances in DL have proven the effectiveness of several architectures to learn new classes using small datasets, a famous sub-field known as Few-Shot Learning (FSL) [89]. FSL can not only solve the recognition problem of new classes that did not appear in the training but also solve the problem of the neural network, which difficult to converge due to the small number of experimental samples, thereby improving the accuracy of small datasets recognition.

The FSL methods used for image classification include model initialization, metric learning and data generate methods. The initialization method focuses on the adjustable parameters in the network so that a new class of classifier can be learned from a limited set of examples [90]–[92]. The aim of metric learning methods is learning to compare. It means that once a network learns to compare classes, it will be able to learn new classes from few labeled samples [93]. Finally, generate data methods, the methods learn a generator from the data in the base classes, and use that generator to generate data for new classes.

FSL solutions for plant leaves classification have been introduced recently, for example, Hu *et al.* [94] present a low shot learning method for tea leaf disease identification, used the improved conditional deep convolutional generative adversarial networks (C-DCGAN) for data augmentation. And the average identification accuracy of the proposed method was 90%. Wang and Wang [95] proposed a few-shot learning method based on the Siamese network with contrastive loss and kNN classifiers to solve plant leaf classification problem with a small sample. Das and Lee [96] proposed a two-stage multilayer neural network for the few-shot recognition of new categories and a detailed mathematical theory derivation process.

Aiming at the problem of too few samples in the training set, Li *et al.* [97] proposed a one-shot learning method for the first time and used Bayesian functions to build a network. Subsequently, many DL architectures for one-shot learning tasks were proposed and achieved remarkable results. The author verified in his work [98] that using FSL can transfer knowledge from a clear source domain (colon tissue) to a more general domain (composed of colon, lung, and breast tissues) by using few training images. Experimental results showed that the FSL can obtain an average accuracy of 90% with only 60 training images, which was better than fine-grained transfer learning (73%). Zhong *et al.* [99] proposed a generative model based on conditional adversarial auto-encoder (CAAE), which was used to perform generalized one-shot and few-shot learning in the case of few or even zero training samples to solve the problem of citrus diseases identification.

Ren *et al.* [100] proposed a plant disease identification method based on one-shot learning for the small sample problem of plant leaf diseases. Taking 8 kinds of plant disease with a small number of samples in the public dataset P1antVillage as the identification object, the focal loss function (FL) was used to train the plant disease classifier based on the relation network. The results showed that the recognition accuracy of the method in 5-way and 1-shot tasks reached 89.90%, which was 4.69% higher than the original relation network model. At the same time, compared with matching network and transfer learning, the improved method had increased the recognition accuracy on the experimental dataset by 25.02% and 41.90%, respectively.

Argüeso *et al.* [101] taken 38 kinds of plant disease images in the public dataset P1antVillage as the identification object, Siamese networks, and triplet loss was used and compared to classical fine-tuning transfer learning. The median accuracy was 55.5 % learning for 1 image per class. Median accuracy were 80.0 % and 90.0 % for 15 and 80 images per class. The FSL method outperformed the classical fine-tuning transfer learning which had an accuracy of 18.0 % and 72.0 % for 1 and 80 images per class, respectively. The author Wu [32] took 3 kinds of tea leaf diseases as the research object, segmented the lesions and expanded the dataset of the segmented lesions at first, and then used the combination of depth transfer and Cayley-Klein metric to realize the identification of tea diseases, and a result of 100% recognition accuracy was achieved.

Table 5 offers a brief overview of recent research works in plant leaf disease detection based on small samples.

V. HYPER-SPECTRAL IMAGING(HSI) WITH DL MODELS

Plants may be affected by multiple pathogens at the same time during the growth process, and some different pathogens may produce similar symptoms and signs [102], [103] and the symptoms are not obvious at the early stage of plant diseases, which makes it easy to use naked eyes or simple computer vision has become very difficult to detect plant diseases.

The electromagnetic spectrum range of hyperspectral sensors is mainly concentrated in the visible and near-infrared (400 ∼ 1000nm), and sometimes includes shortwave infrared (SWIR, 1000 ∼ 2500nm). This sensor can obtain spectral information from hundreds of narrow spectral bands [104]. These narrow bands are highly sensitive to subtle plant leaf changes caused by diseases and can distinguish different types of diseases so that early asymptomatic detection can be carried out. Therefore, HSI is the focus of recent research, for the early detection of plant diseases. For example, the review [105] provided an overview of advanced hyperspectral technologies for plant disease detection.

TABLE 5. A brief overview of recent research works in plant leaf disease detection based on small samples.

Reference	Object	DL frame	Dataset	Sample size	Metric
[98],2020	Colon, lung, and breast tissues	Few-shot, one-shot Learning	Medical center	60	Average accuracy
[100],2019	Plant	One-shot Learning	P1antVillage	20 per class	N-way, K-shot accuracy
[101],2020	Plant	Few-Shot Learning	PlantVillage	1 per class, 15 per class	N-way, K-shot accuracy
[96],2019	Others	Few-Shot Learning	Omniglot, MiniImagenet, CUB-200,CIFAR-100	5~15per class	N-way, K-shot accuracy
[99],2020	Citrus	One-shot, few-shot Learning	Self-acquired wild +network	1160	Top-1 harmonic mean accuracy
[32],2020	Tea leaf	CDCGAN-GPVGG16	Self-acquired wild	40 per class	Accuracy

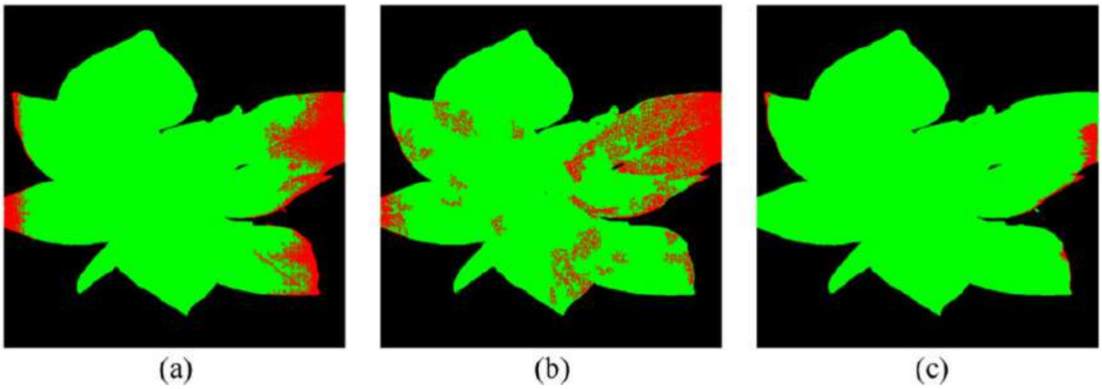


FIGURE 5. Comparison of segmentation results of a typical healthy plant. (a)direct CNN model. (b)AC-GAN model. (c)the proposed OR-AC-GAN model.

Xie *et al.* [106] investigated the feasibility of using a hyperspectral imaging technique to identify two kinds of diseases on tomato leaves. Both imaging information and spectral information were investigated in the study. The ELM model was established to identify the diseased samples and the successive projection algorithm (SPA) was applied to select useful wavelengths. The classification accuracy was 97.1% of the SPA-ELM model on the testing set. The early hyperspectral images of cucumber downy mildew in greenhouses collected infield, it is influenced by environmental illumination and difficult to extract effective features from them. Qin *et al.* [107] proposed a novel method of extracting feature bands based on disease difference information. Which improved combining adaptive weighting algorithm (CARS) and successive projection algorithm (SPA), and an early detection model of cucumber downy mildew was established. For the hyperspectral image of healthy cucumber leaves and the daily hyperspectral images within 12 days of infection, the detection rate of 100% can be obtained from the 2∼12 day of infection, and the detection rate of the test set for 1 day of infection reached 95.8%. Abdulridha *et al.* [108] used hyper-spectral imaging technology combined with the MLP classification method, had an accuracy of 99% for the four stages of tomato bacterial spot disease and bacterial target spot disease (healthy asymptomatic stage, early stage, and late-stage). Yuan *et al.* [109] proposed a method for detecting tea tree anthracnose based on hyperspectral imaging. Through spectral sensitivity analysis, the disease sensitive bands were determined, and two new disease indexes were established using these bands: tea tree anthracnose ratio index (TARI) and tea tree anthracnose normalized index (TANI). A method combined unsupervised classification and adaptive two-dimensional threshold detection is proposed, based on a set of optimized spectral features. The results showed that the overall accuracy of identifying diseases was 98% at the leaf level, and the overall accuracy of identifying diseases was 94% at the pixel level. In [110], a detailed review of DL with the HSI technique was provided. In order to avoid the over-fitting and improve accuracy, a detailed comparison was provided between several DL models like 1D/2D-CNN (2D-CNN better result) LSTM/GRU (both faced over-fitting), 2D-CNN-LSTM/GRU (still over-fitting). Therefore, a novel hybrid method (2D-CNN-BidLSTM/GRU) which from a convolutional and bidirectional gated recurrent network was proposed for the hyperspectral images. The model resolved the problem of over-fitting and achieved 75% F1-score and 73% accuracy for wheat disease detection [111]. In [112], the author developed a supervised 3D-CNN model to learn the spectral and spatial information of hyperspectral images for the classification of healthy and charcoal rot infected samples. A visualization method based on a saliency map was used to identify the classification accuracy of hyper-spectral wavelengths. The importance of wavelength can be inferred by analyzing the size of the gradient distribution of the

TABLE 6. A brief overview of recent research works in plant leaf disease detection using hyperspectral images.

Reference	Object	DL frame	Dataset	Sample size	Metric
[117],2019	Tomato	OR-AC-GAN	Cultivate-self and inoculate the virus, Spectra were collected from day 5,7,13 after inoculation	60	Accuracy, False positive rate
[116],2019	Corn	CNN	Hyper-spectral camera acquisition	10	Accuracy
[108],2019	Tomato	MLP,STDA	LAB acquired and UAV acquired	35	Accuracy
[112],2019	Soybean	3D DCNN	Cultivate-self and inoculate the virus	111	Accuracy
[115],2020	Soybean	CNN-SVM	Cultivate-self and inoculate the virus, Spectra were collected from day 2 to 15 after inoculation	480	Accuracy

saliency map of the image on the hyper-spectral wavelength. Based on the hyper-spectral imaging of the inoculated and simulated inoculated stem images, the classification accuracy of the model 3D-CNN was 95.73%, and the F1 score of the infection category was 87%. For the detection of potato virus, DL was used to describe the hyperspectral images and achieved acceptable values of precision (78%) and recall (88%) [113]. In [114], developed a DL model of multiple Inception-ResNet, which uses both spatial and spectral data on hyperspectral UAV images to detect the yellow rust in wheat. The model achieved an accuracy of 85%, which was quite a lot higher than the RF-classifier (77%). Gui *et al.* [115] divided the early soybean mosaic virus disease (SMV) into 0, 1, and 2 degrees according to its severity. In the case of a small number of experimental soybean samples, they proposed a novel SMV early detection method which combined convolutional neural network and a support vector machine (CNN-SVM), and achieved an accuracy rate of 96.67% on the training set and 94.17% on the testing set. The literature [116] taken corn seedlings after cold stress as the research object, extract the spectral curve of the comprehensive evaluation index of cold damage based on hyper-spectral images, and used DL to construct a corn seedling damage detection model. According to [117], a novel hyper-spectral analysis proximal sensing method based on generative adversarial nets (GANs), named as outlier removal auxiliary classifier generative adversarial nets (OR-AC-GAN) was proposed in order to detect tomato plant disease before its clear symptoms appeared. The classification accuracy achieved 96.25% before visible symptoms show up at plant leaf level (as shown in Fig. 5).

Table 6 offers a brief overview of recent research works in plant leaf disease detection using hyperspectral images.

VI. CONCLUSION AND FUTURE DIRECTIONS

In this paper, we have introduced the basic knowledge of deep learning and presented a comprehensive review of recent research work done in plant leaf disease recognition using deep learning. Provided sufficient data is available for training, deep learning techniques are capable of recognizing plant leaf diseases with high accuracy. The importance of collecting large datasets with high variability, data augmentation, transfer learning, and visualization of CNN activation maps in improving classification accuracy, and the importance of small sample plant leaf disease detection and the importance of hyper-spectral imaging for early detection of plant disease have been discussed. At the same time, there are also some inadequacies.

Most of the DL frameworks proposed in the literature have good detection effects on their datasets, but the effects are not good on other datasets, that is the model has poor robustness. Therefore, better robustness DL models are needed to adapt the diverse disease datasets.

In most of the researches, the PlantVillage dataset was used to evaluate the performance of the DL models. Although this dataset has a lot of images of several plant species with their diseases, it was taken in the lab. Therefore, it is expected to establish a large dataset of plant diseases in real conditions.

Although some studies are using hyperspectral images of diseased leaves, and some DL frameworks are used for early detection of plant leaves diseases, problems that affect the widespread use of HSI in the early detection of plant diseases remain to be resolved. That is, for early plant disease detection, it is difficult to obtain the labeled datasets, and even experienced experts cannot mark where the invisible disease symptoms are, and define purely invisible disease pixels, which is very important for HSI to detect plant disease.

REFERENCES

[1] F. Fina, P. Birch, R. Young, J. Obu, B. Faithpraise, and C. Chatwin, ''Automatic plant pest detection and recognition using k-means clustering algorithm and correspondence filters,'' *Int. J. Adv. Biotechnol. Res.*, vol. 4, no. 2, pp. 189–199, Jul. 2013.

[2] M. A. Ebrahimi, M. H. Khoshtaghaza, S. Minaei, and B. Jamshidi, ''Vision-based pest detection based on SVM classification method,'' *Comput. Electron. Agricult.*, vol. 137, pp. 52–58, May 2017.

[3] S. R. Dubey and A. S. Jalal, ''Adapted approach for fruit disease identification using images,'' *Int. J. Comput. Vis. Image Process.*, vol. 2, no. 3, pp. 44–58, Jul. 2012.

[4] A.-L. Chai, B.-J. Li, Y.-X. Shi, Z.-X. Cen, H.-Y. Huang, and J. Liu, ''Recognition of tomato foliage disease based on computer vision technology,'' *Acta Horticulturae Sinica*, vol. 37, no. 9, pp. 1423–1430, Sep. 2010.

[5] Z. R. Li and D. J. He, ''Research on identify technologies of apple's disease based on mobile photograph image analysis,'' *Comput. Eng. Des.*, vol. 31, no. 13, pp. 3051–3053 and 3095, Jul. 2010.

[6] Z.-X. Guan, J. Tang, B.-J. Yang, Y.-F. Zhou, D.-Y. Fan, and Q. Yao, ''Study on recognition method of rice disease based on image,'' *Chin. J. Rice Sci.*, vol. 24, no. 5, pp. 497–502, May 2010.

[7] J. G. A. Barbedo, ''Factors influencing the use of deep learning for plant disease recognition,'' *Biosyst. Eng.*, vol. 172, pp. 84–91, Aug. 2018.

[8] A. Kamilaris and F. X. Prenafeta-Boldú, ''Deep learning in agriculture: A survey,'' *Comput. Electron. Agricult.*, vol. 147, pp. 70–90, Apr. 2018.

[9] G. L. Grinblat, L. C. Uzal, M. G. Larese, and P. M. Granitto, ''Deep learning for plant identification using vein morphological patterns,'' *Comput. Electron. Agricult.*, vol. 127, pp. 418–424, Sep. 2016.

[10] S. P. Mohanty, D. P. Hughes, and M. Salathé, ''Using deep learning for image-based plant disease detection,'' *Frontiers Plant Sci.*, vol. 7, p. 1419, Sep. 2016.

[11] J. Ma, K. Du, F. Zheng, L. Zhang, Z. Gong, and Z. Sun, "A recognition method for cucumber diseases using leaf symptom images based on deep convolutional neural network," *Comput. Electron. Agricult.*, vol. 154, pp. 18–24, Nov. 2018.

[12] Y. Kawasaki, H. Uga, S. Kagiwada, and H. Iyatomi, "Basic study of automated diagnosis of viral plant diseases using convolutional neural networks," in *Proc. Int. Symp. Vis. Comput.*, Las Vegas, NV, USA, Dec. 2015, pp. 638–645.

[13] Y. Kessentini, M. D. Besbes, S. Ammar, and A. Chabbouh, "A two-stage deep neural network for multi-norm license plate detection and recognition," *Expert Syst. Appl.*, vol. 136, pp. 159–170, Dec. 2019.

[14] A. K. Singh, B. Ganapathysubramanian, S. Sarkar, and A. Singh, "Deep learning for plant stress phenotyping: Trends and future perspectives," *Trends Plant Sci.*, vol. 23, no. 10, pp. 883–898, Oct. 2018.

[15] V. Singh, N. Sharma, and S. Singh, "A review of imaging techniques for plant disease detection," *Artif. Intell. Agricult.*, vol. 4, pp. 229–242, Oct. 2020.

[16] M. H. Saleem, J. Potgieter, and K. M. Arif, "Plant disease detection and classification by deep learning," *Plants*, vol. 8, no. 11, pp. 468–489, Oct. 2019.

[17] L. C. Ngugi, M. Abelwahab, and M. Abo-Zahhad, "Recent advances in image processing techniques for automated leaf pest and disease recognition—A review," *Inf. Process. Agricult.*, vol. 180, pp. 26–50, Apr. 2020.

[18] W. S. McCulloch and W. Pitts, "A logical calculus of the ideas immanent in nervous activity," *Bull. Math. Biophys.*, vol. 5, no. 4, pp. 115–133, Dec. 1943.

[19] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, Sep. 2012, pp. 1097–1105.

[20] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Sacramento, CA, USA, Jun. 2016, pp. 770–778.

[21] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Repr.*, London, U.K., Apr. 2014, pp. 1–14.

[22] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Sacramento, CA, USA, Jun. 2015, pp. 1–9.

[23] J. Amara, B. Bouaziz, and A. Algergawy, "A deep learning-based approach for banana leaf diseases classification," in *Proc. Datenbanksys. Für Bus., Technol. Web (BTW)*, Workshopband, CA, USA, Jul. 2017, pp. 1–24.

[24] M. Rußwurm and M. Körner, "Multi-temporal land cover classification with long short-term memory neural networks," in *Proc. Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.*, Hannover, Germany, Jun. 2017, pp. 551–558.

[25] M. Türkoglu and D. Hanbay, "Plant disease and pest detection using deep learning-based features," *TURKISH J. Electr. Eng. Comput. Sci.*, vol. 27, no. 3, pp. 1636–1651, May 2019.

[26] A. K. Mortensen, M. Dyrmann, H. Karstoft, R. N. Jørgensen, and R. Gislum, "Semantic segmentation of mixed crops using deep convolutional neural network," in *Proc. CIGR-AgEng Conf.*, Aarhus, Denmark, Jun. 2016, pp. 26–29.

[27] M. Dyrmann, H. Karstoft, and H. S. Midtiby, "Plant species classification using deep convolutional neural network," *Biosystems Eng.*, vol. 151, pp. 72–80, Nov. 2016.

[28] W. Xinshao and C. Cheng, "Weed seeds classification based on PCANet deep learning baseline," in *Proc. Asia–Pacific Signal Inf. Process. Assoc. Annu. Summit Conf. (APSIPA)*, Hong Kong, Dec. 2015, pp. 408–415.

[29] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, Quebec City, QC, Canada, Jun. 2014, pp. 2672–2680.

[30] H. Nazki, S. Yoon, A. Fuentes, and D. S. Park, "Unsupervised image translation using adversarial networks for improved plant disease recognition," *Comput. Electron. Agricult.*, vol. 168, Jan. 2020, Art. no. 105117.

[31] Y. Tian, G. Yang, Z. Wang, E. Li, and Z. Liang, "Detection of apple lesions in orchards based on deep learning methods of CycleGAN and YOLOV3-dense," *J. Sensors*, vol. 2019, pp. 1–13, Apr. 2019.

[32] H.-Y. Wu, "Identification of tea leaf's diseases in natural scene images based on low shot learning," M.S. thesis, Dept. Inf. Eng., Anhui Univ., Hefei, China, 2020.

[33] Q. Wu, Y. Chen, and J. Meng, "DCGAN-based data augmentation for tomato leaf disease identification," *IEEE Access*, vol. 8, pp. 98716–98728, 2020.

[34] B. Liu, C. Tan, S. Li, J. He, and H. Wang, "A data augmentation method based on generative adversarial networks for grape leaf disease identification," *IEEE Access*, vol. 8, pp. 102188–102198, 2020.

[35] M. Brahimi, K. Boukhalfa, and A. Moussaoui, "Deep learning for tomato diseases: Classification and symptoms visualization," *Appl. Artif. Intell.*, vol. 31, no. 4, pp. 299–315, May 2017.

[36] M. Brahimi, M. Arsenovic, S. Laraba, S. Sladojevic, K. Boukhalfa, and A. Moussaoui, "Deep learning for plant diseases: Detection and saliency map visualisation," in *Human and Machine Learning*. Boston, MA, USA: Springer, Jun. 2018, pp. 93–117. [Online]. Available: https://link.springer.com/book/10.1007/978

[37] A. C. Cruz, A. Luvisi, L. De Bellis, and Y. Ampatzidis, "Vision-based plant disease detection system using transfer and deep learning," in *Proc. Asabe Annu. Int. Meeting*, Spokane, WA, USA, Jul. 2017, pp. 16–19.

[38] M. Brahimi, S. Mahmoudi, K. Boukhalfa, and A. Moussaoui, "Deep interpretable architecture for plant diseases classification," in *Proc. Signal Process., Algorithms, Archit., Arrangements, Appl. (SPA)*, Poznan, Poland, Sep. 2019, pp. 111–116.

[39] C. DeChant, T. Wiesner-Hanks, S. Chen, E. L. Stewart, J. Yosinski, M. A. Gore, R. J. Nelson, and H. Lipson, "Automated identification of northern leaf blight-infected maize plants from field imagery using deep learning," *Phytopathology*, vol. 107, no. 11, pp. 1426–1432, Nov. 2017.

[40] J. Lu, J. Hu, G. Zhao, F. Mei, and C. Zhang, "An in-field automatic wheat disease diagnosis system," *Comput. Electron. Agricult.*, vol. 142, pp. 369–379, Nov. 2017.

[41] J. G. Ha, H. Moon, J. T. Kwak, S. I. Hassan, M. Dang, O. N. Lee, and H. Y. Park, "Deep convolutional neural network for classifying Fusarium wilt of radish from unmanned aerial vehicles," *J. Appl. Remote Sens.*, vol. 11, no. 4, p. 1, Dec. 2017.

[42] J. G. A. Barbedo, "Plant disease identification from individual lesions and spots using deep learning," *Biosyst. Eng.*, vol. 180, pp. 96–107, Apr. 2019.

[43] S. Ghosal, D. Blystone, A. K. Singh, B. Ganapathysubramanian, A. Singh, and S. Sarkar, "An explainable deep machine vision framework for plant stress phenotyping," *Proc. Nat. Acad. Sci. USA*, vol. 115, no. 18, pp. 4613–4618, Apr. 2018.

[44] Y. Lu, S. Yi, N. Zeng, Y. Liu, and Y. Zhang, "Identification of rice diseases using deep convolutional neural networks," *Neurocomputing*, vol. 267, pp. 378–384, Dec. 2017.

[45] A. Picon, A. Alvarez-Gila, M. Seitz, A. Ortiz-Barredo, J. Echazarra, and A. Johannes, "Deep convolutional neural networks for mobile capture device-based crop disease classification in the wild," *Comput. Electron. Agricult.*, vol. 161, pp. 280–290, Jun. 2019.

[46] A. Johannes, A. Picon, A. Alvarez-Gila, J. Echazarra, S. Rodriguez-Vaamonde, A. D. Navajas, and A. Ortiz-Barredo, "Automatic plant disease diagnosis using mobile capture devices, applied on a wheat use case," *Comput. Electron. Agricult.*, vol. 138, pp. 200–209, Jun. 2017.

[47] M. A. Khan, T. Akram, M. Sharif, M. Awais, K. Javed, H. Ali, and T. Saba, "CCDF: Automatic system for segmentation and recognition of fruit crops diseases based on correlation coefficient and deep CNN features," *Comput. Electron. Agricult.*, vol. 155, pp. 220–236, Dec. 2018.

[48] M. Kerkech, A. Hafiane, and R. Canals, "Deep leaning approach with colorimetric spaces and vegetation indices for vine diseases detection in UAV images," *Comput. Electron. Agricult.*, vol. 155, pp. 237–243, Dec. 2018.

[49] K. Nagasubramanian, A. K. Singh, A. Singh, S. Sarkar, and B. Ganapathysubramanian, "Usefulness of interpretability methods to explain deep learning based plant stress phenotyping," *Comput. Sci.*, vol. 4, pp. 18–32, Jul. 2020.

[50] Y. Sun, Z. Jiang, L. Zhang, W. Dong, and Y. Rao, "SLIC_SVM based leaf diseases saliency map extraction of tea plant," *Comput. Electron. Agricult.*, vol. 157, pp. 102–109, Feb. 2019.

[51] Z.-W. Hu, H. Yang, J.-M. Huang, and Q.-Q. Xie, "Fine-grained tomato disease recognition based on attention residual mechanism," *J. South China Agricult. Univ.*, vol. 40, no. 6, pp. 124–132, Jul. 2019.

[52] S. H. Lee, H. Goëau, P. Bonnet, and A. Joly, "New perspectives on plant disease characterization based on deep learning," *Comput. Electron. Agricult.*, vol. 170, Mar. 2020, Art. no. 105220.

[53] R. Qiu, C. Yang, A. Moghimi, M. Zhang, B. J. Steffenson, and C. D. Hirsch, "Detection of fusarium head blight in wheat using a deep neural network and color imaging," *Remote Sens.*, vol. 11, no. 22, Nov. 2019, Art. no. 2658.

[54] I. Ahmad, M. Hamid, S. Yousaf, S. T. Shah, and M. O. Ahmad, "Optimizing pretrained convolutional neural networks for tomato leaf disease detection," *Complexity*, vol. 2020, pp. 1–6, Sep. 2020, doi: 10.1155/2020/8812019.

[55] C. Bi, J. Wang, Y. Duan, B. Fu, J.-R. Kang, and Y. Shi, "MobileNet based apple leaf diseases identification," *Mobile Netw. Appl.*, vol. 10, pp. 1–9, Aug. 2020, doi: 10.1007/s11036-020-01640-1.

[56] F. Jiang, Y. Lu, Y. Chen, D. Cai, and G. Li, "Image recognition of four rice leaf diseases based on deep learning and support vector machine," *Comput. Electron. Agricult.*, vol. 179, Dec. 2020, Art. no. 105824.

[57] W.-J. Liang, H. Zhang, G.-F. Zhang, and H.-X. Cao, "Rice blast disease recognition using a deep convolutional neural network," *Sci. Rep.*, vol. 9, no. 1, pp. 1–10, Feb. 2019.

[58] J.-P. Huang, J.-X. Chen, K.-X. Li, J.-Y. Li, and H. Liu, "Identification of multiple plant leaf diseases using neural architecture search," *Trans. Chin. Soc. Agricult. Eng.*, vol. 36, no. 16, pp. 166–173, Aug. 2020.

[59] M.-S. Long, C.-J. OuYang, H. Liu, and Q. Fu, "Image recognition of camellia oleifera diseases based on convolutional neural network & transfer learning," *Trans. Chin. Soc. Agricult. Eng.*, vol. 34, no. 18, pp. 194–201, Sep. 2018.

[60] J.-H. Xu, M.-Y. Shao, Y.-C. Wang, and W.-T. Han, "Recognition of corn leaf spot and rust based on transfer learning with convolutional neural network," *Trans. Chin. Soc. Agricult. Mach.*, vol. 51, no. 2, pp. 230–236, Feb. 2020.

[61] R. Thapa, N. Snavely, S. Belongie, and A. Khan, "The plant pathology 2020 challenge dataset to classify foliar disease of apples," *Comput. Sci.*, vol. 3, Feb. 2020, Art. no. 11958.

[62] K.-Z. Li, J.-H. Lin, J.-R. Liu, and Y.-D. Zhao, "Using deep learning for image-based different degrees of ginkgo leaf disease classification," *Information*, vol. 11, no. 2, pp. 95–104, Feb. 2020.

[63] B. Liu, Y. Zhang, D.-J. He, and Y.-X. Li, "Identification of apple leaf diseases based on deep convolutional neural networks," *Symmetry*, vol. 10, no. 1, pp. 11–19, Dec. 2018.

[64] C.-S. Wang, J. Zhou, H.-R. Wu, G.-F. Teng, C.-J. Zhao, and J.-X. Li, "Identification of vegetable leaf diseases based on improved multi-scale ResNet," *Trans. Chin. Soc. Agricult. Eng.*, vol. 36, no. 20, pp. 209–217, Doc. 2020.

[65] S.-G. Ren, F.-W. Jia, X.-J. Gu, P.-S. Yuan, W. Xue, and H.-L. Xu, "Recognition and segmentation model of tomato leaf diseases based on deconvolution-guiding," *Trans. Chin. Soc. Agricult. Eng.*, vol. 36, no. 2, pp. 186–195, Jan. 2020.

[66] X. Q. Guo, T. J. Fan, and X. Shu, "Tomato leaf diseases recognition based on improved multi-scale AlexNet," *Trans. Chin. Soc. Agricult. Eng.*, vol. 35, no. 13, pp. 162–169, Jul. 2019.

[67] X.-P. Fan, J.-P. Zhou, and Y. Xu, "Recognition of field maize leaf diseases based on improved regional convolutional neural network," *J. South China Agricult. Univ.*, vol. 41, no. 6, pp. 82–91, Jun. 2020.

[68] Z. Wang, S.-W. Zhang, and X.-F. Wang, "Method for segmentation of cucumber leaf lesions based on improved full convolution neural network," *Jiangsu J. Agricult. Sci.*, vol. 35, no. 5, pp. 1054–1060, May 2019.

[69] A. Picon, M. Seitz, A. Alvarez-Gila, P. Mohnke, A. Ortiz-Barredo, and J. Echazarra, "Crop conditional convolutional neural networks for massive multi-crop plant disease classification over cell phone acquired images taken on real field conditions," *Comput. Electron. Agricult.*, vol. 167, Dec. 2019, Art. no. 105093.

[70] J. Chen, J. Chen, D. Zhang, Y. Sun, and Y. A. Nanehkaran, "Using deep transfer learning for image-based plant disease identification," *Comput. Electron. Agricult.*, vol. 173, Jun. 2020, Art. no. 105393.

[71] S. Zhang, S. Zhang, C. Zhang, X. Wang, and Y. Shi, "Cucumber leaf disease identification with global pooling dilated convolutional neural network," *Comput. Electron. Agricult.*, vol. 162, pp. 422–430, Jul. 2019.

[72] X. He, S.-Q. Li, and B. Liu, "Grape leaf disease identification based on multi-scale residual network," *Comput. Eng.*, vol. 46, no. 4, pp. 1–8, Apr. 2020.

[73] M. Agarwal, A. Singh, S. Arjaria, A. Sinha, and S. Gupta, "ToLeD: Tomato leaf disease detection using convolution neural network," *Procedia Comput. Sci.*, vol. 167, pp. 293–301, Jan. 2020.

[74] A. Fuentes, S. Yoon, S. Kim, and D. Park, "A robust deep-learning-based detector for real-time tomato plant diseases and pests recognition," *Sensors*, vol. 17, no. 9, p. 2022, Sep. 2017.

[75] A. F. Fuentes, S. Yoon, J. Lee, and D. S. Park, "High-performance deep neural network-based tomato plant diseases and pests diagnosis system with refinement filter bank," *Frontiers Plant Sci.*, vol. 9, Aug. 2018, Art. no. 1162.

[76] P. Jiang, Y. Chen, B. Liu, D. He, and C. Liang, "Real-time detection of apple leaf diseases using deep learning approach based on improved convolutional neural networks," *IEEE Access*, vol. 7, pp. 59069–59080, 2019.

[77] J.-H. Li, L.-J. Lin, and K. Tian, "Detection of leaf diseases of balsam pear in the field based on improved faster R-CNN," *Trans. Chin. Soc. Agricult. Eng.*, vol. 36, no. 12, pp. 179–185, Jun. 2020.

[78] X.-R. Li, S.-Q. Li, and B. Liu, "Apple leaf disease detection method based on improved faster R-CNN," *Comput. Eng.*, vol. 46, no. 11, pp. 59–64, Nov. 2020.

[79] D. Li, R. Wang, C. Xie, L. Liu, J. Zhang, R. Li, F. Wang, M. Zhou, and W. Liu, "A recognition method for rice plant diseases and pests video detection based on deep convolutional neural network," *Sensors*, vol. 20, no. 3, Jan. 2020, Art. no. 578.

[80] W. Zhen, Z. Shanwen, and Z. Baoping, "Crop diseases leaf segmentation method based on cascade convolutional neural network," *Comput. Eng. Appl.*, vol. 56, no. 15, pp. 242–250, Aug. 2020.

[81] M. M. Ozguven and K. Adem, "Automatic detection and classification of leaf spot disease in sugar beet using deep learning algorithms," *Phys. A, Stat. Mech. Appl.*, vol. 535, Dec. 2019, Art. no. 122537.

[82] X.-D. Yu, M.-J. Yang, H.-Q. Zhang, D. Li, Y.-Q. Tang, and X. Yu, "Research and application of crop diseases detection method based on transfer learning," *Trans. Chin. Soc. Agricult. Eng.*, vol. 51, no. 10, pp. 252–258, May 2020.

[83] X.-Z. Li, Y. Xu, Z.-H. Wu, Z. Gao, and L. Liu, "Recognition system of tomato leaf disease based on attentional neural network," *Jiangsu J. Agricult. Sci.*, vol. 36, no. 3, pp. 561–568, Mar. 2020.

[84] F.-Q. Jiang, C. Li, D.-W. Yu, M. Sun, and E.-B. Zhang, "Design and experiment of tobacco leaf grade recognition system based on caffe," *J. Chin. Agricult. Mech.*, vol. 40, no. 1, pp. 126–131, Jan. 2019.

[85] M.-M. Zhou, "Apple foliage diseases recognition in Android system with transfer learning-based," M.S. thesis, Dept. Inf. Eng., Northwest A&F Univ., Yangling, China, 2019.

[86] Y. Liu, Q. Feng, and S.-Z. Wang, "Plant disease identification method based on lightweight CNN and mobile application," *Trans. Chin. Soc. Agricult. Eng.*, vol. 35, no. 17, pp. 194–204, Jun. 2019.

[87] J. G. M. Esgario, R. A. Krohling, and J. A. Ventura, "Deep learning for classification and severity estimation of coffee leaf biotic stress," *Comput. Electron. Agricult.*, vol. 169, Feb. 2020, Art. no. 105162.

[88] Y. Xiong, L. Liang, L. Wang, J. She, and M. Wu, "Identification of cash crop diseases using automatic image segmentation algorithm and deep learning with expanded dataset," *Comput. Electron. Agricult.*, vol. 177, Oct. 2020, Art. no. 105712.

[89] H. Larochelle, "Few-shot learning," in *Computer Vision: A Reference Guide*, 2nd ed., K. Ikeuchi, Ed. Tokyo, Tokyo: Univ. Tokyo Press, 2014. [Online]. Available: http://www.doc88.com/

[90] C. Finn, K. Xu, and S. Levine, "Probabilistic model-agnostic meta-learning," *Comput. Sci.*, vol. 2, pp. 9516–9527, Oct. 2018.

[91] A. A. Rusu, D. Rao, J. Sygnowski, O. Vinyals, R. Pascanu, S. Osindero, and R. Hadsell, "Meta-learning with latent embedding optimization," *Comput. Sci.*, vol. 3, pp. 18–27, Mar. 2018.

[92] A. Nichol and J. Schulman, "Reptile: A scalable meta learning algorithm," *Comput. Sci.*, vol. 2, no. 2, pp. 1–8, Mar. 2018.

[93] G. Koch, "Siamese neural networks for one-shot image recognition," M.S. thesis, Dept. Comput. Sci., Toronto Univ., Toronto, ON, Canada, 2015.

[94] G. Hu, H. Wu, Y. Zhang, and M. Wan, "A low shot learning method for tea leaf's disease identification," *Comput. Electron. Agricult.*, vol. 163, Aug. 2019, Art. no. 104852.

[95] B. Wang and D. Wang, "Plant leaves classification: A few-shot learning method based on Siamese network," *IEEE Access*, vol. 7, pp. 151754–151763, 2019.

[96] D. Das and C. S. G. Lee, "A two-stage approach to few-shot learning for image recognition," *IEEE Trans. Image Process.*, vol. 29, no. 5, pp. 3336–3350, Dec. 2020.

[97] L. Fei-Fei, R. Fergus, and P. Perona, "One-shot learning of object categories," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 4, pp. 594–611, Apr. 2006.

[98] A. Medela, A. Picon, C. L. Saratxaga, O. Belar, V. Cabezón, R. Cicchi, R. Bilbao, and B. Glover, "Few shot learning in histopathological images: Reducing the need of labeled data on biological datasets," in *Proc. IEEE 16th Int. Symp. Biomed. Imag. (ISBI)*, Venice, Italy, Apr. 2019, pp. 1860–1864.
[99] F. Zhong, Z. Chen, Y. Zhang, and F. Xia, "Zero- and few-shot learning for diseases recognition of citrus aurantium L. Using conditional adversarial autoencoders," *Comput. Electron. Agricult.*, vol. 179, Dec. 2020, Art. no. 105828.
[100] S.-N. Ren, Y. Sun, H.-Y. Zhang, and L.-X. Guo, "Plant disease identification for small sample based on one-shot learning," *Jiangsu J. Agricult. Sci.*, vol. 35, no. 5, pp. 1061–1067, May 2019.
[101] D. Argüeso, A. Picon, U. Irusta, A. Medela, M. G. San-Emeterio, A. Bereciartua, and A. Alvarez-Gila, "Few-shot learning approach for plant disease classification using images taken in the field," *Comput. Electron. Agricult.*, vol. 175, Aug. 2020, Art. no. 105542.
[102] P. Baranowski, M. Jedryczka, W. Mazurek, D. Babula-Skowronska, A. Siedliska, and J. Kaczmarek, "Hyperspectral and thermal imaging of oilseed rape (Brassica napus) response to fungal species of the genus alternaria," *PLoS ONE*, vol. 10, no. 3, Mar. 2015, Art. no. e0122913.
[103] S. Graeff, J. Link, and W. Claupein, "Identification of powdery mildew (Erysiphe graminis sp. Tritici) and take-all disease (Gaeumannomyces graminis sp. Tritici) in wheat (Triticum aestivum L.) by means of leaf reflectance measurements," *Open Life Sci.*, vol. 1, no. 2, pp. 275–288, Jun. 2006.
[104] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.
[105] N. Zhang, G. Yang, Y. Pan, X. Yang, L. Chen, and C. Zhao, "A review of advanced technologies and development for hyperspectral-based plant disease detection in the past three decades," *Remote Sens.*, vol. 12, no. 19, Sep. 2020, Art. no. 3188.
[106] C. Xie, Y. Shao, X. Li, and Y. He, "Detection of early blight and late blight diseases on tomato leaves using hyperspectral imaging," *Sci. Rep.*, vol. 5, no. 1, pp. 1–11, Nov. 2015.
[107] L.-F. Qin, X. Zhang, and X.-Q. Zhang, "Early detection of cucumber downy mildew in greenhouse by hyperspectral disease differential feature extraction," *Trans. Chin. Soc. Agricult. Mach.*, vol. 51, no. 11, pp. 212–220, Nov. 2020.
[108] J. Abdulridha, Y. Ampatzidis, S. C. Kakarla, and P. Roberts, "Detection of target spot and bacterial spot diseases in tomato using UAV-based and benchtop-based hyperspectral imaging techniques," *Precis. Agricult.*, vol. 21, no. 5, pp. 955–978, Oct. 2020.
[109] L. Yuan, P. Yan, W. Han, Y. Huang, B. Wang, J. Zhang, H. Zhang, and Z. Bao, "Detection of anthracnose in tea plants based on hyperspectral imaging," *Comput. Electron. Agricult.*, vol. 167, Dec. 2019, Art. no. 105039.
[110] A. Signoroni, M. Savardi, A. Baronio, and S. Benini, "Deep learning meets hyperspectral image analysis: A multidisciplinary review," *J. Imag.*, vol. 5, no. 5, p. 52, May 2019.
[111] X. Jin, L. Jie, S. Wang, H.-J. Qi, and S.-W. Li, "Classifying wheat hyperspectral pixels of healthy heads and Fusarium head blight disease using a deep neural network in the wild field," *Remote Sens.*, vol. 10, no. 3, Mar. 2018, Art. no. 395.
[112] K. Nagasubramanian, S. Jones, A. K. Singh, S. Sarkar, A. Singh, and B. Ganapathysubramanian, "Plant disease identification using explainable 3D deep learning on hyperspectral images," *Plant Methods*, vol. 15, no. 1, pp. 1–10, Aug. 2019.
[113] G. Polder, P. M. Blok, H. A. C. de Villiers, J. M. van der Wolf, and J. Kamp, "Potato virus Y detection in seed potatoes using deep learning on hyperspectral images," *Frontiers Plant Sci.*, vol. 10, Mar. 2019, Art. no. 209.
[114] X. Zhang, L. Han, Y. Dong, Y. Shi, W. Huang, L. Han, P. González-Moreno, H. Ma, H. Ye, and T. Sobeih, "A deep learning-based approach for automated yellow rust disease detection from high-resolution hyperspectral UAV images," *Remote Sens.*, vol. 11, no. 13, Jun. 2019, Art. no. 1554.
[115] J. Gui, J. Fei, Z. Wu, X. Fu, and A. Diakite, "Grading method of soybean mosaic disease based on hyperspectral imaging technology," *Inf. Process. Agricult.*, vol. 160, no. 7, pp. 1–6, Nov. 2020.
[116] W. Yang, C. Yang, Z. Hao, C. Xie, and M. Li, "Diagnosis of plant cold damage based on hyperspectral imaging and convolutional neural network," *IEEE Access*, vol. 7, pp. 118239–118248, 2019.
[117] D. Wang, R. Vinson, M. Holmes, G. Seibel, A. Bechar, S. Nof, and Y. Tao, "Early detection of tomato spotted wilt virus by hyperspectral imaging and outlier removal auxiliary classifier generative adversarial nets (OR-AC-GAN)," *Sci. Rep.*, vol. 9, no. 1, pp. 1–14, Mar. 2019.

LILI LI received the B.S. degree from Datong University, China, in 2012, and the M.S. degree from Chongqing University, China, in 2015. She is currently pursuing the Ph.D. degree with the College of Agricultural Engineering, Shanxi Agricultural University. She is currently working with Shanxi Agricultural University. Her research interest includes applications of computer vision in agriculture, using deep learning methods to study plant leaf disease detection in complex background.

SHUJUAN ZHANG received the Ph.D. degree in agricultural mechanization engineering from Zhejiang University. She is currently a Professor and the Doctoral Director with the College of Agricultural Engineering, Shanxi Agricultural University. She has published over 60 articles in her research-related fields. Her major research interests include digital agriculture information collection technology and equipment, technology and equipment for harvesting, primary processing, and non-destructive testing of agricultural products.

BIN WANG received the B.S. and M.S. degrees from Shanxi Agricultural University, China, in, 2011and 2015, respectively, where he is currently pursuing the Ph.D. degree with the College of Agricultural Engineering. His research interests include hyperspectral technology and nondestructive testing technology.

IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 4 February 2024, accepted 26 February 2024, date of publication 4 March 2024, date of current version 14 March 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3373001

SURVEY

Plant Leaf Disease Detection, Classification, and Diagnosis Using Computer Vision and Artificial Intelligence: A Review

ANUJA BHARGAVA[1], AASHEESH SHUKLA[1], OM PRAKASH GOSWAMI[2], MOHAMMED H. ALSHARIF[3], PEERAPONG UTHANSAKUL[4], (Member, IEEE), AND MONTHIPPA UTHANSAKUL[4], (Member, IEEE)

[1]Department of Electronics and Communication Engineering, GLA University, Mathura 281406, India
[2]Department of Electronics, Somaiya Vidyavihar University, Mumbai 400077, India
[3]Department of Electrical Engineering, College of Electronics and Information Engineering, Sejong University, Seoul 05006, Republic of Korea
[4]School of Telecommunication Engineering, Suranaree University of Technology, Nakhon Ratchasima 30000, Thailand

Corresponding authors: Mohammed H. Alsharif (malsharif@sejong.ac.kr) and Peerapong Uthansakul (uthansakul@sut.ac.th)

This work was supported in part by the Suranaree University of Technology (SUT); in part by Thailand Science Research and Innovation (TSRI); and in part by the National Science, Research and Innovation Fund (NSRF).

ABSTRACT Agriculture is the ultimate imperative and primary source of origin to furnish domestic income for multifarious countries. The disease caused in plants due to various pathogens like viruses, fungi, and bacteria is liable for considerable monetary losses in the agriculture corporation across the world. The security of crops concerning quality and quantity is crucial to monitor disease in plants. Thus, recognition of plant disease is essential. The plant disease syndrome is noticeable in distinct parts of plants. Nonetheless, commonly the infection is detected in distinct leaves of plants. Computer vision, deep learning, few-shot learning, and soft computing techniques are utilized by various investigators to automatically identify the disease in plants via leaf images. These techniques also benefit farmers in achieving expeditious and appropriate actions to avoid a reduction in the quality and quantity of crops. The application of these techniques in the recognition of disease can avert the disadvantage of origin by a factious selection of disease features, extraction of features, and boost the speed of technology and efficiency of research. Also, certain molecular techniques have been established to prevent and mitigate the pathogenic threat. Hence, this review helps the investigator to automatically detect disease in plants using machine learning, deep learning and few shot learning and provide certain diagnosis techniques to prevent disease. Moreover, some of the future works in the classification of disease are also discussed.

INDEX TERMS Deep learning, diagnosis, image processing, machine learning, plant disease.

I. INTRODUCTION

The agriculture and food organization of the United Nations disclosed that the total number of starved people across the world has been growing deliberately since 2015 [1]. The ongoing estimation reveals that around 680 million people are starved and they comprise under 9% of the population around the world. This indicates an expansion of 10 million in a year and approximately 120 million in 10 years. Around more than 85% of people confide in agriculture in the world [2]. Accordingly, the food industry and agriculture demand adequate farming mechanisms. Further, living creatures accept oxygen from plants in the balancing environment by photosynthesis. The plant disease that affects the leaves may lead to death or bad health of the plant and abort to implementation of plant food. Also, plant disease has a gigantic effect on developing food crops. In 1845 potato (Irish famine) resulted in 1.2 million deaths [3]. Some of the laboratory approaches used for plant disease are immunosorbent enzymes, isothermal amplification, and polymerase reaction. Hence, early

The associate editor coordinating the review of this manuscript and approving it for publication was Luca Barletta.

detection, management, and prevention of disease in plants are precisely essential. Nonetheless, disease detection of plants in the enormous field is a very complicated task that involves trained manpower and optical examination of leaves [4]. Farmers observe the symptoms of diseases on plant leaves with the naked eye and diagnose plant diseases based on experience, which is laborious, time-consuming, and requires special skills [5]. The objective of a disease detection system is to support non-expert users, i.e. non-pathologist and non-botanists. Therefore, various automated techniques have been developed using image processing, machine learning, deep learning, few short learning are discussed in this review paper. The agriculture practical applications are not sufficient using conventional ML approaches due to robustness and laboratory conditions [6], [7]. The deep learning in recent times has built exceptional in the classification of images for plant disease [8], [9]. The approach required a large number of samples to meet the challenges in time. Also, all the images are fundamentally annotated by pathologists and ethnologists. Since these efforts are laborious and time-consuming, deep learning applications in disease detection are costly [10], [11]. FSL facilitates learning from machines using a limited number of labeled databases [12], [13]. The experiment's purpose and problem complexity determine the number of shots required. Various pathogens [14], [15] cause the disease a diagnosis by DNA (Deoxyribose nucleic acid), PCR (Polymerase chain reaction), MPG (Modified panchayat mixture), ELISA (Enzyme-linked immunosorbent assay), FISH (Fluorescence in situ hybridization) and IF (Immunofluorescence) method. Therefore, the objective of this review paper is to give a comparative analysis of machine learning, deep learning, few short learnings in plant disease detection and also review various segmentation, feature extraction, and classification techniques. Also, certain molecular diagnosis tools are reported.

II. PHYTOPATHOLOGY

The study of plant pathogens, responsible pathogens their disease, mechanism, control methods of disease and to diminish their brunt on plants is referred to as Phytopathology [16]. As a result, it is a system for dealing with a plant's life cycle. Phytopathology is a Greek letter word that means plant (Phyto), disease (Patho), and knowledge (Logo). The basic objectives of phytopathology are a study of the source, and causes of plant disease as biotic or abiotic (etiology), a mechanism study of development for disease (pathogenesis), the interaction between plant disease and pathogen (epidemiology), and development management for reduction of radiation and losses control management as shown in Fig. 1.

Phytopathology comprises of subdomain under agriculture science and consists of elemental learning of microbiology, physiology, nematology, virology, anatomy, bacteriology, mycology, genetic engineering, botany, meteorology, climatology, and molecular biology indicated in Fig. 2.

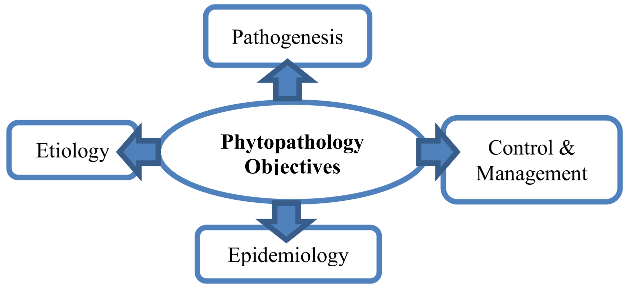


FIGURE 1. Phytopathology objectives.

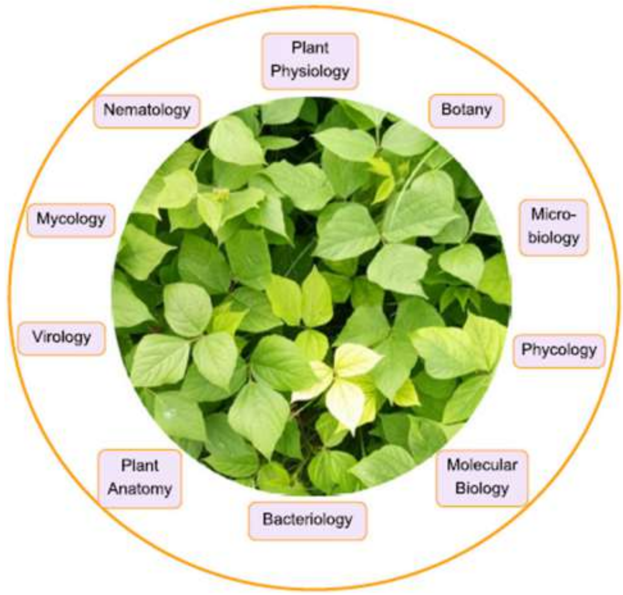


FIGURE 2. Subdomains of phytopathology [16].

III. PLANT DISEASE/TYPES AND SYMPTOMS

The behavior or physiology of the plants in terms of abnormality results in plant disease. It is due to biotic or abiotic agents as indicated in Fig. 3 [16]. Biotic disease is generated from infectious agents whereas abiotic disease is generated from non-infectious agents. The abiotic disease is usually avoidable due to its non-transmissible lesser hazardous nature. Thus, in this manuscript, biotic disease is examined.

- Bacterial Disease- Plant bacterial disease occurs due to water-soaked results in a little green blemish. These lesions augment and then emerge as dead dry blemishes (spots) represented in Fig. 4(a). Example: The foliage consists of water-soaked black blemish or brown leaf spot or halo yellow with equivalent size. The blemish appears as dappled beneath dry conditions. Generally, Bacterial wilt occurs in brinjal crops due to which the whole plant falls [17].
- Viral Disease- Plant viral disease is the most burdensome disease to investigate among all the plant diseases. The virus observed on the plant has no indication or it looks similar to herbicide bruise as well as nutrient

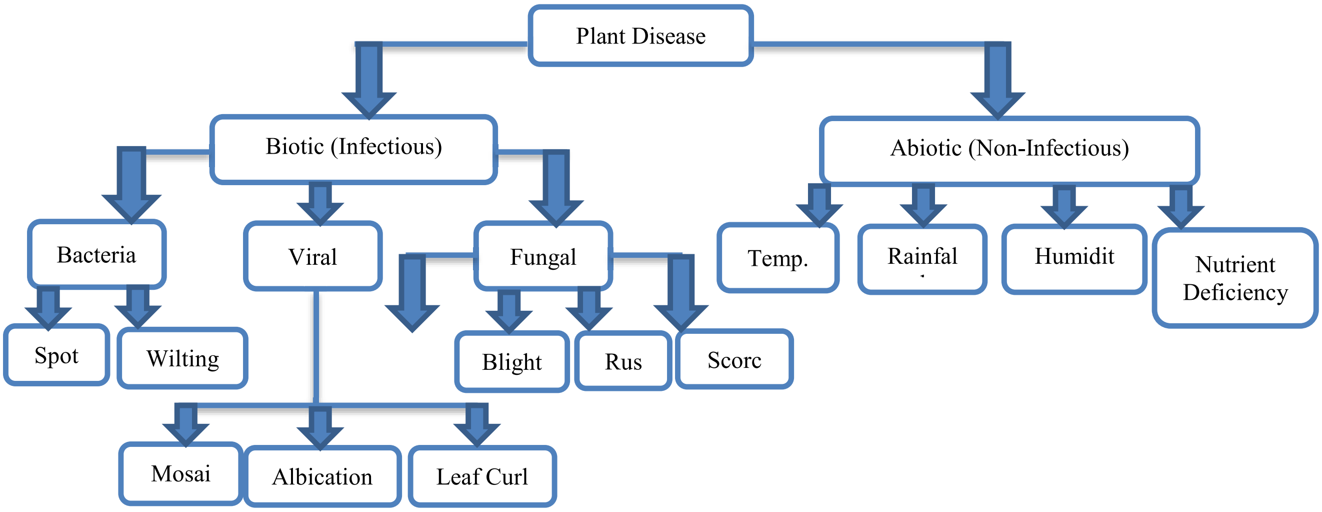


FIGURE 3. Classification of plant disease in distinct categories [16].

FIGURE 4. (a) Bacterial blemish [19] (b) Viral Mosaic [20] (c) Late Blight [22] (d) Early Blight [22] (e) Rust.

TABLE 1. Distinct disease in different plants.

Author's	Plant Name	Bacterial Disease	Viral Disease	Fungal Disease
Zhang et al. 2019 [19] Kianat et al. 2021 [20] Agarwal et al. 2021 [21]	Cucumber	Brown blemish, Angular Blemish, Target blemish	Mosaic, Yellow blemish	Black blemish, Gray mold
Shrivastava et al. 2019 [22] Chen et al. 2021 [23]	Rice	Streak, Blight	Black Dwarf Streaked	Smut False
Sun et al. 2021 [24]	Maize	Streak, Stalk	Crimson, Dwarf	Rust
Ferentinos 2018 [25] Abbas et al. 2021 [26]	Tomato	Canker	Curl leaf yellow	Late/ Early Blight

failure [17]. The most commonly found viral diseases are in beetles, leafhoppers, aphids, and whiteflies such as mosaic viral disease indicated in green or yellow stripes for foliage as shown in Fig. 4 (b).

- Fungal Disease- Plants fungal disease affects various components of plants i.e., sclerotium wilt, common crown rot, stem rust, eyespot (sheath or stem), rust, blight (leaves), ergot (spikes) and carnal bunt, black point (seeds). Phytophthora fungus caused by late blight emerges on former leaves such as gray-green blemish, waterlog shown in Fig. 4(c). This fungus is caused due to climate change in the form of wet and dry. As the late blight disease is cultivated, these blemishes get dark and fungus in white form grows on the surface [18]. Alternaria fungus caused by early blight, emerges on former leaves in the form of tinny brown blemish with bull's eye arrangement in the form of concentric rings as shown in Fig. 4(d). Rust fungus develops on ripe plant leaves on the curtailed surface as shown in Figure 4(e). This blemish becomes black after green-yellow.

TABLE 2. Distinct plants, their disease and responsible pathogen [16].

Plant	Disease	Pathogen	Symptoms
Apple	Scab	Pomi Spilocaea	Brown-Gray on leaf
	Rot	Malorum Sphaeropsis	Dark Brown on leaf
	Rust	Sporangium	Yellow pale on leaf
Cherry	Mildew	Clandestina	Gray powder on leaf
Corn	Gray Spot	Cercospora	Rectangle lesions
	Rust	Sorghi puccinia	Red pustules on leaf
	Light blight	Tutcica setosphaeria	Elliptical lesions
Grape	Rot	Bidwellii guignardia	Red borders on leaf
	Measles	Aleophilum	Necrotic stripping
	Isariopsis blight	Angulata brachypus	Coalesce lesions
Peach	Spot	Arboricola Xanthomonas	Clustered lesions
Potato	Early blight	Solani Alternaria	Brown lesion
	Late blight	Infestans phytophthora	Dark greeb spot
Tomato	Septoria spot	Lycopersici	Foliage
	Mosaic	Mosaic virus	Mottle green leaf
Orange	Green Citrus	Bacteria Motile	Precipitate Demolition
Strawberry	Scorch Fungus	Diplocarpon	Brown edges
Squash	Mildew	Xanthii podosphaers	White powder

The above-described and mentioned disease symptoms are distinct and very less compared to other conventional diseases in plants. That means the equivalent disease symptom may arise due to infectious or non-infectious diseases. Some of the distinct diseases in plants are indicated in Table 1.

Therefore, accurate diagnosis of plant disease is troublesome for the identification of pathogens using particular symptoms.

The artificial intelligence techniques, for detecting the performance of disease detection in plants essentially depend on the feature extraction as well as a classification for disease [27], [28], [29]. These extracted selected features reveal several disease symptoms. Table 2 represents a manifestation of several typical diseases in plants with specific symptoms. The following Table will help the investigator to identify the accurate features which would further lead to high classification accuracy.

IV. PLANT DISEASE DETECTION SYSTEM

The artificial intelligence techniques are addressed to enhance agriculture production by bestowing in plant disease monitoring. Various review research papers have been published [30], [31], some of them focus on particular approaches while others are centered on particular diseases. A detailed summarization of plant disease detection, classification, and diagnosis is not available. Therefore, this review focuses on distinct techniques used by distinct researchers that utilize machine learning (ML), deep learning (DL), few-shot learning (FSL), and soft computing techniques with image processing for RGB as well as hyperspectral images. Also, certain molecular techniques have been established to prevent and mitigate the pathogenic threat.

A. MACHINE LEARNING USING IMAGE PROCESSING

The categorization of disease detection in plants using machine learning with image processing is done in the following sequential steps: Image Acquisition, Image Preprocessing, Image Segmentation, Feature Extraction and selection, and Classification [32]. In this, firstly images of plants (roots, branches, stems & leaves) are captured. Next, image pre-processing i.e., noise removal or blur or any other distortion is removed using techniques like scaling, stretching, smoothening, transformation, and rotation. Next, segmentation to extract the infected region using techniques like thresholding, and clustering is done. Then this segmented region is used to extract features for classification. In this section, the requirement of all these steps is mentioned and the approaches proposed in the literature are also summarized.

1) IMAGE ACQUISITION

The initial gait for any machine learning system is image acquisition. This incorporates image capture or redeems images from repositories. The image quality highly leans on the camera and its acclimatization which leads to disease detection accuracy of the system [33], [34]. The captured image might dwell in the undesired background, adumbration, and noise. To improve the accuracy of the system or to extract the affected part firstly background removal and noise removal are required. Subsequently, other than RGB images from normal cameras, certain specific cameras are used to capture hyperspectral, thermal, and fluorescent images. Various researchers use distinct datasets for plant disease detection systems as indicated in Table 3. The presence of shadow, light, insufficient lighting, and intricated background is burdensome due to uncontrolled/real-time images as compared to laboratory surroundings conditions. The quality of the image captured robustly depends upon the techniques and equipment used. Consequently, the system performance is influenced by the image acquisition stage.

2) IMAGE PRE-PROCESSING

The crucial step in a machine learning system is image processing. It helps to improve the quality of the image that is degraded due to distortions, turbulence, or shadow [56]. Majorly, the dataset is captured in uncontrolled environments or real-time situations. Therefore, the preprocessing step is necessary before feature extraction enlarges the estimated accuracy of the system which results in reduced processing time. The process also reduces the processing time by applying operations such as crop, and resize. The literature utilizes image enhancement, cropping, and elimination of background as extensively used preprocessing techniques. These operations applicability depends on the image quality. Various researchers use unique techniques for preprocessing are indicated in Table 4. Image preprocessing along with augmentation is used to create more images of the data on the existing data for several deep learning techniques that require large amounts of image data. The processing augmentation techniques are noise injection, flipping of image, gamma correction, scaling, rotation, resize, cropping, random shift, zoom, and image transformation such as contrast enhancement and brightness.

3) IMAGE SEGMENTATION

The region of interest or undesired region can be segregated using image segmentation. This approach intends to separate the part having a deformity. This analysis of the representation of the image is simple and further meaningful to discriminate the affected and unaffected parts. It is the most important technique among machine learning techniques as critical issues and techniques in the analysis of an image. The several challenges faced such as erroneous boundaries of the affected region, shadow, illumination, and complex background [74]. The segmentation process is classified into two main divisions: first is conventional methods such as thresholding, region growing and edge detection other is computational techniques such as fuzzy logic, genetic algorithm, and neural network. Generally, computational techniques perform better than conventional techniques. The segmentation process is crucial for the extraction of features. Various researchers use unique techniques for segmentation are indicated in Table 5.

TABLE 3. Details of the dataset available used by various researchers.

	Dataset Name	Authors
Open accessible dataset	APS image dataset	Mohanty et al. 2016 [33]
	Plant Village Image dataset	Mohanty et al. 2016 [33]
	Computers and Optics in Food Inspection (Cofi) laboratory image dataset	Arnal Barbedo et al. 2019 [34]
	Digipathos images (PDDB)	Arnal Barbedo et al. 2019 [34]
	IRRI Dataset	Bashir et al. 2019 [35]
	INIBAP leaf Dataset	Camargo & Smith 2009 [36]
Self-created dataset	-	Karadag et al. 2018 [37]
	-	Coulibaly et al. 2019 [38]
	-	Pantazi et al. 2019 [39]
	-	Fuentes et al. 2017 [40]
	-	Shrivastava and Hooda 2014 [41]
	RoCoLe	J. Parraga Alava. et al. 2019 [42]
		K. Tian, et al. 2019 [43]
Multiple Crop dataset	Citrus dataset	Zhang and Meng 2011 [44]
	Grapefruit Grove	Pydipati et al. 2006 [45]
	-	Masazhar and Kamal 2018 [46]
University Agriculture	-	Deshapande et al. 2019 [47]
	-	Pujari et al. 2016 [48]
	-	Abed and Esmaeel 2018 [49]
Dataset using spectral devices	-	Azadbakht et al 2019 [50]
	-	Rothe and Kshirsagar 2015 [51]
Hyperspectral imaging	-	Abdulridha et al 2019 [52]
	-	Zhang et al 2018 [53]
Charge Couple device camera	-	Huang 2007 [54]
	-	Yao et al 2009 [55]

TABLE 4. Details of pre-processing technique used by various researchers.

Techniques Used		Author's
Color Space Conversion	Enhancement Filtering, Background reduction	Kaur et al. 2018 [56] Das et al. 2020 [57]
	RGB, HSV, HSI, YIQ, L*a*b, grayscale	Khot et al. 2016 [58], Dhingra et al. 2017 [59], Cruz et al. 2018 [60]
	L a* b*, HSV	Rothe and Kshirsagar 2015 [37], Vidyaraj and Priya 2016 [61], Kaur et al. 2018 [37], Pantazi et al. 2019 [39]
	YCbCr, CIE	Kai et al. 2011 [62], Chaudhary et al. 2012 [63], Joshi and Jadhav 2017 [64]
	H, I3a, I3b	Camargo and Smith 2009a [36]
	CIELuv	Ganeshan et al. 2017 [65]
	La*b*, Luv, YCbCr	Meunkaewjinda et al. 2008 [66]
Image Enhancement Techniques	Denoising using Mean and median filtering	Yao et al. 2009 [41], Rao & Kulkarni 2020 [67]
	Sharpening using Gaussian and Laplacian filtering	Asfarian et al. 2013 [68], Abed and Esmaeel 2018 [49]
	Illumination variation using histogram equalization	Dange and Sayyad 2015 [69], Khirade and Patil 2015 [70], Malika and Vasanthi 2017 [71]
	Augmentation	Sladojevic et al. 2016 [72], Goncharov et al. 2019 [73]

TABLE 5. Details of segmentation technique used by various researchers.

Techniques Used		Author's
Edge-based segmentation	Canny edge detection, Sobel Operator, Prewitt Operator	Pydipati et al. 2006 [45], Anthonys and Wickramarachchi 2009 [75], Bankar et al. 2014 [76], Shinde et al. 2015 [77]
Thresholding Techniques	Otsu Thresholding, Adaptative Thresholding, Entropy Thresholding	Khirade and Patil 2015 [70], Pujari et al. 2016 [48], Cruz et al. 2018 [60], Das et al. 2020 [57]
Region Growing	Local Threshold	Pang et al. 2011 [78], Singh et al. 2015a [79]
Clustering	k-means	Rastogi et al. 2015 [80], Jadhav and Patil 2016 [81], Zhang et al. 2017b [82], Kaur et al. 2018b [56], Bashir et al. 2019 [35] Harakamanavar et al. 2022 [83] Zhang et al. 2018 [84]
	Fuzzy c-means	Jagtap and Hambarde 2014 [85], Bai et al. 2017 [86] Sahu & Pandey 2023 [87]
Grab Cut	-	Pantazi et al. 2019 [39]
Genetic Algorithm	-	Singh et al. 2015b [88], Singh and Misra 2017 [89]

4) FEATURE EXTRACTION

To discriminate one part of an image from another part, feature attribute extraction is one of the most important techniques in computer vision/ machine learning systems before classification. The features are advantageous to analyze objects and to allow the class label to an object. Generally, in the literature color, shape, and texture features are utilized for recognition. The classification of plant disease detection systems is enormously dependent upon the techniques used for image characteristic extraction [90]. From the literature, the spotted area segmentation and texture, color & shape feature extraction are of huge importance. Several plant disease detection researchers have used distinct convenient feature extraction techniques depending on texture, color & shape. Therefore, determining the correct features is a major issue in the detection of disease due to similarity in disease characteristics.

In computer vision/ machine learning applications, the selection of relevant features is also important to prevent overfitting and huge computational costs. Therefore, in computer vision and machine learning applications feature selection approaches are selected to find better relevant features from the extracted features. Some of the techniques such as PCA, genetic algorithm, and particle swarm optimization are used for feature selection. The literature utilizes various feature extraction techniques indicated in Table 6. By examining these approaches, an attempt is made to find the better-suited technique.

TABLE 6. Details of feature extraction technique used by various researchers.

Techniques Used		Author's
Feature Descriptor	GLCM, Wavelet Transform, Haralick feature, Gabor Transform, Local Binary Patterns	Mokhtar et al. 2015 [91],
	SURF	Bhagat & Kumar 2023 [92]
Texture Feature	GLCM Features	Bashish et al. 2011 [93], Tian et al. 2014 [94], Mainkar et al. 2015 [95], Islam et al. 2017 [96] Abed and Esmaeel 2018 [49], Sharif et al. 2018 [97], Deshapande et al. 2019 [47],
	Scale-invariant feature transform	Dandawate and Kokare 2015 [98], Mohan et al. 2016 [99], Ramesh et al. 2018 [100]
	Local Binary Pattern	Waghmare et al. 2016 [101], Singh et al. 2019 [102]
	Gabor Filter Transform	Gulhane & Gurjar 2011 [103] Prasad et al. 2012 [104] Kulkarni & R.K. 2012 [105] Jolly & Raman 2016 [106] Kaur et al. 2018b [56] Rao & Kulkarni 2020 [67]
	Wavelet Transform	Sabrol & Kumar 2016a [107] Gawali et al. 2017 [108]
	Histogram of Oriented Gradient	Dalal et al. 2005 [109] Bai et al. 2009 [110] Sannakki et al. 2013 [111] Pires et al. 2016 [112] Ramesh et al. 2018 [100] Kusumo et al. 2018 [113] Zamani et al. 2022 [114]
Color Feature	Color co-occurrence matrix	Pydipati et al. 2006 [45] Kai et al. 2011 [62] Revathi and Hemalatha 2014b [115] Ramakrishnan & Sahaya 2015 [116] Chouhan et al. 2019 [90]
	Color histogram	Caglayan et al. 2013 [117] Ramesh et al. 2018 [118]
Shape Feature	-	Anthonys & Wickramarachchi 2009 [75] Camargo & Smith 2009b [119] Wang et al. 2012 [120] Phadikar et al.2013 [121] Joshi & Jadhav 2017 [122] Sengar et al. 2018 [123] Sahu & Pandey 2023 [87]

The present scenario of literature during the last 12 years using different feature extraction techniques for distinct crops is represented in Fig 5. There are still various techniques to consider for plant disease detection using machine learning and computer vision techniques. The crop selection for such research is utilized by the availability of experts and data sets.

5) CLASSIFICATION

The utmost extrusive stage in computer vision/ machine learning is classification. The attainment of this step builds upon antecedent steps such as acquisition, preprocessing, and feature extraction/ selection. The manuscript is focused on disease detection in plants, the above steps are used for detection based on category and symptoms. In this system, a dataset is trained first and then used to classify the test image as healthy/defective. "Machine learning (ML) is an application of artificial intelligence (AI) which provides you a system an ability to automatically learn, improve from experiences without being explicitly programmed and which can make decisions [124]." ML techniques have been broadly distinguished as Supervised ML and Unsupervised ML. During the training process the labeled and unlabeled data work for supervised and unsupervised techniques respectively. The semi-supervised ML is another special supervised learning that works on both labeled and unlabeled data during the training process. In the literature, various classification techniques are utilized as indicated in Fig. 6.

Generally, support vector machine (SVM), artificial neural network (ANN), and k-nearest neighbor (k-NN) have been adapted for machine learning techniques [32]. Also, in a few studies vegetation index and fuzzy logic feature-based techniques are utilized. Distinct models of ANN like feed-forward NN, error backpropagation, multilayer perceptron, probabilistic neural network, and self-organizing map have been widely utilized for plant disease detection. Table 7 indicates the quality investigation of plant disease detection based on distinct classification techniques used by distinct researchers.

B. DEEP LEARNING USING IMAGE PROCESSING

Deep learning (DL) was first proposed in 1943 [146] as a hierarchy of machine learning for the detection of objects [147], [148], [149], [150], classification of images as well as processing of natural language [151], [152], [153]. DL forms high-level features by combining low-level features for distributed features discovering and sample data attributes. The ability of generalization and accuracy are enhanced compared to traditional ML techniques for target detection and image recognition. The development of DL is in three phases. Phase 1, during 1943-1969 was a linear model of neural network for classification. Phase 2, during 1986-1988 was a non-linear mapping of multi-layer perceptron for the BP (Backpropagation) algorithm. Phase 3, from 2006 to the present, has a ReLu activation function, ImageNet recognition, and deep learning model-AlexNet. The DL methods are

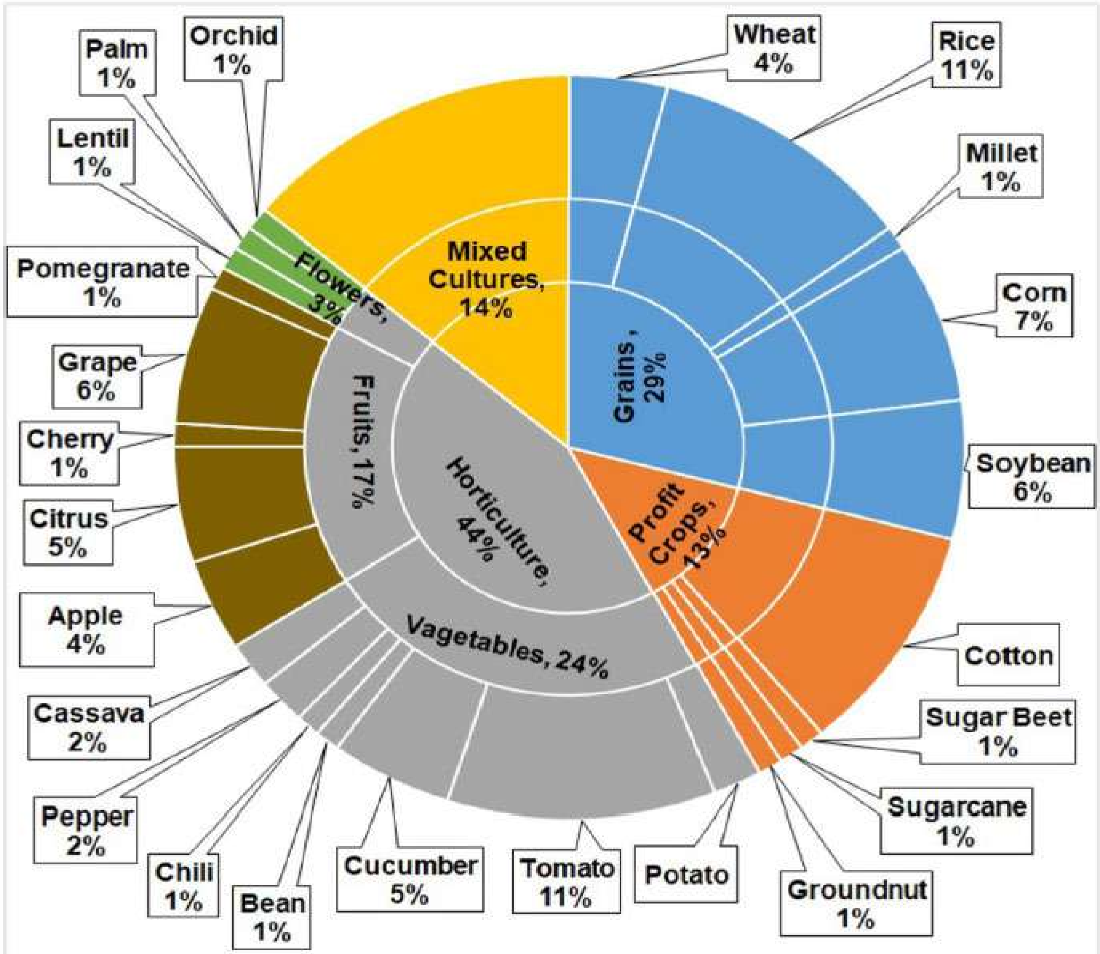


FIGURE 5. Different feature extraction techniques for distinct crops during last 12 years [16].

used for selecting features automatically. The techniques propose high-level features from the combination of low-level features. Subsequently, multifold advanced DL techniques proposed are shown in Fig. 7. Presently, the preeminent neural networks are convolutional neural networks, multilayer perceptron, and recurrent neural networks. The CNN establishes a classification model that incorporates AlexNet, GoogleNet, VGGNet, MobileNet, ResNet, and EfficientNet. AlexNet is the early model to adopt GPU devices for network training. The local normalization and rectified linear units are used as activation functions. Then VGG & GoogleNet use a few kernels with low memory [154]. In 2015, ResNet proposed by Microsoft uses connections and balance blocks [155]. In 2017, MobileNet was proposed by Google teams for embedded and mobile applications [156]. In 2019, EfficientNet was introduced by Google teams that use efficient coefficients to scale width, depth, and resolution. The deep network may not be used in plant disease detection since the basic models like VGG16 and AlexNet can fit the substantial accuracy needed. The C/C++, Python programming language can be used in the DL model. The various interface programming application such as deployment of the network, model support, and duplication of code is provided by the DL framework [157]. Currently, the widely used frameworks are Café, PyTorch, and Keras. The elemental steps of DL are indicated in Fig. 8 where CNN (Convolutional Neural Network) is used to incorporate features via convolutional, max-pooling, and fully connected layers. The features are extracted via the convolutional layer whereas texture and edges are extracted via shallow convolutional layer. The semantic information and complex texture features are extracted using the middle layer. The high-level semantic features are extracted using a deep layer. The max pool layer is used to hold the critical information in an image. Lastly, the fully connected layer is used to classify high-level features. Therefore, the implementation of DL requires the fundamental steps which are discussed in detail in the subsequent section.

1) IMAGE ACQUISITION

The data acquired must be accurate to obtain a perfect model. The DL database consists of training, validation, and test

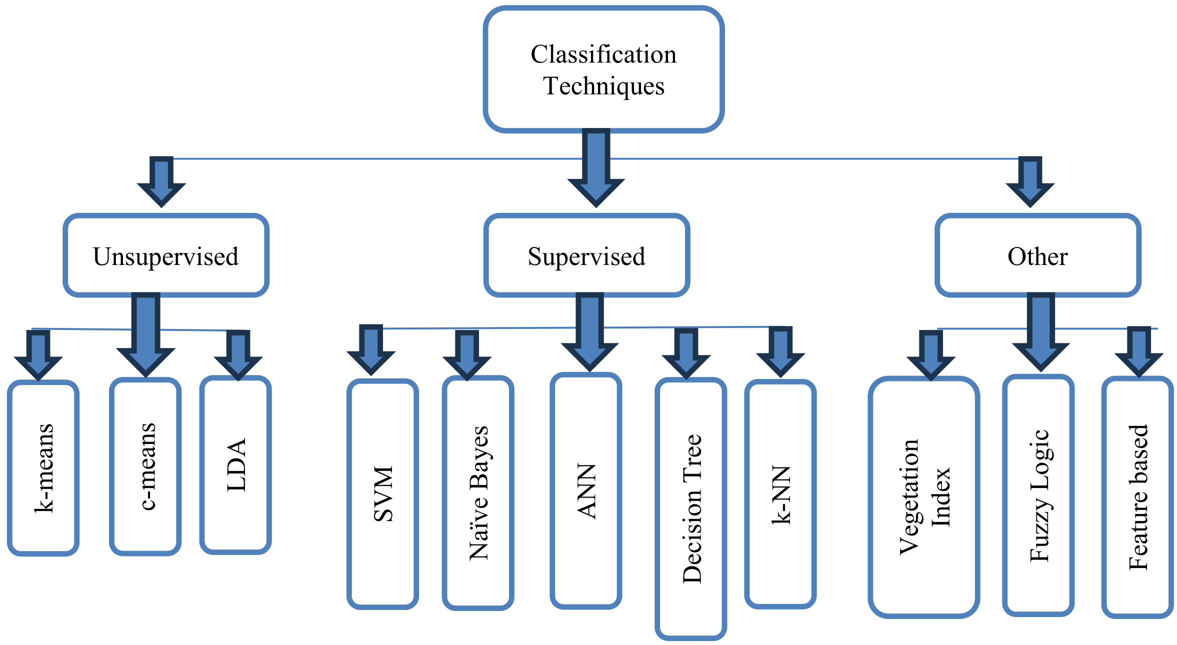


FIGURE 6. Distinct classifiers are analyzed in research for the detection of disease in plants.

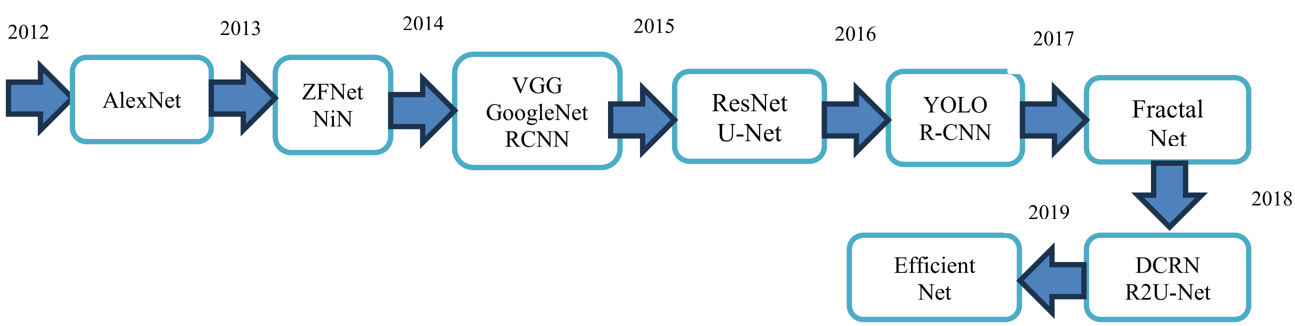


FIGURE 7. Hierarchy of DL techniques.

sets. The model is learned by the training set, hyperparameters are adjusted during validation and the performance evaluation is done by the test set [147], [158]. Various publicly available datasets are indicated in Table 8. These publicly available dataset is from two websites BIFROST (https://datasets.bifrost.ai/, accessed on 15 November 2023) and Kaggle (https://www.kaggle.com/datasets, accessed on 12 November 2023) which can be used for classification. In the literature of DL technique applied to plant disease classification, the most used public datasets are PlantVillage [33], [159] and Kaggle [160], many others also collect their own datasets [161], [162].

2) IMAGE AUGMENTATION

The plant leaf disease labeling, collection, and detection technique requires a huge number of datasets with financial and manpower resources. It is burdensome to collect short-onset period plants. The recognition rate will be destitute for small datasets in deep learning. Therefore, the expansion of the dataset is essential for plant leaf disease detection with deep learning. The practical application and the basic requirement can be removed using data augmentation. For example, the fundamental manifestation of plant disease is color. The original image color will not change by image enhancement. The two ways for data augmentation are conventional augmentation and generating an adversarial network.

Conventional techniques can boost the dataset via rotation, saturation adjustment, or symmetry of the mirror. Some of the advisory use augmentation techniques are AugMix [163], Fast AutoAugment [162], CutMix [164], population-based augmentation [165] and Rand Augment [166]. The flaws of this technique are deficient diversity, unevenness, and meager quality. The data augmentation technique is used by various researchers to improve the efficiency of the system as mentioned in Table 9.

TABLE 7. Details of classification technique used by various researchers.

	Authors	Dataset	Feature Extraction	Classifier	Accuracy
Rice	Joshi & Jadhav 2017 [122]	Agriculture Research (115)	Color & Shape	MD & k-NN	88.15%
	Mohan et al. 2016 [99]	60	Haar & SIFT	AdaBoost, k-NN & SVM	93.33%
	Zhang et al. 2018a [53]	-	Color	Vegetation Index	63.00%
	Bashir et al. 2019 [35]	APS (440)	SIFT	SVM	94.16%
	Shrivastava & Pradhan 2021 [125]	Real Field Images	-	SVM	94.65%
	Rath & Meher 2019 [126]	Real Field Images	-	Radial Basis NN	95.00%
Wheat & Corn	Luo et al. 2015 [127]	Chinese Academy (744)	Histogram	Histogram	94.44%
	Azadbakht et al. 2019 [50]	Hyperspectral data	Index based	Regression	95.00%
	Kusumo et al. 2019 [113]	Plant Village (3823)	SIFT, SURF	SVM, DT, RF, Naïve Bayes	87.00%
	Deshapande et al. 2019 [47]	Agriculture University Dharwad (200)	First Order histogram & GLCM	k-NN, SVM	88.00%
Soyabean	Gharge & Singh 2016 [128]	IPM Database (300)	GLCM	EBPNN	93.30%
	Pires et al. 2016 [112]	Federal (1200)	SIFT, SURF, HOG	SVM	96.25%
	Kaur et al. 2018b [56]	Plant Village (4775)	Color, Texture, Shape	SVM	84.00%
Millet	Caulibaly et al. 2019 [129]	Self (124)	Transfer Learning	VGG16	89.00%
Cotton	Rothe & Kshirsagar 2015 [51]	Self A460 Camera	Hus moments	EBPNN	85.52%
	Sivasangari & Indira 2015 [130]	Self	Color, Shape	SVM	99.30%
Sugar beet	Hallau et al. 2017 [131]	Self (1400)	Texture	SVM	82.00%
Groundnut	Ramkrishnan & Sahaya 2015 [116]	Self	CCM features	EBPNN	97.41%
Cane	Pujari et al. 2016 [48]	Self (9912)	RGB Color	SVM & EBPNN	92.00%
Mix	Sladojevic et al. 2016 [72]	Internet (33469)	-	CNN	95.80%
	Ferentinos 2018 [25]	Plant Village & Self (87848)	Transfer Learning	Alexnet, VGG	99.53%
	Arnal Barbedo 2019 [34]	Self (1575)	Transfer Learning	GoogLeNet	100.00%
	Pantazi et al. 2019 [39]	-	LBP	SVM	95.00%
	Rao & Kulkarni 2020 [67]	Plant Village	Gabor, Curvelet	Fuzzy Logic	90.00%
	Ahmed & Yadav 2022 [132]	Self-Created	GLCM	Random Forest	-

TABLE 7. *(Continued.)* **Details of classification technique used by various researchers.**

	Sahu & Pandey 2023 [87]	Plant Vilaage	Fuzzy c means	HRF SVM	-
Apple	Samajpati & Degadwala 2016 [133]	Self (80)	Color, Histogram, LBP, Gabor	Random forest	95.00%
	Jolly & Raman 2016 [106]	Self (320)	Haarlick & LBP	SVM	96.00%
Citrus	Sharif et al. 2018 [97]	Image gallery dataset (1000)	Color, Texture, Geometrical	Multiclass SVM	95.80%
Cheery	Sengar et al. 2018 [123]	Plant Village	Lesion Area	-	99.00%
Grape	Waghmare et al. 2016 [101]	Self (450)	HSV, LBP	Multiclass SVM	89.30%
	Cruz et al. 2018 [60]	Self (272)	-	AlexNet, Inception-v3	98.00%
	Goncharov et al. 2019 [73]	Plant Village (2986)	-	Siamese	92.00%
	Javidam et al. 2023 [134]	Self	GLCM	SVM	98.97%
	Shantkumari et al. 2023 [135]	Plant Village	-	Improved k-NN	-
Pomegranate	Khot et al. 2016 [58]	-	HSI	Minimum distance classifier	-
Palm Oil	Masazhar & Kamal 2018 [46]	-	GLCM	Multiclass SVM	95.00%
Lentil	Singh et al 2019 [102]	300	LBP	Visual Examination	-
Potato	Islam et al. 2017 [96]	Plant Village (300)	Color, Texture	Multiclass SVM	95.00%
	Patil et al. 2017 [136]	Plant Village (8920)	Texture	SVM, ANN, RF	92.00%
	Verma et al 2020 [137]	Plant Village	-	Capsule Network	91.83%
Bean	Abed & Esmaeel 2018 [49]	100	GLCM	SVM	100.00%
Cucumber	Zhang et al. 2017b [82]	Self (300)	PHOG	SVM	91.48%
	Krishna Kumar & Narayan 2019 [138]	Real Field Images	-	k-means SVM	86.00%
Tomato	Raza et al. 2015 [139]	Self (71)	Pixel Value	SVM	90.00%
	Sabrol & Kumar 2016a [107]	Self (180)	Color moments, Histogram	ANFIS, FF-BPNN	87.20%
	Fuentes et al. 2017 [40]	Self (5000)	DWT, Haar Wavelet	ResNet-50, VGG16	83.06%
	Ashqar & AbuNaser 2018 [140]	Plant Village (9000)	Transfer Learning	CNN	99.84%
	Karadag et al. (2018) [37]	ARI (80)	Wavelet	ANN, NB, k-NN	84.00%
	Das et al. 2020 [57]	Self-Created	Mean, Entropy	SVM, k-NN	87.60%

TABLE 7. *(Continued.)* **Details of classification technique used by various researchers.**

Panigrahi et al. 2020 [141]	Self-Created	-	RF	79.23%
Sujatha et al. 2021 [142]	Self-Created	-	SVM	87.00%
Zamain et al. 2022 [114]	Plant Village	PCA	SVM	-
Harakamanavar et al. 2022 [83]	-	k- means	SVM	88.00%
Rahman et al. 2023 [143]	Self-Created	GLCM	SVM	88.00%
Bhatia et al. 2020 [144]	Mildew dataset	-	Extreme Learning	89.19%
Bhatia et al. 2021 [145]	Mildew dataset	-	SVM Logistic Regression	92.73%

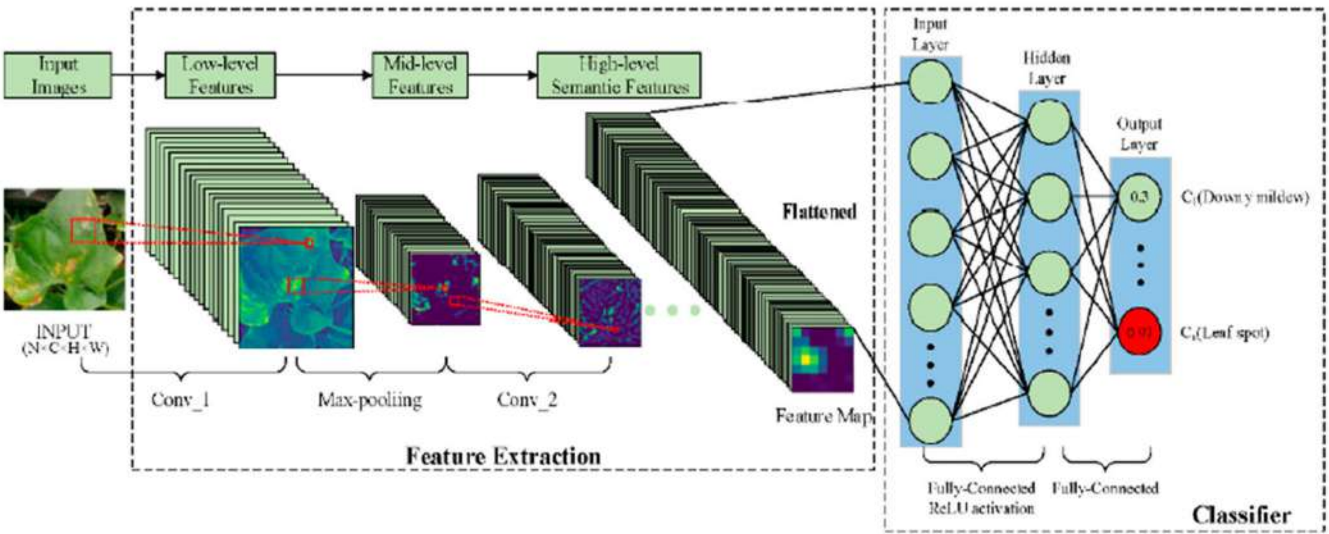


FIGURE 8. Elemental steps of deep learning.

TABLE 8. Bulky publicly available dataset used in deep learning.

Dataset Name	No. of images	Classes	Source
New Plant Disease Dataset	87000	38	Kaggle
Plant Village Dataset	162916	38	Kaggle
Flowers Recognition	4242	4	Kaggle
Plant Seedings Dataset	5539	12	BIFROST
Weed Detection in Soybean Crops	15336	4	Kaggle
Plant Pathology Challenge	3651	4	Kaggle

TABLE 9. Various literature regarding data augmentation.

Authors	Dataset	Techniques	Accuracy increases by
Bin et al. 2017 [168]	1053 to 13689	PCA	4.00%
Lin et al. 2019 [162]	10820 to 32460	Radial blur	3.15%
Ferentinos 2018 [25]	54309 to 87848	Cropping, Resizing	5.23%
Arnal Barbedo 2019 [34]	1567 to 46409	Segmentation	12.00%
Fuente et al. 2017 [48]	5000 to 43398	Resizing, Cropping	3.87%
Srdjan et al. 2016 [169]	4483 to 33469	Rotation	3.00%

Goodfellow et al. 2014 [167] propose a generating model known as Generate Adversarial Network (GAN). Consequently, huge variations of GAN have developed separately like CGAN, DCGAN, LAPGAN, PGGAN, InfoGAN, F-GAN, WGAN, LeakGAN, SeqGAN. The objective is to generate samples of a dataset with the equivalent attributes. This model primarily contains a generator and discriminator as shown in Fig. 9. The GAN technique is used by various researchers to improve the efficiency of the system as mentioned in Table 10.

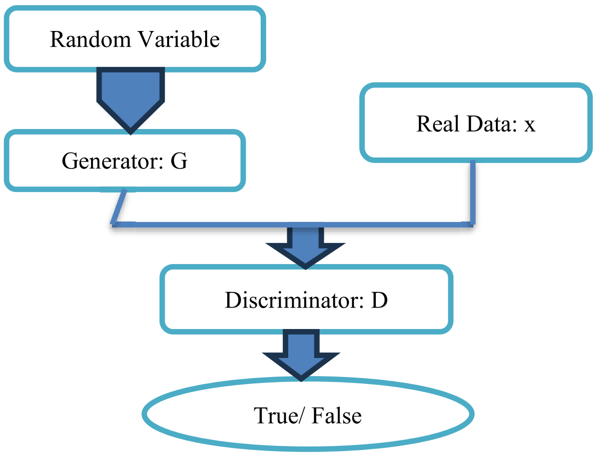


FIGURE 9. Generate adversarial network architecture.

TABLE 10. Various literature regarding data augmentation.

Authors	Dataset	Techniques	Accuracy increases by
Nazki et al. 2020 [170]	2789	ARGAN	5.20%
Tian et al. 2019 [171]	TeaImage	CycleGAN	28.00%
Wu et al. 2020 [172]	GoogleNet	DCGAN	94.33%
Liu et al. 2020 [173]	Grape	LeafGAN	-

3) IMAGE CLASSIFICATION

Recently, several fruitful applications of DL for the classification of disease in plants have been established. Nonetheless, transparency and interpretability are curtailments in classification using DL. The mechanism consists of a black box beyond one minutiae or explanation. The disease detection system must reveal the disease, and symptoms as well as achieve high accuracy. Various researchers utilize distinct techniques using DL as indicated in Table 11.

C. HYPERSPECTRAL IMAGING USING IMAGE PROCESSING

The disease in plants is burdensome sometimes via computer vision systems or naked eyes due to pathogens' growing process [221]. The hyperspectral imaging sensor is primarily fixed in the electromagnetic spectrum range of infrared and visible (400 nm ∼ 2500 nm). These sensors attain shadowy information from a wide range of bands [222]. This range is eminently sensitive to leaf variation caused by distinct leaf disease such that antecedent pathology detection can be done. Consequently, hyperspectral imaging targeted plant disease detection at the early stage. Wang et al. 2019 [149] propose a GAN for hyperspectral imaging as OR-AC-GAN (Outlier Removal—Auxiliary Classifier-GAN) for tomato leaf disease detection and achieve 96.25% accuracy as shown in

TABLE 11. Details of DL technique used by various researchers.

Authors	Dataset	DL Technique	Accuracy
Mohanty et al. 2016 [33]	54306 images, 26 diseases	AlexNet, GoogLeNet	99.35%
Fuentes et al. 2017 [40]	5000 images,	VGGNet, ResNet	83.00%
Bin et al. 2017 [168]	13689 images, 4 diseases	AlexNet	97.62%
Brahimi et al. 2017 [159]	14828 images, 9 diseases	GoogLeNet	99.18%
Ale et al. 2019 [174]	87848 images	VGG	99.53%
Darwaish et al. 2019 [175]	15408 images, 3 diseases	VGG16 & VGG19	98.20%
Yuwana et al. 2019 [176]	5632 images,	MCT	-
Chen et al. 2020 [177]	900 images, 2 categories	VGGNet, Inception	92.00%
Mishra et al. 2020 [178]	-	DCNN	88.46%
Ashok et al. 2020 [179]	Self-Created	AlexNet	95.75%
Trivedi et al. 2020 [180]	Plant Village	CNN	95.81%
Jasmin & Tuwaijari 2020 [181]	Plant Village	CNN	98.29%
Panchal et al. 2021 [182]	Plant Village	CNN	-
Chug et al. 2023 [183]	IARI Dataset	Hybrid DL	100% 99.64%
Wang et al. 2019[184]	Internet Image	RCNN with VGG16	98.00%
Picon et al. [185]	Real Field	ResNet50	86.00%
GadeKallu et al. 2021 [186]	Plant Village	DNN	97.00%
Nitish et al. 2020 [187]	Plant Village	ResNet50	93.40%
Verma et al. 2020 [188]	Plant Village	AlexNet	94.00%
Wspanialy & Moussan 2020 [189]	Plant Village	ResNet, U-Net	98.00%
Karthik et al. 2020 [190]	Plant Village	CNN	99.60%
Ahmad et al. 2020 [191]	Laboratory & Field Dataset	ResNet, InceptionV3	98.00%
Lee et al. 2020 [192]	Plant Village	GoogleNet, VGG16	98.40%
Atila et al. 2021 [193]	Plant Village	EfficientNet	91.20%

TABLE 11. ***(Continued.)*** **Details of DL technique used by various researchers.**

Agarwal et al. 2021 [194]	Plant Village	CNN	98.46%
Ngugi et al. 2020 [195]	Real Condition Dataset	CNN	93.75%
Chen et al. 2021 [196]	Plant Village	Enhanced ANN & CNN	93.45%
Krishnan et al. 2022 [197]		CNN with Fuzzy	99.16%
Singh et al. 2022 [198]	Plant Village	CNN AlexNet	98.83%
Elaraby et al. 2022 [199]		CNN with PSO	
Ashwin Kumar et al. 2022 [200]	Plant Village	MobileNet	98.70%
Pandian et al. 2022 [201]	Self-Created	DCNN	98.41%
Mahum et al. 2022 [202]	Plant Village	DenseNet201	97.20%
Haridson et al. 2022 [203]	-	ReLu Softmax	91.45%
Algani et al. 2022 [204]	Plant Village	CNN	-
Huang et al. 2022 [205]	Self-Created	Fully Convolutional	97.59%
Yu et al. 2022 [206]	Plant Village	Inception Convolutional	98.17%
Khan et al. 2022 [207]	Self- Created	Entropy ELM	98.40%
Yu et al. 2022 [208]	Apple	MSO ResNet50	95.70%
Khan et al. 2022 [209]	Self-Created	DL	88.00%
Algani et al. 2023 [210]	-	ACOCNN	99.98%
Shoaib et al. 2023 [211]	2300 images	ResNet	99.62%
Singh et al. 2023 [212]	Self-Created	CNN	99.39%
Sharma et al. 2023 [213]	Self-Created	DLMN Net	96.56%
Daniya & Vigneshwari 2023 [214]	Self-Created	RHGSO	93.04%
Arun et al. 2023 [215]	Plant Village	Concatenated DL	98.14%
Pal & Kumar 2023 [216]	Plant Village	Inception Visual	-
Reddy et al. 2023 [217]	Plant Village	ResNet50	99.73%
Kaya et al. 2023 [218]	Plant Village	Dense Net	98.17%
Thai et al. 2023 [219]	Cassas leaf	Pruning Network	-
Thakur et al. 2022 [220]	Plant Village	CNN	99.16%

Fig. 10. Various researchers use distinct techniques for HSI plant disease detection are illustrated in Table 12.

TABLE 12. **Details of hyperspectral image classification used by various researchers.**

Authors	Disease	Algorithm	Accuracy
Moshou et al. 2005 [223]	Rust	QDA	94.50%
Mahlein et al. 2012 [224]	Mildew powdery	SAM	98.90%
Huang et al. 2012 [225]	Rust	Linear regression	-
Nagasubramanian et al. 2017 [226]	Rot	3D CNN	95.73%
Alisaac et al. 2018 [227]	Head blight	SVM	99.00%
Abdulridha et al. 2019 [228]	Canker	RBF	96.00%
Ngyugen et al. 2020 [229]	-	MLP	99.00%
Feng et al. 2021 [230]	Rot	Deep domain	88.00%
Wan et al. 2022 [231]	Pathogen infected	-	-
Bhujade & Sambhe 2022 [232]	-	DL	-
Cui et al. 2022 [233]	Valsa Canker	DC-CNN	98.00%
Wu et al. 2022 [234]	Plant Village	Adaptive Coherence	97.41%
Fajardo et al. 2022 [235]	Banana Leaf	Logistic Regression	-
Yong et al. 2023 [236]	Basal Rot	RCNN	91.93%
Wu et al. 2023 [237]	Rust	ELM	96.67%

An extensive number of modern developments in different pathogen systems using distinct group of extremely sensitive sensors and various data analysis pipeline have been summarized in Fig 11 and Table 13. It overviews the current sensors technologies used for automatic identification and detection of plants. These sensors can be implemented in agriculture applications of plants. Depending on the scale, distinct platforms of plants can be managed and observed.

RGB Imaging: RGB (Red Green Blue) is the simple source og digital images for disease detection and identification. The technical parameters such as photo sensor for light sensitivity, optical focus and spatial resolution has been improved significantly [238].

Multi and Hyperspectral Sensors: Multispectral sensors assess the spectral information of object in various wavebands [239]. It also provide RGB and near infrared wavebands [240]. Hyperspectral sensors provides spatial and spectral information from the object. The spatial resolution depends upon the object and the sensor [241].

Thermal Sensors: Infrared thermography (IRT) assess plant temperature and is correlated with the status of plant water, the microclimate in crop stand [242]. IRT can be used in plant science at distinct spatial and temporal scales from small scale applications [238].

Fluorescence Sensors: Chlorophyll fluorescence sensors are active sensors with laser light or LED source that assess photo synethitic transfer [243]. It combine with image analysis technique have been useful for quantification and discrimination of fungal diseases.

TABLE 13. Details of OPTICAL SENSORS used by various researchers.

Authors	Sensors	Crop	Disease
Neumann et al . 2014 [244]	RGB	Sugar beet	Angular Bacateria
Bergstrasser et al. 2015 [245]	Spectral Sensors	Sugar beet	Leaf Spot
Wahabzada et al. 2015 [246]	Spectral Sensors	Barley	Brown Rust
Polder et al. 2014 [247]	Spectral Sensors	Tulip	Virus
Berdugi et al. 2014 [248]	Thermal Sensor	Cucumber	Mildew
Gomez 2014 [249]	Thermal Sensor	Rosa	Mildew
Bauriegel et al 2014 [250]	Fluorscence	Lettuce	Mildew
Rousseau et al. 2013 [251]	Fluorscence	Bean	Bacterial Blight

TABLE 14. Details of FEW SHOT image classification used by various researchers.

Authors	DL Frame	Dataset	Sample
Medela et al. 2020 [255]	One-shot learning	Medical	60
Ren et al. 2019 [256]	One-shot learning	PlantVillage	20
Huang et al. 2020 [257]	Few-shot learning	PlantVillage	1
Nagasubramanian et al. 2017 [258]	Few-shot learning	MiniImageNet	15
Alisaac et al. 2018 [259]	One-shot learning	Self-Acquired	1160
Abdulridha et al. 2019 [260]	CDCGAN	Self-Acquired	40
Li & Yang 2021 [261]	Few Shot	Self-Created	-
Li & Chao 2021 [262]	Few Shot	Plant Village	10
Lang 2021 [263]	Few Shot	Cotton leaf	-
Wang et al. 2022[264]	Few Shot	Self-Created	-
Whusquiza et al. 2022 [265]	Few Shot	Self-Created	-
Garg & Singh 2023 [266]	Few Shot	Plant Doc	30
Liu et al. 2023 [267]	Few Shot	Plant Village	-
Li et al. 2023 [268]	Few Shot	Plant Village	10

D. FEW SHOT LEARNING (FSL) USING IMAGE PROCESSING

A new model of machine learning that facilitates learning from machines using a limited number of labeled databases is known as FSL [252]. The experiment's purpose and problem complexity determine the number of shots required. FSL uses datasets with prior knowledge for training. This process includes data, models, and optimization to tune the samples. The training phase leads to creating source tasks activated by obtaining multiple tasks from single or multiple datasets as represented in Fig. 12. The machine learning model is combined with these source tasks to train the system. The most commonly available dataset for plant recognition systems consists of several thousands of images. The performance of the system depends on the quantity as well as the quality of images. Therefore, FSL is the best possible feasible solution for a confined number of samples [253], [254]. Various researchers use distinct techniques for small samples-based plant leaf disease detection are illustrated in Table 14.

Various applications for FSL to abate parameters are mentioned below.

i. **Embedding-** It features an extraction technique also known as data transformation benefitted for low dimension reduction (higher dimensional map to lower dimension). This includes pre-trained CNN with ImageNet for extraction of features (reduce training time) and SVM for classification. SVM uses distinct kernels to handle complex systems [268].
ii. **Multitask Learning-** The process includes a shared model to train multiple tasks generally classified as hard parameter sharing and soft parameter sharing. In HPS certain parameters are shared to the entire task and other parameters learn precise features. In SPS each parameter is shared with individual tasks. This multitask learning incorporates all the models that can be trained for all the tasks rather than a single task [269].
iii. **Transfer Learning**- The process was initiated to accomplish a diminished training period by reusing gathered knowledge [270].
iv. **Meta-Learning**- This refers to optimization and performance improvement via multiple-task experience. It is also known as support sets that learn from multiple sources instead of a single source (conventional machine) alongside FS to adequately develop a model for clarifying query sets.

The meta-learning and multitasking learning are generally unstable and marginally upgraded by a few proportions. Transfer learning is the most superior and stable to train the model from scratch. These approaches are computationally costly compared to the degree of performance for enhancement. The future work led to enhancing existing FSL algorithms to cooperate with plant disease characteristics recognition. Further comparative and comprehensive studies are done for the performance evaluation of each FSL approach.

E. MOLECULAR DIAGNOSIS TECHNIQUES

Pest and disease may heighten the total food production losses by 35%. Globally, under quarantined pathogens, some of the austere plant diseases are nematodes, parasitic, oomycetes, viruses, bacteria, and fungi.

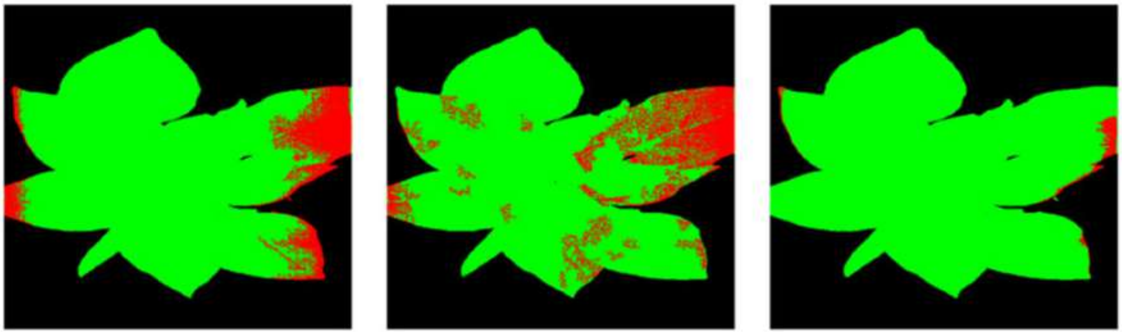


FIGURE 10. Segmentation comparison (a) Direct CNN (b) AC-GAN (c) OR-AC-GAN.

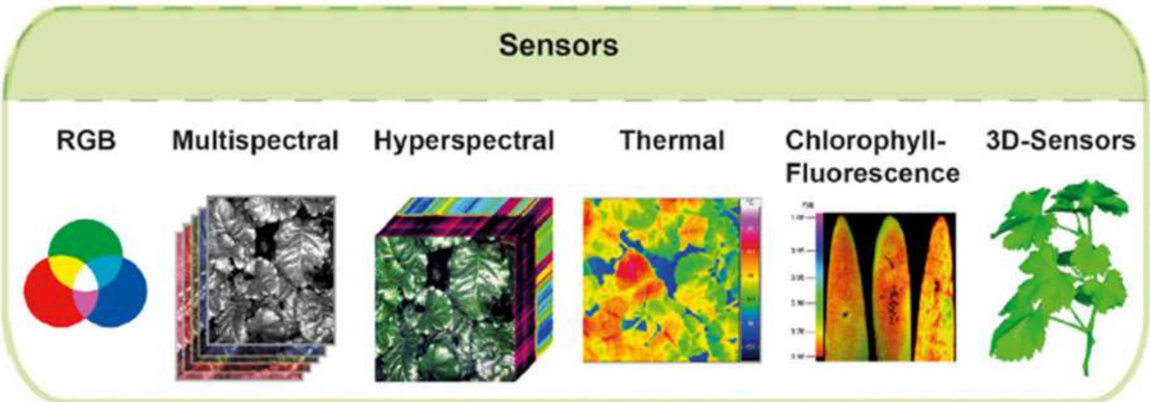


FIGURE 11. Overview of sensor technologies [238].

FIGURE 12. Approach to imitate multiple sources.

Fang and Ramasamy 2015 [15] reported early disease diagnosis is critical to overcome outbreaks of disease. Scientists utilize laboratories and certain advanced facilities compared with traditional techniques resulting in reliable diagnosis methods. Nonetheless, these approaches performed better only with severe infection in plants. Therefore, to bridge this gap certain adequate molecular diagnosis techniques like management control on decisions or tools for plant disease overcome the traditional techniques. The emerging molecular technologies to diagnose plants due to the necessity of managing diseases are as follows.

i. **Enzyme-linked immunosorbent assay (ELISA):** This approach is widely employed as a diagnosis device in plant and chemical pathology. It identifies and detects the particular substance in the sample, and provokes a color change that expands antibodies with enzymes [271]. ELISA is not very sensitive compared to other methods [272] but is labor-intensive and sensitive [273]. Despite antibodies like polyclonal being adequate to distinguish pathogens, nevertheless, these are not ample specific. One other antibiotic monoclonal is innumerable specific but further costly. As ELISA is tremendously sensitive to viruses results in the successful development of antibodies [274]. The technique will be inadequate to detect bacteria due to low sensitivity [15]. Therefore, pathogens bacteria, and fungi for rice disease are not effective using ELISA. Also due to inadequacy of specificity, ELISA is not able

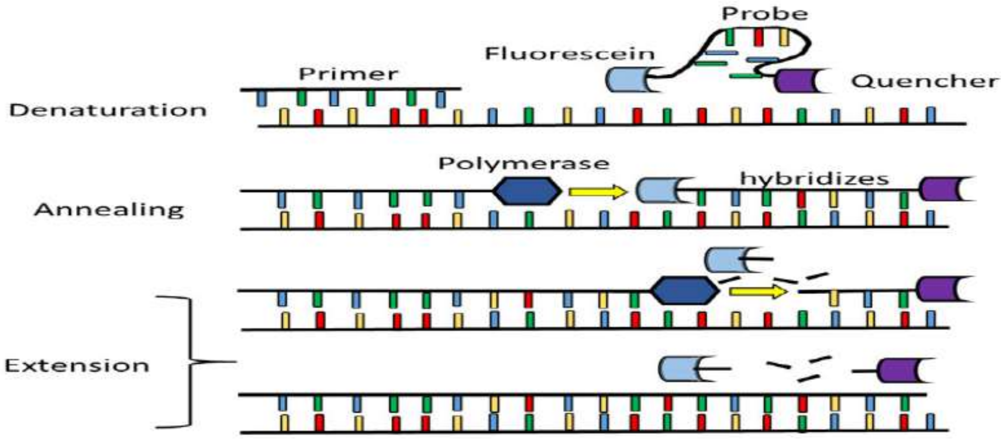


FIGURE 13. Illustrative real-time PCR schematic [286].

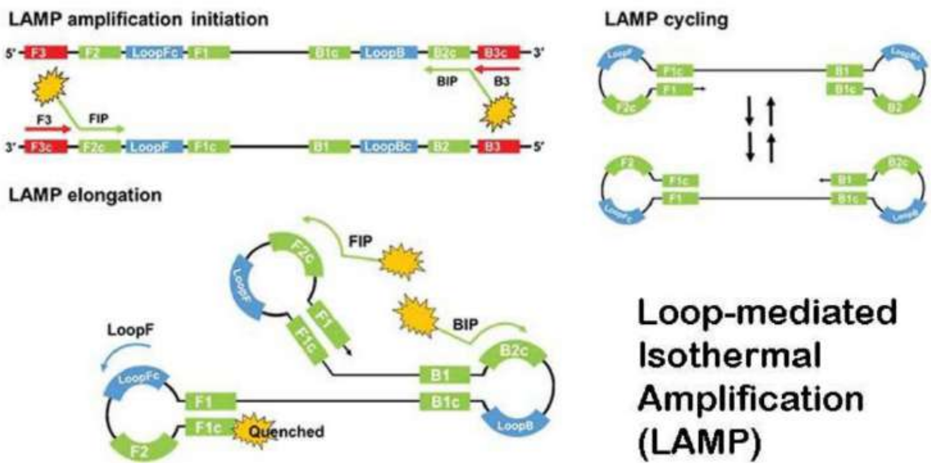


FIGURE 14. Amplification of LAMP [283].

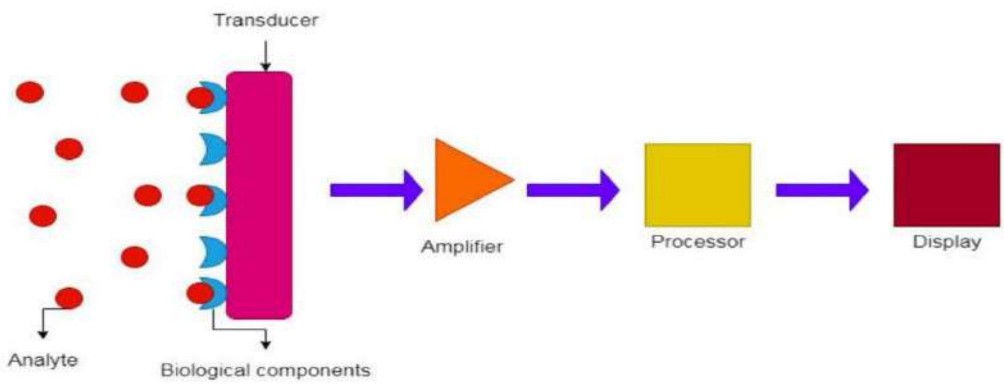


FIGURE 15. Biosensors & its process schematic illustration [286].

to discriminate several strains with luminous specific symptoms [275].

ii. **Conventional PCR**: The most valuable achievement in molecular biology is the Polymerase Chain Reaction (PCR) established in the mid-80s [276]. The technique requires various components and reagents which incorporate monovalent cation, primers, buffer solution, triphosphate deoxynucleotide, and polymers.

The technique anticipates electrophoresis used for the detection of plant pathogens. The nucleic acid evocation technique affects the PCR sensitivity and (Dioxyribo Nucleic Acid) DNA target variability [277]. The highly sensitive nature of PCR results in DNA amplification. The PCR limits contamination susceptibility and inadequacy of robustness for a long time with false positive results.

iii. **Real-Time PCR**: A significantly revolutionary technique developed by Mullis Karry in 80 is also known as a quantitative polymerase chain reaction. This technique diminishes the cross-contamination risk with most precipitate-sensitive detection [278]. The infected tissues and vectors can easily detect DNA pathogens using RT-PCR [279]. This technique points exclusively to known genes [280] with sequence data indicated in Fig. 13.

iv. **Lamp:** The most common alternative to RTPCR is a lamp due to its cost-effectiveness, simplicity, rapidity, and practicality. Under isothermal elaboration, lamps recognize HBV (Hepatitis B Virus) using certain specific sequences of elaborated DNA [281]. Lamp contains four distinct primers two for inner and outer each (Forward inner primer, backward inner primer, forward outer primer, backward outer primer) as indicated in Fig. 14. A loop is formed for high displacement and auto strand DNA for 60 minutes at 65 degrees C isothermal situation [282]. As indicated in the figure the process of corporate initial material production, amplification, recycling, and elongation [283]. Lamp detects amplicon using the detection of colorimetric, fluorimeter, and turbidimeter [284]. The technique incorporates the formation of primer to cultivate the 65-degree C temperature [285]. The technique is highly sensitive for elaboration therefore proper practice and handling are enforced to escape the contamination risk.

v. **Biosensor**: An ample range of challenges and problems in pharmacology medicine, security, food safety, and agriculture can be resolved using biosensors. Clark and Lyons [287] develop biosensors via measurement of glucose using hydrogen peroxide or oxygen electrochemical detection. Vigneshvar et al. 2016 further evolved biosensors into innovative approaches like nanotechnology, and electrochemistry.

"According to International Union of Pure and Applied Chemistry recommendations (1999), a biosensor is described as a device that deploys the integration of specific biochemical reactions mediated by isolated enzymes, immune systems, tissues, organelles or whole cells to detect chemical compounds, commonly by electrical, thermal or optical signals". These analytical appliances are constituted of the transducer coupled with a biorecognition component and disciple into the analytical signal as shown in Fig. 15. The advantage of the device includes cost effectiveness, sensitivity, and simplicity.

vi. **Next Generation Sequencing (NGS):** Generally, the sequencing mechanism depends on molecular detection approaches [288]. The first generation of commercial and laboratory sequencing was employed due to poor radioactivity and improved efficiency [289]. The approach is ineffective and tedious for large sequences and efficient for short sequences. The NGS is also known as a parallel massive sequence capture sequence in less time, low cost, and high throughput [290]. The unique biology molecular opportunities are incorporated via NGS such as mela genomics [291], single-cell sequencing, sew protein [292], and whole genome sequencing [293]. The approach limits to diagnosis of helm pathogen [294], [295]. Advanced NGS adopts colony sequencing, parallel massive sequencing, and pyrosequencing by legation nucleotide [287].

V. LIMITATIONS

While plant disease detection has been analyzed automatically, there are several limitations in each step or process. The samples are very difficult to obtain for particular diseases [37], [38], [39]. It is observed that system performance is immensely influenced by the type of dataset used i.e. laboratory controlled or real-time dataset. The data obtained in an uncontrolled environment results in increased system complexity but it is of huge significance in agriculture development and more adequate in today's research. Several difficulties have been observed during the extraction of features [106]; similarity in the infected area leads first to the extraction of inappropriate features and then false classification based on irrelevant feature matching. Thus, it is necessary to select a set of features because each feature has a different importance level. Various classification techniques used for plant disease detection are SVM, ANN, Naïve Bayes, backpropagation neural network, decision tree, and k-nearest neighbor [123], [124], [125], [126]. Despite these classifiers convolutional neural networks have been explored for huge samples (thousands and more) more accurately. However, the overfitting of CNN is seen as a major issue in systems about plant disease detection using deep neural networks [195]. Various approaches have been observed which can improve the efficiency of convolutional neural network-based classification systems for multiple cultures.

According to the literature, the proposed system has a set of specifications to be fulfilled necessarily, if any specification is unconsidered then the system proposed provides inaccurate disease detection. Therefore, a versatile system must be designed by a researcher with an adjustable set of specifications instead of a constant. The actual use of machine learning has been disturbed due to overfitting. Hence a highly generalized and adaptable system is to be developed for plant disease detection in various cultures. Machine learning is the most popular domain due to the availability of techniques and tools, but these resources must not be compromised for accuracy.

VI. CHALLENGES

From the literature review, it can be found that one of the challenges facing plant disease detection is the lack of experts to annotate accurately. The issue arises when experts cannot differentiate correctly between dead plants and disease-infected plants. This process requires experienced and expert professionals to detect diseases in plants that are costly and difficult, especially for rare or new diseases. Furthermore, using DL techniques for modeling, tuning, and resource training is another important challenge. The shallow architectures are best suited for small datasets. The most recent models for disease detection offer another angle to consider in building a model for plant disease detection. The ML, DL, and FSL are to be considered to enhance the detection models. This paper recommends that future research on plant disease detection, classification, and quantification of disease will improve mart agriculture.

Various issues and factors that may affect plant disease classification and identification are listed below:

(i) The plant disease detection system performance mainly depends on the feature extraction and classification technique used [144].

(ii) Most of the literature reviewed utilizes Plant Village Dataset i.e. laboratory images rather than real time images. The classifier performance heavily depends on the dataset used for testing and training [144].

(iii) The real field images may have complex backgrounds; therefore, segmentation of affected areas is difficult which affects the performance of the system [296].

(iv) The plant has a deficiency in nutrients [74] and contamination at an early stage of development [8].

(v) The infected area estimation and severity management with disease detection may be used to control pesticides [188].

(vi) The real-time illness efficiency on constrained devices could be considered [189].

(vii) Various diseases may create distinct manifestations simultaneously. Therefore, it will be difficult to identify and combine hybrid symptoms.

(viii) The hyperparameters tuning and selection have the potential to have a powerful influence on performance [196].

(ix) Due to the characteristic of disease uniformity and attribute selection, disease identification systems face significant challenges [297].

VII. CONCLUSION

The manifestation of imminent pathogens plants extends to become a considerable threat to food safety, the ecosystem, and the global economy. Moreover, crucial factors like mobility globalization, vectors, changes in climate, and evolution in pathogens emboldened the advancement of invasive pathogen plants. An adequate automated approach is eminently desired to bury losses in agriculture. Therefore, an automated plant disease recognition and classification is reviewed using machine learning, deep learning, and few-shot learning. The review presents various well-known approaches such as acquisition, preprocessing, segmentation, feature extraction, and classification. Despite RGB images some of the studies use hyperspectral imaging for plant leaves. The hyperspectral does not require a labeled database and microscopic symptoms are easily detected. These automated techniques contribute a manifestation of research in accessible time. Also, the diagnosis molecular tools with the latest techniques for disease detection are reviewed. The entire techniques indicated in this paper are gratuitous to the sensitive, specific, and rapid detection of plants. Moreover, in the future to detect disease in plants, the combination of severe side program and client mobile as well as electrophysiology would be exceptional future research guidance.

REFERENCES

[1] *The State of Food Security and Nutrition in the World 2023*, FAO; IFAD; UNICEF; WFP; WHO, Geneva, Switzerland, Jul. 2023.

[2] T. D. March. *State of Agriculture in India*. Accessed: Aug. 9, 2023. [Online]. Available: https://prsindia.org/files/policy/policy_analytical_reports/State%20of%20Agriculture%20in%20India.pdf

[3] D. P. Hughes and M. Salathe, "An open access repository of images on plant health to enable the development of mobile disease diagnostics," 2015, *arXiv:1511.08060*.

[4] S. Verma, A. Chug, and A. P. Singh, "Prediction models for identification and diagnosis of tomato plant diseases," in *Proc. Int. Conf. Adv. Comput., Commun. Informat. (ICACCI)*, Sep. 2018, pp. 1557–1563.

[5] S. Sankaran, A. Mishra, R. Ehsani, and C. Davis, "A review of advanced techniques for detecting plant diseases," *Comput. Electron. Agricult.*, vol. 72, no. 1, pp. 1–13, Jun. 2010.

[6] C. Janiesch, P. Zschech, and K. Heinrich, "Machine learning and deep learning," *Electron. Markets*, vol. 31, no. 3, pp. 685–695, Apr. 2021, doi: 10.1007/s12525-021-00475-2.

[7] K. Kc, Z. Yin, D. Li, and Z. Wu, "Impacts of background removal on convolutional neural networks for plant disease classification in-situ," *Agriculture*, vol. 11, no. 9, p. 827, Aug. 2021, doi: 10.3390/agriculture11090827.

[8] J. Liu and X. Wang, "Plant diseases and pests detection based on deep learning: A review," *Plant Methods*, vol. 17, no. 1, pp. 1–18, Dec. 2021.

[9] L. C. Ngugi, M. Abelwahab, and M. Abo-Zahhad, "Recent advances in image processing techniques for automated leaf pest and disease recognition—A review," *Inf. Process. Agricult.*, vol. 8, no. 1, pp. 27–51, Mar. 2021.

[10] P. Chinmayi, L. Agilandeeswari, and P. Manoharan, "Survey of image processing techniques in medical image analysis: Challenges and methodologies," in *Proc. Int. Conf. Soft Comput. Pattern Recognit.*, Aug. 2017, pp. 460–471, doi: 10.1007/978-3-319-60618-7_45.

[11] I. A. Adeyanju, O. O. Bello, and M. A. Adegboye, "Machine learning methods for sign language recognition: A critical review and analysis," *Intell. Syst. Appl.*, vol. 12, Nov. 2021, Art. no. 200056, doi: 10.1016/j.iswa.2021.200056.

[12] M. Fink, "Object classification from a single example utilizing class relevance metrics," in *Proc. 17th Int. Conf. Neural Inf. Process. Syst.*, 2004, pp. 449–456.

[13] L. Fei-Fei, R. Fergus, and P. Perona, "One-shot learning of object categories," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 4, pp. 594–611, Apr. 2006.

[14] F. Martinelli, R. Scalenghe, S. Davino, S. Panno, G. Scuderi, P. Ruisi, P. Villa, D. Stroppiana, M. Boschetti, L. R. Goulart, and C. E. Davis, "Advanced methods of plant disease detection. A review," *Agron. Sustain. Dev.*, vol. 35, no. 1, pp. 1–25, 2015.

[15] Y. Fang and R. Ramasamy, "Current and prospective methods for plant disease detection," *Biosensors*, vol. 5, no. 3, pp. 537–561, Aug. 2015.

[16] V. K. Vishnoi, K. Kumar, and B. Kumar, "Plant disease detection using computational intelligence and image processing," *J. Plant Diseases Protection*, vol. 128, no. 1, pp. 19–53, Aug. 2020, doi: 10.1007/s41348-020-00368-0.

[17] *Evaluations of Brinjal Germplasm for Resistance to Fusarium Wilt Disease*. Accessed: Aug. 9, 2023. [Online]. Available: https://www.ijsrp.org/research-paper-0717.php?rp=P676604

[18] P. Adhikari, Y. Oh, and D. Panthee, "Current status of early blight resistance in tomato: An update," *Int. J. Mol. Sci.*, vol. 18, no. 10, p. 2019, Sep. 2017.

[19] S. Zhang, S. Zhang, C. Zhang, X. Wang, and Y. Shi, "Cucumber leaf disease identification with global pooling dilated convolutional neural network," *Comput. Electron. Agricult.*, vol. 162, pp. 422–430, Jul. 2019.

[20] J. Kianat, M. A. Khan, M. Sharif, T. Akram, A. Rehman, and T. Saba, "A joint framework of feature reduction and robust feature selection for cucumber leaf diseases recognition," *Optik*, vol. 240, Aug. 2021, Art. no. 166566.

[21] M. Agarwal, S. Gupta, and K. K. Biswas, "A new Conv2D model with modified ReLU activation function for identification of disease type and severity in cucumber plant," *Sustain. Computing: Informat. Syst.*, vol. 30, Jun. 2021, Art. no. 100473.

[22] V. K. Shrivastava, M. K. Pradhan, S. Minz, and M. P. Thakur, "Rice plant disease classification using transfer learning of deep convolution neural network," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 631–635, Jul. 2019.

[23] J. Chen, D. Zhang, A. Zeb, and Y. A. Nanehkaran, "Identification of rice plant diseases using lightweight attention networks," *Exp. Syst. Appl.*, vol. 169, May 2021, Art. no. 114514.

[24] H. Sun, L. Zhai, F. Teng, Z. Li, and Z. Zhang, "qRgls1. 06, a major QTL conferring resistance to gray leaf spot disease in maize," *Crop. J.*, vol. 9, pp. 342–350, Apr. 2021.

[25] K. P. Ferentinos, "Deep learning models for plant disease detection and diagnosis," *Comput. Electron. Agricult.*, vol. 145, pp. 311–318, Feb. 2018.

[26] A. Abbas, S. Jain, M. Gour, and S. Vankudothu, "Tomato plant disease detection using transfer learning with C-GAN synthetic images," *Comput. Electron. Agricult.*, vol. 187, Aug. 2021, Art. no. 106279.

[27] H. R. Kappali, K. M. Sadyojatha, and S. K. Prashanthi, "Computer vision and machine learning in paddy diseases identification and classification: A review," *Indian J. Agricult. Res.*, vol. 10, pp. 1–5, Mar. 2023. [Online]. Available: https://www.arccjournals.com/journal/indian-journal-of-agricultural-search/A-6061

[28] D. S. Joseph, P. M. Pawar, and R. Pramanik, "Intelligent plant disease diagnosis using convolutional neural network: A review," *Multimedia Tools Appl.*, vol. 82, no. 14, pp. 21415–21481, Oct. 2022, doi: 10.1007/s11042-022-14004-6.

[29] F. Ahmed, M. Taimur Ahad, and Y. Rayhan Emon, "Machine learning-based tea leaf disease detection: A comprehensive review," 2023, *arXiv:2311.03240*.

[30] S. Mustofa, M. M. H. Munna, Y. R. Emon, G. Rabbany, and M. T. Ahad, "A comprehensive review on plant leaf disease detection using deep learning," 2023, *arXiv:2308.14087*.

[31] L. Goel and J. Nagpal, "A systematic review of recent machine learning techniques for plant disease identification and classification," *IETE Tech. Rev.*, vol. 40, no. 3, pp. 423–439, Sep. 2022, doi: 10.1080/02564602.2022.2121772.

[32] A. Bhargava and A. Bansal, "Fruits and vegetables quality evaluation using computer vision: A review," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 33, no. 3, pp. 243–257, Mar. 2021.

[33] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers Plant Sci.*, vol. 7, p. 1419, Sep. 2016.

[34] J. G. Arnal Barbedo, "Plant disease identification from individual lesions and spots using deep learning," *Biosyst. Eng.*, vol. 180, pp. 96–107, Apr. 2019.

[35] K. Bashir, M. Rehman, and M. Bari, "Detection and classification of rice diseases: An automated approach using textural features," *Mehran Univ. Res. J. Eng. Technol.*, vol. 38, no. 1, pp. 239–250, Jan. 2019.

[36] A. Camargo and J. S. Smith, "An image-processing based algorithm to automatically identify plant disease visual symptoms," *Biosyst. Eng.*, vol. 102, no. 1, pp. 9–21, Jan. 2009.

[37] K. Karadag, M. E. Tenekeci, R. Tasaltin, and A. Bilgili, "Detection of pepper fusarium disease using machine learning algorithms based on spectral reflectance," *Sustain. Comput., Informat. Syst.*, vol. 28, Dec. 2020, Art. no. 100299.

[38] S. Coulibaly, B. Kamsu-Foguem, D. Kamissoko, and D. Traore, "Deep neural networks with transfer learning in millet crop images," *Comput. Ind.*, vol. 108, pp. 115–120, Jun. 2019.

[39] X. E. Pantazi, D. Moshou, and A. A. Tamouridou, "Automated leaf disease detection in different crop species through image features analysis and one class classifiers," *Comput. Electron. Agricult.*, vol. 156, pp. 96–104, Jan. 2019.

[40] A. Fuentes, S. Yoon, S. Kim, and D. Park, "A robust deep-learning-based detector for real-time tomato plant diseases and pests recognition," *Sensors*, vol. 17, no. 9, p. 2022, Sep. 2017.

[41] S. Shrivastava and D. S. Hooda, "Automatic Brown spot and frog eye detection from the image captured in the field," *Amer. J. Intell. Syst.*, vol. 4, no. 4, pp. 131–134, 2014.

[42] J. Parraga-Alava, K. Cusme, A. Loor, and E. Santander, "RoCoLe: A robusta coffee leaf images dataset for evaluation of machine learning based methods in plant diseases recognition," *Data Brief*, vol. 25, Aug. 2019, Art. no. 104414, doi: 10.1016/j.dib.2019.104414.

[43] K. Tian, J. Li, J. Zeng, A. Evans, and L. Zhang, "Segmentation of tomato leaf images based on adaptive clustering number of K-means algorithm," *Comput. Electron. Agricult.*, vol. 165, Oct. 2019, Art. no. 104962, doi: 10.1016/j.compag.2019.104962.

[44] M. Zhang and Q. Meng, "Automatic citrus canker detection from leaf images captured in field," *Pattern Recognit. Lett.*, vol. 32, no. 15, pp. 2036–2046, 2011.

[45] R. Pydipati, T. F. Burks, and W. S. Lee, "Identification of citrus disease using color texture features and discriminant analysis," *Comput. Electron. Agricult.*, vol. 52, nos. 1–2, pp. 49–59, Jun. 2006.

[46] A. N. I. Masazhar and M. M. Kamal, "Digital image processing technique for palm oil leaf disease detection using multiclass SVM classifier," in *Proc. IEEE 4th Int. Conf. Smart Instrum., Meas. Appl. (ICSIMA)*, Nov. 2017, pp. 1–6.

[47] A. S. Deshapande, S. G. Giraddi, K. G. Karibasappa, and S. D. Desai, "Fungal disease detection in maize leaves using Haar wavelet features," in *Information and Communication Technology for Intelligent Systems*. Singapore: Springer, 2019, pp. 275–286.

[48] J. D. Pujari, R. Yakkundimath, and A. S. Byadgi, "SVM and ANN based classification of plant diseases using feature reduction technique," *Int. J. Interact. Multimedia Artif. Intell.*, vol. 3, no. 7, p. 6, 2016.

[49] S. Abed and A. A. Esmaeel, "A novel approach to classify and detect bean diseases based on image processing," in *Proc. IEEE Symp. Comput. Appl. Ind. Electron. (ISCAIE)*, Apr. 2018, pp. 297–302.

[50] M. Azadbakht, D. Ashourloo, H. Aghighi, S. Radiom, and A. Alimohammadi, "Wheat leaf rust detection at canopy scale under different LAI levels using machine learning techniques," *Comput. Electron. Agricult.*, vol. 156, pp. 119–128, Jan. 2019.

[51] P. R. Rothe and R. V. Kshirsagar, "Cotton leaf disease identification using pattern recognition techniques," in *Proc. Int. Conf. Pervasive Comput. (ICPC)*, Jan. 2015, pp. 1–6.

[52] J. Abdulridha, Y. Ampatzidis, S. C. Kakarla, and P. Roberts, "Detection of target spot and bacterial spot diseases in tomato using UAV-based and benchtop-based hyperspectral imaging techniques," *Precis. Agricult.*, vol. 21, no. 5, pp. 955–978, Oct. 2020.

[53] D. Zhang, X. Zhou, J. Zhang, Y. Lan, C. Xu, and D. Liang, "Detection of rice sheath blight using an unmanned aerial system with high-resolution color and multispectral imaging," *PLoS ONE*, vol. 13, no. 5, May 2018, Art. no. e0187470.

[54] K.-Y. Huang, "Application of artificial neural network for detecting phalaenopsis seedling diseases using color and texture features," *Comput. Electron. Agricult.*, vol. 57, no. 1, pp. 3–11, May 2007.

[55] Q. Yao, Z. Guan, Y. Zhou, J. Tang, Y. Hu, and B. Yang, "Application of support vector machine for detecting rice diseases using shape and color texture features," in *Proc. Int. Conf. Eng. Comput.*, May 2009, pp. 79–83.

[56] S. Kaur, S. Pandey, and S. Goel, "Semi-automatic leaf disease detection and classification system for soybean culture," *IET Image Process.*, vol. 12, no. 6, pp. 1038–1048, Jun. 2018.

[57] M. Francisco, F. Ribeiro, J. Metrôlho, and R. Dionísio, "Algorithms and models for automatic detection and classification of diseases and pests in agricultural crops: A systematic review," *Appl. Sci.*, vol. 13, no. 8, p. 4720, Apr. 2023.

[58] R. B. Dhanakotti et al., "Structural and magnetic properties of cobalt-doped iron oxide nanoparticles prepared by solution combustion method for biomedical applications," *Int. J. Nanomedicine*, p. 189, Oct. 2015, doi: 10.2147/ijn.s82210.

[59] G. Dhingra, V. Kumar, and H. D. Joshi, "Study of digital image processing techniques for leaf disease detection and classification," *Multimedia Tools Appl.*, vol. 77, no. 15, pp. 19951–20000, Aug. 2018.

[60] A. Cruz, Y. Ampatzidis, R. Pierro, A. Materazzi, A. Panattoni, L. De Bellis, and A. Luvisi, "Detection of grapevine yellows symptoms in *Vitis vinifera* L. With artificial intelligence," *Comput. Electron. Agricult.*, vol. 157, pp. 63–76, Feb. 2019.

[61] K. Vidyaraj and S. Priya, "Developing an algorithm for tomato leaf disease detection and classification," *Int. J. Innov. Res. Elect. Electron. Instrum. Control Eng.*, vol. 3, no. 1, Feb. 2016.

[62] S. Kai, L. Zhikun, S. Hang, and G. Chunhong, "A research of maize disease image recognition of corn based on BP networks," in *Proc. 3rd Int. Conf. Measuring Technol. Mechatronics Autom.*, vol. 1, Jan. 2011, pp. 246–249.

[63] P. Chaudhary, A. K. Chaudhari, A. Cheeran, and S. Godara, "Color transform based approach for disease spot detection on plant leaf," *Int. J. Comput. Sci. Telecommun.*, vol. 3, no. 6, pp. 65–70, 2012.

[64] A. A. Joshi and B. D. Jadhav, "Monitoring and controlling rice diseases using image processing techniques," in *Proc. Int. Conf. Comput., Analytics Secur. Trends (CAST)*, Dec. 2016, pp. 471–476.

[65] P. Ganesan, G. Sajiv, and L. M. Leo, "CIELuv color space for identification and segmentation of disease affected plant leaves using fuzzy based approach," in *Proc. 3rd Int. Conf. Sci. Technol. Eng. Manag. (ICONSTEM)*, Mar. 2017, pp. 889–894.

[66] A. Meunkaewjinda, P. Kumsawat, K. Attakitmongcol, and A. Srikaew, "Grape leaf disease detection from color imagery using hybrid intelligent system," in *Proc. 5th Int. Conf. Electr. Engineering/Electronics, Comput., Telecommun. Inf. Technol.*, May 2008, pp. 513–516.

[67] A. Rao and S. B. Kulkarni, "A hybrid approach for plant leaf disease detection and classification using digital image processing methods," *Int. J. Electr. Eng. Educ.*, vol. 12, 2020, Art. no. 002072092095312.

[68] A. Asfarian, Y. Herdiyeni, A. Rauf, and K. H. Mutaqin, "Paddy diseases identification with texture analysis using fractal descriptors based on Fourier spectrum," in *Proc. Int. Conf. Comput., Control, Informat. Appl. (ICINA)*, Nov. 2013, pp. 77–81.

[69] K. Thangadurai and K. Padmavathi, "Computer visionimage enhancement for plant leaves disease detection," in *Proc. World Congr. Comput. Commun. Technol.*, Feb. 2014, pp. 173–175.

[70] S. D. Khirade and A. B. Patil, "Plant disease detection using image processing," in *Proc. Int. Conf. Comput. Commun. Control Autom.*, Feb. 2015, pp. 768–771.

[71] E. Albahari, H. Madzin, R. Wirza, O. K. Rahmat, and R. Mas, "Image enhancement techniques on plant leaf and seed disease detection," *Int. J. Innov. Res. Comput. Commun. Eng.*, vol. 5, no. 1, pp. 109–116, 2019.

[72] S. Sladojevic, M. Arsenovic, A. Anderla, D. Culibrk, and D. Stefanovic, "Deep neural networks based recognition of plant diseases by leaf image classification," *Comput. Intell. Neurosci.*, vol. 2016, pp. 1–11, May 2016.

[73] P. Goncharov, G. Ososkov, A. Nechaevskiy, A. Uzhinskiy, and I. Nestsiarenia, "Disease detection on the plant leaves by deep learning," in *Advances in Neural Computation, Machine Learning, and Cognitive Research II*. Cham, Switzerland: Springer, 2019, pp. 151–159.

[74] J. G. A. Barbedo, "A review on the main challenges in automatic plant disease identification based on visible range images," *Biosystems Eng.*, vol. 144, pp. 52–60, Apr. 2016.

[75] G. Anthonys and N. Wickramarachchi, "An image recognition system for crop disease identification of paddy fields in Sri Lanka," in *Proc. Int. Conf. Ind. Inf. Syst. (ICIIS)*, Sri Lanka, Dec. 2009, pp. 403–407.

[76] S. Bankar, A. Dube, P. Kadam, and S. Deokule, "Plant disease detection techniques using Canny edge detection & color histogram in image processing," *Int. J. Comput. Sci. Inf. Technol.*, vol. 5, no. 2, pp. 1165–1168, 2014.

[77] S. B. Jadhav, V. R. Udup, and S. B. Patil, "Soybean leaf disease detection and severity measurement using multiclass SVM and KNN classifier," *Int. J. Electr. Comput. Eng. (IJECE)*, vol. 9, no. 5, p. 4077, Oct. 2019.

[78] J. Pang, Z.-Y. Bai, J.-C. Lai, and S.-K. Li, "Automatic segmentation of crop leaf spot disease images by integrating local threshold and seeded region growing," in *Proc. Int. Conf. Image Anal. Signal Process.*, Oct. 2011, pp. 590–594.

[79] V. Singh, S. Gupta, and S. Saini, "A methodological survey of image segmentation using soft computing techniques," in *Proc. Int. Conf. Adv. Comput. Eng. Appl.*, Mar. 2015, pp. 419–422.

[80] A. Rastogi, R. Arora, and S. Sharma, "Leaf disease detection and grading using computer vision technology & fuzzy logic," in *Proc. 2nd Int. Conf. Signal Process. Integr. Netw. (SPIN)*, Feb. 2015, pp. 500–505.

[81] S. B. Jadhav and S. B. Patil, "Grading of soybean leaf disease based on segmented image using k-means clustering," *IAES Int. J. Artif. Intell. (IJ-AI)*, vol. 5, no. 1, p. 13, Mar. 2016.

[82] S. Zhang, Y. Zhu, Z. You, and X. Wu, "Fusion of superpixel, expectation maximization and PHOG for recognizing cucumber diseases," *Comput. Electron. Agricult.*, vol. 140, pp. 338–347, Aug. 2017.

[83] S. S. Harakannanavar, J. M. Rudagi, V. I. Puranikmath, A. Siddiqua, and R. Pramodhini, "Plant leaf disease detection using computer vision and machine learning algorithms," *Global Transitions Proc.*, vol. 3, no. 1, pp. 305–310, Jun. 2022.

[84] S. Zhang, H. Wang, W. Huang, and Z. You, "Plant diseased leaf segmentation and recognition by fusion of superpixel, K-means and PHOG," *Optik*, vol. 157, pp. 866–872, Mar. 2018, doi: 10.1016/j.ijleo.2017.11.190.

[85] M. Sachin, B. Jagtap, M. Shailesh, and M. Hambarde. *Agricultural Plant Leaf Disease Detection and Diagnosis Using Image Processing Based on Morphological Feature Extraction*. Accessed: Aug. 9, 2023. [Online]. Available: https://www.iosrjournals.org/iosr-jvlsi/papers/vol4-issue5/Version-1/E04512430.pdf

[86] X. Bai, X. Li, Z. Fu, X. Lv, and L. Zhang, "A fuzzy clustering segmentation method based on neighborhood grayscale information for defining cucumber leaf spot disease images," *Comput. Electron. Agricult.*, vol. 136, pp. 157–165, Apr. 2017.

[87] S. Kumar Sahu and M. Pandey, "An optimal hybrid multiclass SVM for plant leaf disease detection using spatial fuzzy C-means model," *Exp. Syst. Appl.*, vol. 214, Mar. 2023, Art. no. 118989.

[88] V. Singh and A. K. Misra, "Detection of unhealthy region of plant leaves using image processing and genetic algorithm," in *Proc. Int. Conf. Adv. Comput. Eng. Appl.*, Mar. 2015, pp. 1028–1032.

[89] V. Singh and A. K. Misra, "Detection of plant leaf diseases using image segmentation and soft computing techniques," *Inf. Process. Agricult.*, vol. 4, no. 1, pp. 41–49, Mar. 2017.

[90] S. S. Chouhan, U. P. Singh, and S. Jain, "Applications of computer vision in plant pathology: A survey," *Arch. Comput. Methods Eng.*, vol. 27, no. 2, pp. 611–632, Apr. 2020.

[91] U. Mokhtar, N. El Bendary, A. E. Hassenian, E. Emary, M. A. Mahmoud, H. Hefny, and M. F. Tolba, "SVM-based detection of tomato leaves diseases," in *Advances in Intelligent Systems and Computing*. Cham, Switzerland: Springer, 2015, pp. 641–652.

[92] M. Bhagat and D. Kumar, "Efficient feature selection using BoWs and SURF method for leaf disease identification," *Multimedia Tools Appl.*, vol. 82, no. 18, pp. 28187–28211, Feb. 2023, doi: 10.1007/s11042-023-14625-5.

[93] D. Al Bashish, M. Braik, and S. Bani-Ahmad, "Detection and classification of leaf diseases using K-means-based segmentation and neural-networks-based classification," *Inf. Technol. J.*, vol. 10, no. 2, pp. 267–275, Jan. 2011.

[94] Y. Tian, C. Zhao, S. Lu, and X. Guo, "Multiple classifier combination for recognition of wheat leaf diseases," *Intell. Autom. Soft Comput.*, vol. 17, no. 5, pp. 519–529, Jan. 2011.

[95] P. M. Mainkar, S. Ghorpade, and M. Adawadkar, "Plant leaf disease detection and classification using image processing techniques," *Int. J. Innov. Emerg. Res. Eng.*, vol. 2, no. 4, pp. 139–144, 2015.

[96] M. Islam, A. Dinh, K. Wahid, and P. Bhowmik, "Detection of potato diseases using image segmentation and multiclass support vector machine," in *Proc. IEEE 30th Can. Conf. Electr. Comput. Eng. (CCECE)*, Apr. 2017, pp. 1–4.

[97] M. Sharif, M. A. Khan, Z. Iqbal, M. F. Azam, M. I. U. Lali, and M. Y. Javed, "Detection and classification of citrus diseases in agriculture based on optimized weighted segmentation and feature selection," *Comput. Electron. Agricult.*, vol. 150, pp. 220–234, Jul. 2018.

[98] Y. Dandawate and R. Kokare, "An automated approach for classification of plant diseases towards development of futuristic decision support system in Indian perspective," in *Proc. Int. Conf. Adv. Comput., Commun. Informat. (ICACCI)*, Aug. 2015, pp. 794–799.

[99] K. J. Mohan, M. Balasubramanian, and S. Palanivel, "Detection and recognition of diseases from paddy plant leaf images," *Int. J. Comput. Appl.*, vol. 144, no. 12, pp. 34–41, Jun. 2016, doi: 10.5120/ijca2016910505.

[100] S. Ramesh, R. Hebbar, M. Niveditha, R. Pooja, N. Shashank, and P. V. Vinod, "Plant disease detection using machine learning," in *Proc. Int. Conf. Design Innov. 3Cs Compute Communicate Control (ICDIC)*, Apr. 2018, pp. 41–45.

[101] H. Waghmare, R. Kokare, and Y. Dandawate, "Detection and classification of diseases of grape plant using opposite colour local binary pattern feature and machine learning for automated decision support system," in *Proc. 3rd Int. Conf. Signal Process. Integr. Netw. (SPIN)*, Feb. 2016, pp. 513–518.
[102] K. Singh, S. Kumar, and P. Kaur, "Automatic detection of rust disease of lentil by machine learning system using microscopic images," *Int. J. Electr. Comput. Eng. (IJECE)*, vol. 9, no. 1, p. 660, Feb. 2019.
[103] V. A. Gulhane and A. A. Gurjar, "Detection of diseases on cotton leaves and its possible diagnosis," *Int. J. Image Process.*, vol. 5, no. 5, pp. 590–598, 2011. [Online]. Available: https://www.cscjournals.org/library/manuscriptinfo.php?mc=IJIP-478
[104] S. Prasad, P. Kumar, R. Hazra, and A. Kumar, "Plant leaf disease detection using Gabor wavelet transform," in *Swarm, Evolutionary, and Memetic Computing*. Berlin, Germany: Springer, 2012, pp. 372–379.
[105] A. Kulkarni, "Applying image processing technique to detect plant diseases," *Int. J. Modern Eng. Res.*, vol. 2, no. 5, pp. 3661–3664, 2012.
[106] P. Jolly and S. Raman, "Analyzing surface defects in apples using Gabor features," in *Proc. 12th Int. Conf. Signal-Image Technol. Internet-Based Syst. (SITIS)*, Nov. 2016, pp. 178–185.
[107] H. Sabrol and S. Kumar, "Fuzzy and neural network based tomato plant disease classification using natural outdoor images," *Indian J. Sci. Technol.*, vol. 9, no. 44, pp. 1–8, Nov. 2016.
[108] S. M. Kiran and D. N. Chandrappa, "Plant disease identification using discrete wavelet transforms and SVM," *J. Univ. Shanghai Sci. Technol.*, vol. 23, no. 6, pp. 108–114, 2021. [Online]. Available: https://jusst.org/wp-content/uploads/2021/06/Plant-Disease-Identification-Using-Discrete-Wavelet-Transforms-and-SVM-1.pdf
[109] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2005, pp. 886–893. [Online]. Available: http://vision.stanford.edu/teaching/cs231b_spring1213/papers/CVPR05_DalalTriggs.pdf
[110] Y. Bai, L. Guo, L. Jin, and Q. Huang, "A novel feature extraction method using pyramid histogram of orientation gradients for smile recognition," in *Proc. 16th IEEE Int. Conf. Image Process. (ICIP)*, Nov. 2009, pp. 3305–3308.
[111] S. S. Sannakki, V. S. Rajpurohit, V. B. Nargund, and P. Kulkarni, "Diagnosis and classification of grape leaf diseases using neural networks," in *Proc. 4th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, Jul. 2013, pp. 1–5.
[112] R. D. L. Pires, D. N. Gonçalves, J. P. M. Oruě, W. E. S. Kanashiro, J. F. Rodrigues, B. B. Machado, and W. N. Gonçalves, "Local descriptors for soybean disease recognition," *Comput. Electron. Agricult.*, vol. 125, pp. 48–55, Jul. 2016.
[113] B. S. Kusumo, A. Heryana, O. Mahendra, and H. F. Pardede, "Machine learning-based for automatic detection of corn-plant diseases using image processing," in *Proc. Int. Conf. Comput., Control, Informat. Appl. (IC3INA)*, Nov. 2018, pp. 93–97.
[114] A. S. Zamani, L. Anand, K. P. Rane, P. Prabhu, A. M. Buttar, H. Pallathadka, A. Raghuvanshi, and B. N. Dugbakie, "Performance of machine learning and image processing in plant leaf disease detection," *J. Food Qual.*, vol. 2022, pp. 1–7, Apr. 2022.
[115] P. Revathi and M. Hemalatha, "Cotton leaf spot diseases detection utilizing feature selection with skew divergence method," *Int. J. Sci. Eng. Technol.*, vol. 3, no. 1, pp. 22–30, 2014. [Online]. Available: http://www.ijset.com/publication/v3/005.pdf
[116] M. Ramakrishnan, "Groundnut leaf disease detection and classification by using back probagation algorithm," in *Proc. Int. Conf. Commun. Signal Process. (ICCSP)*, Apr. 2015, pp. 0964–0968.
[117] A. Caglayan, O. Guclu, and A. B. Can, "A plant recognition approach using shape and color features in leaf images," in *Image Analysis and Processing—ICIAP*. Berlin, Germany: Springer, 2013, pp. 161–170.
[118] T. Munisami, M. Ramsurn, S. Kishnah, and S. Pudaruth, "Plant leaf recognition using shape features and colour histogram with K-nearest neighbour classifiers," *Proc. Comput. Sci.*, vol. 58, pp. 740–747, Jan. 2015.
[119] A. Camargo and J. S. Smith, "Image pattern classification for the identification of disease causing agents in plants," *Comput. Electron. Agricult.*, vol. 66, no. 2, pp. 121–125, May 2009.
[120] H. Wang, G. Li, Z. Ma, and X. Li, "Application of neural networks to image recognition of plant diseases," in *Proc. Int. Conf. Syst. Informat. (ICSAI)*, May 2012, pp. 2159–2164.
[121] S. Phadikar, J. Sil, and A. K. Das, "Rice diseases classification using feature selection and rule generation techniques," *Comput. Electron. Agricult.*, vol. 90, pp. 76–85, Jan. 2013.
[122] R. R. Patil and S. Kumar, "Rice-fusion: A multimodality data fusion framework for rice disease diagnosis," *IEEE Access*, vol. 10, pp. 5207–5222, 2022.
[123] N. Sengar, M. K. Dutta, and C. M. Travieso, "Computer vision based technique for identification and quantification of powdery mildew disease in cherry leaves," *Computing*, vol. 100, no. 11, pp. 1189–1201, Nov. 2018.
[124] O. Y. Al-Jarrah, A. Siddiqui, M. Elsalamouny, P. D. Yoo, S. Muhaidat, and K. Kim, "Machine-Learning-Based feature selection techniques for large-scale network intrusion detection," in *Proc. IEEE 34th Int. Conf. Distrib. Comput. Syst. Workshops (ICDCSW)*, Jun. 2014, pp. 177–181.
[125] V. K. Shrivastava and M. K. Pradhan, "Rice plant disease classification using color features: A machine learning paradigm," *J. Plant Pathol.*, vol. 103, no. 1, pp. 17–26, Oct. 2020, doi: 10.1007/s42161-020-00683-3.
[126] A. K. Rath and J. K. Meher, "Disease detection in infected plant leaf by computational method," *Arch. Phytopathol. Plant Protection*, vol. 52, nos. 19–20, pp. 1348–1358, Dec. 2019.
[127] J. Luo, S. Geng, C. Xiu, D. Song, and T. Dong, "A curvelet-SC recognition method for maize disease," *J. Electr. Comput. Eng.*, vol. 2015, pp. 1–8, Jan. 2015.
[128] S. Gharge and P. Singh, "Image processing for soybean disease classification and severity estimation," in *Emerging Research in Computing, Information, Communication and Applications*. New Delhi, India: Springer, 2016, pp. 493–500.
[129] A. Kaya, A. S. Keceli, C. Catal, H. Y. Yalic, H. Temucin, and B. Tekinerdogan, "Analysis of transfer learning for deep neural network based plant classification models," *Comput. Electron. Agricult.*, vol. 158, pp. 20–29, Mar. 2019.
[130] A. Sivasangari and K. Priya, "Cotton leaf disease detection and recovery using genetic algorithm," *Int. J. Eng. Res. Gen. Sci.*, vol. 117, pp. 119–123, 2014. [Online]. Available: https://www.acadpubl.eu/jsi/2017-117-20-22/articles/22/23.pdf
[131] L. Hallau, M. Neumann, B. Klatt, B. Kleinhenz, T. Klein, C. Kuhn, M. Röhrig, C. Bauckhage, K. Kersting, A. Mahlein, U. Steiner, and E. Oerke, "Automated identification of sugar beet diseases using smartphones," *Plant Pathol.*, vol. 67, no. 2, pp. 399–410, Feb. 2018.
[132] I. Ahmed and P. K. Yadav, "Plant disease detection using machine learning approaches," *Exp. Syst.*, vol. 40, no. 5, Oct. 2022, doi: 10.1111/exsy.13136.
[133] B. J. Samajpati and S. D. Degadwala, "Hybrid approach for apple fruit diseases detection and classification using random forest classifier," in *Proc. Int. Conf. Commun. Signal Process. (ICCSP)*, Apr. 2016, pp. 1015–1019.
[134] S. M. Javidan, A. Banakar, K. A. Vakilian, and Y. Ampatzidis, "Diagnosis of grape leaf diseases using automatic K-means clustering and machine learning," *Smart Agricult. Technol.*, vol. 3, Feb. 2023, Art. no. 100081, doi: 10.1016/j.atech.2022.100081.
[135] M. Shantkumari and S. V. Uma, "Grape leaf image classification based on machine learning technique for accurate leaf disease detection," *Multimedia Tools Appl.*, vol. 82, no. 1, pp. 1477–1487, Apr. 2022, doi: 10.1007/s11042-022-12976-z.
[136] P. Patil, N. Yaligar, and S. M. Meena, "Comparision of performance of classifiers—SVM, RF and ANN in potato blight disease detection using leaf images," in *Proc. IEEE Int. Conf. Comput. Intell. Comput. Res. (ICCIC)*, Dec. 2017, pp. 1–5.
[137] S. Verma, A. Chug, and A. P. Singh, "Exploring capsule networks for disease classification in plants," *J. Statist. Manag. Syst.*, vol. 23, no. 2, pp. 307–315, Feb. 2020.
[138] A. Krishnakumar and A. Narayanan, "A system for plant disease classification and severity estimation using machine learning techniques," in *Proc. Int. Conf. ISMAC Comput. Vis. Bio-Eng.* (Lecture Notes in Computational Vision and Biomechanics), Jan. 2019, pp. 447–457.
[139] S.-E.-A. Raza, G. Prince, J. P. Clarkson, and N. M. Rajpoot, "Automatic detection of diseased tomato plants using thermal and stereo visible light images," *PLoS ONE*, vol. 10, no. 4, Apr. 2015, Art. no. e0123262.
[140] B. A. M. Ashqar and S. S. Abu-Naser. *Image-Based Tomato Leaves Diseases Detection Using Deep Learning*. Accessed: Aug. 10, 2023. [Online]. Available: http://dstore.alazhar.edu.ps/xDLui/bitstream/handle/123456789/278/ASHITL-3v1.pdf?sequence=1&isAllowed=y

[141] S. Jasrotia, J. Yadav, N. Rajpal, M. Arora, and J. Chaudhary, "Convolutional neural network based maize plant disease identification," *Proc. Comput. Sci.*, vol. 218, pp. 1712–1721, Jan. 2023.

[142] R. Sujatha, J. M. Chatterjee, N. Jhanjhi, and S. N. Brohi, "Performance of deep learning vs machine learning in plant leaf disease detection," *Microprocess. Microsyst.*, vol. 80, Feb. 2021, Art. no. 103615.

[143] S. U. Rahman, F. Alam, N. Ahmad, and S. Arshad, "Image processing based system for the detection, identification and treatment of tomato leaf diseases," *Multimedia Tools Appl.*, vol. 82, no. 6, pp. 9431–9445, Sep. 2022.

[144] A. Bhatia, A. Chug, and A. Prakash Singh, "Application of extreme learning machine in plant disease prediction for highly imbalanced dataset," *J. Statist. Manag. Syst.*, vol. 23, no. 6, pp. 1059–1068, Jul. 2020.

[145] A. Bhatia, A. Chug, and A. P. Singh, "Hybrid SVM-LR classifier for powdery mildew disease prediction in tomato plant," in *Proc. 7th Int. Conf. Signal Process. Integr. Netw.*, Feb. 2020, pp. 218–223. [Online]. Available: https://ieeexplore.ieee.org/document/9071202

[146] W. Mcculloch and W. Pitts, "A logical calculus of the ideas immanent in nervous activity," *Bull. Math. Biol.*, vol. 52, nos. 1–2, pp. 99–115, 1990.

[147] M. Dyrmann, H. Karstoft, and H. S. Midtiby, "Plant species classification using deep convolutional neural network," *Biosystems Eng.*, vol. 151, pp. 72–80, Nov. 2016.

[148] N. Kussul, M. Lavreniuk, S. Skakun, and A. Shelestov, "Deep learning classification of land cover and crop types using remote sensing data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 778–782, May 2017.

[149] R. Wason, "Deep learning: Evolution and expansion," *Cognit. Syst. Res.*, vol. 52, pp. 701–708, Dec. 2018.

[150] G. Geetharamani and A. Pandian, "Identification of plant leaf diseases using a nine-layer deep convolutional neural network," *Comput. Electr. Eng.*, vol. 76, pp. 323–338, Jun. 2019.

[151] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.

[152] E. C. Too, L. Yujian, S. Njuki, and L. Yingchun, "A comparative study of fine-tuning deep learning models for plant disease identification," *Comput. Electron. Agricult.*, vol. 161, pp. 272–279, Jun. 2019.

[153] K. He, X. Zhang, S. Ren, and J. Sun, "Identity mappings in deep residual networks," in *Computer Vision—ECCV*. Cham, Switzerland: Springer, 2016, pp. 630–645.

[154] Z. Gao, Z. Luo, W. Zhang, Z. Lv, and Y. Xu, "Deep learning application in plant stress imaging: A review," *AgriEngineering*, vol. 2, no. 3, pp. 430–446, Jul. 2020.

[155] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," 2015, *arXiv:1512.03385*.

[156] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.

[157] A. Ahmad, D. Saraswat, and A. El Gamal, "A survey on using deep learning techniques for plant disease diagnosis and recommendations for development of appropriate tools," *Smart Agricult. Technol.*, vol. 3, Feb. 2023, Art. no. 100083, doi: 10.1016/j.atech.2022.100083.

[158] M. Türkoglu and D. Hanbay, "Plant disease and pest detection using deep learning-based features," *TURKISH J. Electr. Eng. Comput. Sci.*, vol. 27, no. 3, pp. 1636–1651, May 2019.

[159] M. Brahimi, S. Mahmoudi, K. Boukhalfa, and A. Moussaoui, "Deep interpretable architecture for plant diseases classification," in *Proc. Signal Process., Algorithms, Architectures, Arrangements, Appl.*, Poznan, Poland, Sep. 2019, pp. 111–116.

[160] M. Mehdipour Ghazi, B. Yanikoglu, and E. Aptoula, "Plant identification using deep neural networks via optimization of transfer learning parameters," *Neurocomputing*, vol. 235, pp. 228–235, Apr. 2017.

[161] A. Ramcharan, K. Baranowski, P. McCloskey, B. Ahmed, J. Legg, and D. P. Hughes, "Deep learning for image-based cassava disease detection," *Frontiers Plant Sci.*, vol. 8, p. 1852, Oct. 2017.

[162] Z. Lin, S. Mu, A. Shi, C. Pang, and X. Sun, "A novel method of maize leaf disease image identification based on a multichannel convolutional neural network," *Trans. ASABE*, vol. 61, no. 5, pp. 1461–1474, 2018.

[163] D. Hendrycks, N. Mu, E. D. Cubuk, B. Zoph, J. Gilmer, and B. Lakshminarayanan, "AugMix: A simple data processing method to improve robustness and uncertainty," 2019, *arXiv:1912.02781*.

[164] S. Yun, D. Han, S. Joon Oh, S. Chun, J. Choe, and Y. Yoo, "CutMix: Regularization strategy to train strong classifiers with localizable features," 2019, *arXiv:1905.04899*.

[165] D. Ho, E. Liang, I. Stoica, P. Abbeel, and X. Chen, "Population based augmentation: Efficient learning of augmentation policy schedules," 2019, *arXiv:1905.05393*.

[166] E. D. Cubuk, B. Zoph, J. Shlens, and Q. V. Le, "RandAugment: Practical automated data augmentation with a reduced search space," 2019, *arXiv:1909.13719*.

[167] I. J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 1–9. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2014/file/5ca3e9b122f61f8f06494c97b1afccf3-Paper.pdf

[168] B. Liu, Y. Zhang, D. He, and Y. Li, "Identification of apple leaf diseases based on deep convolutional neural networks," *Symmetry*, vol. 10, no. 1, p. 11, Dec. 2017.

[169] R. Chen, H. Qi, Y. Liang, and M. Yang, "Identification of plant leaf diseases by deep learning based on channel attention and channel pruning," *Frontiers Plant Sci.*, vol. 13, Nov. 2022.

[170] H. Nazki, S. Yoon, A. Fuentes, and D. S. Park, "Unsupervised image translation using adversarial networks for improved plant disease recognition," *Comput. Electron. Agricult.*, vol. 168, Jan. 2020, Art. no. 105117.

[171] Y. Tian, G. Yang, Z. Wang, E. Li, and Z. Liang, "Detection of apple lesions in orchards based on deep learning methods of CycleGAN and YOLOV3-dense," *J. Sensors*, vol. 2019, pp. 1–13, Apr. 2019.

[172] Q. Wu, Y. Chen, and J. Meng, "DCGAN-based data augmentation for tomato leaf disease identification," *IEEE Access*, vol. 8, pp. 98716–98728, 2020.

[173] B. Liu, C. Tan, S. Li, J. He, and H. Wang, "A data augmentation method based on generative adversarial networks for grape leaf disease identification," *IEEE Access*, vol. 8, pp. 102188–102198, 2020.

[174] L. Ale, A. Sheta, L. Li, Y. Wang, and N. Zhang, "Deep learning based plant disease detection for smart agriculture," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, Dec. 2019, pp. 1–6.

[175] A. Darwish, D. Ezzat, and A. E. Hassanien, "An optimized model based on convolutional neural networks and orthogonal learning particle swarm optimization algorithm for plant diseases diagnosis," *Swarm Evol. Comput.*, vol. 52, Feb. 2020, Art. no. 100616.

[176] R. S. Yuwana, E. Suryawati, V. Zilvan, A. Ramdan, H. F. Pardede, and F. Fauziah, "Multi-condition training on deep convolutional neural networks for robust plant diseases detection," in *Proc. Int. Conf. Comput., Control, Informat. Appl. (ICINA)*, Oct. 2019, pp. 30–35.

[177] J. Chen, J. Chen, D. Zhang, Y. Sun, and Y. A. Nanehkaran, "Using deep transfer learning for image-based plant disease identification," *Comput. Electron. Agricult.*, vol. 173, Jun. 2020, Art. no. 105393.

[178] S. Mishra, R. Sachan, and D. Rajpal, "Deep convolutional neural network based detection system for real-time corn plant disease recognition," *Proc. Comput. Sci.*, vol. 167, pp. 2003–2010, Jan. 2020.

[179] N. K. Trivedi, V. Gautam, A. Anand, H. M. Aljahdali, S. G. Villar, D. Anand, N. Goyal, and S. Kadry, "Early detection and classification of tomato leaf disease using high-performance deep neural network," *Sensors*, vol. 21, no. 23, p. 7987, Nov. 2021.

[180] P. Deshwal, K. Sharma, and M. S. Moudgil, "Plant leaf disease detection using machine learning," *Int. J. Res. Appl. Sci. Eng. Technol.*, vol. 11, no. 5, pp. 5928–5932, May 2023.

[181] M. A. Jasim and J. M. Al-Tuwaijari, "Plant leaf diseases detection and classification using image processing and deep learning techniques," in *Proc. Int. Conf. Comput. Sci. Softw. Eng. (CSASE)*, Apr. 2020, pp. 259–265.

[182] A. V. Panchal, S. C. Patel, K. Bagyalakshmi, P. Kumar, I. R. Khan, and M. Soni, "Image-based plant diseases detection using deep learning," *Mater. Today, Proc.*, vol. 80, pp. 3500–3506, Jan. 2023.

[183] A. Chug, A. Bhatia, A. P. Singh, and D. Singh, "A novel framework for image-based plant disease detection using hybrid deep learning approach," *Soft Comput.*, vol. 27, no. 18, pp. 13613–13638, Jun. 2022.

[184] Q. Wang, F. Qi, M. Sun, J. Qu, and J. Xue, "Identification of tomato disease types and detection of infected areas based on deep convolutional neural networks and object detection techniques," *Comput. Intell. Neurosci.*, vol. 2019, pp. 1–15, Dec. 2019.

[185] A. Picon, M. Seitz, A. Alvarez-Gila, P. Mohnke, A. Ortiz-Barredo, and J. Echazarra, "Crop conditional convolutional neural networks for massive multi-crop plant disease classification over cell phone acquired images taken on real field conditions," *Comput. Electron. Agricult.*, vol. 167, Dec. 2019, Art. no. 105093.

[186] T. R. Gadekallu, D. S. Rajput, M. P. K. Reddy, K. Lakshmanna, S. Bhattacharya, S. Singh, A. Jolfaei, and M. Alazab, "A novel PCA-whale optimization-based deep neural network model for classification of tomato plant diseases using GPU," *J. Real-Time Image Process.*, vol. 18, no. 4, pp. 1383–1396, Aug. 2021.

[187] M. Kaushik, P. Prakash, R. Ajay, and S. Veni, "Tomato leaf disease detection using convolutional neural network with data augmentation," in *Proc. 5th Int. Conf. Commun. Electron. Syst. (ICCES)*, Coimbatore, India, Jun. 2020, pp. 1125–1132.

[188] S. Verma, A. Chug, and A. P. Singh, "Application of convolutional neural networks for evaluation of disease severity in tomato plant," *J. Discrete Math. Sci. Cryptogr.*, vol. 23, no. 1, pp. 273–282, Jan. 2020.

[189] P. Wspanialy and M. Moussa, "A detection and severity estimation system for generic diseases of tomato greenhouse plants," *Comput. Electron. Agricult.*, vol. 178, Nov. 2020, Art. no. 105701.

[190] R. Karthik, M. Hariharan, S. Anand, P. Mathikshara, A. Johnson, and R. Menaka, "Attention embedded residual CNN for disease detection in tomato leaves," *Appl. Soft Comput.*, vol. 86, Jan. 2020, Art. no. 105933.

[191] I. Ahmad, M. Hamid, S. Yousaf, S. T. Shah, and M. O. Ahmad, "Optimizing pretrained convolutional neural networks for tomato leaf disease detection," *Complexity*, vol. 2020, pp. 1–6, Sep. 2020.

[192] S. H. Lee, H. Goëau, P. Bonnet, and A. Joly, "New perspectives on plant disease characterization based on deep learning," *Comput. Electron. Agricult.*, vol. 170, Mar. 2020, Art. no. 105220.

[193] Ü. Atila, M. Uçar, K. Akyol, and E. Uçar, "Plant leaf disease classification using EfficientNet deep learning model," *Ecological Informat.*, vol. 61, Mar. 2021, Art. no. 101182.

[194] M. Agarwal, A. Singh, S. Arjaria, A. Sinha, and S. Gupta, "ToLeD: Tomato leaf disease detection using convolution neural network," *Proc. Comput. Sci.*, vol. 167, pp. 293–301, Jan. 2020.

[195] L. C. Ngugi, M. Abdelwahab, and M. Abo-Zahhad, "Tomato leaf segmentation algorithms for mobile phone applications using deep learning," *Comput. Electron. Agricult.*, vol. 178, Nov. 2020, Art. no. 105788.

[196] J. Chen, J. Chen, D. Zhang, Y. A. Nanehkaran, and Y. Sun, "A cognitive vision method for the detection of plant disease images," *Mach. Vis. Appl.*, vol. 32, no. 1, pp. 1–18, Jan. 2021.

[197] V. G. Krishnan, J. Deepa, P. V. Rao, V. Divya, and S. Kaviarasan, "An automated segmentation and classification model for banana leaf disease detection," *J. Appl. Biol. Biotechnol.*, vol. 10, pp. 213–220, Oct. 2022.

[198] R. K. Singh, A. Tiwari, and R. K. Gupta, "Deep transfer modeling for classification of maize plant leaf disease," *Multimedia Tools Appl.*, vol. 81, no. 5, pp. 6051–6067, Feb. 2022.

[199] A. Elaraby, W. Hamdy, and M. Alruwaili, "Optimization of deep learning model for plant disease detection using particle swarm optimizer," *Comput., Mater. Continua*, vol. 71, no. 2, pp. 4019–4031, 2022.

[200] S. Ashwinkumar, S. Rajagopal, V. Manimaran, and B. Jegajothi, "Automated plant leaf disease detection and classification using optimal MobileNet based convolutional neural networks," *Mater. Today, Proc.*, vol. 51, pp. 480–487, Jan. 2022.

[201] J. A. Pandian, K. Kanchanadevi, V. D. Kumar, E. Jasinska, R. Gono, Z. Leonowicz, and M. Jasinski, "A five convolutional layer deep convolutional neural network for plant leaf disease detection," *Electronics*, vol. 11, no. 8, p. 1266, Apr. 2022.

[202] R. Mahum, H. Munir, Z.-U.-N. Mughal, M. Awais, F. S. Khan, M. Saqlain, S. Mahamad, and I. Tlili, "A novel framework for potato leaf disease detection using an efficient deep learning model," *Hum. Ecol. Risk Assessment, Int. J.*, vol. 29, no. 2, pp. 303–326, Apr. 2022, doi: 10.1080/10807039.2022.2064814.

[203] A. Haridasan, J. Thomas, and E. D. Raj, "Deep learning system for paddy plant disease detection and classification," *Environ. Monitor. Assessment*, vol. 195, no. 1, p. 120, Nov. 2022, doi: 10.1007/s10661-022-10656-x.

[204] Y. M. A. Algani, O. J. M. Caro, L. M. R. Bravo, C. Kaur, M. S. Al Ansari, and B. K. Bala, "Leaf disease identification and classification using optimized deep learning," *Meas., Sensors*, vol. 25, Feb. 2023, Art. no. 100643, doi: 10.1016/j.measen.2022.100643.

[205] X. Huang, A. Chen, G. Zhou, X. Zhang, J. Wang, N. Peng, N. Yan, and C. Jiang, "Tomato leaf disease detection system based on FC-SNDPN," *Multimedia Tools Appl.*, vol. 82, no. 2, pp. 2121–2144, Jun. 2022, doi: 10.1007/s11042-021-11790-3.

[206] S. Yu, L. Xie, and Q. Huang, "Inception convolutional vision transformers for plant disease identification," *Internet Things*, vol. 21, Apr. 2023, Art. no. 100650, doi: 10.1016/j.iot.2022.100650.

[207] M. A. Khan, A. Alqahtani, A. Khan, S. Alsubai, A. Binbusayyis, M. M. I. Ch, H.-S. Yong, and J. Cha, "Cucumber leaf diseases recognition using multi level deep entropy-ELM feature selection," *Appl. Sci.*, vol. 12, no. 2, p. 593, Jan. 2022, doi: 10.3390/app12020593.

[208] H. Yu, X. Cheng, C. Chen, A. A. Heidari, J. Liu, Z. Cai, and H. Chen, "Apple leaf disease recognition method with improved residual network," *Multimedia Tools Appl.*, vol. 81, no. 6, pp. 7759–7782, Jan. 2022, doi: 10.1007/s11042-022-11915-2.

[209] A. I. Khan, S. M. K. Quadri, S. Banday, and J. Latief Shah, "Deep diagnosis: A real-time apple leaf disease detection system based on deep learning," *Comput. Electron. Agricult.*, vol. 198, Jul. 2022, Art. no. 107093, doi: 10.1016/j.compag.2022.107093.

[210] *Researchgate.net*. Accessed: Aug. 10, 2023. [Online]. Available: https://www.researchgate.net/publication/366239200_Leaf_disease_identification_and_classification_using_optimized_deep_learning

[211] M. Shoaib, B. Shah, S. Ei-Sappagh, A. Ali, A. Ullah, F. Alenezi, T. Gechev, T. Hussain, and F. Ali, "An advanced deep learning models-based plant disease detection: A review of recent research," *Frontiers Plant Sci.*, vol. 14, Mar. 2023, Art. no. 1158933.

[212] P. Singh, P. Singh, U. Farooq, S. S. Khurana, J. K. Verma, and M. Kumar, "CottonLeafNet: Cotton plant leaf disease detection using deep neural networks," *Multimedia Tools Appl.*, vol. 82, no. 24, pp. 37151–37176, Oct. 2023.

[213] *X-Mol X-Mol.Net*. Accessed: Aug. 10, 2023. [Online]. Available: https://www.x-mol.net/paper/article/1626716389179998208

[214] T. Daniya and S. Vigneshwari, "Rice plant leaf disease detection and classification using optimization enabled deep learning," *J. Environ. Inform.*, vol. 42, p. 25, Sep. 2023.

[215] R. Arumuga Arun and S. Umamaheswari, "Effective multi-crop disease detection using pruned complete concatenated deep learning model," *Exp. Syst. Appl.*, vol. 213, Mar. 2023, Art. no. 118905, doi: 10.1016/j.eswa.2022.118905.

[216] A. Pal and V. Kumar, "AgriDet: Plant leaf disease severity classification using agriculture detection framework," *Eng. Appl. Artif. Intell.*, vol. 119, Mar. 2023, Art. no. 105754, doi: 10.1016/j.engappai.2022.105754.

[217] S. R. G. Reddy, G. P. S. Varma, and R. L. Davuluri, "Resnet-based modified red deer optimization with DLCNN classifier for plant disease identification and classification," *Comput. Electr. Eng.*, vol. 105, Jan. 2023, Art. no. 108492, doi: 10.1016/j.compeleceng.2022.108492.

[218] Y. Kaya and E. Gürsoy, "A novel multi-head CNN design to identify plant diseases using the fusion of RGB images," *Ecological Informat.*, vol. 75, Jul. 2023, Art. no. 101998, doi: 10.1016/j.ecoinf.2023.101998.

[219] H.-T. Thai, K.-H. Le, and N. L.-T. Nguyen, "FormerLeaf: An efficient vision transformer for cassava leaf disease detection," *Comput. Electron. Agricult.*, vol. 204, Jan. 2023, Art. no. 107518, doi: 10.1016/j.compag.2022.107518.

[220] P. S. Thakur, T. Sheorey, and A. Ojha, "VGG-ICNN: A lightweight CNN model for crop disease identification," *Multimedia Tools Appl.*, vol. 82, no. 1, pp. 497–520, Jun. 2022, doi: 10.1007/s11042-022-13144-z.

[221] P. Baranowski, M. Jedryczka, W. Mazurek, D. Babula-Skowronska, A. Siedliska, and J. Kaczmarek, "Hyperspectral and thermal imaging of oilseed rape (*Brassica napus*) response to fungal species of the genus *Alternaria*," *PLoS ONE*, vol. 10, no. 3, Mar. 2015, Art. no. e0122913.

[222] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[223] D. Moshou, C. Bravo, R. Oberti, J. West, L. Bodria, A. McCartney, and H. Ramon, "Plant disease detection based on data fusion of hyper-spectral and multi-spectral fluorescence imaging using Kohonen maps," *Real-Time Imag.*, vol. 11, no. 2, pp. 75–83, Apr. 2005.

[224] A.-K. Mahlein, U. Steiner, C. Hillnhütter, H.-W. Dehne, and E.-C. Oerke, "Hyperspectral imaging for small-scale analysis of symptoms caused by different sugar beet diseases," *Plant Methods*, vol. 8, no. 1, p. 3, 2012.

[225] L. S. Huang, J. L. Zhao, D. Y. Zhang, L. Yuan, Y. Y. Dong, and J. C. Zhang, "Identifying and mapping stripe rust in winter wheat using multi-temporal airborne hyperspectral images," *Int. J. Agricult. Biol.*, vol. 14, pp. 697–704, 2012.

[226] K. Nagasubramanian, S. Jones, A. K. Singh, A. Singh, B. Ganapathysubramanian, and S. Sarkar, "Explaining hyperspectral imaging based plant disease identification: 3D CNN and saliency maps," 2018, *arXiv:1804.08831*. [Online]. Available: http://www.interpretable-ml.org/nips2017workshop/papers/16.pdf

[227] E. Alisaac, J. Behmann, M. T. Kuska, H.-W. Dehne, and A.-K. Mahlein, "Hyperspectral quantification of wheat resistance to fusarium head blight: Comparison of two fusarium species," *Eur. J. Plant Pathol.*, vol. 152, no. 4, pp. 869–884, Dec. 2018.

[228] J. Abdulridha, O. Batuman, and Y. Ampatzidis, "UAV-based remote sensing technique to detect citrus canker disease utilizing hyperspectral imaging and machine learning," *Remote Sens.*, vol. 11, no. 11, p. 1373, Jun. 2019.

[229] C. Nguyen, V. Sagan, M. Maimaitiyiming, M. Maimaitijiang, S. Bhadra, and M. T. Kwasniewski, "Early detection of plant viral disease using hyperspectral imaging and deep learning," *Sensors*, vol. 21, no. 3, p. 742, Jan. 2021.

[230] L. Feng, B. Wu, Y. He, and C. Zhang, "Hyperspectral imaging combined with deep transfer learning for rice disease detection," *Frontiers Plant Sci.*, vol. 12, Sep. 2021, Art. no. 693521.

[231] L. Wan, H. Li, C. Li, A. Wang, Y. Yang, and P. Wang, "Hyperspectral sensing of plant diseases: Principle and methods," *Agronomy*, vol. 12, no. 6, p. 1451, Jun. 2022, doi: 10.3390/agronomy12061451.

[232] V. G. Bhujade and V. Sambhe, "Role of digital, hyper spectral, and SAR images in detection of plant disease with deep learning network," *Multimedia Tools Appl.*, vol. 81, no. 23, pp. 33645–33670, Apr. 2022, doi: 10.1007/s11042-022-13055-z.

[233] R. Cui, J. Li, Y. Wang, S. Fang, K. Yu, and Y. Zhao, "Hyperspectral imaging coupled with dual-channel convolutional neural network for early detection of apple valsa canker," *Comput. Electron. Agricult.*, vol. 202, Nov. 2022, Art. no. 107411, doi: 10.1016/j.compag.2022.107411.

[234] Y. Wu, Y. Cao, and Z. Zhai, "Early detection of bacterial blight in hyperspectral images based on random forest and adaptive coherence estimator," *Sustainability*, vol. 14, no. 20, p. 13168, Oct. 2022, doi: 10.3390/su142013168.

[235] J. Ugarte Fajardo, M. Maridueña-Zavala, J. Cevallos-Cevallos, and D. O. Donoso, "Effective methods based on distinct learning principles for the analysis of hyperspectral images to detect black Sigatoka disease," *Plants*, vol. 11, no. 19, p. 2581, Sep. 2022, doi: 10.3390/plants11192581.

[236] L. Z. Yong, S. Khairunniza-Bejo, M. Jahari, and F. M. Muharam, "Automatic disease detection of basal stem rot using deep learning and hyperspectral imaging," *Agriculture*, vol. 13, no. 1, p. 69, Dec. 2022.

[237] G. Wu, Y. Fang, Q. Jiang, M. Cui, N. Li, Y. Ou, Z. Diao, and B. Zhang, "Early identification of strawberry leaves disease utilizing hyperspectral imaging combing with spectral features, multiple vegetation indices and textural features," *Comput. Electron. Agricult.*, vol. 204, Jan. 2023, Art. no. 107553.

[238] A.-K. Mahlein, "Plant disease detection by imaging sensors—Parallels and specific demands for precision agriculture and plant phenotyping," *Plant Disease*, vol. 100, no. 2, pp. 241–251, Feb. 2016, doi: 10.1094/pdis-03-15-0340-fe.

[239] A.-K. Mahlein, M. T. Kuska, J. Behmann, G. Polder, and A. Walter, "Hyperspectral sensors and imaging technologies in phytopathology: State of the art," *Annu. Rev. Phytopathol.*, vol. 56, no. 1, pp. 535–558, Aug. 2018, doi: 10.1146/annurev-phyto-080417-050100.

[240] S. K. von Bueren, A. Burkart, A. Hueni, U. Rascher, M. P. Tuohy, and I. J. Yule, "Deploying four optical UAV-based sensors over grassland: Challenges and limitations," *Biogeosciences*, vol. 12, no. 1, pp. 163–175, Jan. 2015, doi: 10.5194/bg-12-163-2015.

[241] W. Lv and X. Wang, "Overview of hyperspectral image classification," *J. Sensors*, vol. 2020, pp. 1–13, Jul. 2020, doi: 10.1155/2020/4817234.

[242] J.-H. Lenthe, E.-C. Oerke, and H.-W. Dehne, "Digital infrared thermography for monitoring canopy health of wheat," *Precis. Agricult.*, vol. 8, nos. 1–2, pp. 15–26, Mar. 2007, doi: 10.1007/s11119-006-9025-6.

[243] E. Bauriegel, H. Brabandt, U. Gärber, and W. B. Herppich, "Chlorophyll fluorescence imaging to facilitate breeding of bremia lactucae-resistant lettuce cultivars," *Comput. Electron. Agricult.*, vol. 105, pp. 74–82, Jul. 2014, doi: 10.1016/j.compag.2014.04.010.

[244] M. Neumann, L. Hallau, B. Klatt, K. Kersting, and C. Bauckhage, "Erosion band features for cell phone image based plant disease classification," in *Proc. 22nd Int. Conf. Pattern Recognit.*, Stockholm, Sweden, Aug. 2014, pp. 3315–3320, doi: 10.1109/ICPR.2014.571.

[245] S. Bergstrasser, D. Fanourakis, S. Schmittgen, M. P. Cendrero-Mateo, M. Jansen, H. Scharr, and U. Rascher, "HyperART: Non-invasive quantification of leaf traits using hyperspectral absorption-reflectance-transmittance imaging," *Plant Methods*, vol. 11, pp. 1–17, Dec. 2015.

[246] M. Wahabzada, A.-K. Mahlein, C. Bauckhage, U. Steiner, E.-C. Oerke, and K. Kersting, "Metro maps of plant disease dynamics—Automated mining of differences using hyperspectral images," *PLoS ONE*, vol. 10, no. 1, Jan. 2015, Art. no. e0116902.

[247] G. Polder, N. V. D. Westeringh, J. Kool, H. A. Khan, G. Kootstra, and A. Nieuwenhuizen, "Automatic detection of tulip breaking virus (TBV) using a deep convolutional neural network," *IFAC-PapersOnLine*, vol. 52, no. 30, pp. 12–17, 2019.

[248] E. C. Oerke, A. K. Mahlein, and U. Steiner, "Proximal sensing of plant diseases," in *Detection and Diagnostics of Plant Pathogens*, vol. 5, M. Gullino and P. Bonants, Eds. Dordrecht, The Netherlands: Springer, 2014.

[249] S. G. Caro. (Jan. 21, 2014). *Infection and Spread of Peronospora Sparsa on Rosa SP. (Berk)*. Accessed: Jan. 8, 2024. [Online]. Available: https://bonndoc.ulb.uni-bonn.de/xmlui/handle/20.500.11811/5823

[250] E. Bauriegel and W. Herppich, "Hyperspectral and chlorophyll fluorescence imaging for early detection of plant diseases, with special reference to fusarium spec. Infections on wheat," *Agriculture*, vol. 4, no. 1, pp. 32–57, Mar. 2014, doi: 10.3390/agriculture4010032.

[251] C. Rousseau, E. Belin, E. Bove, D. Rousseau, F. Fabre, R. Berruyer, J. Guillaumès, C. Manceau, M.-A. Jacques, and T. Boureau, "High throughput quantitative phenotyping of plant resistance using chlorophyll fluorescence image analysis," *Plant Methods*, vol. 9, no. 1, p. 17, Dec. 2013, doi: 10.1186/1746-4811-9-17.

[252] Y. Li and X. Chao, "Semi-supervised few-shot learning approach for plant diseases recognition," *Plant Methods*, vol. 17, no. 1, pp. 1–10, Jun. 2021, doi: 10.1186/s13007-021-00770-1.

[253] M. Buda, A. Maki, and M. A. Mazurowski, "A systematic study of the class imbalance problem in convolutional neural networks," *Neural Netw.*, vol. 106, pp. 249–259, Oct. 2018.

[254] J. M. Johnson and T. M. Khoshgoftaar, "Survey on deep learning with class imbalance," *J. Big Data*, vol. 6, no. 1, pp. 1–54, Dec. 2019.

[255] A. Medela, A. Picon, C. L. Saratxaga, O. Belar, V. Cabezon, R. Cicchi, R. Bilbao, and B. Glover, "Few shot learning in histopathological images:Reducing the need of labeled data on biological datasets," in *Proc. IEEE 16th Int. Symp. Biomed. Imag. (ISBI)*, Venice, Italy, Apr. 2019, pp. 1860–1864.

[256] S.-N. Ren, Y. Sun, H.-Y. Zhang, and L.-X. Guo, "Plant disease identification for small sample based on one-shot learning," *Jiangsu J. Agricult. Sci.*, vol. 35, no. 5, pp. 1061–1067, May 2019.

[257] D. Argüeso, A. Picon, U. Irusta, A. Medela, M. G. San-Emeterio, A. Bereciartua, and A. Alvarez-Gila, "Few-shot learning approach for plant disease classification using images taken in the field," *Comput. Electron. Agricult.*, vol. 175, Aug. 2020, Art. no. 105542.

[258] D. Das and C. S. G. Lee, "A two-stage approach to few-shot learning for image recognition," *IEEE Trans. Image Process.*, vol. 29, pp. 3336–3350, 2020.

[259] F. Zhong, Z. Chen, Y. Zhang, and F. Xia, "Zero- and few-shot learning for diseases recognition of citrus aurantium L. Using conditional adversarial autoencoders," *Comput. Electron. Agricult.*, vol. 179, Dec. 2020, Art. no. 105828.

[260] H.-Y. Wu, "Identification of tea leaf's diseases in natural scene images based on low shot learning," M.S. thesis, Dept. Inf. Eng., Anhui Univ., Hefei, China, 2020.

[261] Y. Li and J. Yang, "Meta-learning baselines and database for few-shot classification in agriculture," *Comput. Electron. Agricult.*, vol. 182, Mar. 2021, Art. no. 106055, doi: 10.1016/j.compag.2021.106055.

[262] X. Liang, "Few-shot cotton leaf spots disease classification based on metric learning," *Plant Methods*, vol. 17, no. 1, pp. 1–11, Nov. 2021, doi: 10.1186/s13007-021-00813-7.

[263] S. Wang, Y. Han, J. Chen, X. He, Z. Zhang, X. Liu, and K. Zhang, "Weed density extraction based on few-shot learning through UAV remote sensing RGB and multispectral images in ecological irrigation area," *Frontiers Plant Sci.*, vol. 12, Mar. 2022, Art. no. 735230, doi: 10.3389/fpls.2021.735230.

[264] I. Egusquiza, A. Picon, U. Irusta, A. Bereciartua-Perez, T. Eggers, C. Klukas, E. Aramendi, and R. Navarra-Mestre, "Analysis of few-shot techniques for fungal plant disease classification and evaluation of clustering capabilities over real datasets," *Frontiers Plant Sci.*, vol. 13, Mar. 2022, Art. no. 813237, doi: 10.3389/fpls.2022.813237.

[265] S. Garg and P. Singh, "An aggregated loss function based lightweight few shot model for plant leaf disease classification," *Multimedia Tools Appl.*, vol. 82, no. 15, pp. 23797–23815, Feb. 2023, doi: 10.1007/s11042-023-14372-7.
[266] X. Liu, Z. Ji, Y. Pang, and Z. Han, "Self-taught cross-domain few-shot learning with weakly supervised object localization and task-decomposition," *Knowl.-Based Syst.*, vol. 265, Apr. 2023, Art. no. 110358, doi: 10.1016/j.knosys.2023.110358.
[267] P. Li, F. Liu, L. Jiao, S. Li, L. Li, X. Liu, and X. Huang, "Knowledge transduction for cross-domain few-shot learning," *Pattern Recognit.*, vol. 141, Sep. 2023, Art. no. 109652, doi: 10.1016/j.patcog.2023.109652.
[268] M. A. Chandra and S. S. Bedi, "Survey on SVM and their application in image classification," *Int. J. Inf. Technol.*, vol. 13, no. 5, pp. 1–11, Oct. 2021.
[269] S. Vandenhende, S. Georgoulis, W. Van Gansbeke, M. Proesmans, D. Dai, and L. Van Gool, "Multi-task learning for dense prediction tasks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3614–3633, Jul. 2022. [Online]. Available: https://homes.esat.kuleuven.be/~konijn/publications/2020/SVDH.pdf
[270] S. Bozinovski, "Reminder of the first paper on transfer learning in neural networks, 1976," *Informatica*, vol. 44, no. 3, pp. 291–302, Sep. 2020.
[271] T. Uehara-Ichiki, T. Shiba, K. Matsukura, T. Ueno, M. Hirae, and T. Sasaya, "Detection and diagnosis of rice-infecting viruses," *Frontiers Microbiol.*, vol. 4, p. 289, Oct. 2013.
[272] T. Stackhouse, A. D. Martinez-Espinoza, and M. E. Ali, "Turfgrass disease diagnosis: Past, present, and future," *Plants*, vol. 9, no. 11, p. 1544, Nov. 2020.
[273] S. Sakamoto, W. Putalun, S. Vimolmangkang, W. Phoolcharoen, Y. Shoyama, H. Tanaka, and S. Morimoto, "Enzyme-linked immunosorbent assay for the quantitative/qualitative analysis of plant secondary metabolites," *J. Natural Medicines*, vol. 72, no. 1, pp. 32–42, Jan. 2018.
[274] H. A. McCartney, S. J. Foster, B. A. Fraaije, and E. Ward, "Molecular diagnostics for fungal plant pathogens," *Pest Manag. Sci.*, vol. 59, no. 2, pp. 129–142, Feb. 2003.
[275] S. Loreti, "The diagnosis of plant pathogenic bacteria a state of art," *Frontiers Bioscience*, vol. 10, no. 3, pp. 449–460, 2018.
[276] K. B. Mullis and F. A. Faloona, "Specific synthesis of DNA in vitro via a polymerase-catalyzed chain reaction," in *Methods in Enzymology*, vol. 155, San Diego, CA, USA: Elsevier, 1987, pp. 335–350.
[277] P. Bastien, G. W. Procop, and U. Reischl, "Quantitative real-time PCR is not more sensitive than 'conventional' PCR," *J. Clin. Microbiol.*, vol. 46, no. 6, pp. 1897–1900, Jun. 2008.
[278] *Lsuhsc.edu*. Accessed: Aug. 10, 2023. [Online]. Available: https://nursing.lsuhsc.edu/JBI/docs/JBIBooks/Diagnostic%20Accuracy.pdf
[279] J. M. Crosslin, G. J. Vandemark, and J. E. Munyaneza, "Development of a real-time, quantitative PCR for detection of the Columbia basin potato purple top phytoplasma in plants and beet leafhoppers," *Plant Disease*, vol. 90, no. 5, pp. 663–667, May 2006.
[280] C. J. Smith and A. M. Osborn, "Advantages and limitations of quantitative PCR (Q-PCR)-based approaches in microbial ecology," *FEMS Microbiol. Ecol.*, vol. 67, no. 1, pp. 6–20, Jan. 2009.
[281] T. Notomi, "Loop-mediated isothermal amplification of DNA," *Nucleic Acids Res.*, vol. 28, no. 12, p. 63, Jun. 2000.
[282] E. S. M. Tuppurainen, E. H. Venter, J. L. Shisler, G. Gari, G. A. Mekonnen, N. Juleff, N. A. Lyons, K. De Clercq, C. Upton, T. R. Bowden, S. Babiuk, and L. A. Babiuk, "Review: Capripoxvirus diseases: Current status and opportunities for control," *Transboundary Emerg. Diseases*, vol. 64, no. 3, pp. 729–745, Jun. 2017.
[283] P. Hardinge and J. A. H. Murray, "Reduced false positives and improved reporting of loop-mediated isothermal amplification using quenched fluorescent primers," *Sci. Rep.*, vol. 9, no. 1, pp. 1–13, May 2019.
[284] S. Waliullah, K.-S. Ling, E. J. Cieniewicz, J. E. Oliver, P. Ji, and M. E. Ali, "Development of loop-mediated isothermal amplification assay for rapid detection of cucurbit leaf crumple virus," *Int. J. Mol. Sci.*, vol. 21, no. 5, p. 1756, Mar. 2020.
[285] A. Rani, N. Donovan, and N. Mantri, "Review: The future of plant pathogen diagnostics in a nursery production system," *Biosensors Bioelectron.*, vol. 145, Dec. 2019, Art. no. 111631.
[286] M. M. F. Azizi, N. H. Mardhiah, and H. Y. Lau, "Current and emerging molecular technologies for the diagnosis of plant diseases—An overview," *J. Experim. Biol. Agricult. Sci.*, vol. 10, no. 2, pp. 294–305, Apr. 2022.
[287] L. C. Clark and C. Lyons, "Electrode systems for continuous monitoring in cardiovascular surgery," *Ann. New York Acad. Sci.*, vol. 102, no. 1, pp. 29–45, Oct. 1962.
[288] F. Sanger, S. Nicklen, and A. R. Coulson, "DNA sequencing with chain-terminating inhibitors," *Proc. Nat. Acad. Sci. USA*, vol. 74, no. 12, pp. 5463–5467, Dec. 1977.
[289] L. Liu, Y. Li, S. Li, N. Hu, Y. He, R. Pong, D. Lin, L. Lu, and M. Law, "Comparison of next-generation sequencing systems," *J. Biomed. Biotechnol.*, vol. 2012, pp. 1–11, Oct. 2012.
[290] A. M. Kanzi, J. E. San, B. Chimukangara, E. Wilkinson, M. Fish, V. Ramsuran, and T. de Oliveira, "Next generation sequencing and bioinformatics analysis of family genetic inheritance," *Frontiers Genet.*, vol. 11, Oct. 2020, Art. no. 544162.
[291] J. Qin, R. Li, J. Raes, M. Arumugam, K. S. Burgdorf, C. Manichanh, T. Nielsen, N. Pons, F. Levenez, T. Yamada, and D. R. Mende, "A human gut microbial gene catalogue established by metagenomic sequencing," *Nature*, vol. 464, no. 7285, pp. 59–65, 2010.
[292] S. Darmanis, R. Y. Nong, J. Vänelid, A. Siegbahn, O. Ericsson, S. Fredriksson, C. Bäcklin, M. Gut, S. Heath, I. G. Gut, L. Wallentin, M. G. Gustafsson, M. Kamali-Moghaddam, and U. Landegren, "ProteinSeq: High-performance proteomic analyses by proximity ligation and next generation sequencing," *PLoS ONE*, vol. 6, no. 9, Sep. 2011, Art. no. e25583.
[293] T. M. Walker, C. L. Ip, R. H. Harrell, J. T. Evans, G. Kapatai, M. J. Dedicoat, D. W. Eyre, D. J. Wilson, P. M. Hawkey, D. W. Crook, J. Parkhill, D. Harris, A. S. Walker, R. Bowden, P. Monk, E. G. Smith, and T. E. Peto, "Whole-genome sequencing to delineate mycobacterium tuberculosis outbreaks: A retrospective observational study," *Lancet Infectious Diseases*, vol. 13, no. 2, pp. 137–146, Feb. 2013.
[294] G. A. Díaz-Cruz, C. M. Smith, K. F. Wiebe, S. M. Villanueva, A. R. Klonowski, and B. J. Cassone, "Applications of next-generation sequencing for large-scale pathogen diagnoses in soybean," *Plant Disease*, vol. 103, no. 6, pp. 1075–1083, Jun. 2019.
[295] B. E. Slatko, A. F. Gardner, and F. M. Ausubel, "Overview of next-generation sequencing technologies," *Curr. Protoc. Mol. Biol.*, vol. 122, no. 1, p. e59, 2018.
[296] T. H. Meen, "IoT, communication, and engineering," in *Proc. IEEE Eurasia Conf. IoT, Commun., Eng.*, vol. 3, Yunlin, Taiwan, Oct. 2019.
[297] S. M. M. Hossain, K. Deb, P. K. Dhar, and T. Koshiba, "Plant leaf disease recognition using depth-wise separable convolution-based models," *Symmetry*, vol. 13, no. 3, pp. 511–529, Mar. 2021.

ANUJA BHARGAVA received the Ph.D. degree in electronics and communication from GLA University, Mathura, Uttar Pradesh, in 2020. She is currently an Assistant Professor with the Department of Electronics and Communication, Institute of Engineering and Technology, GLA University. She has more than 14 years of teaching experience. Her current research interests include image processing, machine learning, and computer vision. She has more than 25 publications in international journals and conference of repute.

AASHEESH SHUKLA received the Ph.D. degree in electronics and communication from GLA University, Mathura, Uttar Pradesh, in 2017. He is currently a Professor with the Department of Electronics and Communication, GLA University of Technology and Management. He has more than 15 years of teaching experience. His current research interests include wireless communication, machine learning, and image processing. He has more than 30 publications in international journals and conference of repute.

OM PRAKASH GOSWAMI holds the Ph.D. degree in signal processing from the Netaji Subhas Institute of Technology, University of Delhi, India. He is working as an Assistant Professor with Somaiya Vidyavihar University, Mumbai. His research interests include digital signal processing and fractional-filters. He has co-authored several reputed research papers.

MOHAMMED H. ALSHARIF received the B.Eng. degree in electrical engineering (wireless communication and networking) from the Islamic University of Gaza, Palestine, in 2008, and the M.Sc.Eng. and Ph.D. degrees in electrical engineering (wireless communication and networking) from the National University of Malaysia, Malaysia, in 2012 and 2015, respectively. In 2016, he joined the Faculty of Sejong University, South Korea, as an Assistant Professor with the Department of Electrical Engineering, where he has since been promoted to an Associate Professor. His current research interests include wireless communications and networks, including cutting-edge areas, such as wireless communications, network information theory, the Internet of Things (IoT), green communication, energy-efficient wireless transmission techniques, wireless power transfer, and wireless energy harvesting. His research achievements are reflected in his extensive publication record in top-tier journals in electrical and electronics/communications engineering. His expertise and contributions have also been recognized by leading publishers, such as IEEE, Elsevier, Springer, and MDPI, who has invited him to serve as a guest editor for many special issues.

PEERAPONG UTHANSAKUL (Member, IEEE) received the B.Eng. and M.Eng. degrees in electrical engineering from Chulalongkorn University, Thailand, in 1996 and 1998, respectively, and the Ph.D. degree in information technology and electrical engineering from The University of Queensland, Australia, in 2007. From 1998 to 2001, he was a Telecommunication Engineer in one of the leading telecommunication company of Thailand. He has been an Associate Professor and the Director of the Research and Development Institute, Suranaree University of Technology, Thailand. He has got more than 300 research publications and he is the author/coauthor of various books related to MIMO technologies. He has won various national awards from the government of Thailand due to his contributions and motivation in the field of Science and Technology. His research interests include green communications, wave propagation modeling, wireless communications, MIMO, massive MIMO, brain wave engineering, mobile networks, and the quality of experience (QoE).

MONTHIPPA UTHANSAKUL (Member, IEEE) received the B.E. degree in telecommunication engineering from the Suranaree University of Technology, Thailand, in 1997, the M.E. degree in electrical engineering from Chulalongkorn University, Thailand, in 1999, and the Ph.D. degree in information technology and electrical engineering from The University of Queensland, Australia, in 2007. She is currently an Associate Professor with the Telecommunication School, Suranaree University of Technology. Her research interests include wideband/narrowband smart antennas, automatic switch beam antenna, DOA finder, microwave components, the application of smart antenna, and advance wireless communications. She received the Second Prize of Young Scientist Award from 16th International Conference on Microwaves, Radar, and Wireless Communications, Poland, in 2006.

IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 14 December 2023, accepted 19 January 2024, date of publication 25 January 2024, date of current version 2 February 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3358333

RESEARCH ARTICLE

Real-Time Plant Disease Dataset Development and Detection of Plant Disease Using Deep Learning

DIANA SUSAN JOSEPH[1], (Student Member, IEEE), PRANAV M. PAWAR[1], (Member, IEEE), AND KAUSTUBH CHAKRADEO[2]

[1]Birla Institute of Technology and Science Pilani, Dubai Campus, Dubai, United Arab Emirates
[2]Department of Public Health, University of Copenhagen, 1356 Copenhagen, Denmark

Corresponding author: Pranav M. Pawar (pranav@dubai.bits-pilani.ac.in)

ABSTRACT Agriculture plays a significant role in meeting food needs and providing food security for the increasingly growing global population, which has increased by 0.88% since 2022. Plant diseases can reduce food production and affect food security. Worldwide crop loss due to plant disease is estimated to be around 14.1%. The lack of proper identification of plant disease at the early stages of infection can result in inappropriate disease control measures. Therefore, the automatic identification and diagnosis of plant diseases are highly recommended. Lack of availability of large amounts of data that are not processed to a large extent is one of the main challenges in plant disease diagnosis. In the current manuscript, we developed datasets for food grains specifically for rice, wheat, and maize to address the identified challenges. The developed datasets consider the common diseases (two bacterial diseases and two fungal diseases of rice, four fungal diseases of maize, and four fungal diseases of wheat) that affect crop yields and cause damage to the whole plant. The datasets developed were applied to eight fine-tuned deep learning models with the same training hyperparameters. The experimental results based on eight fine-tuned deep learning models show that, while recognizing maize leaf diseases, the models Xception and MobileNet performed best with a testing accuracy of 0.9580 and 0.9464 respectively. Similarly, while recognizing the wheat leaf diseases, the models MobileNetV2 and MobileNet performed best with a testing accuracy of 0.9632 and 0.9628 respectively. The Xception and Inception V3 models performed best, with a testing accuracy of 0.9728 and 0.9620, respectively, for recognizing rice leaf diseases. The research also proposes a new convolutional neural network (CNN) model trained from scratch on all three food grain datasets developed. The proposed model performs well and shows a testing accuracy of 0.9704, 0.9706, and 0.9808 respectively on the maize, rice, and wheat datasets.

INDEX TERMS Deep learning, convolutional neural network, dataset development, plant disease classification, transfer learning.

I. INTRODUCTION

Plant diseases are among the leading causes of crop loss in many countries. Traditional disease analysis methods involve visual assessment by experts [1], [2]. However, it takes longer to diagnose the disease compared to automation techniques. Experts may only be available in some countries [3]. So, automatic recognition of plant diseases by image analysis is essential to solving this problem. Estimating the disease's severity is equally important to predict yield and suggest treatment [4].

Rice, wheat, and maize are important food grains with many health benefits. They are grown in large quantities and are energy giving food to the world. Food grains are an inevitable source of food for the growing population in many countries. These can be affected by a variety of

The associate editor coordinating the review of this manuscript and approving it for publication was Miaohui Wang.

diseases and pests. The diseases affecting them can further prevent the growth of the crop or affect the yield. Among the different food grains, the diseases affecting the rice, wheat, and maize/corn plants are given importance in this manuscript. Some common diseases affecting rice plants are Rice Blast, Sheath Blight, Bacterial Leaf Streak, Bacterial Leaf Blight, etc. [5]. The wheat plants' diseases include Powdery Mildew, Septoria Leaf Spot, Tanspot, Snowmold, etc. [6]. Charcoal rot, Downey mildew, Commonrust, black bundle disease, etc., are some diseases that commonly affect maize plants.

CNN [7], [8], [9] is a popular method that has been widely used in the field of plant disease detection [10], [11], [12]. The dataset used by the CNN models should be of good quality, and it has to be properly labeled. The performance of the CNN models greatly depends on the quality and amount of data used to train the model. The deep learning based CNN models require a large amount of data for training. It will not perform well if the model is trained using less data.

A dataset is an important component in machine learning or deep learning to learn and evaluate the performance of any algorithm or model. A dataset helps to organize the information collected from different sources and to arrive at the desired outcome. Only a well-organized dataset used for training a machine learning or deep learning model will be able to reflect the quality of the model and its effectiveness in arriving at the target outcome. One of the main challenges in plant disease diagnosis is the lack of availability of large and non-lab datasets. Some of the disadvantages of the existing datasets are most of the datasets available are laboratory processed; they include mostly images taken under controlled settings, a single leaf is mostly considered, they don't include all the symptoms of a particular disease, and complex backgrounds with varying lighting conditions are not considered. These disadvantages can limit the performance of disease detection by a CNN model, as the images can have multiple leaves with different complex backgrounds.

Our work focussed on these challenges, and three datasets for three important food grains, rice, wheat, and maize, were developed. The images in the datasets included all the disease stages early, advancing, and severe of all the diseases considered. The datasets includes processed images, images under complex backgrounds and lighting conditions, and images with multiple leaves under complex backgrounds. The images in the datasets were collected from internet sources and existing datasets. In the current work, the detection and classification of the diseases considered in the manuscript affecting rice, wheat, and maize were made using CNN models. Pre-trained eight CNN models trained on the ImageNet dataset were fine-tuned, trained, and tested on the datasets developed. A CNN model, namely Maize, Rice, Wheat-CNN model (MRW-CNN), is also proposed in our work to evaluate the performance of the developed datasets. Compared to the existing heavy state-of-the-art models for plant disease diagnosis, the MRW-CNN model has less number of layers, and the experimental results show that it generalizes well on the datasets. As a future enhancement, the model can be experimented with for disease diagnosis of other plant leaf datasets.

The main contributions of the current research work are as follows:

1) A real-time dataset for three important food grains, rice, wheat, and maize, was developed. Four diseases affecting each crop and a healthy class are present in all three datasets.
2) Early, advancing, and severe stages of the diseases were focused on while developing the datasets. Then, all three stages of a particular disease were put together in one class.
3) A hundred images were collected under each class for all five classes in a dataset. The collected images are pre-processed if needed and augmented to increase the size of the dataset.
4) A hundred images under each class of the datasets collected were annotated manually using the MakeSense AI tool for use in object detection models like Mask R-CNN.
5) Eight CNN models were fine-tuned, trained, and tested on the datasets developed. The performance of the fine-tuned CNN models and the proposed CNN model on the developed real-time datasets was evaluated using the accuracy and loss performance graphs, confusion matrices, classification reports, etc., on the test data.
6) A CNN model, namely MRW-CNN, is proposed for detecting maize, rice, and wheat leaf diseases. The proposed model can identify the features of the diseased and healthy maize, rice, and wheat leaves and accurately detect the diseases.
7) As a future extension of the current work, the different stages of the diseases included under a particular disease can be separated into classes like early, advancing, and severe, and predictions on disease stages can be made.

In the current work during the recognition of the maize leaf diseases, the models MobileNet, MobileNetV2, Inception V3, InceptionResNetV2, and Xception performed well with a testing accuracy of above 90%. For rice leaf disease recognition, all the models except ResNet 50 and ResNet 101 gave a testing accuracy above 90%. The models MobileNet, MobileNetV2, Inception V3, and InceptionResNetV2 gave a testing accuracy of above 90% on the wheat leaf dataset. The experimental results demonstrate that when trained from scratch on all the datasets developed, the proposed model gave a testing accuracy above 95%. The proposed model proved to generalize well on all three datasets considered in our work.

The rest of the paper is organized into seven sections. Section II deals with the related works in plant disease diagnosis. It focuses on the literature review of the works concentrating on dataset development for different plants and CNN models used for plant disease diagnosis. Section III deals with the steps in data set development, like data collection, pre-processing, augmentation, and annotation. Section IV discusses the development of the fine-tuned models, like how the CNN models selected are fine-tuned and trained, the basic architecture used, etc. Section V deals with the development of the proposed MRW model. Section VI deals with the results of the experiments conducted on the CNN models with the datasets developed for each food grain, experimental setup, hyperparameters used, visualization of activations, etc. Section VII discusses the case study of the application of the proposed method. Finally, Section VIII deals with the conclusion and future work.

II. RELATED WORKS

The review conducted in [13] mainly focuses on current developments in plant disease diagnosis using techniques like machine learning and deep learning and how these techniques speed up the process of disease detection in plants. The study elaborates on the importance of the detection of pests and diseases in plants, the various datasets used in plant disease detection, performance metrics used to evaluate the plant disease detection models, the challenges in the field, and future enhancements.

A 3DCNN is built to identify plant diseases in [14]; the work focussed on lesion segmentation and plant survival probability. The proposed model was evaluated on the diseases affecting three plants, pepper, potato, and tomato, from the Plant Village dataset. The model achieved a DICE coefficient of 90% and classification accuracies of 91.11%, 93.01%, and 99.04% for pepper, potato, and tomato plants, respectively.

A lightweight CNN model was developed in [15] based on VGG19 for disease diagnosis in peach leaves. Bacteriosis infection is mainly focussed on in the manuscript. The dataset for conducting the experiments was obtained from Farm of Agriculture University Peshawar, Pakistan. A total of 10,000 images were present in the dataset, and augmentation techniques were used to increase its size. The images in the dataset were also pre-processed.

The plant village dataset [16] contains 54,303 healthy and unhealthy images of leaves. The images in the dataset are laboratory processed. The plant village dataset is an open-access repository of images that can help to detect and classify plant diseases. The dataset is divided into 38 categories by species and diseases. Many studies in the field of plant disease diagnosis have made use of the plant village dataset [17], [18], [19], [20].

Temperature and carbon dioxide effect on maydis leaf blight disease affecting maize was analyzed in [21]. The results of the study showed that the severity of the disease can increase with the increase in temperature. Tolerance attributes were calculated for six disease stress based on diseased and non-diseased according to the yield. Wheat disease identification model based on EfficientNet architecture was proposed in [22] for detecting wheat rusts. A dataset consisting of 6556 images was prepared from natural field conditions. Several experiments were conducted using the CNN models VGG19, ResNet152, DenseNet169, Inception-NetV3, MobileNetV2, and eight variants of EfficientNet architecture; the results showed the fine-tuned EfficientNet B4 model proposed in the study performed better with a testing accuracy of 99.35% when compared to the other models.

Leaf tip detection and the number of leaves in a rice plant were determined by training a You Only Look Once (YOLO) deep learning algorithm in [23]. The study in [24] proposed a model that uses machine learning and deep learning advantages. The proposed model includes forty different hybrid deep learning models and five machine-learning techniques. The hyperparameters of the classifiers were optimized using the Optuna framework. A dataset for tomato early blight disease was collected from the Indian Agricultural Research Institute. The proposed model performed well on the dataset (IARI-TomEBD) formed and gave an accuracy of 87.55–100%. Validation of the model was also performed on PlantVillage-TomEBD and PlantVillage-BBLS.

A deep learning approach was proposed in [25] to identify diseases affecting maize crops. A dataset was prepared from fields of ICAR-IIMR, Ludhiana, India. Three different architectures based on the Inception-v3 framework were trained on the dataset prepared. The best performing model obtained an accuracy of 95.99%. A comparison of the state-of-the-art models with the model that performed best was done.

Analysis of Rice Blast disease by deep learning based model is done in [26] using remote sensing images. The proposed model obtained a training accuracy of 90.02% and a validation accuracy of 85.33%. Moderate resolution imaging spectroradiometer is used to establish the spatial distribution of leaf blasts. A deep learning based approach was used in [27] to detect tomato plant diseases. Two state-of-the-art semantic segmentation models, U-Net and Modified U-Net, were used to detect and segment the disease affected regions. Different U-Net versions were looked into for selecting the optimum one. Different classification methods were used in the study; both binary and multiclass classification of tomato diseases were done.

A dataset was developed by collecting 500 images of maize leaves from Google websites and the plant village dataset in [28]. The different periods of occurrence of maize diseases are included in the dataset. The size of the dataset was increased to 3060 images by applying augmentation techniques. There were eight categories of diseased leaves in the dataset. The study proposed two improved GoogLeNet and Cifar10 models for plant disease recognition. Eight kinds of maize diseases were identified.

A dataset of maize leaves was developed with the help of an Unmanned Aerial Vehicle (UAV) in [29]. The dataset contains 6267 images, where 3741 images are with lesions and 2526 are without lesions. The maize leaf images in the dataset are field images of the disease Northern Leaf Blight.

A CNN model ResNet-34 pre-trained on the ImageNet dataset was used in the study for training. The CNN model trained was used to generate heat maps that showed the disease regions.

The NLB dataset is used in [30]. It has three different parts. The first part is hand-held. The second part is the boom set. The last data set is an unmanned drone set. The hand-held part has more clarity; thus, the hand-held part of the dataset is used in this study. The dataset contains 1019 images with different angles and backgrounds and 7669 annotations. Augmentation techniques were applied to the dataset, and the size of the dataset was increased to 8152 images. A multi-scale feature fusion instance detection method was proposed, based on CNN (ResNet-101), to recognize the disease maize leaf blight.

A dataset of maize leaves was formed in [31] by collecting some images from the corn plantations in Raebareli and Sultanpur district, the rest from the Plant Village dataset. The dataset had three categories of images: Rust, Northern Leaf Blight, and Healthy. An Agriculture Scientist from India was consulted to label the images captured. A model was formed based on DCNN to identify two diseases affecting the maize plants.

The citrus dataset developed in [32] contains 759 images of leaves and citrus fruits, which are healthy and diseased. The diseases considered in the dataset are black spot, canker, scab, greening, and melanose affecting the citrus plants. The images in the dataset were manually captured using a DSLR with the help of an expert.

A real-time dataset of wheat diseases was developed in [33]. The dataset peers into most challenges for wheat disease diagnosis, specifically complex backgrounds, different capture conditions, etc. A model using deep learning and multiple instances learning for the real-time detection of wheat diseases was proposed in the study.

A dataset of wheat leaves was developed in [34] to detect the Fusarium Head Blight disease (FHB). The images in the dataset were collected from the field. Twelve images were collected initially, and later, it was increased to 2829 by applying data augmentation techniques. In this study, a method was proposed that uses Mask-RCNN along with color images to detect FHB.

The rice diseases image database was created with 500 common rice disease images of leaves and stems collected from the experimental field in [35]. This work trains the CNNs to identify ten diseases affecting rice plants. A real-time dataset of rice leaves was developed in [36]. The dataset contains a total of 8911 images. The rice leaf images included in the database contain both diseased and healthy images. The diseases considered are Rice blast, Red blight, Stripe blight, and Sheath blight. CNN was used for feature extraction, and d support vector machine (SVM) was used for predicting diseases.

PlantDoc, a Visual Plant Disease Detection dataset, was developed in [37]. The dataset contains 2,598 data points across 13 plant species. It includes 17 classes of diseases. The images were annotated using 300 human hours. The images in the dataset were downloaded from Google Images and Ecosia. Three models were used in the study to show the efficacy of the proposed dataset. An increase in classification accuracy by 31% was reported in their work.

From the literature survey conducted, it can be concluded that most of the datasets developed do not cover all of the underlying challenges existing in the field of plant disease diagnosis, like complex backgrounds, the variations of disease symptoms according to climate, the occurrence of multiple diseases on plants, etc. Only a few datasets are publicly available for further research or study. Some concentrate on the stages of the different diseases, but mainly, it is limited to one or two plants. Therefore, there is a need to develop a dataset that will address most of the underlying challenges in the field. The current manuscript mainly concentrates on developing a dataset of good quality for food grains rice, wheat, and maize. The dataset's primary focus is on real-life leaf images of the food grains considered. For example, Common Rust affecting the maize plants; for this particular disease, real-life images of all the stages are considered early, middle, and advancing and are included under CommonRust disease class, and the same for all diseases in the dataset developed.

III. PLANT DISEASE DATASET DEVELOPMENT

A real-time plant disease dataset for three important food grains, rice, wheat, and maize, based on leaf images, was developed in the current study. Diseases that cause great yield loss and have similar symptoms, many samples available, etc., were selected to form the dataset. Four diseases were selected for each food grain. For the diseases considered under each food grain, while collecting the images from internet sources, the leaf images for different stages of a particular disease, specifically early, advancing, and severe, were collected and put under the same disease class. More details on the dataset developed and the samples collected are available in Table 1, Table 2, and Table 3. In the current work, the diseased and healthy leaf images collected for each food grain are pre-processed, augmented, and then applied to deep learning based CNN models for detecting and classifying diseases. The results obtained are discussed in Section VI. Later, for the future extension of the work to be done, the images were manually annotated after the data collection and pre-processing steps for use in object detection models. Around 1500 images were annotated for all three food grains.

The overall process involved in developing the plant disease dataset for rice, wheat, and maize food grains in the current work is shown in Fig. 1. In Fig. 1, the different processes involved in developing the datasets are shown. The figure mainly shows the food grains and their stages selected in our work, the sources from which the datasets' images are collected, and the pre-processing and augmentation techniques used for processing the collected images. The different steps in developing the datasets are detailed in Subsections A, B, C, and D.

TABLE 1. Details of maize leaf disease dataset.

Class	Plant	Disease	Caused by	Type of disease	No of images	Areas affected	Occurrence
C1	Maize	Common Rust	Puccinia sorghi	Fungus	Total: 100 Early: 30 Advancing: 29 Severe: 41	Upper and lower leaf surfaces	Late May-Early July
C2		GreyLeafSpot	Cercospora zeae-maydis, Cercospora zeina.	Fungus	Total: 100 Early: 10 Advancing: 46 Severe: 44	Lower leaves are primarily affected	Every growing season
C3		Healthy	-	-	Total: 100	-	-
C4		Northern Leaf Blight	Exserohilum turcicum	Fungus	Total: 100 Early: 3 Advancing: 57 Severe: 40	Lower leaves are primarily affected	Relatively cool and wet seasons
C5		Southern Rust	Puccinia polysora	Fungus	Total: 100 Early: 17 Advancing: 41 Severe: 42	Primary infection on upper leaves	High humidity and temperatures above 80 F

TABLE 2. Details of rice leaf disease dataset.

Class	Plant	Disease	Caused by	Type of disease	No of images	Areas affected	Occurrence
C1	Rice	Rice Bacterial Leaf Streak	Xanthomonas oryzae pv. oryzicola	Bacteria	Total: 100 Early: 21 Advancing: 52 Severe: 27	Leaves	High temperature and high humidity
C2		Rice Bacterial Blight	Xanthomonas oryzae pv. oryzae.	Bacteria	Total: 100 Early: 8 Advancing: 51 Severe: 41	Leaves	Temperatures (25-34°C) relative humidity above 70%.
C3		Rice Blast	Magnaporthe oryzae	Fungus	Total: 100 Early: 21 Advancing: 39 Severe: 40	Leaves	Cooler temperatures
C4		Rice Brown Spot	Bipolarisoryzae	Fungus	Total: 100 Early: 4 Advancing: 65 Severe: 31	Leaves	High relative humidity (86-100%) and temperature between 16 and 36°C.
C5		Healthy	-	-	Total: 100	-	-

TABLE 3. Details of wheat leaf disease dataset.

Class	Plant	Disease	Caused by	Type of disease	No of images	Areas affected	Occurrence
C1	Wheat	LeafRust	Puccinia triticina	Fungus	Total: 100 Early: 10 Advancing: 36 Severe: 54	Primary infection on lower leaves	Moderate nights and warm days
C2		Powderymildew	Blumeria graminis f. sp. tritici	Fungus	Total:100 Early: 6 Advancing: 45 Severe: 49	Most prevalent on lower leaves	High humidity and temperatures (15-22C)
C3		StripeRust	Puccinia striiformis	Fungus	Total:100 Early: 9 Advancing: 46 Severe: 46	Leaves	Preferred by temperature (2-23C) and nonlimiting moisture
C4		Tanspot	Pyrenophora tritici-repentis	Fungus	Total:100 Early: 9 Advancing: 62 Severe: 29	Upper and lower leaves	
C5		Healthy	-	-	Total:100	-	-

A. DATA COLLECTION

The dataset includes real-time images collected from Google websites, Ecosia, Bing images, Flickr, etc. The images were manually selected and filtered. The dataset also comprises a few processed images for the healthy maize, rice, and wheat classes. Few images were taken from the plant village [16]

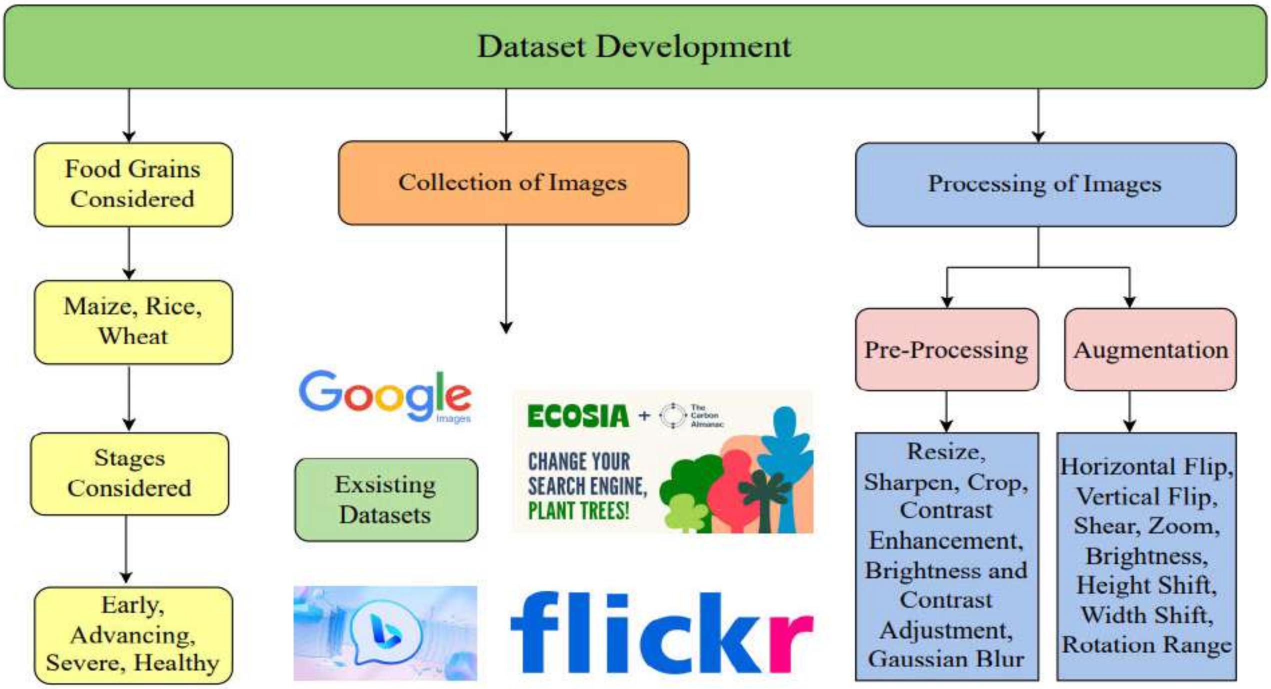


FIGURE 1. Dataset development methodology.

dataset for the healthy class of maize, and the healthy class of rice and wheat images were downloaded from the Kaggle website [38], [39]. The dataset was created by downloading images from the internet. A total of 5 classes were considered for each plant, including the healthy class, a different class from the diseased class. For all the diseased classes, images for all the stages of the disease were attempted to be collected, like images of early infection, advancing of the disease stage, and finally, the severe stage.

CNN models trained on real-life images are believed to perform better when compared to laboratory processed images. Therefore real-life images were given more importance while developing the dataset when compared to processed images. The crops considered for developing the dataset are rice, wheat, and maize. Four diseases affecting each of these plants were assessed for developing the dataset. The diseases considered for rice plants are Rice Bacterial Blight, Rice Bacterial Leaf Streak, Rice Blast, and Rice Brown Spot.

Similarly, the diseases considered for the wheat plant are Leaf Rust, Powdery Mildew, Stripe Rust, and Tan Spot. The diseases looked into for maize plants are Common Rust, Grey Leaf Spot, Northern Leaf Blight, and Southern Rust. Table 1, Table 2, and Table 3 give detailed descriptions regarding the type of disease, the number of images collected under each class and the number of images collected for early, advancing, and severe stages of the disease in each class, the areas of the leaves affected, and the conditions favorable for the disease to occur and advance; for the datasets developed for maize, rice, and wheat plants. The whole process of data collection is shown in Fig. 2. The samples of a few raw images collected for each food grain rice, wheat, and maize for the development of the datasets are shown in Fig. 3, Fig. 4, and Fig. 5.

B. DATA PRE-PROCESSING

Data pre-processing helps enhance some features of the image data required for processing. It can help in improving the accuracy and reliability of a dataset. Some images downloaded from the internet for the datasets were pre-processed before training the model. The software tools used for pre-processing the images were ImageJ [67], [68], and Python code. The purpose of pre-processing was to remove foreign objects from the images. The pre-processing steps applied to some images were resize, crop, sharpen, contrast enhancement, brightness and contrast adjustment, gaussian blur, etc. Python code was used to resize all the images in the dataset. The images were resized to 224 × 224. Some images were cropped using the ImageJ software and Python code to select the diseased areas of the leaves other than the background.

Some images in the dataset were manually selected and sharpened to visually focus on the diseased areas of the leaves more clearly. Similarly, other pre-processing techniques like contrast enhancement of the images to increase or decrease the contrast, brightness adjustments to increase or decrease the brightness of specific images, and gaussian blur to blur certain portions of the image were done using the

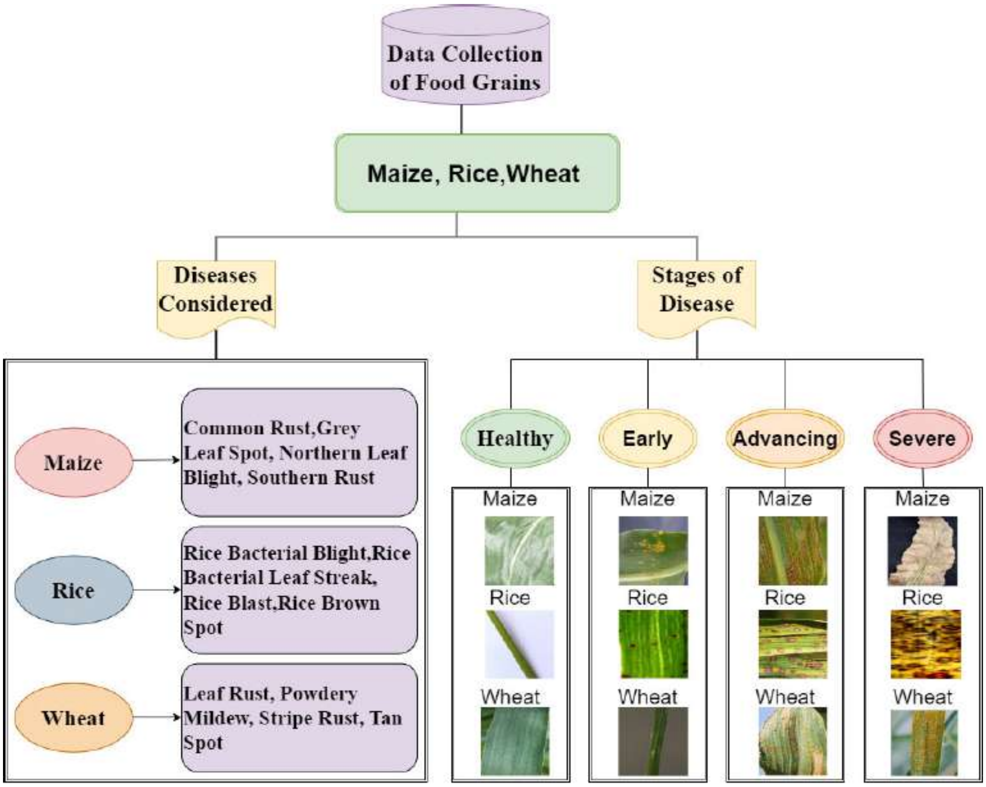
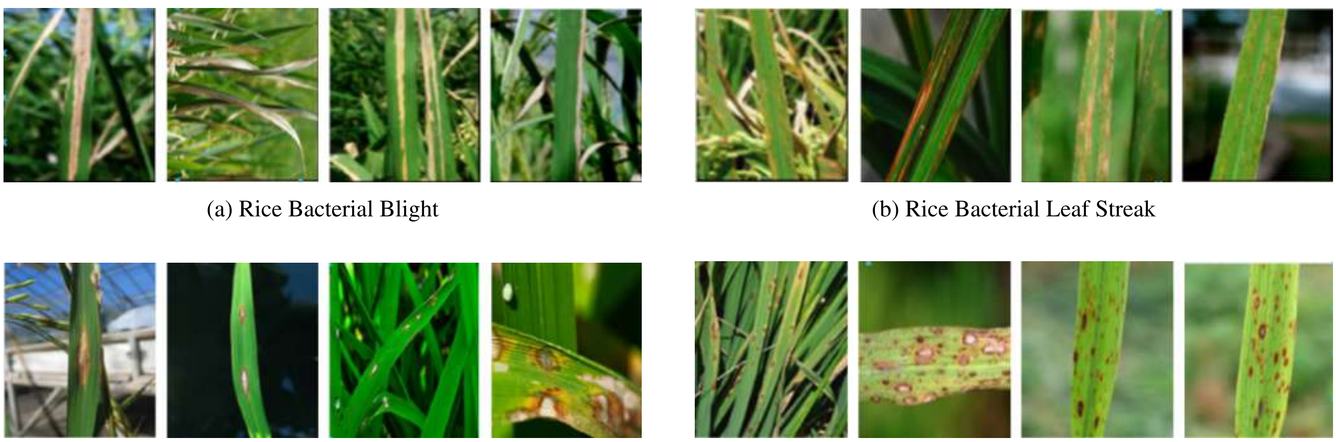


FIGURE 2. Data collection.



(a) Rice Bacterial Blight

(b) Rice Bacterial Leaf Streak

(c) Rice Blast

(d) Rice Brown Spot

FIGURE 3. Samples of a few raw diseased images used to form the rice leaf dataset [40], [41].

ImageJ software. Examples of a few pre-processed images of Common Rust disease affecting the maize plant from the datasets are shown in Fig. 6.

C. DATA AUGMENTATION

A total of hundred images were collected under each class for the plants considered in this manuscript. The number of images under each class was insufficient to train a deep learning model. Therefore, data augmentation techniques were applied to the dataset to increase the size of the dataset. Python code was used to augment all the images in the dataset. Offline augmentation was done to all the images and saved to a folder. Then, the original images were combined with the augmented images of each class to form the dataset.

The images in all three datasets collected were augmented by applying specific augmentation techniques like horizontal and vertical flipping of the images, rotating the images by 90 degrees, applying a height shift range of 0.2, applying a width shift range of 0.2, zooming and shearing of the images by a range of 0.2, a brightness range of 0.2 to 0.8 was

(a) LeafRust (b) PowderyMildew

(c) StripeRust (d) TanSpot

FIGURE 4. Samples of a few raw diseased images used to form the wheat leaf dataset [42], [43], [44], [45], [47], [48], [49], [50], [51], [52], [53].

(a) CommonRust (b) GreyLeafSpot

(c) NorthernLeafBlight (d) SouthernRust

FIGURE 5. Samples of a few raw diseased images used to form the maize leaf dataset [54], [55], [56], [57], [59], [60], [61], [62], [63], [64], [65], [66].

applied to the images. Examples of a few augmented images of Tanspot and Powdery Mildew diseases from the datasets are shown in Fig. 6. Data augmentation techniques helped increase the model's ability to generalize and reduced the problem of overfitting.

D. DATA ANNOTATION

Data annotation can help to understand the meaning of data more precisely, increasing the performance of the deep learning models. The Make Sense AI [69] Tool was used in the current work to annotate the images in the dataset under each class. The Make Sense AI is an open-source and free to use annotation tool. It runs on a web browser. The different stages of a particular disease in the dataset were given importance during labeling. The whole process of data annotation is shown in Fig. 7. In the figure, the images of Common Rust disease affecting the maize plants are loaded into the Make Sense AI Tools web browser. The labels to be assigned to the images are mentioned. Each image uploaded is selected individually, and the polygon bounding box is selected to annotate the images. After annotating all the images uploaded to the web browser, the labels can be exported in YOLO, VOC, XML, VGG, JSON, and CSV formats. Since the polygon bounding box was used for annotation, the labels were exported in VGG and JSON formats.

IV. DEVELOPMENT OF FINE-TUNED MODELS

The collected, pre-processed, and augmented datasets for each food grain are split 80% for training and 10% each for validation and testing. The datasets for each food grain had a total of around 5000 images under each class after augmentation. The size of all three datasets was increased from 1500 images to 25000 images after augmentation. After splitting the dataset into training, validation, and testing, each class contains 4000 images for training and 500 images

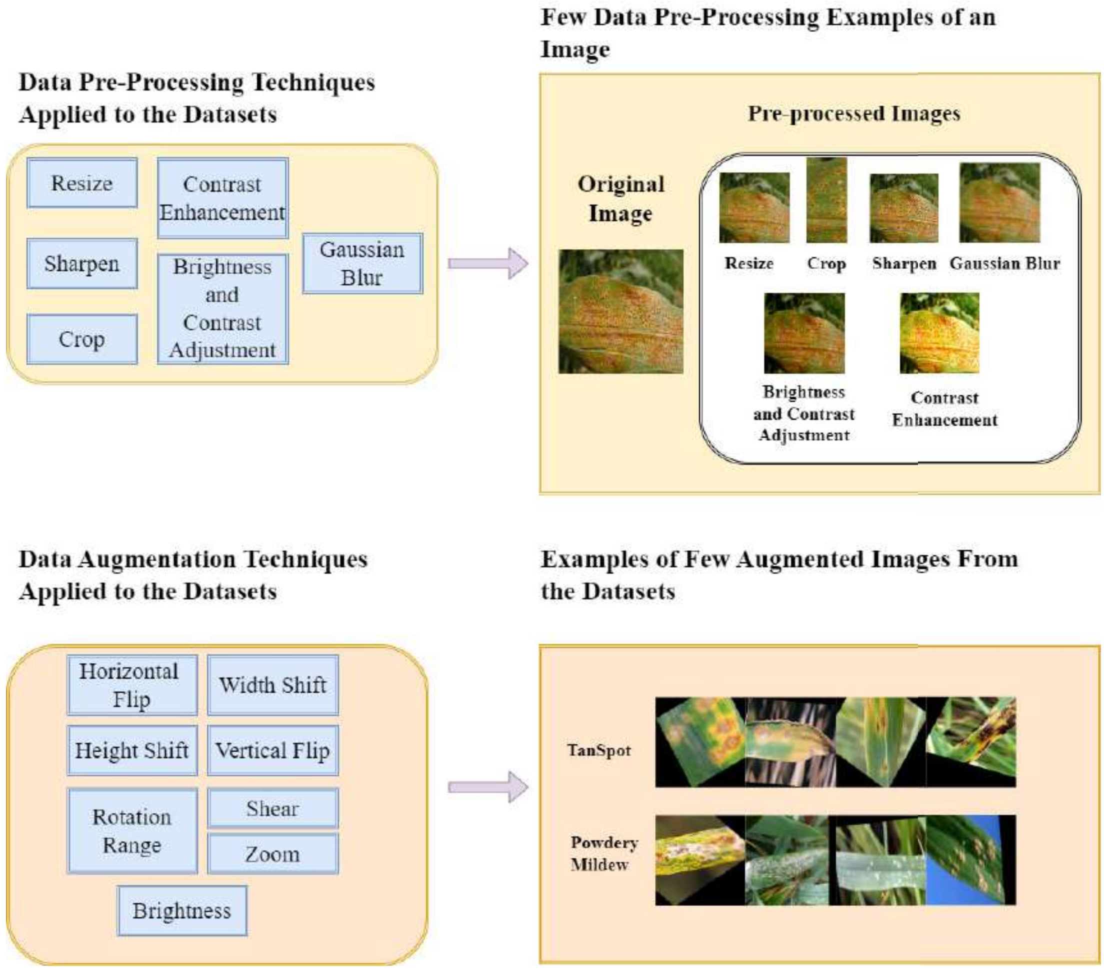


FIGURE 6. Data pre-processing and data augmentation.

Algorithm 1 Fine-Tuning of CNN Models

Input: Pre-trained models with top layers removed

Output: A fine-tuned model for detection and classification of diseases from the dataset developed

Load pre-trained models for fine-tuning

$m \leftarrow loadedmodel(pre - trained = True)$

while $m.toplayer = False$ **do**

  Set $layers.trainable = False$

  $x \leftarrow dropout(0.2)(m.output)$

  $x \leftarrow addBachNormalization(x)$ {x is output from previous layer}

  $x \leftarrow addFlattenlayer(x)$

  $x \leftarrow addDenselayer(x)$

  $l2regulizer \leftarrow 0.01$

  $x \leftarrow dropout(0.2)(x)$

  $predictions \leftarrow outputclasses(activation = softmax)$

  $finetunedmodel \leftarrow (m.input, predictions)$

end while

each for testing and validation. The number of images in all classes was balanced in all three datasets. The dataset developed for each plant was applied to various CNN based deep learning models like VGG16, ResNet 50, ResNet 101, Inception V3, InceptionResNetV2, MobileNetV2, and MobileNet. Pre-trained models trained on the ImageNet dataset [70] were downloaded from Keras and used. The concepts of transfer learning and fine-tuning were made use of for the training. Algorithm 1 discusses how the pre-trained models trained on the ImageNet dataset were fine-tuned. The steps in the algorithm include the following: the models were downloaded from Keras. The top layers from the eight models downloaded were removed. The layers trainable is set to False. After which, a few layers like Dropout, BachNormalization, Flatten, Dense layer, and the Softmax layer for the output classes were added.

A Dropout layer was added to nullify the effect of some neurons on the next layer. The BachNormalization layer works differently during training and inference. The layer can help the model to generalize well by normalizing its inputs in batches. The Bach Normalization layer can help smoothen the loss function by optimizing the model parameters and increasing the models' training speed. A Flatten layer was added to convert the inputs to one-dimensional. The Dense layer was added after the Flatten layer with a kernel-regularizer l2 of 0.01. Again, a Dropout of 0.2 was added.

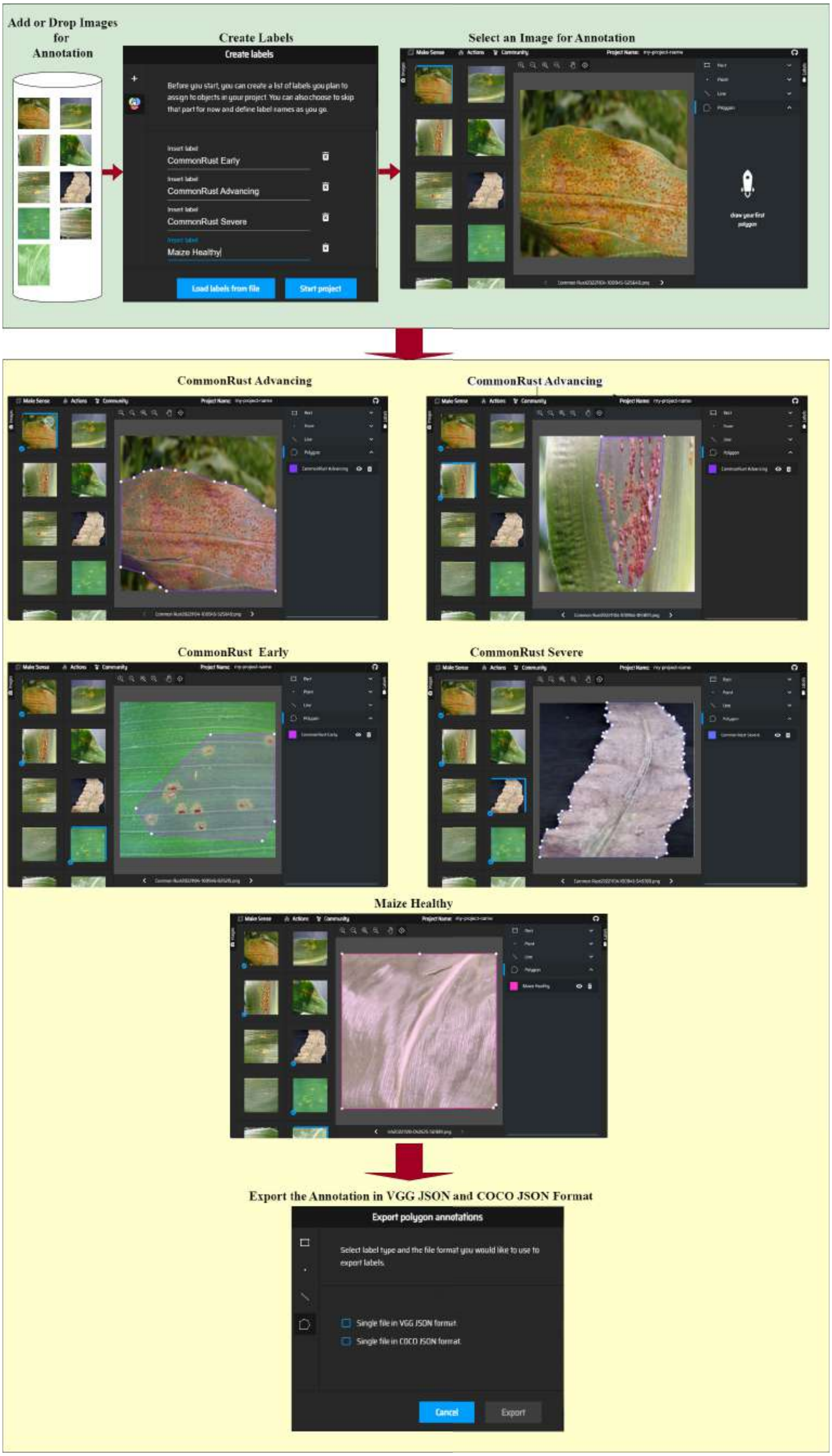


FIGURE 7. Example of annotation of commonrust disease stages and healthy stage of maize using the makesense AI tool.

The Softmax layer was added to get the output classes. Finally, a fine-tuned model was obtained where the input is the model's input, and the output is the predictions. After conducting several experiments, a Dropout of 0.2 and kernel-regularizer l2 and Bach Normalization layer, Early stopping was decided to be added to the pre-trained models to prevent over-fitting. Early Stopping automatically terminates the training when a performance measure does not improve further. The model architecture used is shown in Fig. 8.

Fig. 8 contains mainly two phases: the training and testing phases. In the training phase, the input images from either of the three datasets, which are pre-processed and augmented, are given as input to the state-of-the-art deep learning models. The datasets are split into 80% for training, 10% for validation, and 10% for testing before being given to the CNN models. The pre-trained models ResNet 50, ResNet 101, MobileNetV2, MobileNet, Inception V3, InceptionResNetV2, VGG 16, and Xception, trained on the ImageNet dataset, are selected in the current work for training on the datasets developed. The pre-trained CNN models selected are fine-tuned by removing the top layers and adding certain layers, as mentioned in Algorithm 2. The CNN models fine-tuned are trained on the training data and validated on the validation data. Examples of feature maps from a few layers of the InceptionResNetV2 model after training on the input images are also shown in Fig. 8.

In the testing phase, the trained CNN models are tested on the testing data, and predictions are made; the testing data is the data the model has not seen before. Each dataset developed for the food grains rice, wheat, and maize has five classes each. Therefore, a total of 15 output classes are shown in Fig. 8.

V. PROPOSED CNN MODEL OF MAIZE, RICE AND WHEAT (MRW-CNN) LEAF DISEASES

Fig. 9 shows the architecture of the proposed CNN model that has been used to classify the diseases affecting the food grains considered in our manuscript. The model consists of five Convolution layers, each followed by BatchNormalization and Rectified Linear Unit (ReLU). After the convolution layers, a Global Average Pooling layer is added, followed by a Dense layer with a Softmax activation function that acts as the output layer. The model is trained using Stochastic gradient descent (SGD). In the proposed CNN model, the convolution layers were used to extract the features from the training images; batch normalization is used to normalize the outputs from each layer during training for every batch. The layer helps to address the internal covariate shift problem. The activation function ReLU does not activate all the neurons at the same time; this helps to make the computation easier.

The problem of overfitting can be reduced by using the Global Average Pooling layer. The images can be down sampled using average pooling. The Dense layer is used for predicting the output class of the input image. The output of the Dense layer is used to evaluate the performance of the proposed model. The training and testing of the proposed CNN model was done by splitting the dataset developed for the three food grains into training, validation, and testing datasets. The datasets were split into 80% for training and 10% each for validation and testing. In the training phase, the input images from the training set that are pre-processed and augmented are given to the proposed model for training. The model is then validated on the validation dataset. In the testing phase, the trained MRW-CNN model is tested on the testing data which the model has not seen during training, predictions are then made. The training, validation, and testing datasets contain samples from all classes in the dataset. A total of five output classes are available for each of the datasets developed in our work.

VI. RESULTS

A. EXPERIMENT SETUP

Eight pre-trained models trained on the Imagenet dataset were downloaded from Keras and fine-tuned. GPU (NVIDIA ®Quadro ®P1000,4GB,4mDP) was used to train all fine-tuned models. Tensorflow, Keras, CUDNN, etc., were used for software implementation. The proposed MRW-CNN model was trained using the T4 GPU available in Google CoLab.

B. HYPERPARAMETERS

All the fine-tuned CNN based deep learning models are optimized by the SGD algorithm. The learning rate for all the models was set to 0.001, and a batch size 32 was chosen. Dropout operation was added to all the pre-trained models on the ImageNet dataset to prevent over-fitting and improve the generalization of the models. The MRW-CNN model is optimized by the SGD, a learning rate of 0.001 and batch size of 32 were used.

C. CONFUSION MATRIX

A confusion matrix can greatly help in visually estimating the performance of a model. The infected images of food grains considered in the current manuscript can be easily confused with multiple classes since most of the diseases selected for the development of the dataset have similar symptoms. The infected leaf images at different backgrounds can lead to confusion and complexity, which can result in performance deterioration.

The confusion matrix [71] is an m*m matrix that can be used to measure the performance of the classification models. In a confusion matrix, the X-axis contains the predicted values, and Y-axis includes the actual values. The True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) for each class in a multi-class classification can be computed from the confusion matrix, and from these, the values the precision, recall and F1-Score can be calculated per-class. The precision measures out of all predicted positives how many are actually positive. It can be

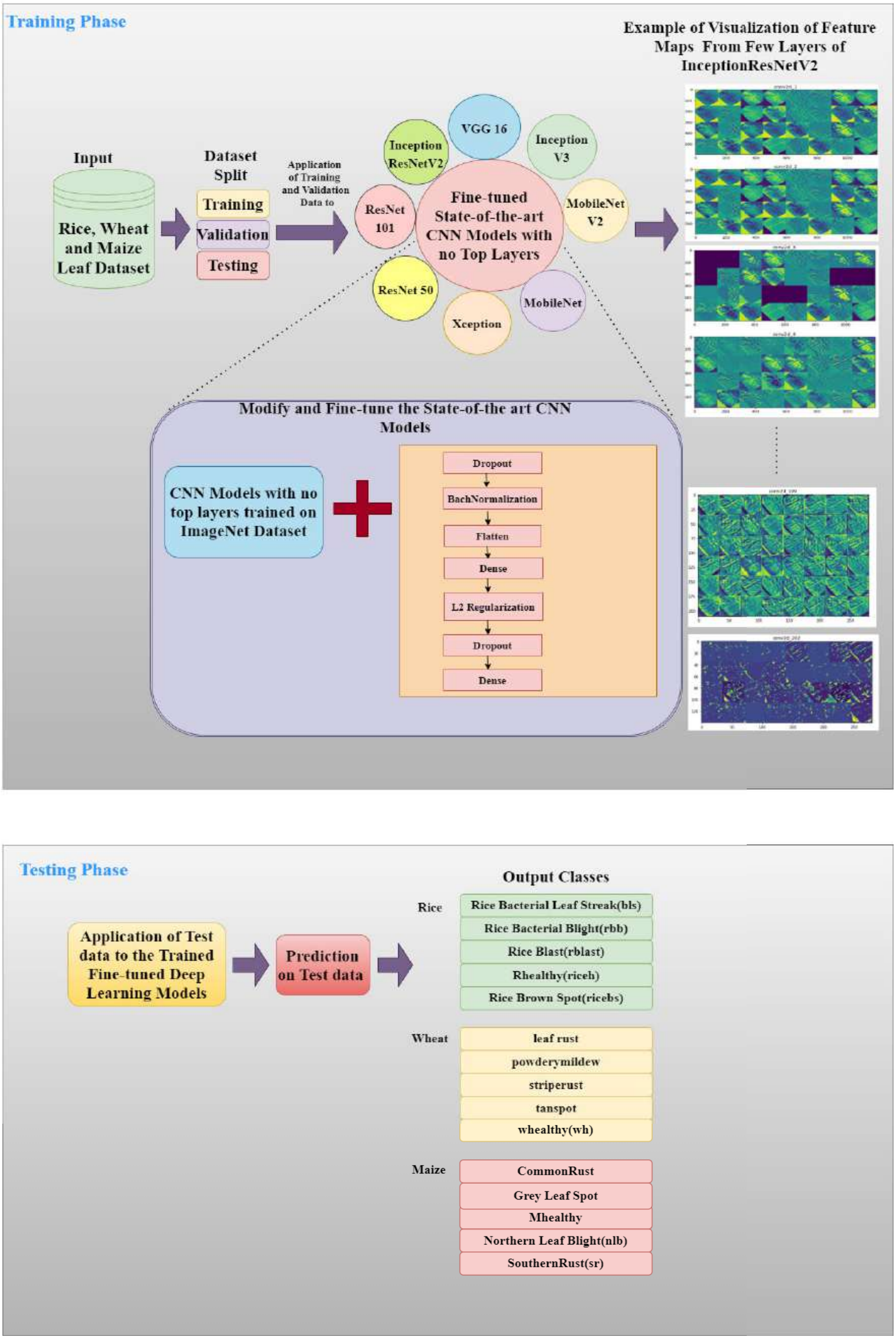


FIGURE 8. Architecture of Fine-tuned CNN Models.

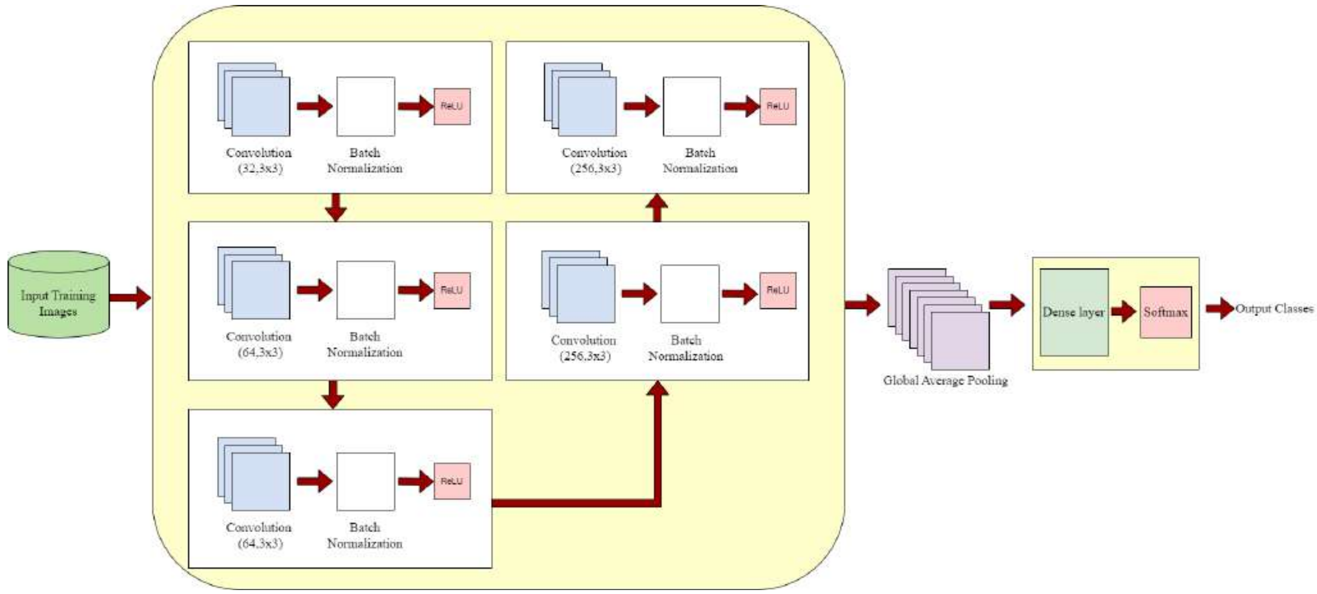


FIGURE 9. Architecture of proposed CNN model MRW-CNN.

calculated per-class using the equation (1)

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

Recall measures how many positive records are predicted correctly. It can be calculated per-class using equation (2)

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

F1-Score is the harmonic mean of precision and recall it can be calculated per-class as expressed in equation (3)

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{3}$$

D. CLASSIFICATION REPORT

The classification report is a performance evaluation metric that displays precision, recall,f1-score, accuracy, macro avg, weighted avg, and support for each class. It is used to measure the performance of the trained models on the test data. The accuracy is calculated by dividing the correct predictions by the total predictions from the confusion matrix obtained, as expressed in equation (4)

$$Accuracy = \frac{CorrectPredictions}{TotalPredictions} \tag{4}$$

Macro average recall scores are calculated by taking the arithmetic mean of all the per-class recall scores [72] as expressed in equation (5).

$$RecallMacroAvg = \sum_{i=1}^{n} \frac{Recall_i}{n} \tag{5}$$

Macro average precision scores are calculated by taking the arithmetic mean of all the per-class precision scores as expressed in equation (6)

$$PrecisionMacroAvg = \sum_{i=1}^{n} \frac{Precision_i}{n} \tag{6}$$

Macro average F1 scores are calculated by taking the arithmetic mean of all the per-class F1 scores as expressed in equation (7)

$$F1 - scoreMacroAvg = \sum_{i=1}^{n} \frac{F1 - score_i}{n} \tag{7}$$

The weighted-averaged F1-score [73] is calculated using equation (8) and equation (9) by taking the mean of all per-class F1 scores for a N-class dataset, while considering each class's support.

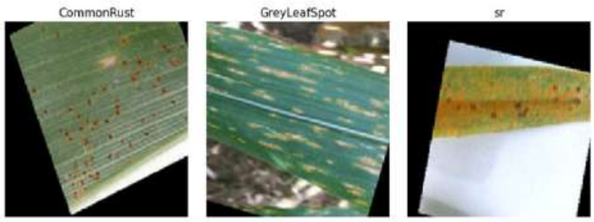
$$Weighted\ F1 - score = \sum_{i=1}^{N} W_i \times F1 - score_i \tag{8}$$

$$where,\ W_i = \frac{No\ of\ samples\ in\ each\ class_i}{Total\ number\ of\ samples} \tag{9}$$

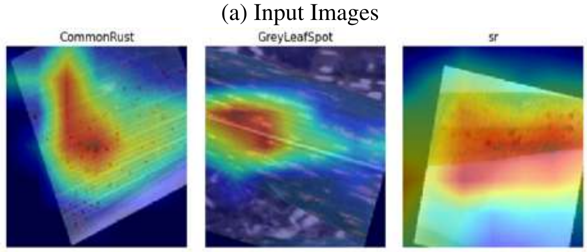
In a similar way, the weighted average scores for precision and recall can also be computed.

E. VISUALIZATION OF ACTIVATIONS

In some cases, the models during training are trained on some background or irrelevant features, which gives a satisfactory model accuracy. Such situations can be checked by generating activation maps. The activation maps can help determine if the models focus on the right features. For example, the output of the visualization of activations created by Grad-cam [74] for the MobileNet model given three input images from the maize leaf dataset is as shown in Fig.10. The figure shows that the model focuses on the suitable feature for all the



(a) Input Images



(b) Activation maps generated from last convolutional layer

FIGURE 10. Activation maps generated by Grad-CAM given three input images (a) input images (b) activation maps generated from last convolutional layer.

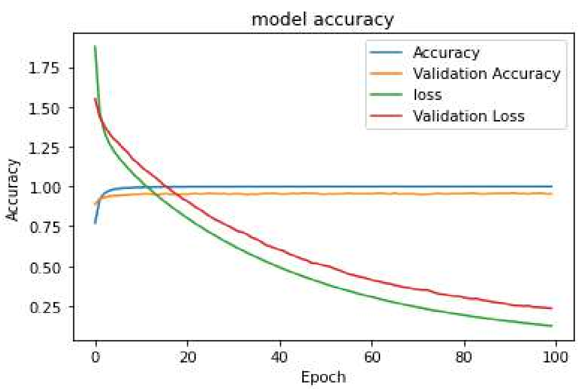


FIGURE 11. Accuracy and loss performance of the model xception on the maize leaf dataset.

input images from the maize leaf dataset, the images from three different classes CommonRust, GreyLeafSpot, and sr were given as input to the trained MobileNet model.

F. MAIZE

The performance of different models on the maize dataset is shown in Table 4. The maize dataset includes five classes, including the healthy class, for analysis. Xception, MobileNet, MobileNetV2, Inception V3, and InceptionResNetV2, models trained on the maize leaf dataset, showed better performance when compared to the other fine-tuned models. Xception showed a validation accuracy of 0.9608 and a testing accuracy of 0.9580. The InceptionV3 model showed a validation accuracy of 0.9468 and a testing accuracy of 0.9448. MobileNet showed a validation accuracy of 0.9484 and a testing accuracy of 0.9464. The MobileNetV2 model showed a validation accuracy of 0.9412 and a testing accuracy of 0.9404. The InceptionResNetV2 model showed

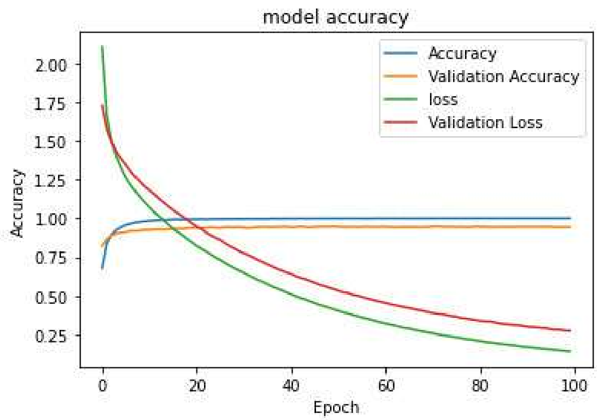


FIGURE 12. Accuracy and loss performance of the model MobileNet on the maize leaf dataset.

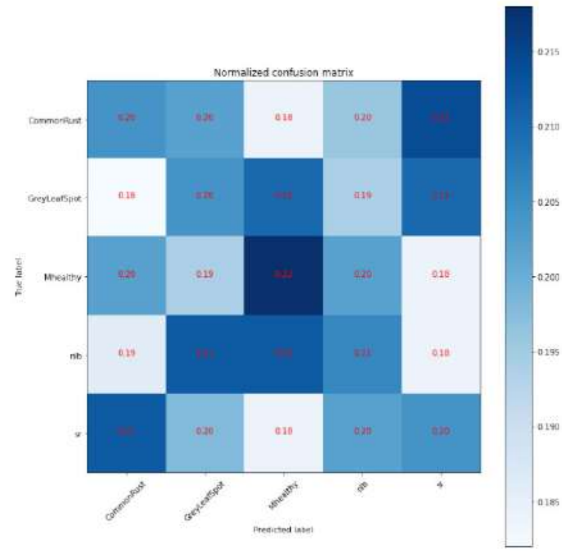


FIGURE 13. Normalized confusion matrix of the model xception on the maize leaf dataset.

a validation accuracy of 0.9336 and a testing accuracy of 0.9312. The fine-tuned Xception model performed best with a testing accuracy of 0.9580. The graphs showing the accuracy and loss performance of the models Xception and MobileNet showing better testing accuracy when compared to other fine-tuned models on the maize leaf dataset are shown in Fig. 11 and Fig. 12. The normalized confusion matrix of the models Xception and MobileNet on the maize leaf dataset is shown in Fig. 13 and Fig. 14. The performance of the models can be evaluated from the normalized confusion matrix by observing the diagonal elements. By referring to the color bar, the darker the color blue of the diagonal elements, the better the performance of the model.

In Fig. 13, for the Xception model on the maize leaf dataset, when observing the diagonal values, the Mhealthy gives a better value when compared to all the other classes. Confusion among classes can also be observed. For example,

TABLE 4. Application of state-of-the-art models on the maize leaf dataset.

Fine-tuned CNN Models	Training Accuracy	Validation Accuracy	Testing Accuracy	Optimizer	Learning Rate	No of Epochs
ResNet50	0.5709	0.5748	0.5604	SGD	0.001	100
ResNet 101	0.5939	0.5532	0.5180	SGD	0.001	100
MobileNetV2	0.9995	0.9412	0.9404	SGD	0.001	100
MobileNet	0.9991	0.9484	0.9464	SGD	0.001	100
Inception V3	0.9995	0.9468	0.9448	SGD	0.001	100
InceptionResNetV2	0.9994	0.9336	0.9312	SGD	0.001	100
Xception	0.9995	0.9608	0.9580	SGD	0.001	100
VGG16	0.9922	0.8976	0.8828	SGD	0.001	100
MRW-CNN(Proposed Model)	0.9518	0.9656	0.9704	SGD	0.001	100

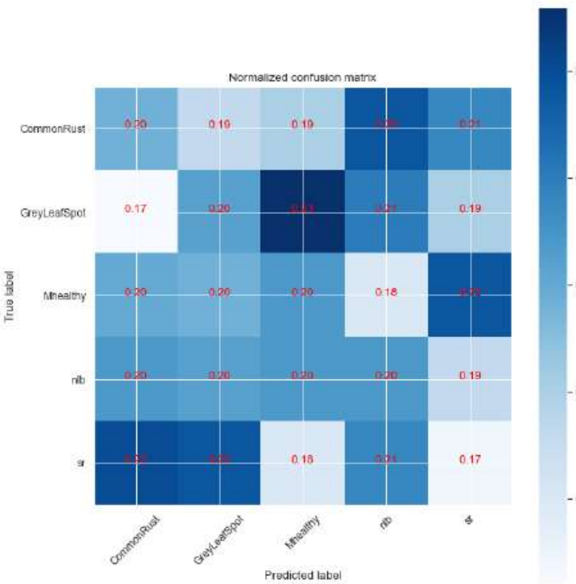


FIGURE 14. Normalized confusion matrix of the model MobileNet on the maize leaf dataset.

TABLE 5. Classification report of xception model on the maize leaf dataset.

	precision	recall	f1-score	support
CommonRust	0.21	0.21	0.21	500
GreyLeafSpot	0.20	0.21	0.20	500
Mhealthy	0.20	0.20	0.20	500
nlb	0.19	0.19	0.19	500
sr	0.19	0.19	0.19	500
accuracy			0.20	2500
macro avg	0.20	0.20	0.20	2500
weighted avg	0.20	0.20	0.20	2500

TABLE 6. Classification report of mobilenet model on the maize leaf dataset.

	precision	recall	f1-score	support
CommonRust	0.22	0.22	0.22	500
GreyLeafSpot	0.19	0.20	0.20	500
Mhealthy	0.20	0.20	0.20	500
nlb	0.21	0.22	0.22	500
sr	0.19	0.18	0.18	500
accuracy			0.20	2500
macro avg	0.20	0.20	0.20	2500
weighted avg	0.20	0.20	0.20	2500

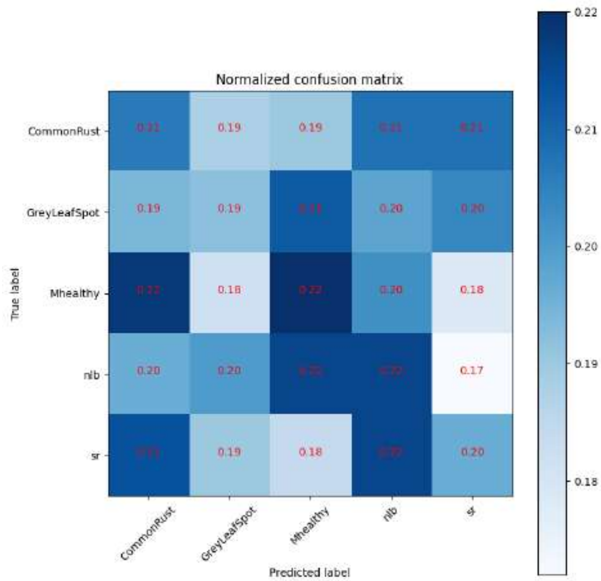


FIGURE 15. Normalized confusion matrix of the model MRW-CNN on the maize leaf dataset.

CommonRust and sr, nlb and GreyLeafSpot, nlb, Mhealthy, etc. In Fig. 14 for the MobileNet model on the maize leaf dataset, it can be observed that the diagonal values for all classes are nearly the same except for one. Therefore, it can be assumed that all classes perform equally the same except for the disease SouthernRust (sr). Some confusion among classes can also be observed; for example, GreyLeafSpot is highly confused with the class Mhealthy. There are also confusions between the classes sr and CommonRust and also sr and GreyLeafSpot, etc.

The proposed MRW-CNN model trained from scratch on the maize leaf dataset performed well; it showed a validation

TABLE 7. Classification report of MRW-CNN model on the maize leaf dataset.

	precision	recall	f1-score	support
CommonRust	0.21	0.21	0.20	500
GreyLeafSpot	0.21	0.20	0.21	500
Mhealthy	0.20	0.20	0.20	500
nlb	0.19	0.20	0.19	500
sr	0.19	0.19	0.19	500
accuracy			0.20	2500
macro avg	0.20	0.20	0.20	2500
weighted avg	0.20	0.20	0.20	2500

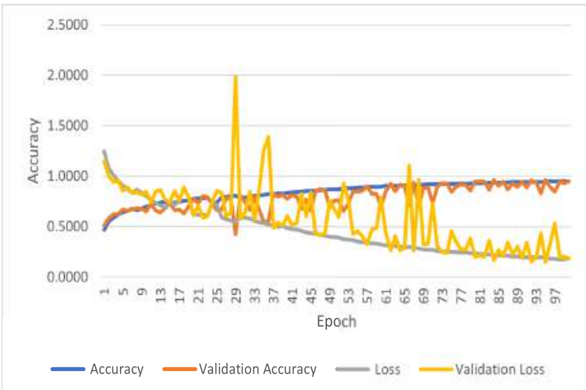


FIGURE 16. Accuracy and loss performance of the model MRW-CNN on the maize leaf dataset.

accuracy of 0.9656 and a 2% increase in testing accuracy of 0.9704 compared to the fine-tuned models. The normalized confusion matrix of the MRW-CNN model on the maize leaf dataset is shown in Fig. 15. In Fig. 15, when observing the diagonal elements, the Mhealthy and nlb classes of maize performed equally well when compared to the other classes. Confusion among classes can also be observed; for example, the class nlb is highly confused with Mhealthy, the Mhealthy class is confused with CommonRust, etc. The accuracy and loss performance of the MRW-CNN model on the maize leaf dataset is shown in Fig. 16.

The classification report of the Xception, MobileNet and the proposed MRW-CNN model on the maize leaf dataset is shown in Table 5, Table 6 and Table 7. The tables show the precision, recall, F1-score, and support scores for each class in the maize leaf dataset on the test data. The macro avg, micro average, and accuracy are also calculated. More details regarding how these values can be calculated in mentioned in Subsection D of Section VI.

G. RICE

The performance of the state-of-the-art deep learning models on the rice leaf dataset is shown in Table 8. Out of all eight fine-tuned models trained on the rice leaf dataset, the models Xception, MobileNet, Inception V3, and MobileNetV2

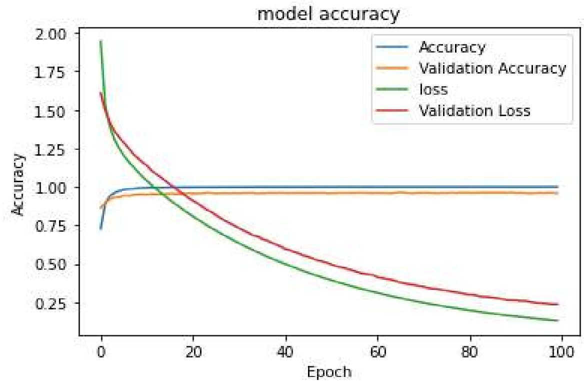


FIGURE 17. Accuracy and loss performance of the model inception V3 on the rice leaf dataset.

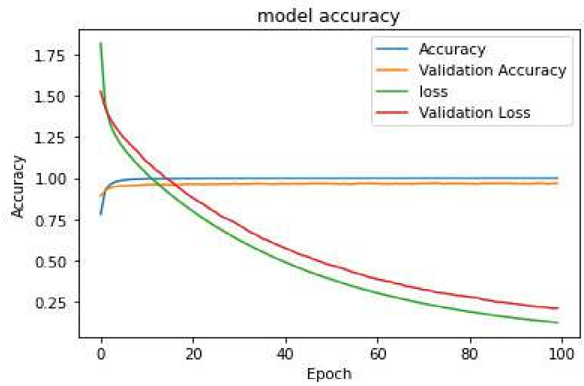


FIGURE 18. Accuracy and loss performance of the model xception on the rice leaf dataset.

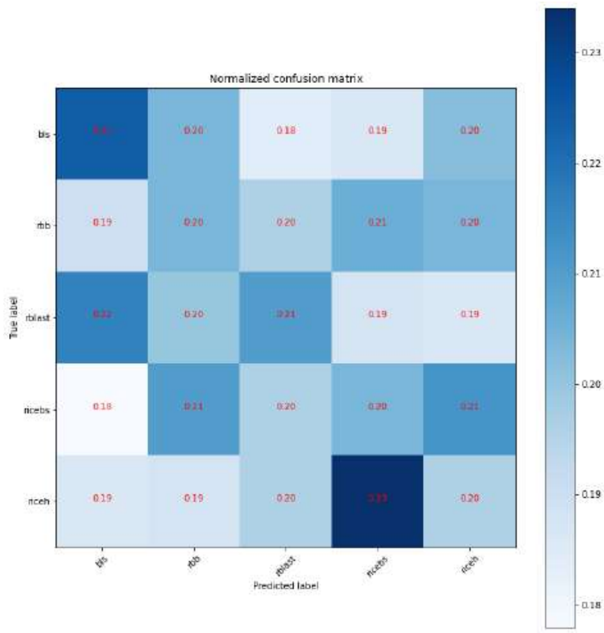


FIGURE 19. Normalized confusion matrix of the model inception V3 on the rice leaf dataset.

showed validation and testing accuracy above 95% when compared to all the other models trained. The Xception model performed best when compared to all the other models.

TABLE 8. Application of the state-of-the-art models on the rice leaf dataset.

Fine-tuned CNN Models	Training Accuracy	Validation Accuracy	Testing Accuracy	Optimizer	Learning Rate	No of Epochs
ResNet 50	0.6657	0.6184	0.6232	SGD	0.001	100
ResNet 101	0.7075	0.6188	0.6068	SGD	0.001	100
MobileNetV2	0.9991	0.9624	0.9584	SGD	0.001	100
MobileNet	0.9994	0.9676	0.9572	SGD	0.001	100
InceptionV3	0.9998	0.9628	0.9620	SGD	0.001	100
InceptionResNetV2	0.9996	0.9492	0.9532	SGD	0.001	100
VGG16	0.9941	0.9148	0.9120	SGD	0.001	100
Xception	0.9999	0.9708	0.9728	SGD	0.001	100
MRW-CNN(Proposed Model)	0.9504	0.9729	0.9706	SGD	0.001	100

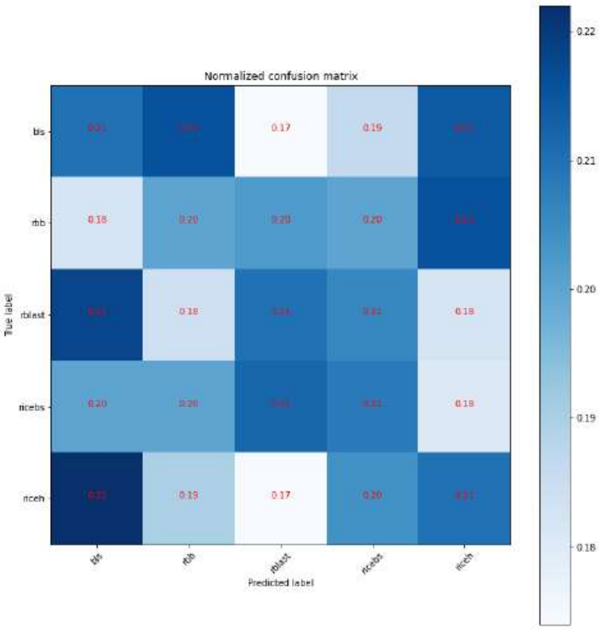


FIGURE 20. Normalized confusion matrix of the model xception on the rice leaf dataset.

TABLE 9. Classification report of xception model on the rice leaf dataset.

	precision	recall	f1-score	support
bls	0.21	0.22	0.22	500
rbb	0.21	0.20	0.21	500
rblast	0.19	0.18	0.19	500
ricebs	0.19	0.19	0.19	500
riceh	0.20	0.20	0.20	500
accuracy			0.20	2500
macro avg	0.20	0.20	0.20	2500
weighted avg	0.20	0.20	0.20	2500

TABLE 10. Classification report of inception v3 model on the rice leaf dataset.

	precision	recall	f1-score	support
bls	0.20	0.21	0.20	500
rbb	0.20	0.20	0.20	500
rblast	0.19	0.19	0.19	500
ricebs	0.19	0.19	0.19	500
riceh	0.20	0.20	0.20	500
accuracy			0.20	2500
macro avg	0.20	0.20	0.20	2500
weighted avg	0.20	0.20	0.20	2500

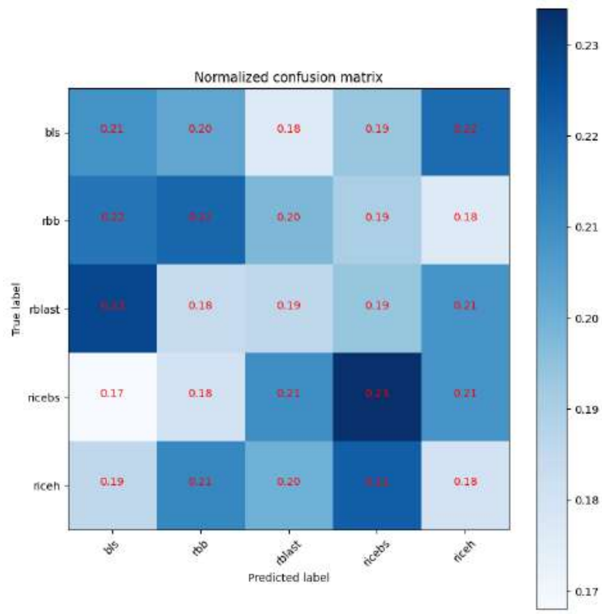


FIGURE 21. Normalized confusion matrix of the model MRW-CNN on the rice leaf dataset.

The MobileNetV2 model showed a validation accuracy of 0.9624 and a testing accuracy of 0.9584. The MobileNet model showed a validation accuracy of 0.9676 and a testing accuracy of 0.9572.

The Inception V3 model showed a validation accuracy of 0.9628 and a testing accuracy of 0.9620. The InceptionResNetV2 showed a validation accuracy of 0.9492 and a testing accuracy of 0.9532. The VGG16 model showed a validation accuracy of 0.9148 and a testing accuracy of 0.9120. The Xception model showed a validation accuracy of 0.9708 and a testing accuracy of 0.9728. The graphs

TABLE 11. Classification Report of MRW-CNN model on the rice leaf dataset.

	precision	recall	f1-score	support
bls	0.22	0.21	0.21	500
rbb	0.20	0.20	0.20	500
rblast	0.20	0.19	0.20	500
ricebs	0.22	0.23	0.23	500
riceh	0.20	0.20	0.20	500
accuracy			0.21	2500
macro avg	0.21	0.21	0.21	2500
weighted avg	0.21	0.21	0.21	2500

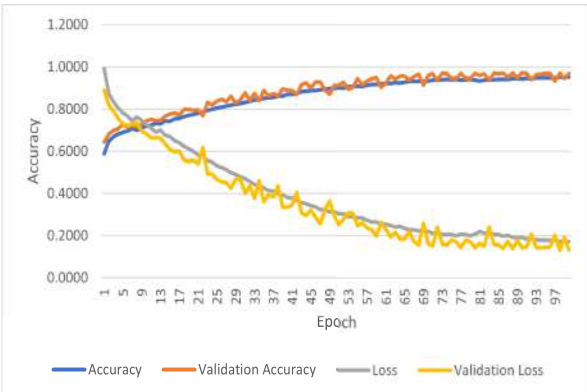


FIGURE 22. Accuracy and loss performance of the model MRW-CNN on the rice leaf dataset.

showing the accuracy and loss performance of the fine-tuned models trained on the rice leaf dataset with a testing accuracy of above 96% are shown in Fig. 17 and Fig. 18. The normalized confusion matrix of the fine-tuned models that showed a testing accuracy of above 96% on the rice leaf dataset is shown in Fig. 19 and Fig. 20.

Fig. 19 shows the normalized confusion matrix of the model Inception V3 on the rice leaf dataset. It can be observed from the confusion matrix that the class bls performs slightly better when compared to the other classes. Some classes are confused with each other, like rblast and bls, riceh and ricebs, etc. Similarly, Fig. 20 shows the normalized confusion matrix for the Xception model on the rice leaf dataset. It can be observed from the figure that all the classes perform equally the same except for rbb. Some classes that are highly confused with each other are rblast and bls, riceh, and bls, etc. The proposed MRW-CNN model trained from scratch performed equally well on the rice leaf dataset with a validation accuracy of 0.9729 and a testing accuracy of 0.9706 when compared with the fine-tuned models. The normalized confusion matrix of the MRW-CNN model on the rice leaf dataset is shown in Fig. 21. In Fig. 21 from the normalized confusion matrix, it can be observed that class ricebs performs slightly better when compared to the other classes. Confusion among classes

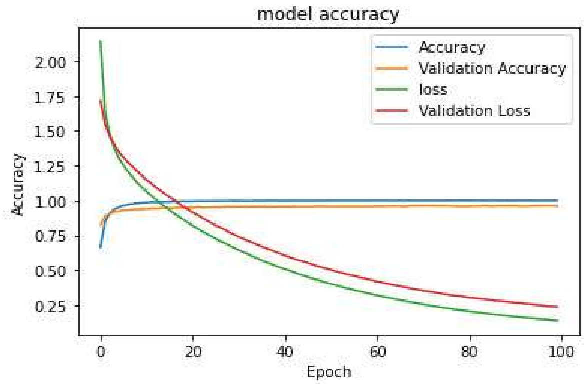


FIGURE 23. Accuracy and loss performance of the model mobilenet on the wheat leaf dataset.

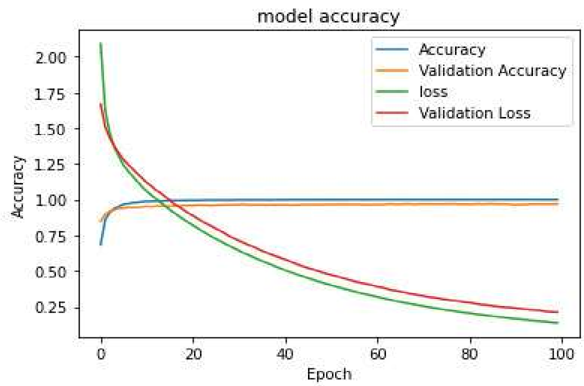


FIGURE 24. Accuracy and loss performance of the model mobilenetv2 on the wheat leaf dataset.

can also be observed, for example, rblast and bls, riceh and ricebs, etc. The accuracy and loss performance of the MRW model on the rice leaf dataset is shown in Fig. 22. The classification report of the Xception, Inception V3, and the MRW-CNN model on the rice leaf dataset is shown in Table 9, Table 10, and Table 11. The classification report shows the precision, recall, F1-Score, support scores for all the classes in the rice leaf dataset for the test images. It also calculates the macro average, accuracy, and weighted average.

H. WHEAT

The performance of the state-of-the-art deep learning models on the wheat leaf dataset is shown in Table 12. Out of all eight models trained on the wheat leaf dataset, the models MobileNetV2, MobileNet, Inception V3, and InceptionResNetV2 showed a validation accuracy above 90% when compared to all the other models trained. The MobileNetV2 model showed a validation accuracy of 0.9724 and a testing accuracy of 0.9632. The MobileNet model showed a validation accuracy of 0.9632 and a testing accuracy of 0.9628. The Inception V3 model showed a validation accuracy of 0.9496 and a testing accuracy of 0.9508. The InceptionResNetV2 model showed a validation

TABLE 12. Application of the state-of-the-art models on the wheat leaf dataset.

Fine-tuned CNN Models	Training Accuracy	Validation Accuracy	Testing Accuracy	Optimizer	Learning Rate	No of Epochs
ResNet50	0.6028	0.5492	0.5536	SGD	0.001	100
ResNet 101	0.6382	0.5320	0.5299	SGD	0.001	100
MobileNetV2	0.9992	0.9724	0.9632	SGD	0.001	100
MobileNet	0.9995	0.9632	0.9628	SGD	0.001	100
InceptionV3	0.9996	0.9496	0.9508	SGD	0.001	100
InceptionResNetV2	0.9995	0.9516	0.9488	SGD	0.001	100
VGG16	0.9937	0.8972	0.8884	SGD	0.001	100
Xception	0.7377	0.4184	0.4018	SGD	0.001	100
MRW-CNN(Proposed Model)	0.9733	0.9793	0.9808	SGD	0.001	100

TABLE 13. Classification report of mobilenet model on the wheat leaf dataset.

	precision	recall	f1-score	support
leafrust	0.21	0.21	0.21	500
powderymildew	0.21	0.22	0.22	500
striperust	0.20	0.20	0.20	500
tanspot	0.20	0.20	0.20	500
wh	0.20	0.20	0.20	500
accuracy			0.20	2500
macro avg	0.20	0.20	0.20	2500
weighted avg	0.20	0.20	0.20	2500

TABLE 14. Classification report of MobileNetV2 model on the wheat leaf dataset.

	precision	recall	f1-score	support
leafrust	0.20	0.20	0.20	500
powderymildew	0.21	0.22	0.21	500
striperust	0.20	0.20	0.20	500
tanspot	0.20	0.20	0.20	500
wh	0.20	0.20	0.20	500
accuracy			0.20	2500
macro avg	0.20	0.20	0.20	2500
weighted avg	0.20	0.20	0.20	2500

TABLE 15. Classification report of MRW-CNN model on the wheat leaf dataset.

	precision	recall	f1-score	support
leafrust	0.21	0.22	0.21	500
powderymildew	0.21	0.20	0.21	500
striperust	0.20	0.20	0.20	500
tanspot	0.19	0.19	0.19	500
wh	0.20	0.20	0.20	500
accuracy			0.20	2500
macro avg	0.20	0.20	0.20	2500
weighted avg	0.20	0.20	0.20	2500

accuracy of 0.9516 and a testing accuracy of 0.9488. The graphs showing the accuracy and loss performance of the fine-tune models trained on the wheat leaf dataset with a testing accuracy of above 96% are shown in Fig. 23 and Fig. 24.

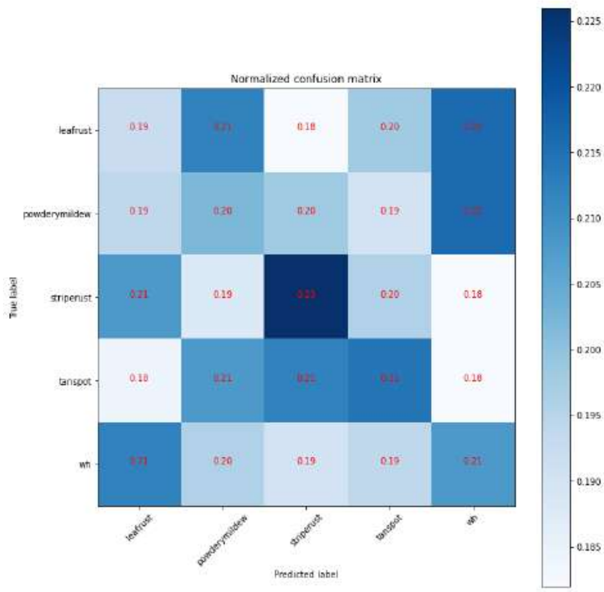


FIGURE 25. Normalized confusion matrix of the model mobilenet on the wheat leaf dataset.

The normalized confusion matrix of the fine-tuned models that showed a testing accuracy of above 96% on the wheat leaf dataset are shown in Fig. 25 and Fig. 26. Fig. 25 shows the normalized confusion matrix of the MobileNet model on the wheat leaf dataset. From the confusion matrix, it can be observed that the striperust class performed best when compared to all the other classes in the test data. Confusions among classes also were observed, for example, leafrust and wh, powderymildew and wh, etc. Fig. 26 shows the normalized confusion matrix of the MobileNetV2 model on the wheat leaf dataset.

The stripe rust class performed well when compared to others. Some confusing classes were wh and leafrust, powderymildew and wh, leafrust and wh, etc. The proposed MRW-CNN model trained from scratch on the wheat leaf dataset performed well; it showed a validation accuracy of 0.9793 and a 2% increase in testing accuracy of 0.9808 when compared to the fine-tuned models. The normalized confusion matrix of the MRW-CNN model on the wheat leaf

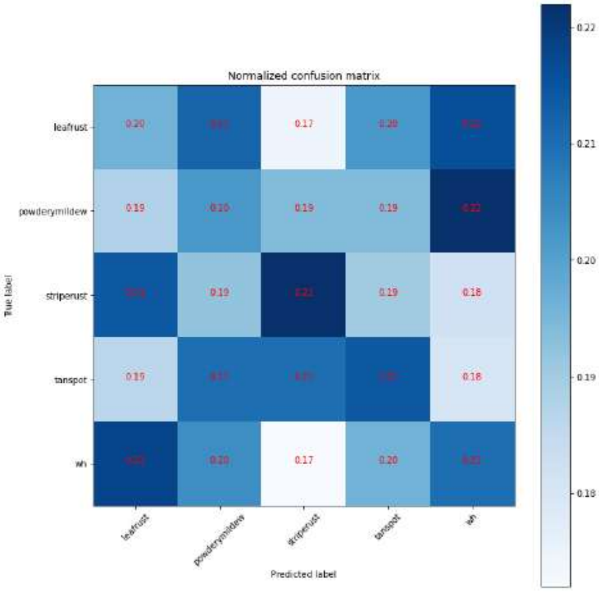


FIGURE 26. **Normalized confusion matrix of the model mobilenetv2 on the wheat leaf dataset.**

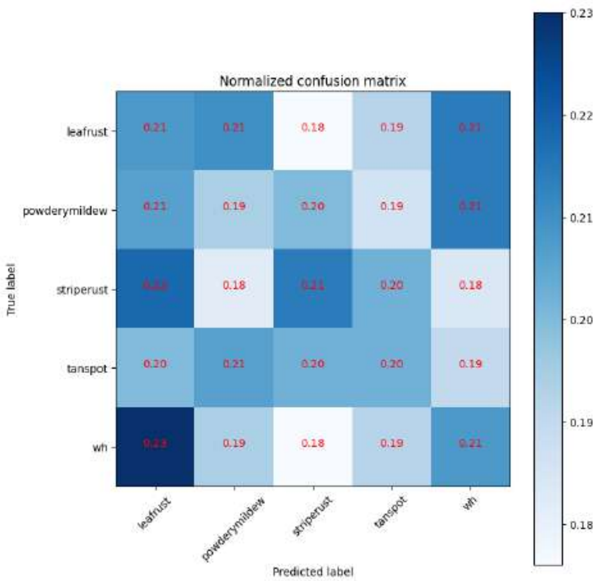


FIGURE 27. **Normalized confusion matrix of the model MRW-CNN on the wheat leaf dataset.**

dataset is shown in Fig. 27. In Fig. 27, when observing the diagonal elements, the leafrust, striperust, and wh class of wheat performed equally well when compared to the other classes. Confusion among classes can also be observed; for example, the class wh is highly confused with leafrust, the leafrust class is confused with wh, etc. The accuracy and loss performance of the MRW-CNN model on the wheat leaf dataset is shown in Fig. 28. The classification report of the MobileNet, MobileNetV2, and the MRW-CNN model on the wheat leaf dataset is shown in Table 13, Table 14 and Table 15. The tables show the precision, recall, F1-score and

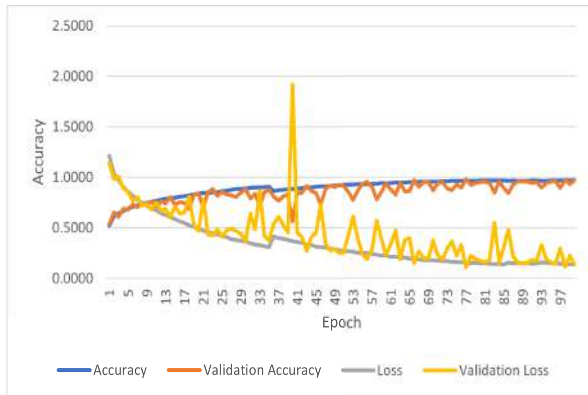


FIGURE 28. **Accuracy and loss performance of the model MRW-CNN on the wheat leaf dataset.**

support scores for each class in the wheat leaf dataset on the test data. The macro avg, micro average, and accuracy are also calculated.

VII. CASE STUDY OF THE APPLICATION OF THE PROPOSED METHOD

Computer vision technology will be a need in autonomous futuristic farms. CNN based computer vision tasks have shown great performance in recent years. Therefore, there is an increased need to concentrate on the practical applications of the system developed. If deployed properly, the proposed method can help reduce crop loss and increase yield. A plant protection system APP based on food grains can be developed using the proposed system and the datasets developed. It can be useful for detecting the diseases affecting the food grains of rice, wheat, and maize.

Depending on the disease detected, appropriate curable measures can be suggested. Identifying the disease at the correct stage can help the users decide on the amount of pesticide to be used and in what amount. The APP can be trained to identify diseases with similar symptoms by asking users additional questions about the weather conditions, the location of the disease on the leaf, etc. By knowing more about the weather conditions and the location of the disease. It becomes easy to distinguish between diseases showing similar symptoms to some extent.

Additional information regarding the weather conditions that can favor the spread or advancement of the disease can be given to the users. Suggestions on the pesticides that can be used and the amount can also be made available. Small-scale growers are the people who benefit more from these kinds of solutions available for diagnosing plant diseases when they are not able to get help from experts all the time. The overall architecture of the plant protection system that can be developed is shown in Fig. 29.

Fig. 29 discusses the possible use cases the user can give the plant disease protection system. In case 1, the user provides a diseased leaf to the system to detect the disease affecting the leaf along with its stage. After the detection

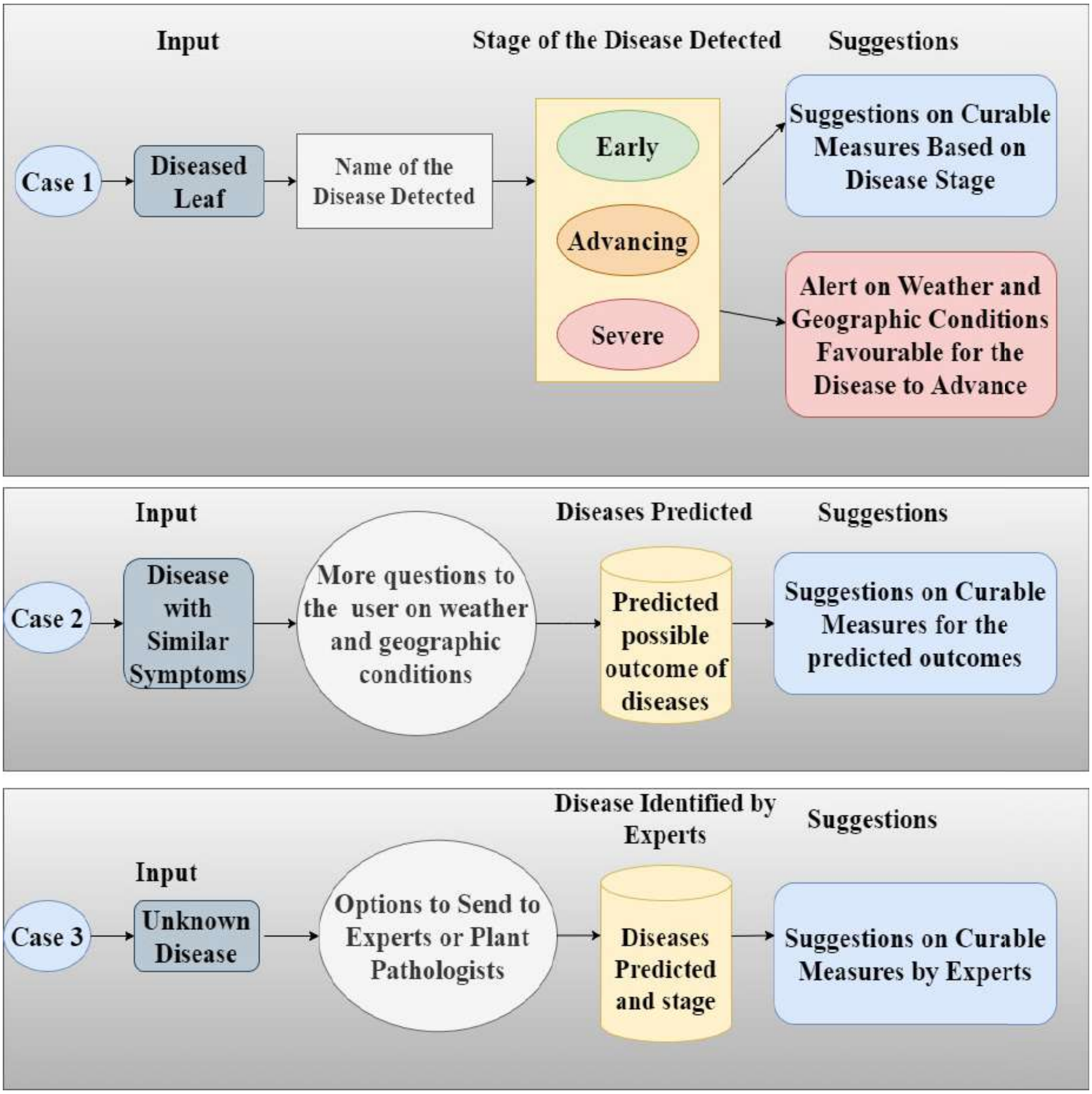


FIGURE 29. Application of the proposed system.

of the disease and its stage. The system also provides suggestions regarding the curable measures and alerts on the weather conditions that can be favorable for advancing the disease. In case 2, when a diseased leaf has similar symptoms of one or two diseases, more questions can be asked to the users about the weather and geographical conditions. Based on this information, possible predictions of diseases can be made by the system. Also, suggestions for curable measures for these predicted outcomes can be made.

In case 3, when the system encounters an unknown disease, options can be given to the user to send it to an expert or plant pathologist for analysis. The system learns from the disease identified by the experts and their suggestions. The system learns from the communication between the users or growers and the experts.

VIII. CONCLUSION AND FUTURE WORK

A. CONCLUSION

In the current work, real-time datasets for the food grains, rice, wheat, and maize were formed by considering images from different datasets and internet sources. Importance was given to real-life images of different disease severity levels, which is important for farmers or growers to make appropriate decisions at the early stages of infection. Identification of disease at the early stage of infection can reduce yield loss and financial loss.

According to experimental results, CNN-based deep learning models pre-trained on the ImageNet dataset, with deep transfer learning strategy, downloaded from Keras and fine-tuned, and the proposed MRW-CNN model performs well in identifying and classifying rice, maize, and wheat diseases considered in the current work. All the models were trained for 100 epochs.

During the development of the dataset, real-life images were concentrated on more, rather than processed images. Selected images from the datasets were pre-processed to remove foreign objects if present. The number of images collected under each class was 100 which was not enough for training a deep learning model. So, the technique of data augmentation was applied to all images per class to increase the size of the dataset. The original images were combined with the augmented images, and then the dataset formed was splitted 80% for training and 10% each for validation and testing for all three food grains: rice, wheat, and maize. The datasets were then applied to different fine-tuned CNN models and the proposed MRW-CNN model. The models were evaluated using performance metrics like accuracy, f1-score, precision, recall, etc.

B. FUTURE WORK

It can be observed that the datasets we developed have some limitations. Although most of the images in the dataset are real-life, some are pre-processed. In real-life scenarios, more factors will influence the results than those concentrated on developing the datasets in the current work. Therefore, as future work, more importance can be given to factors like how the impact of climate change can affect plant diseases collection of leaf diseased images during different climatic conditions.

Some of the other future works can be annotated images from the dataset can be used in object detection models like MaskRCNN to detect disease severity levels. A CNN based deep learning system can be developed to detect and classify diseases in food grains; the dataset can be further extended by adding images of more food grains, images of multiple diseases on the same leaf, etc.

REFERENCES

[1] J. G. A. Barbedo, "A review on the main challenges in automatic plant disease identification based on visible range images," *Biosystems Eng.*, vol. 144, pp. 52–60, Apr. 2016.

[2] D. S. Joseph, P. M. Pawar, and R. Pramanik, "Intelligent plant disease diagnosis using convolutional neural network: A review," *Multimedia Tools Appl.*, vol. 82, no. 14, pp. 21415–21481, Jun. 2023.

[3] J. Chen, J. Chen, D. Zhang, Y. Sun, and Y. A. Nanehkaran, "Using deep transfer learning for image-based plant disease identification," *Comput. Electron. Agricult.*, vol. 173, Jun. 2020, Art. no. 105393.

[4] G. Wang, Y. Sun, and J. Wang, "Automatic image-based plant disease severity estimation using deep learning," *Comput. Intell. Neurosci.*, vol. 2017, pp. 1–8, Jul. 2017.

[5] O. Getch. *Rice Knowledge Bank: Your Information Source for Rice Farming*. Accessed: Jan. 30, 2023. [Online]. Available: http://www.knowledgebank.irri.org/step-by-step-production/growth/pests-and-diseases/diseases

[6] E. D. Wolf and J. P. Shroyer. *Wheat Disease Identification*. Accessed: Nov. 25, 2022. [Online]. Available: https://www.bookstore.ksre.ksu.edu/pubs/MF2994.pdf

[7] S. Mahadik, P. M. Pawar, and R. Muthalagu, "Efficient intelligent intrusion detection system for heterogeneous Internet of Things (HetIoT)," *J. Netw. Syst. Manag.*, vol. 31, no. 1, p. 2, Jan. 2023.

[8] K. Syama, J. A. A. Jothi, and N. Khanna, "Automatic disease prediction from human gut metagenomic data using boosting GraphSAGE," *BMC Bioinf.*, vol. 24, no. 1, p. 126, Mar. 2023.

[9] H. Hussain and P. S. Tamizharasan, "The impact of cascaded optimizations in CNN models and end-device deployment," in *Proc. 20th ACM Conf. Embedded Networked Sensor Syst.*, J. Gummeson, S. I. Lee, J. Gao, and G. Xing, Eds. Boston, MA, USA, 2022, pp. 954–961.

[10] G. Zhou, W. Zhang, A. Chen, M. He, and X. Ma, "Rapid detection of rice disease based on FCM-KM and faster R-CNN fusion," *IEEE Access*, vol. 7, pp. 143190–143206, 2019.

[11] Z. Libo, H. Tian, G. Chunyun, and M. Elhoseny, "Real-time detection of Cole diseases and insect pests in wireless sensor networks," *J. Intell. Fuzzy Syst.*, vol. 37, no. 3, pp. 3513–3524, Oct. 2019.

[12] P. Jiang, Y. Chen, B. Liu, D. He, and C. Liang, "Real-time detection of apple leaf diseases using deep learning approach based on improved convolutional neural networks," *IEEE Access*, vol. 7, pp. 59069–59080, 2019.

[13] M. Shoaib, B. Shah, S. Ei-Sappagh, A. Ali, A. Ullah, F. Alenezi, T. Gechev, T. Hussain, and F. Ali, "An advanced deep learning models-based plant disease detection: A review of recent research," *Frontiers Plant Sci.*, vol. 14, Mar. 2023, Art. no. 1158933.

[14] M. Shoaib, B. Shah, T. Hussain, A. Ali, A. Ullah, F. Alenezi, T. Gechev, F. Ali, and I. Syed, "A deep learning-based model for plant lesion segmentation, subtype identification, and survival probability estimation," *Frontiers Plant Sci.*, vol. 13, Dec. 2022, Art. no. 1095547.

[15] M. Akbar, M. Ullah, B. Shah, R. U. Khan, T. Hussain, F. Ali, F. Alenezi, I. Syed, and K. S. Kwak, "An effective deep learning approach for the classification of bacteriosis in peach leave," *Frontiers Plant Sci.*, vol. 13, p. 4723, Nov. 2022.

[16] D. P. Hughes and M. Salathé, "An open access repository of images on plant health to enable the development of mobile disease diagnostics through machine learning and crowdsourcing," 2015, *arXiv:1511.08060*.

[17] Q. Zeng, X. Ma, B. Cheng, E. Zhou, and W. Pang, "GANs-based data augmentation for citrus disease severity detection using deep learning," *IEEE Access*, vol. 8, pp. 172882–172891, 2020.

[18] J. Ma, K. Du, F. Zheng, L. Zhang, Z. Gong, and Z. Sun, "A recognition method for cucumber diseases using leaf symptom images based on deep convolutional neural network," *Comput. Electron. Agricult.*, vol. 154, pp. 18–24, Nov. 2018.

[19] Q. Wu, Y. Chen, and J. Meng, "DCGAN-based data augmentation for tomato leaf disease identification," *IEEE Access*, vol. 8, pp. 98716–98728, 2020.

[20] E. C. Too, L. Yujian, S. Njuki, and L. Yingchun, "A comparative study of fine-tuning deep learning models for plant disease identification," *Comput. Electron. Agricult.*, vol. 161, pp. 272–279, Jun. 2019.

[21] R. Kumar, U. Mina, R. Gogoi, A. Bhatia, and R. C. Harit, "Effect of elevated temperature and carbon dioxide levels on maydis leaf blight disease tolerance attributes in maize," *Agricult., Ecosystems Environ.*, vol. 231, pp. 98–104, Sep. 2016.

[22] S. Nigam, R. Jain, S. Marwaha, A. Arora, M. A. Haque, A. Dheeraj, and V. K. Singh, "Deep transfer learning model for disease identification in wheat crop," *Ecological Informat.*, vol. 75, Jul. 2023, Art. no. 102068.

[23] M. K. Vishal, D. Tamboli, A. Patil, R. Saluja, B. Banerjee, A. Sethi, D. Raju, S. Kumar, R. N. Sahoo, V. Chinnusamy, and J. Adinarayana, "Image-based phenotyping of diverse rice (*Oryza sativa* L.) genotypes," 2020, *arXiv:2004.02498*.

[24] A. Chug, A. Bhatia, A. P. Singh, and D. Singh, "A novel framework for image-based plant disease detection using hybrid deep learning approach," *Soft Comput.*, vol. 27, no. 18, pp. 13613–13638, Sep. 2023.

[25] M. A. Haque, S. Marwaha, C. K. Deb, S. Nigam, A. Arora, K. S. Hooda, P. L. Soujanya, S. K. Aggarwal, B. Lall, M. Kumar, S. Islam, M. Panwar, P. Kumar, and R. C. Agrawal, "Deep learning-based approach for identification of diseases of maize crop," *Sci. Rep.*, vol. 12, no. 1, p. 6334, Apr. 2022.

[26] S. Das, A. Biswas, and P. Sinha, "Deep learning analysis of rice blast disease using remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[27] M. Shoaib, T. Hussain, B. Shah, I. Ullah, S. M. Shah, F. Ali, and S. H. Park, "Deep learning-based segmentation and classification of leaf images for detection of tomato plant disease," *Frontiers Plant Sci.*, vol. 13, Oct. 2022, Art. no. 1031748.

[28] X. Zhang, Y. Qiao, F. Meng, C. Fan, and M. Zhang, "Identification of maize leaf diseases using improved deep convolutional neural networks," *IEEE Access*, vol. 6, pp. 30370–30377, 2018.

[29] H. Wu, T. Wiesner-Hanks, E. L. Stewart, C. DeChant, N. Kaczmar, M. A. Gore, R. J. Nelson, and H. Lipson, "Autonomous detection of plant disease symptoms directly from aerial imagery," *Plant Phenome J.*, vol. 2, no. 1, pp. 1–9, Jan. 2019.

[30] J. Sun, Y. Yang, X. He, and X. Wu, "Northern maize leaf blight detection under complex field environment based on deep learning," *IEEE Access*, vol. 8, pp. 33679–33688, 2020.

[31] S. Mishra, R. Sachan, and D. Rajpal, "Deep convolutional neural network based detection system for real-time corn plant disease recognition," *Proc. Comput. Sci.*, vol. 167, pp. 2003–2010, Jan. 2020.

[32] H. T. Rauf, B. A. Saleem, M. I. U. Lali, M. A. Khan, M. Sharif, and S. A. C. Bukhari, "A citrus fruits and leaves dataset for detection and classification of citrus diseases through machine learning," *Data Brief*, vol. 26, Oct. 2019, Art. no. 104340.

[33] J. Lu, J. Hu, G. Zhao, F. Mei, and C. Zhang, "An in-field automatic wheat disease diagnosis system," *Comput. Electron. Agricult.*, vol. 142, pp. 369–379, Nov. 2017.

[34] R. Qiu, C. Yang, A. Moghimi, M. Zhang, B. J. Steffenson, and C. D. Hirsch, "Detection of fusarium head blight in wheat using a deep neural network and color imaging," *Remote Sens.*, vol. 11, no. 22, p. 2658, Nov. 2019.

[35] Y. Lu, S. Yi, N. Zeng, Y. Liu, and Y. Zhang, "Identification of rice diseases using deep convolutional neural networks," *Neurocomputing*, vol. 267, pp. 378–384, Dec. 2017.

[36] F. Jiang, Y. Lu, Y. Chen, D. Cai, and G. Li, "Image recognition of four rice leaf diseases based on deep learning and support vector machine," *Comput. Electron. Agricult.*, vol. 179, Dec. 2020, Art. no. 105824.

[37] D. P. Singh, N. Jain, P. Jain, P. Kayal, S. Kumawat, and N. Batra, "PlantDoc: A dataset for visual plant disease detection," in *Proc. 7th ACM IKDD CoDS 25th COMAD*, 2019, pp. 249–253.

[38] S. Riyaz. *Rice Leafs: An Image Collection Four Rice Diseases*. Accessed: Jan. 30, 2023. [Online]. Available: https://www.kaggle.com/datasets/shayanriyaz/riceleafs

[39] O. Getch. *Wheat Leaf Dataset: Disease Affected and Healthy Wheat Leaf*. Accessed: Jan. 30, 2023. [Online]. Available: https://www.kaggle.com/datasets/olyadgetch/wheat-leaf-dataset

[40] *Rice Knowledge Bank*. Accessed: Oct. 12, 2022. [Online]. Available: http://www.knowledgebank.irri.org/decision-tools/rice-doctor/rice-doctor-fact-sheets

[41] D. Groth. *Brown Spot of Rice*. Accessed: Oct. 6, 2022. [Online]. Available: https://www.forestryimages.org/browse/detail.cfm?imgnum=5410673

[42] *Leaf Rust*. Accessed: Oct. 12, 2022. [Online]. Available: https://agritech.tnau.ac.in/crop_protection/wheat/crop_prot_crop

[43] *Leaf Rust*. Accessed: Oct. 12, 2022. [Online]. Available: https://www.cropscience.bayer.co.nz/pests/diseases/leaf-rust—wheat

[44] *Leaf Rust of Wheat*. Accessed: Oct. 12, 2022. [Online]. Available: https://extensionaus.com.au/FieldCropDiseasesVic/docs/identification-management-of-field-crop-diseases-in-victoria/foliar-diseases-of-wheat/leaf-rust-of-wheat/

[45] *Powdery Mildew*. Accessed: Oct. 12, 2022. [Online]. Available: https://www.cropscience.bayer.co.nz/pests/diseases/powdery-mildew—wheat

[46] *New Permits Aid Control of Powdery Mildew in Wheat*. Accessed: Oct. 12, 2022. [Online]. Available: https://www.graincentral.com/cropping/new-permits-aid-control-of-powdery-mildew-in-wheat/

[47] *Scout Wheat Fields for Early Disease Detection*. Accessed: Oct. 12, 2022. [Online]. Available: https://cropwatch.unl.edu/2021/scout-wheat-fields-early-disease-detection

[48] *Powdery Mildew Symptoms*. Accessed: Oct. 12, 2022. [Online]. Available: https://doraagri.com/powdery-mildew-on-plants/

[49] *Stripe Rust (Yellow Rust) of Wheat*. Accessed: Oct. 12, 2022. [Online]. Available: https://extension.uga.edu/publications/detail.html?number=C960&

[50] *Tan Spot*. Accessed: Oct. 12, 2022. [Online]. Available: https://andersonscanada.com/2017/05/17/wheat-leaf-diseases-2/

[51] *Tan Spot*. Accessed: Oct. 12, 2022. [Online]. Available: https://extension.okstate.edu/programs/digital-diagnostics/plant-diseases/tan-spot.html

[52] *Tan Spot of Wheat*. Accessed: Oct. 12, 2022. [Online]. Available: https://extensionpublications.unl.edu/assets/html/g429/build/g429.htm

[53] *Tan Spot*. Accessed: Oct. 12, 2022. [Online]. Available: https://cropscience.bayer.co.uk/threats/diseases/cereal-diseases/tan-spot/

[54] *Common Rust of Corn*. Accessed: Oct. 7, 2022. [Online]. Available: https://ohioline.osu.edu/factsheet/plpath-cer-02

[55] *Common Corn Rust Puccinia Sorghi Schwein*. Accessed: Oct. 7, 2022. [Online]. Available: https://www.forestryimages.org/browse/detail.cfm?imgnum=1538028

[56] A. Sisson. *Common Corn Rust Puccinia Sorghi Schwein*. Accessed: Oct. 7, 2022. [Online]. Available: https://www.forestryimages.org/browse/detail.cfm?imgnum=5608271

[57] G. Holmes. *Common Corn Rust Puccinia Sorghi Schwein*. Accessed: Oct. 7, 2022. [Online]. Available: https://www.forestryimages.org/browse/detail.cfm?imgnum=5606719

[58] A. Robertson. *Gray Leaf Spot*. Accessed: Oct. 7, 2022. [Online]. Available: https://crops.extension.iastate.edu/cropnews/2018/06/weather-conditions-brown-spot-and-node-rot-and-gray-leaf-spot

[59] *Northern Corn Leaf Blight, Grey Leaf Spot Top Ontario Corn Diseases*. Accessed: Nov. 22, 2022. [Online]. Available: https://www.topcropmanager.com/northerncornleafblightgreyleafspottop-ontariocorndiseases19935/

[60] *Managing Northern Corn Leaf Blight*. Accessed: Oct. 7, 2022. [Online]. Available: https://www.cropscience.bayer.ca/articles/2021/northern-corn-leaf-blight

[61] Chroanch. *Northern Corn Leaf Blight*. Accessed: Oct. 7, 2022. [Online]. Available: https://commons.wikimedia.org/Northern_corn_leaf_blight.JPG

[62] D. Mueller. *Northern Corn Leaf Blight of Corn*. Accessed: Oct. 7, 2022. [Online]. Available: https://cropprotectionnetwork.org/encyclopedia/northern-corn-leaf-blight-of-corn

[63] A. Robertson. *Northern Corn Leaf Blight of Corn*. Accessed: Oct. 7, 2022. [Online]. Available: https://cropprotectionnetwork.org/encyclopedia/northern-corn-leaf-blight-of-corn

[64] *Southern Rust*. Accessed: Oct. 7, 2022. [Online]. Available: https://cropwatch.unl.edu/plantdisease/corn/southern-rust

[65] *Southern Rust of Corn*. Accessed: Oct. 7, 2022. [Online]. Available: https://www.pioneer.com/us/agronomy/southern_rust_cropfocus.html

[66] *Mississippi Crop Situation*. Accessed: Oct. 7, 2022. [Online]. Available: https://www.mississippi-crops.com/2012/07/21/southern-corn-rust-is-a-late-fungicide-application-warranted/

[67] C. A. Schneider, W. S. Rasband, and K. W. Eliceiri, "NIH image to ImageJ: 25 years of image analysis," *Nature Methods*, vol. 9, no. 7, pp. 671–675, Jul. 2012.

[68] *Imagej*. Accessed: Oct. 6, 2022. [Online]. Available: https://imagej.net/software/imagej/

[69] *Make Sense Ai Tool*. Accessed: Oct. 6, 2022. [Online]. Available: https://www.makesense.ai/

[70] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei, "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.

[71] I. Chelliah. *Confusion Matrix for Multiclass Classification*. Accessed: Feb. 21, 2023. [Online]. Available: https://medium.com/mlearning-ai/confusion-matrix-for-multiclass-classification-f25ed7173e66

[72] *Micro-Average & Macro-Average Scoring Metrics—Python*. Accessed: Feb. 21, 2023. [Online]. Available: https://vitalflux.com/micro-average-macro-average-scoring-metrics-multi-class-classification/

[73] *F1 Score in Machine Learning: Intro & Calculation*. Accessed: Feb. 21, 2023. [Online]. Available: https://www.v7labs.com/blog/f1-score-guide

[74] R. R. Selvaraju, P. Chattopadhyay, M. Elhoseiny, T. Sharma, D. Batra, D. Parikh, and S. Lee, "Choose your neuron: Incorporating domain knowledge through neuron-importance," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 526–541.

DIANA SUSAN JOSEPH (Student Member, IEEE) received the B.Tech. degree in computer science and engineering from the Rajagiri School of Engineering and Technology, in 2008, and the M.Tech. degree in computer and communication from Karunya University, Coimbatore, in 2011. She is currently pursuing the Ph.D. degree in computer science and engineering with the Birla Institute of Technology and Science, Dubai. She was an Assistant Professor with the Department of Computer Science and Engineering, in 2012. Her research interests include image processing, machine learning, and deep learning.

PRANAV M. PAWAR (Member, IEEE) received the Graduate degree in computer engineering from Dr. Babasaheb Ambedkar Technological University, Maharashtra, India, in 2005, the master's degree in computer engineering from Pune University, in 2007, and the Ph.D. degree in wireless communication from Aalborg University, Denmark, in 2016. His Ph.D. thesis received nomination for Best Thesis Award from Aalborg University. Currently, he is an Assistant Professor with the Department of Computer Science, Birla Institute of Technology and Science, Dubai, before to BITS, he was a Postdoctoral Fellow with Bar-Ilan University, Israel, from March 2019 to October 2020, in an area of wireless communication and deep leaning. He was recipient of Outstanding Postdoctoral Fellowship from Israel Planning and Budgeting Committee. He was an Associate Professor with MIT ADT University, Pune, from 2018 to 2019, and also as an Associate Professor with the Department of Information Technology, STES's Shrimati Kashibai Navale College of Engineering, Pune, from 2008 to 2018. From 2006 to 2007, he was a System Executive with POS-IPC, Pune, India. He received Recognition from Infosys Technologies Ltd., for contribution in Campus Connect Program and also received different funding for research and attending conferences at international level. He has published more than 40 papers at national and international level. He is also IBM DB2 and IBM RAD certified professional and completed NPTEL certification in different subjects. His research interests include energy efficient MAC for WSN, QoS in WSN, wireless security, green technology, computer architecture, database management systems, and bioinformatics.

KAUSTUBH CHAKRADEO received the Bachelor of Engineering degree in information technology from Savitribai Phule Pune University (formerly Pune University), in 2019, and the Master of Science degree in artificial intelligence from Radboud University, The Netherlands, in 2022. He is currently pursuing the Ph.D. degree with the Section of Epidemiology, Department of Public Health, University of Copenhagen. His research interests include computer vision, deep learning, AI ethics, and infectious disease modeling.